

# Applied Statistics (MATH10096)

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# 1 Preliminaries

This course builds on Statistics (Year 2) (MATH08051) and on Statistical Methodology (MATH10095). In particular, it assumes that you are familiar with the key ideas underlying parametric statistical inference.

- *Statistics* is about using data to extract meaningful information about the system generating the data, when the data generating process is such that the data vary randomly from one replication of the process to the next, even if the underlying system remains the same.
- A *statistical model* is a simplified mathematical representation of the data generating process (a sort of mathematical cartoon of the system that we wish to learn about). It typically depends on known quantities, such as other variables measured alongside the data, and some unknown parameters:  $\beta$ , say. At this point, we quote Sir David Cox, in a comment published in the *Journal of the Royal Statistical Society, Ser. A*, **158**, 455–456 (1995):

*“The very word model implies simplification and idealization. The idea that complex physical, biological or sociological systems can be exactly described by a few formulae is patently absurd. The construction of idealized representations that capture important stable aspects of such systems is, however, a vital part of general scientific analysis.”*

There is also the well-known saying:

*“Essentially all models are wrong but some are useful. However, the approximate nature of the model must always be borne in mind.”*

This sentence is from the following book: Box, G. E. P. and Draper, N. R. (1987). *Empirical Model-Building and Response Surfaces*. Wiley.

- A key point about a statistical model is that *if* we knew the values of  $\beta$ , then the model should be able to simulate data that ‘look like’ the observed data. In principle, given parameter values, a statistical model also allows us to assign relative probabilities to observing one data set, as opposed to another.
- Once a model has been specified, a major goal of statistics is to make *inferences* about  $\beta$  from the observed data. In particular, we seek to answer the following questions:
  1. What value of  $\beta$  is most consistent with the data?
  2. What range of values of  $\beta$  is consistent with the data?
  3. Is a pre-specified value of, or restriction on,  $\beta$  consistent with the data?
  4. Are there any values of  $\beta$  for which the model is consistent with the data at all?
- The answers to these questions are provided by
  1. *Parameter estimation*, especially maximum likelihood estimation, which seeks the value  $\hat{\beta}$  that makes the observed data as probable as possible under the model.
  2. *Confidence interval estimation*, which uses the data to construct intervals that contain the true parameter values with a specified probability over replication of the data generating process.
  3. *Hypothesis testing*, which assesses the plausibility of a hypothesis, by computing a measure of how improbable the observed data are under that hypothesis: the *p-value*. A low p-value suggests that the data would be improbable if the hypothesis were true.
  4. *Model checking*: could the model have generated the observed data at all? If not, the conclusions drawn from items 1–3 are unreliable. Useful checking is often done graphically.

The course assumes familiarity with random variables, probability density functions, cumulative distribution functions, and quantile functions (the inverse of cumulative distribution functions). It also assumes that, if given  $\hat{\beta} \sim N(\beta, \sigma_{\hat{\beta}}^2)$  or  $(\hat{\beta} - \beta)/\hat{\sigma}_{\hat{\beta}} \sim t_d$  along with any necessary values, you are able to test hypotheses about  $\beta$  and construct confidence intervals for  $\beta$ . Familiarity with maximum likelihood estimation is also assumed. Additionally, concepts from matrix algebra will be used, particularly in the study of the linear model.

**Topics to be covered** include:

- A thorough and in-depth **study of linear models** (LMs), which serves as the foundation for all subsequent developments.
- **Generalised linear models** (GLMs): a *generalisation* of linear regression models that allows the response variable to follow a distribution other than the normal, by modelling a transformed expectation of the response variable. These models can handle, for example:
  - *Binary* response variables (e.g., success/ failure, yes/no), using logistic regression.
  - *Count* response variables (e.g., the number of road deaths in a given year), using Poisson regression.
  - *non-negative* continuous response variables (e.g., insurance claim sizes), using (for instance) a gamma regression model.
- **Mixed effects models**: LMs and GLMs typically assume independence between observations. However, in many real-world applications, observations are not independent. For instance, in education research, students within the same school or classroom may perform similarly due to a shared environment or teaching methods. Mixed effects models allow us to analyse data of this type.

## Useful books

No specific textbook is required for this course, as the lecture notes are self-contained, provided the necessary prerequisites are met. However, the following books are recommended for further reading (listed alphabetically by first author). They are not mandatory and may include material beyond the scope of the course.

- Agresti, A. (2015). *Foundations of Linear and Generalized Linear Models*, Wiley.
- Dobson, A. J., Barnett, A. G. (2018). *An Introduction to Generalized Linear Models*, 4th edition, CRC Press.
- Fahrmeir, L., Kneib, T., Lang, S., and Marx, B. D. (2022). *Regression: Models, Methods and Applications*, 2nd edition, Springer.
- Faraway, J. (2014). *Linear Models with R*, 2nd edition, CRC Press.
- Faraway, J. (2016). *Extending the Linear Model with R: Generalized Linear, Mixed Effects, and Nonparametric Regression*, 2nd edition, CRC Press.
- McCullagh, P., and Nelder, J. A. (1989). *Generalized Linear Models*, 2nd edition, Chapman & Hall/CRC.
- Roback, P., and Legler, J. (2020). *Beyond Multiple Linear Regression: Applied Generalized Linear Models And Multilevel Models in R*, CRC Press.
- Wood, S. N. (2017). *Generalized Additive Models: An Introduction with R*, 2nd edition, CRC Press.

A significant portion of the material in these lecture notes is based on the final reference listed.

## 2 Introduction

How old is the universe? Hubble's law, named after Edwin Hubble, is a fundamental principle in physical cosmology that describes the expansion of the universe. It states that the recessional velocity of a galaxy (i.e., how fast it is moving away from an observer located on or near Earth) is directly proportional to its distance from Earth. In other words, the farther away a galaxy is, the faster it moves away from us. Mathematically:

$$y = \beta x,$$

where  $y$  is the recessional velocity of a galaxy,  $x$  is the galaxy's distance from Earth, and  $\beta$  is the 'Hubble constant' (in standard astrophysical notation  $y \equiv v$ ,  $x \equiv d$  and  $\beta \equiv H_0$ ).  $\beta^{-1}$  gives the approximate age of the universe, but  $\beta$  is unknown and must somehow be estimated from observations of  $y$  and  $x$ , made for a variety of galaxies at different distances from us. The following link provides an interesting explanation of the Hubble's law:

[news.uchicago.edu/explainer/hubble-constant-explained](https://news.uchicago.edu/explainer/hubble-constant-explained)

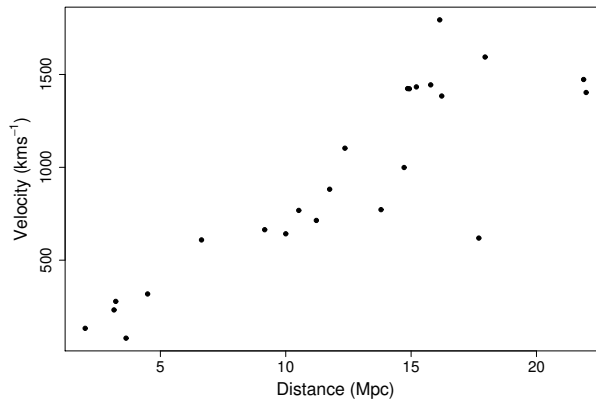


Figure 1: A Hubble diagram showing the relationship between distance,  $x$  (measured in megaparsecs), and recessional velocity,  $y$  (measured in kilometers per second), for 24 galaxies containing Cepheid stars. The data are from the Hubble Space Telescope key project to measure the Hubble constant.

Figure 1 plots recessional velocity against distance for 24 galaxies, according to measurements made using the Hubble Space Telescope.<sup>1</sup> It is clear from the figure that the observed data do not follow Hubble's law exactly, but given the measurement process, it would be surprising if they did. Given the apparent variability, what can be inferred from these data? In particular:

- (i) What value of  $\beta$  is most consistent with the data?
- (ii) What range of  $\beta$  values is consistent with the data?
- (iii) Are some particular, theoretically derived, values of  $\beta$  consistent with the data?

Statistics is about trying to answer these three sorts of questions through, respectively, point estimates, confidence intervals, and hypothesis tests.

One way to proceed is to formulate a linear statistical model of the way that the data were generated, and to use this as the basis for inference. Specifically, suppose that, rather than being governed directly by Hubble's law,

<sup>1</sup>Velocities are assessed by measuring the Doppler effect red shift in the spectrum of light observed from the galaxies concerned, although some correction for 'local' velocity components is required. Distance measurement is much less direct, and is based on the 1912 discovery, by Henrietta Leavitt, of a relationship between the period of a certain class of variable stars, known as the Cepheids, and their luminosity. The intensity of Cepheids varies regularly with a period of between 1.5 and something over 50 days, and the mean intensity increases predictably with period. This means that, if you can find a Cepheid, you can tell how far away it is, by comparing its apparent brightness to its period predicted intensity.

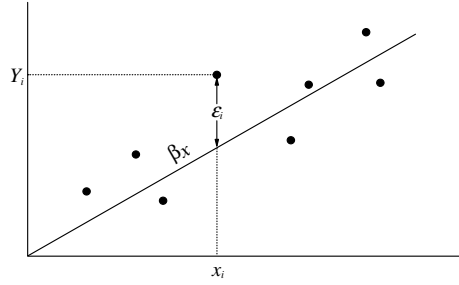


Figure 2: Schematic illustration of a simple linear model with one explanatory variable.

the observed velocity is given by Hubble's constant multiplied by the observed distance plus a 'random variability' term. That is

$$y_i = \beta x_i + \epsilon_i, \quad i = 1 \dots 24, \quad (1)$$

where the  $\epsilon_i$  terms are independent random variables such that  $\mathbb{E}(\epsilon_i) = 0$  and  $\mathbb{E}(\epsilon_i^2) = \sigma^2$ , thus implying that  $\text{var}(\epsilon_i) = \sigma^2$ . The random component of the model is intended to capture the fact that if we gathered a replicate set of data, for a new set of galaxies, Hubble's law would not change, but the apparent random variation from it would be different, as a result of different measurement errors. Notice that it is not implied that these errors are completely unpredictable: their mean and variance are assumed to be fixed, it is only their particular values, for any particular galaxy, that are not known.

## 2.1 A simple linear model

This section develops statistical methods for a simple linear model of the form (1). This allows the key concepts of linear modelling to be reviewed without the distraction of any mathematical difficulty.

Formally, consider  $n$  pairs of observations,  $(x_i, y_i)$ , where  $y_i$  is an observation on random variable,  $Y_i$ , with expectation,  $\mu_i \equiv \mathbb{E}(Y_i)$ . Suppose that an appropriate model for the relationship between  $x$  and  $y$  is:

$$Y_i = \mu_i + \epsilon_i, \quad \text{where } \mu_i = \beta x_i. \quad (2)$$

Here  $\beta$  is an unknown parameter and the  $\epsilon_i$  are mutually independent zero mean random variables, each with the same variance  $\sigma^2$ . So the model says that  $Y$  is given by  $x$  multiplied by a constant plus a random term.  $Y$  is an example of a *response (or outcome or dependent) variable*, while  $x$  is an example of a *predictor (or covariate or independent) variable*<sup>2</sup>. Figure 2 illustrates this model for a case where  $n = 8$ .

**Important note:** In this regression model we are not including an intercept term, which represents the point at which the regression line crosses the  $y$ -axis. In this case, because we have no intercept the regression line crosses the  $y$ -axis at the origin. And this makes sense, as when the distance is zero, the corresponding recessional velocity is also zero. In general, by default, linear regression models should include an intercept, unless theory or prior knowledge suggests that the relationship between the two variables passes through the origin. Omitting the intercept when it should be included can lead to biased estimates of the regression coefficients and, consequently, incorrect conclusions.

### 2.1.1 Simple least squares estimation

How can  $\beta$ , in model (2), be estimated from the  $\{(x_i, y_i)\}_{i=1}^n$  data? A sensible approach is to choose a value of  $\beta$  that makes the model fit the data well. To do this, we need to define a measure of how well (or how poorly) a model with a given value of  $\beta$  fits the data. One such measure is the **residual sum of squares (RSS)** of the model:

$$\mathcal{S} = \sum_{i=1}^n (y_i - \mu_i)^2.$$

<sup>2</sup>Throughout, we assume that the predictor(s) are fixed (deterministic), whereas the response is random.

For the current model, where  $\mu_i = \beta x_i$ , this becomes

$$\mathcal{S} = \sum_{i=1}^n (y_i - \beta x_i)^2.$$

If we have chosen a good value of  $\beta$ , close to the true value, then the model predicted  $\mu_i$  should be relatively close to the  $y_i$ , so that  $\mathcal{S}$  should be small, whereas poor choices will lead to  $\mu_i$  far from their corresponding  $y_i$ , and high values of  $\mathcal{S}$ . Hence,  $\beta$  can be estimated by minimizing  $\mathcal{S}$  with respect to (w.r.t.)  $\beta$  and this is known as the method of *least squares*.

To minimize  $\mathcal{S}$ , for the current model, differentiate w.r.t.  $\beta$ :

$$\frac{\partial \mathcal{S}}{\partial \beta} = - \sum_{i=1}^n 2x_i(y_i - \beta x_i)$$

and set the result to zero to find  $\hat{\beta}$ , the least squares estimate of  $\beta$ :

$$- \sum_{i=1}^n 2x_i(y_i - \hat{\beta} x_i) = 0 \Rightarrow \sum_{i=1}^n x_i y_i - \hat{\beta} \sum_{i=1}^n x_i^2 = 0 \Rightarrow \hat{\beta} = \sum_{i=1}^n x_i y_i / \sum_{i=1}^n x_i^2.^3$$

## 2.2 Sampling properties of $\hat{\beta}$

To evaluate the reliability of the least squares estimate,  $\hat{\beta}$ , it is useful to consider the sampling properties of  $\hat{\beta}$ . That is, we examine properties of the distribution of  $\hat{\beta}$  values, which would be obtained from repeated independent replications of the data used for estimation. To do this, it is helpful to introduce the concept of an *estimator*, which is obtained by replacing the observations,  $y_i$ , in the expression for  $\hat{\beta}$  with the corresponding random variables,  $Y_i$ , yielding

$$\hat{\beta} = \sum_{i=1}^n x_i Y_i / \sum_{i=1}^n x_i^2.$$

Clearly, the *estimator*,  $\hat{\beta}$ , is a random variable and we can therefore discuss its distribution<sup>4</sup>. For now, consider only the first two moments of that distribution.

The expected value of  $\hat{\beta}$  is obtained as follows:

$$\mathbb{E}(\hat{\beta}) = \mathbb{E} \left( \sum_{i=1}^n x_i Y_i / \sum_{i=1}^n x_i^2 \right) = \sum_{i=1}^n x_i \mathbb{E}(Y_i) / \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i^2 \beta / \sum_{i=1}^n x_i^2 = \beta.$$

So  $\hat{\beta}$  is an unbiased estimator — its expected value is equal to the true value of the parameter that it is supposed to estimate.

Unbiasedness is a reassuring property, but knowing that an estimator is correct on average does not tell us much about how good any particular estimate is likely to be. To assess this, we also need to know how much estimates vary from one replicated data set to the next—that is, we need to know the variance of the estimator.

From general probability theory we know that if  $Y_1, Y_2, \dots, Y_n$  are *independent* random variables and  $a_1, a_2, \dots, a_n$  are real constants then

$$\text{var} \left( \sum_i a_i Y_i \right) = \sum_i a_i^2 \text{var}(Y_i).$$

But we can write

$$\hat{\beta} = \sum_i a_i Y_i \text{ where } a_i = x_i / \sum_i x_i^2,$$

<sup>3</sup> $\partial^2 \mathcal{S} / \partial \beta^2 = 2 \sum_{i=1}^n x_i^2 > 0$ , confirming that the solution corresponds to a minimum of  $\mathcal{S}$ .

<sup>4</sup>For notational simplicity, we do not explicitly write  $\hat{\beta}(\mathbf{y})$  and  $\hat{\beta}(\mathbf{Y})$  to distinguish between the estimate and the estimator, respectively.

and from the original model specification we have that  $\text{var}(Y_i) = \sigma^2$ , for all  $i = 1, \dots, n$ . Hence,

$$\text{var}(\hat{\beta}) = \sigma_{\hat{\beta}}^2 = \frac{\sum_i x_i^2 \text{var}(Y_i)}{(\sum_i x_i^2)^2} = \frac{\sigma^2}{\sum_i x_i^2}. \quad (3)$$

In most circumstances  $\sigma^2$  itself is an unknown parameter and must also be estimated. Since  $\sigma^2$  is the variance of the  $\epsilon_i$ , it makes sense to estimate it using the variance of the ‘estimated’  $\epsilon_i$ , the model **residuals**. In general, these are defined as  $\hat{\epsilon}_i = y_i - \hat{\mu}_i$ , where  $\hat{\mu}_i$  are the **fitted values**, the model estimates of  $\mu_i = \mathbb{E}(Y_i)$ . For the current model  $\hat{\mu}_i = \hat{\beta}x_i$ , so  $\hat{\epsilon}_i = y_i - \hat{\beta}x_i$ . An unbiased estimator of  $\sigma^2$  for this model is then:

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_i (Y_i - \hat{\beta}x_i)^2 \quad (4)$$

(proof of unbiasedness is given later for the general case). Plugging this estimate of  $\sigma^2$  into (3) yields an unbiased estimator of the variance of  $\hat{\beta}$ <sup>5</sup>, leading to  $\widehat{\text{var}}(\hat{\beta}) = \hat{\sigma}_{\hat{\beta}}^2 = \hat{\sigma}^2 / \sum_i x_i^2$ .

## 2.3 So how old is the universe?

The least squares calculations derived above are available as part of the statistical package and environment R. The function `lm` fits linear models to data, including the simple example currently under consideration. The Cepheid distance and velocity data shown in Figure 1 are stored in the data frame<sup>6</sup> `hubble` available in the package `gamair`. The following R code fits the model and produces the output shown (and more!).

```
library(gamair); data(hubble)
#To know more about the data
help(hubble)

#Fit simple linear regression model (with no intercept)
hub.mod <- lm(y ~ x - 1, data = hubble)
summary(hub.mod)$coefficients

##      Estimate Std. Error t value      Pr(>|t|)
## x 76.58117    3.964794 19.3153 1.031907e-15
```

The call to `lm` passed two arguments to the function. The first is a *model formula*, `y~x-1`, specifying the model to be fitted: the name of the response variable is to the left of ‘~’ while the predictor variable is specified on the right; the ‘-1’ term indicates that the model has no ‘intercept’ term, i.e. that the model is a straight line through the origin. The second (optional) argument gives the name of the data frame in which the variables are to be found. `lm` takes this information and uses it to fit the model by least squares: the results are returned in a ‘fitted model object’, which in this case has been assigned to an object called `hub.mod` for later examination. ‘<-’ is the assignment operator, and `hub.mod` is created by this assignment (overwriting any previously existing object of this name).

The `summary` function is then used to examine the fitted model object. Only part of its output is shown here and relevant to us so far:  $\hat{\beta}$  and the estimate of the standard error of  $\hat{\beta}$  (the square root of the estimated variance of  $\hat{\beta}$ , i.e.  $\hat{\sigma}_{\hat{\beta}}$ , derived above)<sup>7</sup>. Before using these quantities it is important to check the model assumptions. The way to do this is to examine residual plots.

Recall that the ‘fitted values’ for this model are defined as  $\hat{\mu}_i = \hat{\beta}x_i$ , while the residuals are simply  $\hat{\epsilon}_i = y_i - \hat{\mu}_i$ . A plot of residuals against fitted values is often useful and the following produces the plot in Figure 3(a).

<sup>5</sup> $\mathbb{E}(\widehat{\text{var}}(\hat{\beta})) = \text{var}(\hat{\beta})$ .

<sup>6</sup>A data frame is just a two dimensional array of data in which the values of different variables are stored in different named columns.

<sup>7</sup>The column `t value`, is not relevant for now, but it gives the value of the test statistic associated to testing  $H_0 : \beta = 0$  versus  $H_1 : \beta \neq 0$ . The last column gives the p-value associated with this hypothesis test.

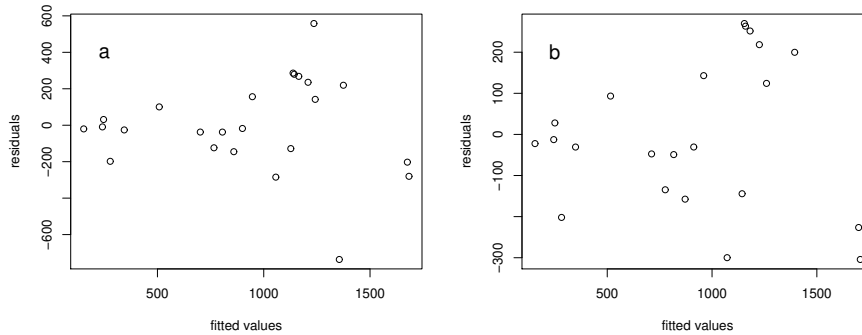


Figure 3: Residuals against fitted values for (a) the model (1) fitted to all the data in figure 1 and (b) the same model fitted to data with two substantial outliers omitted. Note the difference in the  $y$ -axis scales.

```
plot(fitted(hub.mod), residuals(hub.mod),
     xlab="fitted values", ylab="residuals")
abline(h = 0, lty = 2)
```

What we would like to see, in such a plot, is an apparently random scatter of residuals around zero, with no trend in either the mean of the residuals, or their variability, as the fitted values increase. The main problematic feature of Figure 3(a) is the presence of two points with very large magnitude residuals, suggesting a problem with the constant variance assumption. It is probably prudent to repeat the model fit, with and without these points, to check that they are not having undue influence on our conclusions. The following code omits the offending points and produces a new residual plot shown in Figure 3(b).

```
#Extracting the index of the outlying observations
as.numeric(which(residuals(hub.mod) > 400 | residuals(hub.mod) < -600))

## [1] 3 15

#Fitting model without outliers
hub.mod1 <- lm(y ~ x - 1, data = hubble[-c(3, 15), ])
#Another way of extracting the coefficients from the summary output
coef(summary(hub.mod1))

##      Estimate Std. Error t value      Pr(>|t|)
## x 77.67292      2.97028 26.15003 1.659996e-17

plot(fitted(hub.mod1), residuals(hub.mod1),
     xlab="fitted values", ylab="residuals")
abline(h = 0, lty = 2)
```

The omission of the two large outliers has improved the residuals and changed  $\hat{\beta}$  somewhat, but not drastically<sup>8</sup>. The Hubble constant estimates have units of  $(\text{km})\text{s}^{-1} (\text{Mpc})^{-1}$ . A Mega-parsec is  $3.09 \times 10^{19}\text{km}$ , so we need to divide  $\hat{\beta}$  by this amount, in order to obtain Hubble's constant with units of  $\text{s}^{-1}$ . The approximate age of the universe, in seconds, is then given by the reciprocal of  $\hat{\beta}$ . Here are the two possible estimates expressed in years:

<sup>8</sup>Note that, in general, it is not good practice to remove outlying observations unless there is a strong motivation to do so (e.g., clearly impossible values such as a negative glucose level or a cholesterol level of 1000). Any such decision should be made explicit in the analysis, and results should ideally be reported both with and without the outlying observations.

```

hubble.const <- c(coef(hub.mod), coef(hub.mod1))/3.09e19
#Approximate age of the universe in seconds
age <- 1/hubble.const
#Approximate age of the universe in years
age/(60^2*24*365)

##          x          x
## 12794692825 12614854757

```

Both fits give an age of around 13 billion years. We now have an estimate of the age of the universe, but what range of ages would be consistent with these data?

## 2.4 Adding a distributional assumption

So far, our analysis of the simple model has relied on the model equations together with the assumptions of zero-mean, independent error terms with constant variance.

If we wish to go further, and find confidence intervals for  $\beta$ , or test hypotheses related to the model, then an additional distributional assumption will be necessary.

Specifically, we assume that  $\epsilon_i \sim N(0, \sigma^2)$ , independently for  $i = 1, \dots, n$ , which is equivalent to assuming  $Y_i \sim N(\beta x_i, \sigma^2)$ ,  $i = 1, \dots, n$ . Now we have already seen that  $\hat{\beta}$  is just a weighted sum of  $Y_i$ , but the  $Y_i$  are now assumed to be independent normal random variables, and a weighted sum of normal random variables is itself a normal random variable. Hence the estimator,  $\hat{\beta}$ , must be a normal random variable. Since we have already established the mean and variance of  $\hat{\beta}$ , we have that

$$\hat{\beta} \sim N\left(\beta, \left(\sum x_i^2\right)^{-1} \sigma^2\right). \quad (5)$$

### 2.4.1 Testing hypotheses about $\beta$

One thing we might want to do is to try and evaluate the consistency of some hypothesized value of  $\beta$  with the observed data. For example, some Creation Scientists estimate the age of the universe to be 6000 years, based on a reading of the Bible<sup>9</sup>. This would imply that  $\beta$  is approximately equal to  $163 \times 10^6$ , with the exact value being equal to  $3.09 \times 10^{19} / (6000 \times 365 \times 24 \times 60^2)$ . The consistency with data of such a hypothesized value for  $\beta$  can be assessed by considering the probability of observing the estimated value  $\hat{\beta}$ , if the true value of  $\beta$  is the hypothetical one.

Specifically, we can test the null hypothesis,  $H_0 : \beta = \beta_0$ , versus the alternative hypothesis,  $H_1 : \beta \neq \beta_0$ , for some specified value  $\beta_0$ , by examining the probability of getting the observed  $\hat{\beta}$ , or one further from  $\beta_0$ , assuming  $H_0$  to be true. If  $\sigma^2$ , the variance of the error term, were known then we could work directly from (5), as follows.

The probability required is known as the **p-value** of the test. It is

*the probability of obtaining a value of  $\hat{\beta}$  at least as extreme (i.e., at least as favourable to  $H_1$ ) as the one actually observed, if  $H_0$  is actually true.*

Actually, this definition holds for the p-value of any hypothesis test, if the specific ' $\hat{\beta}$ ' is replaced by the general '*a test statistic*'. In this case it helps to distinguish notationally between the estimate,  $\hat{\beta}_{\text{obs}}$ , and estimator  $\hat{\beta}$ . Noting that, under the null hypothesis of  $\beta = \beta_0$ , it holds that  $Z \equiv (\hat{\beta} - \beta_0)/\sigma_{\hat{\beta}} \sim N(0, 1)$ , the p-value is then

$$\begin{aligned}
p &= \Pr \left[ |\hat{\beta} - \beta_0| \geq |\hat{\beta}_{\text{obs}} - \beta_0| \mid H_0 \right] \\
&= \Pr \left[ \left| \frac{\hat{\beta} - \beta_0}{\sigma_{\hat{\beta}}} \right| \geq \left| \frac{\hat{\beta}_{\text{obs}} - \beta_0}{\sigma_{\hat{\beta}}} \right| \mid H_0 \right] \\
&= \Pr[|Z| > |z|]
\end{aligned}$$

<sup>9</sup>This isn't really valid, of course, since the Creation Scientists are not postulating a big bang theory.

where  $z = (\hat{\beta}_{\text{obs}} - \beta_0)/\sigma_{\hat{\beta}}$  and  $\sigma_{\hat{\beta}}^2 = (\sum x_i^2)^{-1}\sigma^2$ . Here  $Z$  is the test statistic (a random variable) and  $z$  is the realisation/observed value of the test statistic. Hence, having formed  $z$ , the p-value is easily worked out, using the cumulative distribution function for the standard normal distribution. Small p-values suggest that the data are inconsistent with  $H_0$ , while large values do not provide evidence against  $H_0$ . The threshold 0.05 is often used as the boundary between ‘small’ and ‘large’ in this context. In reality  $\sigma^2$  is usually unknown. Broadly the same testing procedure can still be adopted, by replacing  $\sigma$  with  $\hat{\sigma}$  (see Equation (4)), but we need to somehow allow for the extra uncertainty that this introduces (unless the sample size is very large). It turns out that if  $H_0 : \beta = \beta_0$  is true then

$$T \equiv \frac{\hat{\beta} - \beta_0}{\hat{\sigma}_{\hat{\beta}}} \sim t_{n-1},$$

where  $n$  is the sample size,  $\hat{\sigma}_{\hat{\beta}}^2 = (\sum x_i^2)^{-1}\hat{\sigma}^2$ , and  $t_{n-1}$  is the t-distribution with  $n - 1$  degrees of freedom. The value  $n - 1$  comes from the number of data minus the number of estimated model parameters, and the result is proven in Section 4. Large magnitude values of  $T$  favour  $H_1$ , so using  $T$  as the test statistic, instead of  $Z$ , we can calculate a p-value by evaluating

$$p = \Pr[|T| > |t|],$$

where  $T \sim t_{n-1}$  and  $t = (\hat{\beta}_{\text{obs}} - \beta_0)/\hat{\sigma}_{\hat{\beta}}$ . Here is some code to evaluate the p-value for  $H_0$  : *the Hubble constant is 163000000*.

```
cs.hubble <- 163000000
#test statistic
t.stat <- (coef(hub.mod1) - cs.hubble)/summary(hub.mod1)$coefficients[2]
t.stat

##           x
## -54876951

#An Alternative way to extract the regression coefficient
t.stat <- (summary(hub.mod1)$coefficients[1] - cs.hubble)/
  summary(hub.mod1)$coefficients[2]
t.stat

## [1] -54876951

#p-value
df <- nrow(hubble[-c(3, 15), ]) - 1
pt(abs(t.stat), df = df, lower = FALSE)*2 # 2 because of |T| in p-value defn.

## [1] 3.906388e-150
```

Based on the observed test statistic,  $t$ , the data would be extremely improbable if  $\beta = 1.63 \times 10^8$ . Therefore, the hypothesized value can be rejected rather firmly (in this case, using the data with the outliers also leads to an extremely low p-value and, consequently, to a rejection of the null hypothesis).

Hypothesis testing is particularly useful when there are good reasons to want to stick with some null hypothesis, until there is good reason to reject it. This is often the case when comparing models of differing complexity: it is often a good idea to retain the simpler model until there is quite firm evidence that it is inadequate. Note one interesting property of hypothesis testing. If we choose to reject a null hypothesis whenever the p-value is less than some fixed level, say  $\alpha$  (often termed the *significance level* of a test), then we will inevitably reject a proportion,  $\alpha$ , of correct null hypotheses. We could try and reduce the probability of such mistakes by making  $\alpha$  very small, but in that case we pay the price of reducing the probability of rejecting  $H_0$  when it is false!

## 2.4.2 Confidence intervals

Having seen how to test whether a *particular* hypothesized value of  $\beta$  is consistent with the data, the question naturally arises of what *range* of values of  $\beta$  would be consistent with the data? To answer this, we need to select

a definition of ‘consistent’: a common choice is to say that any parameter value is consistent with the data if it results in a p-value of  $\geq \alpha^{10}$ , when used as the null value in a hypothesis test.

Sticking with the Hubble constant example, and working at a significance level of  $\alpha = 0.05$ , we would have rejected any hypothesized value  $\beta_0$  for the constant, that resulted in a  $t$  value outside the range  $(-2.08, 2.08)$ , since these values would result in p-values of less than 0.05. The R function `qt` can be used to find such ranges: e.g. `qt(c(0.025, 0.975), df = 21)` returns the range of the middle 95% of  $t_{21}$  random variables. So we would not reject any  $\beta_0$  fulfilling:

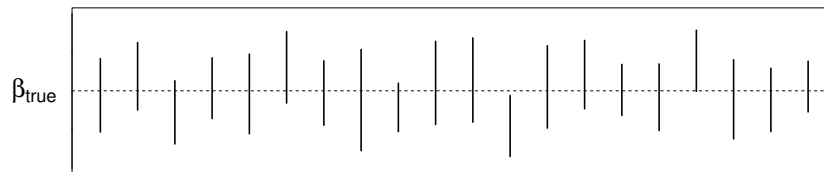
$$-2.08 \leq \frac{\hat{\beta} - \beta_0}{\hat{\sigma}_{\hat{\beta}}} \leq 2.08$$

which we re-arrange to give the interval

$$\hat{\beta} - 2.08\hat{\sigma}_{\hat{\beta}} \leq \beta_0 \leq \hat{\beta} + 2.08\hat{\sigma}_{\hat{\beta}}.$$

Such an interval is known as a 95% confidence interval (CI) for  $\beta$ .

The defining property of a 95% confidence interval is this: if we were to gather an infinite sequence of independent replicate data sets, and calculate 95% confidence intervals for  $\beta$  from each, then 95% of these intervals would include the true  $\beta$ , and 5% would not. It is easy to see how this comes about. By construction, a hypothesis test with a significance level of 5% rejects the correct null hypothesis for 5% of replicate data sets, and does not reject it for the other 95% of replicates. Hence, 5% of 95% confidence intervals must exclude the true parameter, while 95% include it. The following illustrates this, showing intervals for a parameter  $\beta$  computed for 20 replicate datasets.



For the Hubble example, a 95% CI for the constant (in the usual astro-physicists units) is given by:

```
sigb <- summary(hub.mod1)$coefficients[2]
h.ci <- coef(hub.mod1) + qt(c(0.025, 0.975), df = df)*sigb
h.ci
## [1] 71.49588 83.84995
```

This can be converted to a confidence interval for the age of the universe, in years, as follows:

```
h.ci <- h.ci*60^2*24*365.25/3.09e19 # convert to 1/years
sort(1/h.ci)
## [1] 11677548698 13695361072
```

That is, the 95% CI is (11.7,13.7) billion years. Actually this ‘Hubble age’ is the age of the universe if it has been expanding freely, neither being slowed down by gravitation, nor accelerating for reasons not fully understood. In fact this free expansion assumption does not appear to be quite right, so our answer will be out by a bit more than the CI might suggest. But remember this is a very simple model, mainly for illustrative purposes.

<sup>10</sup>E.g.,  $\alpha = 0.05$  for 95% confidence intervals.

### 3 Linear models in general

The simple linear model, introduced above, can be generalized by allowing the response variable to depend on multiple predictor variables (plus an intercept). These extra predictor variables can themselves be transformations of the original predictors. Here are some examples, for each of which a response observation,  $y_i$ , is treated as an observation on a random variable,  $Y_i$ , where  $\mathbb{E}(Y_i) \equiv \mu_i$ , the  $\epsilon_i$  are zero mean random variables, and the  $\beta_j$  are unknown model parameters to be estimated from the data.

1.  $\mu_i = \beta_1 + \beta_2 x_i$ ,  $Y_i = \mu_i + \epsilon_i$ , is a straight line relationship between  $y$  and the predictor variable,  $x$ .
2.  $\mu_i = \beta_1 + \beta_2 x_i + \beta_3 x_i^2 + \beta_4 x_i^3$ ,  $Y_i = \mu_i + \epsilon_i$ , is a cubic model for the relationship between  $y$  and  $x$ .
3.  $\mu_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2} + \beta_4 \log(x_{i1} x_{i2})$ ,  $Y_i = \mu_i + \epsilon_i$ , is a model in which  $y$  depends on predictor variables  $x_1$  and  $x_2$  and on the log of their product.

Each of these is a *linear model* because the  $\epsilon_i$  terms and the model parameters,  $\beta_j$ , enter the model in a linear way. Notice that the predictor variables can enter the model non-linearly, making the linear model very flexible. By contrast, the model

$$Y_i = \beta_1 + \beta_2 x_i^{\beta_3} + \epsilon_i, \quad (6)$$

is not linear in the parameters. However, a model of the form

$$Y_i = \beta_1 x_i^{\beta_2} \eta_i,$$

with multiplicative errors  $\eta_i > 0$ , can be transformed into a linear model by taking logarithms.

Exactly as in the simple Hubble's law regression model discussed earlier, the parameters of linear models can be estimated by finding the  $\beta_j$  values which make the models best fit the observed data, in the sense of minimizing  $\sum_i (y_i - \mu_i)^2$ , recalling that  $\mu_i$  is a function of the  $\beta_j$ s. The theory for doing this will be developed in Section 4, and that development is based entirely on re-writing the linear model using matrices and vectors.

To see how this re-writing is done, consider the straight line model given in Example 1. above. Writing out the  $\mu_i$  equations for all  $n$  pairs,  $(x_i, y_i)$ , results in a large system of linear equations:

$$\begin{aligned} \mu_1 &= \beta_1 + \beta_2 x_1 \\ \mu_2 &= \beta_1 + \beta_2 x_2 \\ \mu_3 &= \beta_1 + \beta_2 x_3 \\ &\vdots \\ \mu_n &= \beta_1 + \beta_2 x_n \end{aligned}$$

which can be re-written in matrix-vector form as

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \vdots \\ \mu_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}.$$

So the model has the general form

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta},$$

i.e. the expected value vector  $\boldsymbol{\mu}$  is given by a **model matrix** (also known as a design matrix),  $\mathbf{X}$ , multiplied by a **parameter vector**,  $\boldsymbol{\beta}$ . All linear models can be written in this general form. Of course since  $y_i = \mu_i + \epsilon_i$ , we can also write the model as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

As a second illustration, the cubic example given above can be written in matrix vector form as

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \cdot \\ \cdot \\ \mu_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ 1 & x_3 & x_3^2 & x_3^3 \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 1 & x_n & x_n^2 & x_n^3 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}.$$

Models in which data are divided into different groups, each of which is assumed to have a different mean, are less obviously of the form  $\mu = \mathbf{X}\beta$ , but in fact they can be written in this way, by use of dummy (indicator) variables. Again, this is most easily seen by example. Consider the model

$$\mu_i = \beta_j \text{ if observation } i \text{ is in group } j,$$

and suppose that there are three groups, each with two observations, so that  $i = 1, \dots, 6$ , and  $j = 1, 2, 3$ . One motivating example is the study of daily sodium intake and blood pressure for patients grouped into three age categories. Each age category represents a group, and there are two patients (observations) in each group. Assuming that observations 1 and 2 belong to group 1, observations 3 and 4 to group 2, and observations 5 and 6 to group 3, the model can be rewritten as

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}.$$

Variables indicating the group to which a response observation belongs, are known as *factor* variables. Somewhat confusingly, the groups themselves are known as *levels* of a factor. So the above model involves one factor, ‘group’, with three levels. Models of this type, involving factors, are commonly used for the analysis of designed experiments. In this case the model matrix depends on the design of the experiment (i.e. on which units belong to which groups), and for this reason the terms ‘design matrix’ and ‘model matrix’ are often used interchangeably. Whatever it is called,  $\mathbf{X}$  is absolutely central to understanding the theory of linear models and generalized linear models.

### 3.1 Notation summary

- A linear model can generally be written as:

$$\mathbf{y} = \mathbf{X}\beta + \epsilon,$$

where

- $\mathbf{y}$  is the *response variable* vector and has dimension  $n \times 1$ .
- $\mathbf{X}$  is the *model matrix*, which has dimension  $n \times p$ , where  $p$  denotes the number of parameters to be estimated including, in general, the intercept.
- $\beta$  is the *parameter vector* (or *coefficient vector*) and has dimension  $p \times 1$ .
- $\epsilon$  is a *random error* vector of dimension  $n \times 1$  with independent components and such that  $\mathbb{E}(\epsilon) = \mathbf{0}$ , with variance–covariance matrix  $\mathbf{V}_\epsilon = \text{var}(\epsilon) = \sigma^2 \mathbf{I}_n$ .
- The *expected value* vector  $\mu = \mathbb{E}(\mathbf{y})$  has dimension  $n \times 1$  and according to the linear model,  $\mu = \mathbf{X}\beta$ .
- Once the model is estimated, the *parameter estimates* are denoted by  $\hat{\beta}$ . This notation will also be used for the *parameter estimator*, the context indicating which is meant.
- The *fitted value* vector is  $\hat{\mu} = \mathbf{X}\hat{\beta}$ .
- The *residual* vector is  $\hat{\epsilon} = \mathbf{y} - \hat{\mu}$ .

## 3.2 R and linear models in general

R has functions for setting up model matrices automatically, on the basis of a *model formula* specifying the structure for the columns of the model matrix. We will now explore one example with more than one variable using the `cars` data frame, which contains data on stopping distances of cars against `speed` when the driver was first signalled to stop. It typically takes some fixed amount of ‘reaction time’, after the stop signal, for the driver to apply the brakes, hence the car will travel a distance proportional to initial speed before beginning to slow at all: so we need a linear dependence on `speed` in the model. The kinetic energy of a car is proportional to the square of its `speed`, and once the brakes are applied they can dissipate this energy (ultimately as heat) at a constant rate per unit distance travelled: hence the braking phase contributes a distance proportional to the square of `speed` to the total stopping distance. That is, a suitable model for the data is:

$$\text{distance}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i,$$

where the  $\epsilon_i$  are i.i.d.  $N(0, \sigma^2)$  random variables. Of course, we would expect the parameter  $\beta_1$  to be zero, given the justification for the model, but there is no harm in leaving it in for the moment. If there is statistical evidence that it is not zero, then this would suggest something wrong with our physical model: so it provides a useful check. Let us first look at the data; to know more about the dataset we can type `help(cars)`.

```
data(cars)
# First rows of the dataset
head(cars)

##    speed dist
## 1      4     2
## 2      4    10
## 3      7     4
## 4      7    22
## 5      8    16
## 6      9    10

# Last rows of the dataset
tail(cars)

##    speed dist
## 45     23    54
## 46     24    70
## 47     24    92
## 48     24    93
## 49     24   120
## 50     25    85

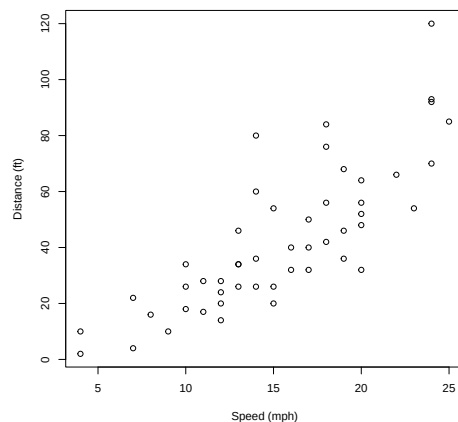
# Structure of the data
str(cars)

## 'data.frame': 50 obs. of  2 variables:
##  $ speed: num  4 4 7 7 8 9 10 10 10 11 ...
##  $ dist : num  2 10 4 22 16 10 18 26 34 17 ...

# Summary of the data
summary(cars)

##      speed      dist
##  Min.   : 4.0    Min.   : 2.00
##  1st Qu.:12.0    1st Qu.: 26.00
##  Median :15.0    Median : 36.00
##  Mean   :15.4    Mean   : 42.98
##  3rd Qu.:19.0    3rd Qu.: 56.00
##  Max.   :25.0    Max.   :120.00
```

```
#Scatter plot of the data
plot(cars$speed, cars$dist,
      xlab = "Speed (mph)", ylab = "Distance (ft)")
```



The R function `model.matrix` can be used to set up the model matrix for this model of the `cars` data, as follows.

```
X <- model.matrix(~ speed + I(speed^2), data = cars)
```

Like `lm`, `model.matrix` takes a model formula and a data frame as arguments. It returns a model matrix. In this case the formula can be *one sided* as the response variable is immaterial when setting up the model matrix. Everything to the right of `~` in the formula specifies the structure of the model matrix. In this case we specify that there should be one column containing the speeds, one column containing the squares of the speeds and, since we did not suppress it, there will also be a column of 1s, corresponding to the model intercept,  $\beta_1$ . Because some arithmetic symbols have special meanings (see later in the notes) within model formulae, we have to ‘protect’ the term `speed^2`, by writing it as `I(speed^2)`, which means that we want `^2` to have its usual arithmetic meaning here. `I()` is the *identity function*: it just returns its argument. Here is what the first rows of `X` looks like...

```
head(X)
```

```
##      (Intercept) speed I(speed^2)
## 1             1     4         16
## 2             1     4         16
## 3             1     7         49
## 4             1     7         49
## 5             1     8         64
## 6             1     9         81
```

Having got R to set up an appropriate model matrix, we *could* use it much as we used the single covariate  $x$  in the Hubble data example...

```
stop.model0 <- lm(dist ~ X - 1, data = cars)
summary(stop.model0)

##
## Call:
## lm(formula = dist ~ X - 1, data = cars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.720  -9.184  -3.188   4.628  45.152
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## X(Intercept)  2.47014    14.81716   0.167   0.868
## Xspeed        0.91329     2.03422   0.449   0.656
## XI(speed^2)   0.09996     0.06597   1.515   0.136
##
## Residual standard error: 15.18 on 47 degrees of freedom
## Multiple R-squared:  0.9133, Adjusted R-squared:  0.9078
## F-statistic: 165.1 on 3 and 47 DF,  p-value: < 2.2e-16
```

Notice the `Coefficients` section of the table, in particular. It now has a row for each model coefficient, with each row having much the same interpretation as we have already covered for the single parameter case. Actually, we would not usually form a model matrix explicitly, and then use `lm` in the way just done. Instead we would supply the more elaborate model formula directly to `lm`, from which it can create the appropriate model matrix automatically. For example:

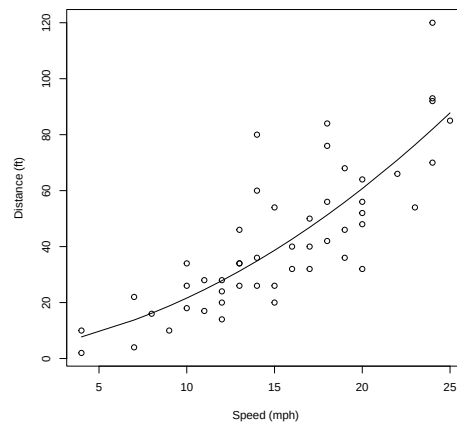
```
stop.model <- lm(dist ~ speed + I(speed^2), data = cars)
summary(stop.model)

##
## Call:
## lm(formula = dist ~ speed + I(speed^2), data = cars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.720  -9.184  -3.188   4.628  45.152
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.47014    14.81716   0.167   0.868
## speed        0.91329     2.03422   0.449   0.656
## I(speed^2)   0.09996     0.06597   1.515   0.136
##
## Residual standard error: 15.18 on 47 degrees of freedom
## Multiple R-squared:  0.6673, Adjusted R-squared:  0.6532
## F-statistic: 47.14 on 2 and 47 DF,  p-value: 5.852e-12
```

The results are, of course, identical to those produced by the more long winded approach. Well, we see that the  $R^2$  and  $F$  statistic values in the summary output of the two models are different. The correct values are those obtained from `stop.model`; this is due to a silly thing that R does behind the scenes (this will be explained carefully in Section 5). In particular, in `stop.model0`, the `lm` function interprets all columns of the model matrix as predictors, including the column corresponding to the intercept. For now, we will proceed. Let us add the fitted values to the scatter plot of the data.

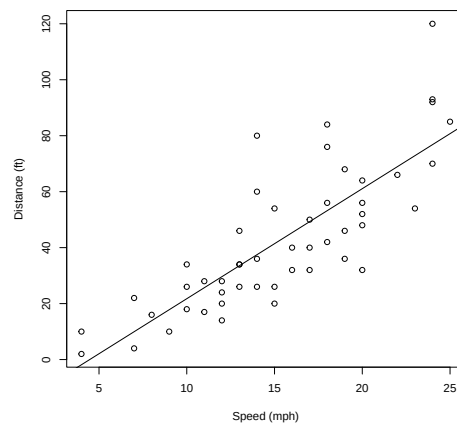
```
plot(cars$speed, cars$dist,
      xlab = "Speed (mph)", ylab = "Distance (ft)")

ss <- sort(cars$speed)
lines(ss, coef(stop.model)[1] + coef(stop.model)[2]*ss+ coef(stop.model)[3]*ss^2)
```



Out of curiosity, let us look at the fitted values for a model without the quadratic term.

```
stop.model2 <- lm(dist ~ speed, data = cars)
plot(cars$speed, cars$dist,
     xlab = "Speed (mph)", ylab = "Distance (ft)")
abline(stop.model2)
```



By looking at the two plots, the quadratic model seems to provide a better fit (later we will learn tools to perform model selection). Let us for a moment go back to the summary output of the quadratic model. Looking at the summary information in more detail, all the p-values for testing the single parameter hypotheses,  $H_0 : \beta_j = 0$ , for  $j = 1, 2, 3$ , are very high. Does this mean that we cannot reject all such hypotheses, and conclude that all the parameter values could really be zero? Absolutely not! Each test is only valid if the other parameters are allowed to take non-zero values: this is because the parameter estimators are not independent and so the p-values cannot be either. It is possible to test that all parameters except the intercept are simultaneously zero, and this is the test that the final p-value on the last line of the summary relates to: clearly such a hypothesis has no support from the data.

The high p-values in the summary do suggest that *something* could be removed from our model, and a sensible approach is to try removing the term with the highest p-value. This is the intercept, and on the physical grounds, given in the model motivation, we might expect the intercept to be zero. Actually, the confidence interval of the intercept is very wide.

```
coef(stop.model)[1] +
  qt(c(0.025, 0.975), df = nrow(cars) - 3) * summary(stop.model)$coefficients[1, 2]
## [1] -27.33815 32.27843
```

```
#Alternatively we can use the function confint
confint(stop.model)[1,]

##      2.5 %      97.5 %
## -27.33815  32.27843
```

Let us then fit the model without an intercept.

```
stop.model3 <- lm(dist ~ speed + I(speed^2) - 1, data = cars)
summary(stop.model3)

##
## Call:
## lm(formula = dist ~ speed + I(speed^2) - 1, data = cars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.836  -9.071  -3.152   4.570  44.986
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## speed          1.23903     0.55997   2.213  0.03171 *
## I(speed^2)     0.09014     0.02939   3.067  0.00355 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.02 on 48 degrees of freedom
## Multiple R-squared:  0.9133, Adjusted R-squared:  0.9097
## F-statistic: 252.8 on 2 and 48 DF,  p-value: < 2.2e-16
```

Now both  $\beta_2$  and  $\beta_3$  have much lower associated p-values: there are no grounds for dropping any more model terms. We should, of course, check residual plots for this model, exactly as was done for the Hubble data, but for the moment we will move on.

## 4 Linear model theory

As in the simple one parameter case, point estimates of the linear model parameters,  $\beta$ , can be obtained by the method of least squares; that is, by minimising the residual sum of squares

$$\mathcal{S}(\beta) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \sum_{j=1}^p \beta_j x_{ij})^2 = \sum_{i=1}^n (y_i - \mu_i)^2, \quad \mu_i = \mathbf{x}_i^\top \beta,$$

where  $x_{i1} = 1, \forall i = 1, \dots, n$ , for design matrices including an intercept term and  $\mathbf{x}_i^\top = (x_{i1}, \dots, x_{ip})$  is the  $i$ th row of the design matrix,  $\mathbf{X}$ , which is of dimension  $n \times p$ . Fitting a model so that we minimise the squared error terms, or equivalently try to make the  $\mu_i$  as close as possible to the  $y_i$ , has intuitive appeal. The following link leads to a *shiny app* allowing users to explore the residual sum of squares, by changing the intercept and slope values, in a simple linear model:

[shinyserv.es/shiny/least-squares](http://shinyserv.es/shiny/least-squares)

From now on, we will not distinguish notationally between the random response vector and its observed value; the context will make clear which is meant.

To use least squares with a linear model, written in general matrix-vector form, first recall the link between the squared Euclidean length of a vector and the sum of squares of its elements. If  $\mathbf{v}$  is any vector, say of dimension  $n \times 1$ , then its squared Euclidean length is  $\|\mathbf{v}\|^2 \equiv \mathbf{v}^\top \mathbf{v} \equiv \sum_{i=1}^n v_i^2$ . Letting

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \beta \\ \mathbf{x}_2^\top \beta \\ \vdots \\ \mathbf{x}_n^\top \beta \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix},$$

we have<sup>11</sup>

$$\begin{aligned} \mathcal{S}(\beta) &= \sum_{i=1}^n (y_i - \mu_i)^2 = \|\mathbf{y} - \boldsymbol{\mu}\|^2 = \|\mathbf{y} - \mathbf{X}\beta\|^2 \\ &= (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) \\ &= (\mathbf{y}^\top - \beta^\top \mathbf{X}^\top) (\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta \\ &= \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\beta + \beta^\top \mathbf{X}^\top \mathbf{X}\beta, \end{aligned}$$

where from the penultimate to the last equality we have exploited the fact that the terms  $\beta^\top \mathbf{X}^\top \mathbf{y}$  and  $\mathbf{y}^\top \mathbf{X}\beta$  (as well as  $\mathbf{y}^\top \mathbf{y}$  and  $\beta^\top \mathbf{X}^\top \mathbf{X}\beta$ ) are scalars. Thus, in particular,  $\beta^\top \mathbf{X}^\top \mathbf{y}$  is equal to its transpose, i.e.,  $\beta^\top \mathbf{X}^\top \mathbf{y} = (\beta^\top \mathbf{X}^\top \mathbf{y})^\top = \mathbf{y}^\top \mathbf{X}\beta$  and therefore

$$-\mathbf{y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{y} = -\mathbf{y}^\top \mathbf{X}\beta - \mathbf{y}^\top \mathbf{X}\beta = -2\mathbf{y}^\top \mathbf{X}\beta.$$

We minimise  $\mathcal{S}(\beta)$  by setting the  $(p \times 1)$  vector of first derivatives  $\frac{\partial \mathcal{S}(\beta)}{\partial \beta}$  to zero and by showing that the  $(p \times p)$  matrix of second derivatives  $\frac{\partial^2 \mathcal{S}(\beta)}{\partial \beta \partial \beta^\top}$  is positive definite<sup>12</sup>. We start by calculating the vector of first derivatives

$$\begin{aligned} \frac{\partial \mathcal{S}(\beta)}{\partial \beta} &= \frac{\partial (\mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\beta + \beta^\top \mathbf{X}^\top \mathbf{X}\beta)}{\partial \beta} \\ &= 0 - 2 \frac{\partial (\mathbf{y}^\top \mathbf{X}\beta)}{\partial \beta} + \frac{\partial (\beta^\top \mathbf{X}^\top \mathbf{X}\beta)}{\partial \beta}. \end{aligned}$$

<sup>11</sup>Let  $\mathbf{A}$  and  $\mathbf{B}$  be matrices of appropriate dimensions. Then,  $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$  and  $(\mathbf{A}\mathbf{B})^\top = \mathbf{B}^\top \mathbf{A}^\top$ .

<sup>12</sup>A symmetric matrix  $\mathbf{A}$  is positive definite if the real number  $\mathbf{v}^\top \mathbf{A} \mathbf{v} > 0$  for every nonzero real vector  $\mathbf{v}$ .

Let us work out the first term  $\frac{\partial(\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}$ . We note that

$$\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta} = \beta_1(y_{11}x_{11} + \dots + y_{n1}x_{n1}) + \dots + \beta_p(y_{1p}x_{1p} + \dots + y_{np}x_{np}),$$

and therefore

$$\frac{\partial(\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \begin{pmatrix} \frac{\partial(\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \beta_1} \\ \vdots \\ \frac{\partial(\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \beta_p} \end{pmatrix} = \begin{pmatrix} y_{11}x_{11} + \dots + y_{n1}x_{n1} \\ \vdots \\ y_{1p}x_{1p} + \dots + y_{np}x_{np} \end{pmatrix} = \mathbf{X}^\top \mathbf{y}.$$

The vector of derivatives  $\frac{\partial(\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}$  is obtained in the same way but the calculations are admittedly tedious. We therefore make use of the following results, assuming that  $\mathbf{A}$  is a matrix and  $\mathbf{u}$  is a vector, both of appropriate dimensions, ensuring that all operations are well-defined:

1.  $\frac{\partial \mathbf{u}^\top \mathbf{A} \mathbf{u}}{\partial \mathbf{u}} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{u} = 2\mathbf{A} \mathbf{u}$ , if  $\mathbf{A}$  is symmetric.
2.  $\frac{\partial \mathbf{A} \mathbf{u}}{\partial \mathbf{u}^\top} = \left( \frac{\partial \mathbf{A} \mathbf{u}}{\partial \mathbf{u}} \right)^\top = \mathbf{A}$ .

Letting  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ , which is symmetric, in item 1. of the differentiation rules, we have that

$$\frac{\partial(\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Putting both terms together leads to

$$\frac{\partial \mathcal{S}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Therefore, the least squares estimator  $\hat{\boldsymbol{\beta}}$  of  $\boldsymbol{\beta}$  will satisfy  $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$ . This system of equations is known as the *normal equations*. Assuming that  $\mathbf{X}$  is full column rank, i.e.,  $\text{rank}(\mathbf{X}) = p$ <sup>13</sup>, the matrix  $\mathbf{X}^\top \mathbf{X}$  is full rank and therefore invertible<sup>14</sup>, the least squares estimator of  $\boldsymbol{\beta}$  is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Let us confirm that  $\hat{\boldsymbol{\beta}}$  is indeed a minimum. Taking second derivatives results in

$$\frac{\partial^2 \mathcal{S}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} = \frac{\partial}{\partial \boldsymbol{\beta}^\top} \left( \frac{\partial \mathcal{S}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \right) = \frac{\partial(-2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}^\top} = 2\mathbf{X}^\top \mathbf{X},$$

where in the last equality we have applied item 2. from the differentiation rule with  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ . Because  $\mathbf{X}$  is full column rank, then the matrix  $\mathbf{X}^\top \mathbf{X}$  is positive definite<sup>15</sup> and  $\hat{\boldsymbol{\beta}}$  is the (unique) minimum. If  $\mathbf{X}$  is not of full rank, then there are many least squares estimators  $\hat{\boldsymbol{\beta}}$  that satisfy the normal equations and we say that  $\boldsymbol{\beta}$  is unidentifiable.

It can be easily shown that  $\hat{\boldsymbol{\beta}}$  is unbiased (has expectation equal to the true  $\boldsymbol{\beta}$ ). Recall that for a matrix of constants  $\mathbf{A}$ ,  $\mathbb{E}(\mathbf{A}\mathbf{y}) = \mathbf{A}\mathbb{E}(\mathbf{y})$ . Since  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is a matrix of constants,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}] = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}(\mathbf{y}).$$

Substituting  $\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$  for  $\mathbf{y}$

$$\begin{aligned} \mathbb{E}(\hat{\boldsymbol{\beta}}) &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbb{E}(\mathbf{X}\boldsymbol{\beta}) + \mathbb{E}(\boldsymbol{\epsilon})) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} + \mathbf{0}_n) \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X})\boldsymbol{\beta} \\ &= \mathbf{I}_p \boldsymbol{\beta} \\ &= \boldsymbol{\beta} \end{aligned}$$

<sup>13</sup>That is, all  $p$  columns of  $\mathbf{X}$  are linearly independent.

<sup>14</sup>Note that  $\mathbf{X}$  is not a square matrix (its dimension is  $n \times p$ ) but  $\mathbf{X}^\top \mathbf{X}$  is a  $p \times p$  square matrix.

<sup>15</sup>**Theorem:** Let  $\mathbf{B}$  be an  $n \times p$  matrix. If  $\text{rank}(\mathbf{B}) = p$ , then the matrix  $\mathbf{B}^\top \mathbf{B}$  is positive definite.

The least squares estimator  $\hat{\beta}$  is unbiased for  $\beta$  as long as  $\mathbb{E}(\epsilon) = \mathbf{0}_n$ . To study the variance (or variance-covariance) matrix of  $\hat{\beta}$ , we will need the assumption that  $\text{var}(\epsilon) = \sigma^2 \mathbf{I}_n$  (i.e., that the error terms have constant variance and are independent of each other and thus uncorrelated). Recall also that for a constant matrix  $\mathbf{A}$  and a constant vector  $\mathbf{b}$ ,  $\text{var}(\mathbf{A}\mathbf{y} + \mathbf{b}) = \mathbf{A}\text{var}(\mathbf{y})\mathbf{A}^\top$ . We have<sup>16</sup>

$$\begin{aligned}\mathbf{V}_{\hat{\beta}} &= \text{var}(\hat{\beta}) = \text{var}\{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}\} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{var}(\mathbf{y}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{var}(\mathbf{y}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{var}(\mathbf{y}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{var}(\mathbf{y}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{var}(\mathbf{X}\beta + \epsilon) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{var}(\epsilon) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \sigma^2 \mathbf{I}_n \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 \mathbf{I}_p (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\end{aligned}$$

Note that  $\mathbf{V}_{\hat{\beta}}$  is the  $p \times p$  variance-covariance matrix of the vector  $\hat{\beta}$ . Specifically, the  $j$ th diagonal entry is  $\text{var}(\hat{\beta}_j)$  and the  $(j, j')$ th entry is  $\text{cov}(\hat{\beta}_j, \hat{\beta}_{j'})$ , for  $j = 1, \dots, p$ , and  $j' = 1, \dots, p$ .

In numerical linear algebra, there is a well-known rule of thumb that we should avoid explicitly computing matrix inverses whenever possible. One way to avoid explicit matrix inversion when calculating the least squares estimates of  $\beta$  is to use the so-called QR decomposition, a method that will be introduced in the next section. In fact, this is the method implemented by the `lm` function. To illustrate why matrix inversion should be avoided, we simulate data from an admittedly extreme example, chosen to highlight numerical instability. In particular, we simulate 50 observations from the following linear model

$$y_i = 1 + x_i + x_i^2 + x_i^3 + \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, 1), \quad i = 1, \dots, 50,$$

where  $x_1, \dots, x_{50}$  are a sequence of 50 evenly spaced values between 1 and 500.

```
# Number of observations
n <- 50
# One predictor only
x <- seq(1, 500, len = n)
# Design matrix (polynomial regression)
X <- cbind(1, x, x^2, x^3) #Thus p = 4
colnames(X) <- c("Intercept", "x", "x2", "x3")
# True parameter vector
beta <- matrix(c(1, 1, 1, 1), nrow = 4, ncol = 1)
# Fixing the seed so that results are reproducible
set.seed(1)
# Simulating the error terms
eps <- rnorm(n) # Equivalent to rnorm(n, mean = 0, sd = 1)
# Simulating the response data
y <- X %*% beta + eps

# OLS estimate of beta
solve(t(X) %*% X) %*% t(X) %*% y

## Error in `solve.default()`:
## ! system is computationally singular: reciprocal condition number = 2.93617e-17
```

<sup>16</sup>Note that  $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}$  for a square full-rank matrix  $\mathbf{A}$ .

```
# An equivalent but usually better way of computing OLS estimates
solve(crossprod(X)) %*% crossprod(X, y)

## Error in `solve.default()`:
## ! system is computationally singular: reciprocal condition number = 2.93617e-17
```

Oh, we got an error! Although  $\mathbf{X}^T \mathbf{X}$  is theoretically invertible, explicitly forming and inverting it can be numerically unstable when the columns of  $\mathbf{X}$  are on very different numerical scales (for example, when some columns contain values orders of magnitude larger than others) and/or are highly correlated.

Let us use the `lm` function to see what happens, that is, whether it also returns an error or successfully computes the least squares estimate.

```
lm_fit <- lm(y ~ X - 1)
lm_fit$coefficients

## XIntercept      Xx      Xx2      Xx3
## 0.9038372 1.0066440 0.9999622 1.0000001
```

We did not get an error and the estimated values are very close to the true values used to simulate the data. Let us then learn the method `lm` is using.

## 4.1 QR decomposition

In this section we introduce the QR method for finding the least squares estimator  $\hat{\beta}$ . As with any real matrix, the design matrix  $\mathbf{X}$  can always be decomposed as

$$\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} = \mathbf{Q}\mathbf{R}, \quad (7)$$

where  $\mathbf{R}$  is a  $p \times p$  upper triangular matrix<sup>17</sup>,  $\mathbf{0}$  is an  $(n - p) \times p$  matrix of zeroes, and  $\mathbf{Q}$  is an  $n \times n$  orthogonal matrix, the first  $p$  columns of which form  $\mathbf{Q}$ . Recall that an orthogonal matrix  $\mathbf{Q}$  is a square matrix whose columns and rows are orthogonal unit vectors. It therefore holds that  $\mathbf{Q}\mathbf{Q}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$ . Note also that  $\mathbf{Q}$  is an  $n \times p$  matrix with orthonormal columns, but non-orthonormal rows, so that  $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_p$  but  $\mathbf{Q}\mathbf{Q}^T \neq \mathbf{I}_n$ . Further note that orthogonal matrices rotate/reflect vectors, but do not change their length. Hence, multiplying  $\mathbf{y} - \mathbf{X}\beta$  by  $\mathbf{Q}^T$  implies that<sup>18</sup>

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 = \|\mathbf{Q}^T(\mathbf{y} - \mathbf{X}\beta)\|^2 = \|\mathbf{Q}^T\mathbf{y} - \mathbf{Q}^T\mathbf{X}\beta\|^2 = \left\| \mathbf{Q}^T\mathbf{y} - \mathbf{Q}^T\mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} \beta \right\|^2 = \left\| \mathbf{Q}^T\mathbf{y} - \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} \beta \right\|^2.$$

Note that  $\mathbf{Q}^T\mathbf{y}$  is an  $(n \times 1)$  vector and defining  $\mathbf{f}$  as a  $p \times 1$  dimensional vector and  $\mathbf{r}$  as a  $(n - p) \times 1$  vector, so that

$$\mathbf{Q}^T\mathbf{y} = \begin{pmatrix} [\mathbf{Q}^T\mathbf{y}]_{1:p} \\ [\mathbf{Q}^T\mathbf{y}]_{(p+1):n} \end{pmatrix} \equiv \begin{pmatrix} \mathbf{f} \\ \mathbf{r} \end{pmatrix},$$

yields

$$\|\mathbf{y} - \mathbf{X}\beta\|^2 = \left\| \begin{pmatrix} \mathbf{f} \\ \mathbf{r} \end{pmatrix} - \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} \beta \right\|^2 = \left\| \begin{pmatrix} \mathbf{f} \\ \mathbf{r} \end{pmatrix} - \begin{pmatrix} \mathbf{R}\beta \\ \mathbf{0}_{n-p} \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} \mathbf{f} - \mathbf{R}\beta \\ \mathbf{r} \end{pmatrix} \right\|^2 = \|\mathbf{f} - \mathbf{R}\beta\|^2 + \|\mathbf{r}\|^2.$$

If the final equality is not obvious, recall that  $\|\mathbf{x}\|^2 = \sum_i x_i^2$ , so if  $\mathbf{x} = \begin{pmatrix} \mathbf{v} \\ \mathbf{w} \end{pmatrix}$ ,  $\|\mathbf{x}\|^2 = \sum_i v_i^2 + \sum_i w_i^2 = \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2$ .

<sup>17</sup>That is,  $R_{ij} = 0$  if  $i > j$ .

<sup>18</sup> $\|\mathbf{Q}^T(\mathbf{y} - \mathbf{X}\beta)\|^2 = [\mathbf{Q}^T(\mathbf{y} - \mathbf{X}\beta)]^T [\mathbf{Q}^T(\mathbf{y} - \mathbf{X}\beta)] = (\mathbf{y} - \mathbf{X}\beta)^T \mathbf{Q}\mathbf{Q}^T(\mathbf{y} - \mathbf{X}\beta) = (\mathbf{y} - \mathbf{X}\beta)^T \mathbf{I}_n(\mathbf{y} - \mathbf{X}\beta) = \|\mathbf{y} - \mathbf{X}\beta\|^2$

Recall also that our goal is to find the vector  $\beta$  that minimises  $\|\mathbf{y} - \mathbf{X}\beta\|^2$  or, equivalently, the vector  $\beta$  that minimises  $\|\mathbf{f} - \mathbf{R}\beta\|^2 + \|\mathbf{r}\|^2$ . The length of  $\mathbf{r}$  does not depend on  $\beta$  and  $\|\mathbf{f} - \mathbf{R}\beta\|^2$  can be reduced to zero by choosing  $\beta$  so that  $\mathbf{R}\beta$  equals  $\mathbf{f}$ . That is, the least squares estimator satisfies

$$\mathbf{R}\hat{\beta} = \mathbf{f}. \quad (8)$$

Assuming that  $\mathbf{X}$  is full column rank and hence  $\mathbf{R}$  also has full rank, we have

$$\hat{\beta} = \mathbf{R}^{-1}\mathbf{f} = \mathbf{R}^{-1}[\mathbf{Q}^\top \mathbf{y}]_{1:p} = \mathbf{R}^{-1}\mathbf{Q}^\top \mathbf{y}, \quad (9)$$

since  $\mathbf{Q}$  is by construction formed by the first  $p$  columns of  $\mathbf{Q}$ . Note that because  $\mathbf{R}$  is an upper triangular matrix, the system of equations in (8) can be easily solved by back substitution, thus allowing to easily find  $\hat{\beta}$  without the need to invert any matrix. Further note that inverting  $\mathbf{R}$  is numerically more stable than inverting  $\mathbf{X}^\top \mathbf{X}$ .

Notice that

$$\|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 = \|\mathbf{f} - \mathbf{R}\hat{\beta}\|^2 + \|\mathbf{r}\|^2 = \|\mathbf{f} - \mathbf{R}\mathbf{R}^{-1}\mathbf{f}\|^2 + \|\mathbf{r}\|^2,$$

and thus  $\|\mathbf{r}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2$  is the *residual sum of squares* for the model fit.

Note that the estimator we have obtained before,  $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  is obviously equivalent to  $\hat{\beta} = \mathbf{R}^{-1} \mathbf{f}$ <sup>19</sup>

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \\ &= ((\mathbf{Q}\mathbf{R})^\top \mathbf{Q}\mathbf{R})^{-1} (\mathbf{Q}\mathbf{R})^\top \mathbf{y} \\ &= (\mathbf{R}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{R})^{-1} (\mathbf{Q}\mathbf{R})^\top \mathbf{y} \\ &= (\mathbf{R}^\top \mathbf{R})^{-1} (\mathbf{Q}\mathbf{R})^\top \mathbf{y} \\ &= \mathbf{R}^{-1} (\mathbf{R}^\top)^{-1} \mathbf{R}^\top \mathbf{Q}^\top \mathbf{y} \\ &= \mathbf{R}^{-1} \mathbf{Q}^\top \mathbf{y} \\ &= \mathbf{R}^{-1} \mathbf{f}. \end{aligned}$$

Of course, the same equivalence also holds for the covariance matrix. Note that the covariance matrix of  $\mathbf{f} = \mathbf{Q}^\top \mathbf{y}$  is

$$\text{var}(\mathbf{f}) = \text{var}(\mathbf{Q}^\top \mathbf{y}) = \mathbf{Q}^\top \text{var}(\mathbf{y}) \mathbf{Q} = \sigma^2 \mathbf{Q}^\top \mathbf{I}_n \mathbf{Q} = \sigma^2 \mathbf{I}_p,$$

which implies that

$$\mathbf{V}_{\hat{\beta}} = \text{var}(\hat{\beta}) = \text{var}(\mathbf{R}^{-1} \mathbf{f}) = \mathbf{R}^{-1} \text{var}(\mathbf{f}) \mathbf{R}^{-\top} = \mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2.$$

Remember that we have found before that  $\mathbf{V}_{\hat{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \sigma^2$ . Finally, we only need to notice that

$$\begin{aligned} \mathbf{V}_{\hat{\beta}} &= (\mathbf{X}^\top \mathbf{X})^{-1} \sigma^2 = (\mathbf{R}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{R})^{-1} \sigma^2 = (\mathbf{R}^\top \mathbf{R})^{-1} \sigma^2 \\ &= \mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2. \end{aligned}$$

A natural question that arises after all this is: how do we determine  $\mathbf{Q}$  and  $\mathbf{R}$ ? Three methods for finding  $\mathbf{Q}$  and  $\mathbf{R}$  are associated with the names of:

- Householder:  $\mathbf{Q}$  is a product of reflections.
- Givens:  $\mathbf{Q}$  is a product of rotations.
- Gram-Schmidt: successive orthogonalization.

In practice we will use the `lm` function, which utilises the QR decomposition behind the scenes. There is also the `qr` function that implements the QR decomposition. Let us now get an ‘hands-on’ understanding of the QR decomposition. To do so, we revisit the simulated data example which we used to illustrate how directly inverting  $\mathbf{X}^\top \mathbf{X}$  can be problematic. In the code below `Q_orthogonal` denotes  $\mathbf{Q}$  and `Q` denotes the matrix  $\mathbf{Q}$ , which is formed by the first  $p$  rows of  $\mathbf{Q}$ .

<sup>19</sup>If  $\mathbf{A}$  and  $\mathbf{B}$  are two square invertible matrices of the same dimension, then  $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ .

```

# QR decomposition of matrix X
qrx <- qr(X)
# n times n orthogonal matrix Q
Q_orthogonal <- qr.Q(qrx, complete = TRUE)
# R matrix
R <- qr.R(qrx)
# Q matrix (of dimension n times p)
Q <- qr.Q(qrx)
# Oh, R it is upper triangular (note that here p = 4)
R

##      Intercept      x      x2      x3
## [1,] -7.071068 -1771.302 -596425.3 -225914269
## [2,]  0.000000  1039.159  520618.6  235997829
## [3,]  0.000000   0.000 -136509.6 -102586942
## [4,]  0.000000   0.000   0.0  17591856

# Let us look at properties of Q_orthogonal
dim(Q_orthogonal)

## [1] 50 50

Qort_Qort_t <- Q_orthogonal%*%t(Q_orthogonal) #identity matrix
#alternatively the above could be computed as tcrossprod(Q_orthogonal, Q_orthogonal)

Qort_t_Qort <- t(Q_orthogonal)%*%Q_orthogonal #identity matrix
#alternatively the above could be computed as crossprod(Q_orthogonal, Q_orthogonal)

# Now check the properties of Q
Q_t_Q <- crossprod(Q, Q) #identity matrix
Q_t_Q

##      [,1]      [,2]      [,3]      [,4]
## [1,] 1.000000e+00 5.551115e-17 5.551115e-17 8.326673e-17
## [2,] 5.551115e-17 1.000000e+00 0.000000e+00 3.469447e-18
## [3,] 5.551115e-17 0.000000e+00 1.000000e+00 0.000000e+00
## [4,] 8.326673e-17 3.469447e-18 0.000000e+00 1.000000e+00

Q_Q_t <- tcrossprod(Q, Q) #not an identity matrix

# Determine beta hat through the system R \beta = Q_t y
betahat <- backsolve(R, crossprod(Q, y))
betahat

##      [,1]
## [1,] 0.9038372
## [2,] 1.0066440
## [3,] 0.9999622
## [4,] 1.0000001

# Directly through the function qr.solve
qr.solve(X, y)

##      [,1]
## Intercept 0.9038372
## x         1.0066440
## x2        0.9999622
## x3        1.0000001

```

```

# Compare with lm results
lm_fit$coefficients

## XIntercept      Xx      Xx2      Xx3
## 0.9038372  1.0066440  0.9999622  1.0000001

# Let us obtain f and r as defined above in the text
f <- t(Q)%*%y
# Or alternatively
p <- 4
aux <- t(Q_orthogonal)%*%y
f_alt <- aux[1:p]
f; f_alt

##           [,1]
## [1,] -226512473
## [2,]  236519487
## [3,] -102723452
## [4,]   17591857
## [1] -226512473  236519487 -102723452   17591857

r <- aux[(p + 1):n]

# Least squares estimates
backsolve(R, f)

##           [,1]
## [1,] 0.9038372
## [2,] 1.0066440
## [3,] 0.9999622
## [4,] 1.0000001

# Alternatively
solve(R, f) # help(solve)

##           [,1]
## Intercept 0.9038372
## x         1.0066440
## x2        0.9999622
## x3        1.0000001

# check that the norm of vector r indeed is the residual sum of squares (rss)
r_norm <- as.numeric(t(r)%*%r)
rss <- sum((y - X%*%betahat)^2)
r_norm; rss

## [1] 32.83243
## [1] 32.83243

sum(lm_fit$residuals^2)

## [1] 32.83243

```

## 4.2 Checking

Further inference, beyond simply estimating  $\beta$ , requires that the assumptions of independence and constant variance of the  $\epsilon_i$  hold. Consequently, we need some way of checking these assumptions if our inference is to be

sound. Formally, we will also assume normality of the  $\epsilon_i$ , but in practice this assumption is often reasonably robust due to the central limit theorem.

The most useful checks of model assumptions are often graphical, as these tend to indicate not just that an assumption is violated, but also *how* it is violated. This in turn may allow us to fix the problem. Once we have estimated a linear model, all information about model fit is contained in the model *residuals*

$$\hat{\epsilon}_i = y_i - \hat{\mu}_i, \quad i = 1, \dots, n,$$

where  $\hat{\mu} = \mathbf{X}\hat{\beta}$ . To check the model assumptions the  $\hat{\epsilon}_i$  are usually plotted against the model predictor variables and the *fitted values*,  $\hat{\mu}_i$ . Systematic pattern in the mean of the residuals often results from something wrong with how the dependence on a predictor variable has been specified. Systematic pattern in the variability of the residuals indicates violation of the constant variance assumption. This is most often manifested in the plot of  $\hat{\epsilon}_i$  against  $\hat{\mu}_i$ . We will revisit this in more detail in the next section.

### 4.3 Further inference results: intervals and testing

Having covered estimation and checking, we can now consider how to answer the inferential questions:

1. What range of parameter values are consistent with the data?
2. Are some pre-specified values (or restrictions) for the parameters consistent with the data?

In the specific case of linear models, we want to find confidence intervals for the  $\beta_j$  and to *test hypotheses* about the  $\beta_j$ . For example, we often want to test the null hypothesis that one or several elements of  $\beta$  are zero, corresponding to terms being omitted from the linear model.

As we have already seen in the case of a simple linear model, confidence intervals and hypothesis tests are probabilistic concepts, so at this stage we need to turn our linear model into a fully probabilistic model, by specifying a distribution for the  $\epsilon_i$ . We assume the errors  $\epsilon_i$  are independent and identically distributed as  $N(0, \sigma^2)$ , for all  $i = 1, \dots, n$ , or, equivalently, that  $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$  which, in turn, implies that  $\mathbf{y} \sim N(\mathbf{X}\beta, \sigma^2 \mathbf{I}_n)$ , and because  $\hat{\beta}$  is a linear transformation of  $\mathbf{y}$ , it follows that  $\hat{\beta}$  is also normally distributed

$$\hat{\beta} \sim N_p(\beta, \mathbf{V}_{\hat{\beta}}), \text{ where } \mathbf{V}_{\hat{\beta}} = \mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2. \quad (10)$$

This is not yet in the most directly usable form: it gives the distribution of  $\hat{\beta}$  in terms of the unknowns  $\beta$  and  $\sigma^2$ . As we saw in Section 2.4.1, if we want to make inferences about  $\beta_j$  it is helpful if we can start from a *pivotal* statistic, i.e., a function of the data and unobservable parameters whose distribution does not depend on any unknown parameters.

Note that, as a result of (10), each individual regression coefficient follows a univariate normal distribution

$$\hat{\beta}_j \sim N(\beta_j, \sigma_{\hat{\beta}_j}^2), \quad \sigma_{\hat{\beta}_j}^2 = [\mathbf{V}_{\hat{\beta}}]_{jj} = [\mathbf{R}^{-1} \mathbf{R}^{-\top}]_{jj} \sigma^2, \quad j = 1, \dots, p.$$

Further note that, in general,  $\hat{\beta}_j$  and  $\hat{\beta}_{j'}$  are not independent for  $j \neq j'$ , since the off-diagonal entries of  $\mathbf{V}_{\hat{\beta}}$  are typically nonzero.

#### 4.3.1 $(\hat{\beta}_j - \beta_j) / \hat{\sigma}_{\hat{\beta}_j} \sim t_{n-p}$

First, consider the distribution of  $\mathbf{Q}^\top \mathbf{y}$ . Since it is a linear transformation of the normal random vector  $\mathbf{y}$ , it must also be a normal random vector, and its covariance matrix is

$$\mathbf{V}_{\mathbf{Q}^\top \mathbf{y}} = \text{var}(\mathbf{Q}^\top \mathbf{y}) = \mathbf{Q}^\top \mathbf{I} \mathbf{Q} \sigma^2 = \mathbf{I}_n \sigma^2.$$

Note that the off-diagonal entries of  $\mathbf{V}_{\mathbf{Q}^\top \mathbf{y}}$  are zero, meaning that the covariance between any two elements of  $\mathbf{Q}^\top \mathbf{y}$  is zero. Since, for the normal distribution only, zero covariance implies independence, the elements of  $\mathbf{Q}^\top \mathbf{y}$  are mutually independent. Furthermore,

$$\mathbb{E} \begin{pmatrix} \mathbf{f} \\ \mathbf{r} \end{pmatrix} = \mathbb{E}(\mathbf{Q}^\top \mathbf{y}) = \mathbf{Q}^\top \mathbb{E}(\mathbf{y}) = \mathbf{Q}^\top \mathbf{X} \beta = \mathbf{Q}^\top \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} \beta = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} \beta = \begin{pmatrix} \mathbf{R} \beta \\ \mathbf{0} \end{pmatrix}$$

$$\Rightarrow \mathbb{E}(\mathbf{f}) = \mathbf{R}\boldsymbol{\beta} \text{ and } \mathbb{E}(\mathbf{r}) = \mathbf{0}_{n-p}.$$

So we have that

$$\mathbf{f} \sim N(\mathbf{R}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_p) \text{ and } \mathbf{r} \sim N(\mathbf{0}_{n-p}, \sigma^2 \mathbf{I}_{n-p}),$$

with both vectors independent of each other (as, by construction, they are the constituents of  $\mathbf{Q}^\top \mathbf{y}$  whose elements are mutually independent).

Because the  $n - p$  elements of  $\mathbf{r}$  are i.i.d.  $N(0, \sigma^2)$  random variables and, as such, the random variable  $\frac{r_i}{\sigma}$ , for  $i = 1, \dots, n - p$ , also follows a normal distribution with mean zero but with variance equal to one<sup>20</sup>, leads to

$$\frac{1}{\sigma^2} \|\mathbf{r}\|^2 = \frac{1}{\sigma^2} \sum_{i=1}^{n-p} r_i^2 = \sum_{i=1}^{n-p} \left( \frac{r_i}{\sigma} \right)^2 \sim \chi_{n-p}^2.$$

Recall that because  $r_i/\sigma \sim N(0, 1)$ , then  $(r_i/\sigma)^2 \sim \chi_1^2$  and because all  $r_i$ 's are independent of each other for  $i = 1, \dots, n - p$ , then  $\sum_{i=1}^{n-p} r_i^2/\sigma^2 \sim \chi_{n-p}^2$ . The mean of a  $\chi_{n-p}^2$  random variable is  $n - p$ , so this result is sufficient (but not necessary; see Workshop sheet 1) to imply that

$$\hat{\sigma}^2 = \|\mathbf{r}\|^2/(n - p) \quad (11)$$

is an unbiased estimator of  $\sigma^2$ . To see why  $\hat{\sigma}^2$  is unbiased observe that

$$\mathbb{E} \left( \frac{1}{\sigma^2} \|\mathbf{r}\|^2 \right) = n - p \Rightarrow \mathbb{E} \left( \frac{\|\mathbf{r}\|^2}{n - p} \right) = \sigma^2 \Rightarrow \mathbb{E}(\hat{\sigma}^2) = \sigma^2.$$

The independence of  $\mathbf{r}$  and  $\mathbf{f}$  also implies that  $\hat{\boldsymbol{\beta}}$  and  $\hat{\sigma}^2$  are independent, as  $\hat{\boldsymbol{\beta}}$  depends only  $\mathbf{f}$  and  $\hat{\sigma}^2$  depends only on  $\mathbf{r}$ .

Now consider a single-parameter estimator,  $\hat{\beta}_j$ , for  $j = 1, \dots, p$ , with standard deviation,  $\sigma_{\hat{\beta}_j}$ , given by the square root of element  $(j, j)$  of  $\mathbf{V}_{\hat{\boldsymbol{\beta}}} = \mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2$ . An unbiased estimator of  $\mathbf{V}_{\hat{\boldsymbol{\beta}}}$  is obtained by replacing  $\sigma^2$  by its unbiased estimator  $\hat{\sigma}^2$  (see Equation (11))<sup>21</sup>

$$\hat{\mathbf{V}}_{\hat{\boldsymbol{\beta}}} = \mathbf{R}^{-1} \mathbf{R}^{-\top} \hat{\sigma}^2 = \left( \frac{1}{\sigma^2} \sigma^2 \right) \mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2 = \frac{1}{\sigma^2} (\mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2) \hat{\sigma}^2 = \frac{1}{\sigma^2} \mathbf{V}_{\hat{\boldsymbol{\beta}}} \hat{\sigma}^2,$$

where the last equality will be helpful to determine the pivotal statistic. An estimator of  $\sigma_{\hat{\beta}_j}$ ,  $\hat{\sigma}_{\hat{\beta}_j}$ , is therefore given by the square root of element  $(j, j)$  of  $\hat{\mathbf{V}}_{\hat{\boldsymbol{\beta}}}$  and it is clear that  $\hat{\sigma}_{\hat{\beta}_j} = \sigma_{\hat{\beta}_j} \hat{\sigma}/\sigma$ . Hence, the pivotal statistic is

$$T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{\hat{\beta}_j}} = \frac{\hat{\beta}_j - \beta_j}{\sigma_{\hat{\beta}_j} \hat{\sigma}/\sigma} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}}{\sqrt{\frac{1}{\sigma^2} \hat{\sigma}^2}} = \frac{(\hat{\beta}_j - \beta_j)/\sigma_{\hat{\beta}_j}}{\sqrt{\frac{1}{\sigma^2} \|\mathbf{r}\|^2/(n - p)}} \sim \frac{N(0, 1)}{\sqrt{\chi_{n-p}^2/(n - p)}} \sim t_{n-p}, \quad (12)$$

where the independence of  $\hat{\beta}_j$  and  $\hat{\sigma}^2$  has been used, along with the definition of a  $t$  random variable (in particular, note that if  $Z_1 \sim N(0, 1)$  and independently  $Z_2 \sim \chi_n^2$ , then  $T = Z_1/\sqrt{Z_2/n} \sim t_n$ ). This result can be used for any  $\beta_j$  exactly as it was used for the single  $\beta$  in Sections 2.4.1 and 2.4.2 (with the implied adjustment of degrees of freedom, of course).

### 4.3.2 Testing $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$

Now consider a more general testing problem. Recall that factor variables classify observations into groups, and in a linear model each factor typically corresponds to several regression parameters (in principle, one for each group). If we want to test whether the factor variable should be included in the model, we must therefore test whether all of its associated parameters could simultaneously be zero. This is a special case of the general null hypothesis

<sup>20</sup>Note that  $\text{var} \left( \frac{r_i}{\sigma} \right) = \frac{1}{\sigma^2} \text{var}(r_i) = \frac{1}{\sigma^2} \sigma^2 = 1$ .

<sup>21</sup> $\mathbb{E}(\hat{\mathbf{V}}_{\hat{\boldsymbol{\beta}}}) = \mathbb{E}(\mathbf{R}^{-1} \mathbf{R}^{-\top} \hat{\sigma}^2) = \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbb{E}(\hat{\sigma}^2) = \mathbf{R}^{-1} \mathbf{R}^{-\top} \sigma^2 = \mathbf{V}_{\hat{\boldsymbol{\beta}}}$ .

$H_0 : \mathbf{C}\beta = \mathbf{d}$  (against  $H_1 : \mathbf{C}\beta \neq \mathbf{d}$ ). Note that approaching such a testing problem with a separate test for each parameter of the factor is a bad idea: how would you go about combining the different p-values, for example?

Suppose that  $\mathbf{C}$  is a  $q \times p$  matrix and  $\mathbf{d}$  a  $q \times 1$  dimensional vector, where  $q < p$ . Under  $H_0$ , we have that  $\mathbf{C}\hat{\beta} - \mathbf{d} \sim N(\mathbf{0}_q, \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)$ , from basic properties of the transformation of normal random vectors. In particular, since  $\hat{\beta} \sim N(\beta, \mathbf{V}_{\hat{\beta}})$ , then  $\mathbf{C}\hat{\beta} \sim N(\mathbf{C}\beta, \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)$  and because under  $H_0$   $\mathbf{C}\beta = \mathbf{d}$ , we finally have that  $\mathbf{C}\hat{\beta} - \mathbf{d} \sim N(\mathbf{0}_q, \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)$ . Any positive definite matrix can be factored into the crossproduct of a triangular matrix with itself: a *Cholesky decomposition*. Forming the Cholesky decomposition  $\mathbf{L}^\top\mathbf{L} = \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top$  (if the design matrix  $\mathbf{X}$  is full column rank, then  $\mathbf{V}_{\hat{\beta}}$  is positive definite), where  $\mathbf{L}$  is a  $q \times q$  upper triangular matrix, it is then easy to show that, under  $H_0$ ,  $\mathbf{L}^{-\top}(\mathbf{C}\hat{\beta} - \mathbf{d}) \sim N(\mathbf{0}_q, \mathbf{I}_q)^{22}$ , so

$$\begin{aligned} (\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d}) &= (\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{L}^\top\mathbf{L})^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\ &= (\mathbf{C}\hat{\beta} - \mathbf{d})^\top \mathbf{L}^{-1} \mathbf{L}^{-\top} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\ &= [\mathbf{L}^{-\top}(\mathbf{C}\hat{\beta} - \mathbf{d})]^\top \mathbf{L}^{-\top} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\ &= \|\mathbf{L}^{-\top}(\mathbf{C}\hat{\beta} - \mathbf{d})\|^2 \\ &= \sum_{i=1}^q ([\mathbf{L}^{-\top}(\mathbf{C}\hat{\beta} - \mathbf{d})]_i)^2 \\ &\sim \sum_{i=1}^q N(0, 1)^2 \\ &\sim \chi_q^2. \end{aligned}$$

As in the previous section, plugging in  $\hat{\mathbf{V}}_{\hat{\beta}} = \mathbf{V}_{\hat{\beta}}\hat{\sigma}^2/\sigma^2$  gives the computable test statistic and its distribution under  $H_0$ :

$$\begin{aligned} F &= \frac{1}{q} (\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\hat{\mathbf{V}}_{\hat{\beta}}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\ &= \frac{1}{q} (\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\mathbf{V}_{\hat{\beta}}(\hat{\sigma}^2/\sigma^2)\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\ &= \frac{\sigma^2}{q\hat{\sigma}^2} (\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\ &= \frac{(\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d})/q}{\hat{\sigma}^2/\sigma^2} \\ &= \frac{(\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d})/q}{\frac{1}{\hat{\sigma}^2} \|\mathbf{r}\|^2/(n-p)} \\ &\sim \frac{\chi_q^2/q}{\chi_{n-p}^2/(n-p)} \sim F_{q, n-p}, \end{aligned} \tag{13}$$

where again the independence between  $\hat{\beta}$  and  $\hat{\sigma}^2$  was used. This result can be used to compute a p-value for the test. Recall that if  $Z_n \sim \chi_n^2$  and, independently,  $Z_m \sim \chi_m^2$ , then  $F = \frac{Z_n/n}{Z_m/m} \sim F_{n,m}$ , a  $F$  distribution with  $n$  and  $m$  degrees of freedom. The  $F$  distribution, defined as the ratio of two chi-squared distributions, is restricted to the positive real line. As with the other tests, we reject the null hypothesis if the observed value of the  $F$  statistic, our test statistic, lies in an ‘extreme’ region of the corresponding  $F$  distribution. This is a one-sided test meaning that we reject  $H_0$  if the  $F$  statistic is larger than the 95% quantile of the corresponding  $F_{q, n-p}$  distribution (if we use a significance level of 5%, of course).

Apart from testing the significance of factor variables, this test for general linear hypotheses has other interesting testing problems as special cases. For instance, consider the following hypothetical model

$$y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \beta_4 x_{3i} + \epsilon_i,$$

---

<sup>22</sup> $\text{var}\{\mathbf{L}^{-\top}(\mathbf{C}\hat{\beta} - \mathbf{d})\} = \mathbf{L}^{-\top} \text{var}(\mathbf{C}\hat{\beta} - \mathbf{d})(\mathbf{L}^{-\top})^\top = \mathbf{L}^{-\top} \mathbf{C}\mathbf{V}_{\hat{\beta}}\mathbf{C}^\top (\mathbf{L}^{-\top})^\top = \mathbf{L}^{-\top} (\mathbf{L}^\top\mathbf{L})(\mathbf{L}^{-\top})^\top = (\mathbf{L}^\top)^{-1} \mathbf{L}^\top \mathbf{L} \mathbf{L}^{-1} = \mathbf{I}_q$

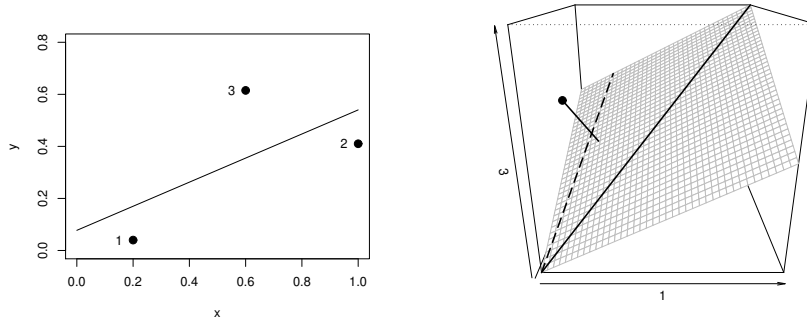


Figure 4: Illustration of the geometry of least squares. Left: a straight line fit to three  $(x, y)$  data points. Right: the space in which the  $y$  coordinates of the data define a single point, while the columns of the model matrix (solid and dashed line) span the subspace shown in grey. The least squares estimate of  $\mathbb{E}(y)$  is the orthogonal projection of the data point onto the model subspace.

and suppose further that we want to test  $H_0 : \beta_3 = \beta_4$  (or, equivalently,  $H_0 : \beta_3 - \beta_4 = 0$ ) against the alternative  $H_1 : \beta_3 \neq \beta_4$ . Then, in this case,  $\mathbf{C} = (0, 0, 1, -1)$  and  $\mathbf{d} = 0$ . Alternatively, suppose that for the same model interest instead lies in testing  $\beta_3 = \beta_4 = 0$ . The question of interest corresponding to this hypothesis is: Can  $y$  be explained by fewer variables? The alternative hypothesis,  $H_1$  is that, at least, one of the two coefficients is different from zero. In this case we have  $\mathbf{d} = (0, 0)^T$  and  $\mathbf{C} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ , leading to

$$\mathbf{C}\boldsymbol{\beta} = \begin{pmatrix} \beta_3 \\ \beta_4 \end{pmatrix} = \mathbf{d} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Finally, note that when the null hypothesis is that a single coefficient is zero, then (13) is equivalent to (the squared version of) (12). In this case,  $\mathbf{d}$  is a scalar, and  $\mathbf{C}$  is a row vector of dimension  $p$  with one of the entries equal to one and all the other  $p - 1$  entries are zero.

When our interest is in testing whether several elements of  $\boldsymbol{\beta}$  could simultaneously be zero, then  $\mathbf{d} = \mathbf{0}$  while  $\mathbf{C}$  is made up of a selection of rows of the  $p \times p$  identity matrix (as we have just illustrated in the example above). In that case let  $\text{RSS}_1$  denote the residual sum of squares for the full model, and  $\text{RSS}_0$  denote the residual sum of squares for the reduced model in which the  $H_0$  is true. It can be shown that the above test statistic then reduces to

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/q}{\text{RSS}_1/(n - p)}.$$

#### 4.4 The geometry of least squares

Least squares estimation of linear models amounts to finding the orthogonal projection of the  $n$ -dimensional response vector  $\mathbf{y}$  onto the  $p$ -dimensional linear subspace spanned by the columns of  $\mathbf{X}$ . The linear model states that  $\mathbb{E}(\mathbf{y}) = \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$  lies in the space spanned by all possible linear combinations of the columns of the  $n \times p$  model matrix  $\mathbf{X}$ , and least squares seeks the point in that space that is closest to  $\mathbf{y}$  in Euclidean distance. That is, the least squares estimate of  $\mathbb{E}(\mathbf{y})$ ,  $\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ , is the orthogonal projection of the data point  $\mathbf{y}$  in the space spanned by all possible linear combinations of the columns of  $\mathbf{X}$ . Figure 4 illustrates this geometry for the model,

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} .05 \\ .40 \\ .65 \end{pmatrix} = \begin{pmatrix} 1 & .1 \\ 1 & 1 \\ 1 & .6 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix}.$$

The following YouTube video provides a clear explanation of the geometric interpretation of ordinary least squares.

[www.youtube.com/watch?v=oWuhZuLOEFY](http://www.youtube.com/watch?v=oWuhZuLOEFY)

Given this view of least squares fitting as orthogonal projection it is interesting to look at the projection matrix that maps the data  $\mathbf{y}$  to the fitted values  $\hat{\boldsymbol{\mu}}$ . We have

$$\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Q}\mathbf{R}\mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y} = \mathbf{Q}\mathbf{Q}^T\mathbf{y}.$$

So the required projection matrix is  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^T$ , which is often known as the *influence matrix*, or *hat matrix*<sup>23</sup> of

<sup>23</sup>It puts 'a hat' on  $\mathbf{y}$ :  $\hat{\mathbf{y}} = \hat{\mathbb{E}}(\mathbf{y}) = \hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{y}$ .

the linear model. In terms of the design matrix  $\mathbf{A} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . One interesting property of  $\mathbf{A}$  is that it is idempotent:  $\mathbf{A}\mathbf{A} = \mathbf{A}$ <sup>24</sup>. In a way this is obvious, as the orthogonal projection of  $\hat{\boldsymbol{\mu}}$  onto the column space of  $\mathbf{X}$  must be  $\hat{\boldsymbol{\mu}}$ . Also,  $\mathbf{A} = \mathbf{A}^\top$ . The hat matrix is important for checking assumptions of the linear model and for model diagnostics (more on this in the following section).

## 4.5 Maximum likelihood estimation of $\boldsymbol{\beta}$

We obtained the least squares estimator of  $\boldsymbol{\beta}$  without making any specific distributional assumptions about the error term  $\epsilon$ . Assuming normally distributed errors,  $\epsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ , the maximum likelihood estimator for the vector of regression coefficients is obtained and coincides with the least squares estimator. Under this assumption,  $\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$ , which yields the likelihood

$$L(\boldsymbol{\beta}, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \right\}.$$

The log likelihood is thus

$$\ell(\boldsymbol{\beta}, \sigma^2) = \log L(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (14)$$

When maximising the log likelihood with respect to  $\boldsymbol{\beta}$ , we can ignore the first two terms in Equation (14) because they are independent of  $\boldsymbol{\beta}$ . Moreover, maximising  $-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$  is equivalent to minimising  $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ , i.e., the least squares criterion. Therefore, the maximum likelihood estimator of  $\boldsymbol{\beta}$  is identical to the least squares estimator.

## 4.6 Further considerations

Before proceeding to the next section, a couple of considerations are worth noting.

- An effect  $\beta$  in a linear model with a single covariate, say  $x_1$ , is usually not the same as the effect, say  $\beta_1$  of the same variable  $x_1$  in a model with multiple covariates. The effect  $\beta$  is a *marginal* effect, ignoring all other potential covariates, whereas  $\beta_1$  is a *conditional* effect, adjusting for the other covariates in the model. While this should be obvious, it is always worth reiterating.
- The theory of linear models presented assumes that design matrix  $\mathbf{X}$  is fixed. If we repeat a study, we expect the response variable observations  $\mathbf{y}$  to change, but if  $\mathbf{X}$  is fixed, then the covariate values should be constant across replications of the study<sup>25</sup>. This situation accurately describes an experiment, where covariates are under the researcher's control. However, in observational studies (common in the social sciences, education, and many applied fields) the covariates typically vary between replications, and hence  $\mathbf{X}$  is random rather than fixed. Fortunately, the statistical theory of linear models, formulated under the assumption of a fixed design, remains valid under mild conditions when  $\mathbf{X}$  is random.

<sup>24</sup>  $\mathbf{A}\mathbf{A} = (\mathbf{Q}\mathbf{Q}^\top)(\mathbf{Q}\mathbf{Q}^\top) = \mathbf{Q}(\mathbf{Q}^\top \mathbf{Q})\mathbf{Q}^\top = \mathbf{Q}\mathbf{I}_p\mathbf{Q}^\top = \mathbf{Q}\mathbf{Q}^\top = \mathbf{A}$

<sup>25</sup> How realistic do you find this assumption in the Hubble's law example?

## 5 Linear models in R

Having covered most of the theory needed to understand linear modelling, it is time to focus on application of that theory, so this section will cover the use of linear models in more detail.

### 5.1 The `cars` data again

We will use the `cars` data again in order to relate the theory covered in Section 4 to practical analysis with R in the context of a now familiar example.

#### 5.1.1 Confidence intervals and hypothesis tests for single parameters

This section will use the result of Section 4.3.1, to find confidence intervals for, and test hypotheses about, model parameters. First refit the full quadratic model:

$$\text{dist}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2),$$

to the data in the `cars` data frame in R.

```
data(cars)
cm <- lm(dist ~ speed + I(speed^2), data = cars)
```

From the resulting fitted model object, `cm`, it is possible to extract all the quantities needed to construct the  $T$  statistics, covered in Section 4.3.1. The following extracts,  $\hat{\beta}$ ,  $\hat{\mathbf{V}}_{\beta}$  and the  $\hat{\sigma}_{\hat{\beta}_j}$ 's respectively:

```
beta.hat <- coef(cm)
# Alternative way of extracting the regression coefficients
# cm$coefficients
# summary(cm)$coefficients[,1]

# Estimated variance-covariance matrix of estimated parameters
V.beta <- vcov(cm)

# Estimated standard deviation of estimated parameters
sigma.beta <- sqrt(diag(V.beta))
```

#### Confidence intervals

Now, using the result from Section 4.3.1, exactly as it was used in Sections 2.4.1 and 2.4.2, we can find confidence intervals for the  $\beta_j$ 's,  $j = 1, 2, 3$ . Let's start by finding a 90% CI for  $\beta_2$ . This will involve using the `qt` function to evaluate the value above which 5% of  $t_{47}$  (note that  $n = 50$  and  $p = 3$ ) distributed random variables lie (a further 5% lying below the negative of that value, by symmetry of the  $t$  distribution). For example

```
df <- nrow(cars) - 3
df

## [1] 47

qt(0.95, df = df)

## [1] 1.677927
```

implies that a  $t_{47}$  r.v. has a probability 0.05 of being greater than 1.678. The following returns the required confidence interval.

```

beta.hat[2] + qt(0.95, df = df)*sigma.beta[2]

##      speed
## 4.32656

beta.hat[2] - qt(0.95, df = df)*sigma.beta[2]

##      speed
## -2.499985

```

That is, a 90% CI for  $\beta_2$  is  $(-2.500, 4.327)$ .

Actually, it is possible to calculate intervals for all three parameters at the same time. The following finds 95% CI's for all the parameters

```

beta.hat + qt(0.975, df = df)*sigma.beta

## (Intercept)      speed  I(speed^2)
## 32.2784284    5.0056113    0.2326702

beta.hat - qt(0.975, df = df)*sigma.beta

## (Intercept)      speed  I(speed^2)
## -27.33815279   -3.17903606   -0.03275162

```

So, for example, a 95% CI for  $\beta_3$  is  $(-0.033, 0.233)$ . Note that the `confint` function calculates these confidence intervals automatically.

```

confint(cm, level = 0.95)

##              2.5 %      97.5 %
## (Intercept) -27.33815279 32.2784284
## speed       -3.17903606  5.0056113
## I(speed^2)   -0.03275162  0.2326702

# Also, for 90% CI for beta2
confint(cm, level = 0.9)[2, ]

##          5 %      95 %
## -2.499985  4.326560

```

## Hypothesis tests

The result from Section 4.3.1 is also useful for testing hypotheses about the values of the  $\beta_j$ . For example, consider testing  $H_0 : \beta_2 = 0$  vs.  $H_1 : \beta_2 \neq 0$ . To test this we first form the t-statistic,  $T = \hat{\beta}_2 / \hat{\sigma}_{\hat{\beta}_2}$ , and compute its observed value,

```

obs_t <- beta.hat[2]/sigma.beta[2]
obs_t

##      speed
## 0.448962

```

If  $H_0$  is true then this should be an observation of a  $t_{47}$  random variable, but if  $H_1$  is true then it should have too large a magnitude for this (basically because  $\hat{\beta}_2$  will not then be ‘close’ to the hypothesized value, zero). As usual we quantify the evidence for or against  $H_0$  using a p-value. This is given, in this case, by the following probability:  $\Pr[|T| > |t|]$ , where  $T \sim t_{47}$  and  $t = 0.449$ . This is easily evaluated in R as follows.

```
p.val <- 2*pt(abs(obs_t), df = df, lower.tail = FALSE)
p.val

##      speed
## 0.6555224
```

A p-value of 0.66 suggests that the data are quite compatible with  $H_0$ . That is, provided that the model contains  $\beta_1$  and  $\beta_3$  terms,  $\beta_2$  seems to be superfluous. However it might be that one of  $\beta_1$  and  $\beta_3$  is even more superfluous, so we should test whether either of those might be zero, before dropping any terms. In fact we can work out equivalent p-values for all 3 parameters simultaneously:

```
obs_t <- beta.hat/sigma.beta
p.val <- 2*pt(abs(obs_t), df = df, lower.tail=F)

#parameter estimates
beta.hat

## (Intercept)      speed  I(speed^2)
##  2.4701378    0.9132876  0.0999593

#estimated standard deviation of beta.hat
sigma.beta

## (Intercept)      speed  I(speed^2)
## 14.81716473    2.03422044  0.06596821

#observed t statistics
obs_t

## (Intercept)      speed  I(speed^2)
##  0.1667079    0.4489620  1.5152647

#p-values
p.val

## (Intercept)      speed  I(speed^2)
##  0.8683151    0.6555224  0.1364024
```

Actually, p-values for testing whether parameters might be zero are very often useful, so R produces them by default in the summary of a fitted linear model.

```
summary(cm)$coefficients

##      Estimate Std. Error  t value Pr(>|t|)
## (Intercept)  2.4701378 14.81716473  0.1667079 0.8683151
## speed        0.9132876  2.03422044  0.4489620 0.6555224
## I(speed^2)   0.0999593  0.06596821  1.5152647 0.1364024
```

Notice how the columns of the Coefficients table contain exactly the values that we have just calculated. These single parameter tests are often used to decide whether to drop a term from the model: usually the term with the highest p-value greater than some threshold, such as 0.05, would be dropped (provided that there are no prior grounds for insisting on its retention in the model). The model is refitted without the term, and the new p-values re-examined to see if any further simplification is possible. This is repeated until all terms have p-values lower than, say, 0.05. Note that the different estimates in  $\hat{\beta}$  are not independent (please see below  $\hat{V}_{\hat{\beta}}$ ). Consequently, dropping one term (setting it to zero) will change the estimates of the other coefficients and, therefore, their p-values. This is why we need to remove one term at a time and refit the model, rather than removing multiple parameters simultaneously, as the p-values are computed under the assumption that the other terms remain in the model.

```
V.beta

##           (Intercept)           speed      I(speed^2)
## (Intercept) 219.5483705 -28.9523122  0.872858710
## speed      -28.9523122   4.1380528 -0.131439753
## I(speed^2)   0.8728587  -0.1314398  0.004351805
```

### Different hypotheses and model selection

Sometimes other types of hypotheses may be tested. For example, it is hard to think of a sensible physical interpretation for a negative  $\beta_1$ . Hence, it might be more appropriate to test  $H_0 : \beta_1 = 0$  vs.  $H_1 : \beta_1 > 0$ .

```
obs_t <- beta.hat[1]/sigma.beta[1]
obs_t

## (Intercept)
## 0.1667079
```

For this one-sided alternative only the upper tail of the  $t_{47}$  distribution is relevant. The p-value is  $\Pr[T > t]$  where  $T \sim t_{47}$  and  $t = 0.167$ .

```
p.val <- pt(obs_t, df = df, lower.tail = F)
p.val

## (Intercept)
## 0.4341575
```

Again, there is no reason to reject the null hypothesis (provided the other terms are in the model), but notice that the p-value is halved by the change in alternative.

Another use of the Section 4.3.1 result is to test whether  $\beta_j$  might have some hypothesized value other than zero. For example we might be able to calculate what  $\beta_3$  ought to be, based on the frictional properties of the brakes and tyres: suppose this calculation says that  $\beta_3 = 0.2$ . We can test this theoretical value, by testing  $H_0 : \beta_3 = 0.2$  vs.  $H_1 : \beta_3 \neq 0.2$ , as follows.

```
obs_t <- (beta.hat[3] - 0.2)/sigma.beta[3]
p.val <- 2*pt(abs(obs_t), df = df, lower.tail = F)
p.val

## I(speed^2)
## 0.136091
```

So not much evidence with which to reject  $H_0$ . However, this test provides a very good example of why it is often a good idea to base statistical conclusions on the simplest model consistent with the data. If we repeat the test, based on a model without an intercept, then things are rather different.

```
cml <- lm(dist ~ speed + I(speed^2) - 1, data = cars)
beta.hat <- coef(cml)
V.beta <- vcov(cml)
sigma.beta <- sqrt(diag(V.beta))
obs_t <- (beta.hat[2] - 0.2)/sigma.beta[2]
p.val <- 2*pt(abs(obs_t), df = (nrow(cars) - 2), lower.tail = F)
p.val

## I(speed^2)
## 0.0004934814
```

The p-value has dropped enormously, and we now have firm evidence to reject  $H_0 : \beta_3 = 0.2$ . What has happened here, is that the parameter estimators are so correlated in the full model, that all the estimators have very high

variance. As a result we have a hard time rejecting a null hypothesis even if it quite a long way from the truth. The redundant parameters cause a loss of precision, and consequent loss of *power* (probability of correctly rejecting  $H_0$ ) when testing. The more parsimonious model gives much better precision, and higher power when testing.

Note however, that there is a price to pay for *selecting* a model statistically and then performing hypothesis tests as if there were no question about the truth of the selected model. By doing this we fail to account for our initial uncertainty about which model was right when performing the hypothesis tests. This tends to mean that our p-values are a bit lower than if we had properly accounted for all uncertainty: i.e. the p-values are really only approximate now.

### 5.1.2 F-ratio tests about several parameters

This section illustrates the use of the F-ratio result from Section 4.3.2, with R. We start by considering again the cars model with the following functional specification

$$\text{dist}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2).$$

Suppose that we had some prior reason to want to test:

$$H_0 : \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad H_1 : \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Note that  $H_1$  states that, at least, one of the parameters is not zero. This is exactly the situation in which the results of Section 4.3.2 are useful. Recall that the result is:

$$F = \frac{(\text{RSS}_0 - \text{RSS}_1)/(p_1 - p_0)}{\hat{\sigma}^2} \sim F_{p_1 - p_0, n - p_1} \text{ if } H_0 \text{ is true.}$$

Here  $p_1 - p_0 = 2$  is the difference in degrees of freedom between the models (difference in number of parameters) and  $\hat{\sigma}^2 = \text{RSS}_1/(n - p_1)$  is the residual variance estimator from the alternative (larger) model. Fitting the null model, it is then straightforward to calculate this  $F$  ratio statistic:

```
cm <- lm(dist ~ speed + I(speed^2), data = cars)
cm0 <- lm(dist ~ I(speed^2) - 1, data = cars)
rss0 <- sum(residuals(cm0)^2)
rss1 <- sum(residuals(cm)^2)
f_stat <- ((rss0 - rss1)/2)/(rss1/47)
f_stat

## [1] 2.412267
```

Note that, as claimed in Section 4.3.2, the  $F$  test statistic in the way we have just computed it is equivalent to

$$F = \frac{1}{q}(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{C}\hat{\mathbf{V}}_{\hat{\boldsymbol{\beta}}}\mathbf{C}^\top)^{-1}(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}),$$

where in this case  $q = 2$  and

$$\mathbf{C} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{d} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

```
beta.hat <- coef(cm)
V.beta <- vcov(cm)
C <- matrix(c(1, 0, 0, 0, 1, 0), nrow = 2, ncol = 3, byrow = T)
d <- matrix(c(0, 0), ncol = 1, nrow = 2)
q <- 2
as.numeric(((t(C%*%beta.hat - d))%*%solve(C%*%V.beta%*%t(C))%*%(C%*%beta.hat - d))/q)

## [1] 2.412267
```

From the result of Section 4.3.2 this ought to be an observation on an  $F_{2,47}$  random variable, if  $H_0$  is true. Otherwise it should be improbably large for an  $F_{2,47}$  random variable. As usual we quantify the compatibility of the test statistic (and hence the data) with the stated distribution by calculating a p-value, which in this case is given by  $\Pr[F > f]$  where  $F \sim F_{2,47}$  and  $f = 2.412$ , which is easily evaluated using the `pf` function.

```
p.val <- pf(f_stat, 2, 47, lower.tail = F)
p.val

## [1] 0.1006278
```

The data provide little evidence to reject  $H_0$ , in this case.

While the above calculations are useful for linking the theory in Section 4.3.2 to what is actually done to use the result for a model and data, there is usually no need to ‘spell out’ the calculations like this. R has built in functions to automate the process. For example the test just conducted can be done using the `anova` function as follows.

```
anova(cm0, cm)

## Analysis of Variance Table
##
## Model 1: dist ~ I(speed^2) - 1
## Model 2: dist ~ speed + I(speed^2)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      49 11936
## 2      47 10825  2    1111.2  2.4123 0.1006
```

The quantities in this *ANOVA table* exactly match the quantities used in the previous, long winded, calculation.

Before proceeding, it is worth mentioning that the last line in the `summary` object provides an  $F$ -statistic where the null hypothesis is that all coefficients, except the intercept, are zero. In this case  $H_0 : \beta_2 = \beta_3 = 0$ .

```
summary(cm)

##
## Call:
## lm(formula = dist ~ speed + I(speed^2), data = cars)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -28.720  -9.184  -3.188   4.628  45.152
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.47014   14.81716   0.167   0.868
## speed         0.91329    2.03422   0.449   0.656
## I(speed^2)    0.09996    0.06597   1.515   0.136
##
## Residual standard error: 15.18 on 47 degrees of freedom
## Multiple R-squared:  0.6673, Adjusted R-squared:  0.6532
## F-statistic: 47.14 on 2 and 47 DF,  p-value: 5.852e-12
```

It is also possible to compare a longer sequence of models using the `anova` command. Each pair in the sequence is compared in exactly the same way as two models would be compared, except that the residual variance estimate in the denominator of each  $F$ -ratio statistic is always taken from the ‘largest’ model (i.e. the one with most parameters).

Finally, the following is possible:

```

anova(cm)

## Analysis of Variance Table
##
## Response: dist
##           Df Sum Sq Mean Sq F value    Pr(>F)
## speed      1 21185.5  21185.5   91.986 1.211e-12 ***
## I(speed^2)  1   528.8    528.8    2.296   0.1364
## Residuals 47 10824.7    230.3
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

This single argument call causes `anova` to successively remove terms from your model, at each stage testing the null hypothesis that the current version of the model is correct, against the alternative that the previous version was right. F-ratio tests are used each time, but again  $\hat{\sigma}^2$  according to the original, larger, model is always used as the denominator of the  $F$  ratio statistic. The tables have to be read from bottom row up. So, the `I(speed^2)` row compares models with and without the dependence on `speed`<sup>2</sup>, and suggests that we could drop it. Turning to the `speed` row, this is about comparing a model with intercept and linear dependence on `speed` to a model with only an intercept: it suggests that we should hang on to the `speed` term. Note that when using the `anova` command with a single argument, the order in which the predictors enter the model matters. Actually this single argument use of `anova` only really makes sense in very specific situations. As we have already seen, if we reduce the `cars` model in a sensible order, we would drop the intercept term, and then nothing else.

### 5.1.3 Checking the model assumptions: residuals

In the previous few sections, models for the `cars` data have been used to test hypotheses and find confidence intervals, but we have not actually checked that the model assumptions are reasonable. If the assumptions are not reasonable, then we have been wasting our time! For this reason, it is usually important to check model assumptions immediately after fitting models, and before performing any analysis that relies on the assumptions being reasonable. The only reason that this was not done above was to ensure that illustration of the practicalities of using the results of Sections 4.3.1 and 4.3.2 followed as soon as possible after those sections.

The assumptions of the linear model are:

1. The error terms have expectation zero. This is equivalent to assuming that the systematic component  $\mathbf{X}\beta$  correctly specifies the expectation  $\mathbb{E}(y)$ .
2. The errors have constant variance (*homoscedasticity*).
3. The errors are independent.
4. The errors follow a normal distribution.

A couple of comments on two of these four assumptions are worth mentioning. Regarding assumption 1., as we learned in Section 3, nonlinear effects of predictors can be accommodated within the framework of linear models; the only requirement is that the resulting model is linear in the parameters. Regarding assumption 4, if the normality assumption does not hold, the usual tests and confidence intervals are not exact. However, we can appeal to asymptotic results (often justified via central limit-type arguments), which imply that these procedures become increasingly accurate approximations as the sample size grows. Hence, the derived results are typically good approximations provided the sample size is sufficiently large and departures from normality are not too severe (e.g., extremely heavy tails, strong skewness, pronounced multimodality).

Although it is not part of regression diagnostics, it is also worthwhile to mention that an even more important assumption is that the data at hand are relevant to the question of interest. Of course, this requires some qualitative judgement and domain knowledge and, in contrast to the four assumptions listed above, cannot be checked through plots.

The errors are not observable but we can examine the residuals,  $\hat{\epsilon}$ . Recall that the residuals,  $\hat{\epsilon}_i$ , are simply the response observation,  $y_i$ , minus the fitted values,  $\hat{\mu}_i = \mathbf{x}_i^\top \hat{\beta}$ , for  $i = 1, \dots, n$ , i.e.

$$\hat{\epsilon} = \mathbf{y} - \hat{\boldsymbol{\mu}}, \text{ where } \hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

The residuals contain all the information left in the data after fitting the model. Note that, provided that  $\mathbb{E}(\epsilon) = \mathbf{0}_n$  so that  $\mathbb{E}(\hat{\beta}) = \beta$ ,

$$\mathbb{E}(\hat{\epsilon}) = \mathbb{E}(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbb{E}(\mathbf{y}) - \mathbf{X}\mathbb{E}(\hat{\beta}) = \mathbf{X}\beta - \mathbf{X}\beta = \mathbf{0}_n.$$

To evaluate the covariance matrix of  $\hat{\epsilon}$  we will make use of the influence matrix  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$ , introduced in Section 4.4, which is of dimension  $n \times n$ . Recall that  $\mathbf{A} = \mathbf{A}^\top$  and  $\mathbf{A}\mathbf{A} = \mathbf{A}$ . Note that we can write the residuals as

$$\hat{\epsilon} = \mathbf{y} - \hat{\boldsymbol{\mu}} = \mathbf{y} - \mathbf{A}\mathbf{y} = (\mathbf{I}_n - \mathbf{A})\mathbf{y}.$$

Under the assumption of independent error terms with constant variance  $\sigma^2$ , which implies that the observations in the response vector  $\mathbf{y}$  are also independent and have the same variance  $\sigma^2$ , we have

$$\begin{aligned} \mathbf{V}_{\hat{\epsilon}} &= \text{var}\{(\mathbf{I}_n - \mathbf{A})\mathbf{y}\} = (\mathbf{I}_n - \mathbf{A})\text{var}(\mathbf{y})(\mathbf{I}_n - \mathbf{A})^\top \\ &= (\mathbf{I}_n - \mathbf{A})(\sigma^2 \mathbf{I}_n)(\mathbf{I}_n - \mathbf{A})^\top \\ &= (\mathbf{I}_n - \mathbf{A})(\mathbf{I}_n - \mathbf{A})^\top \sigma^2 \\ &= (\mathbf{I}_n - \mathbf{A})(\mathbf{I}_n - \mathbf{A}) \sigma^2 \\ &= (\mathbf{I}_n - \mathbf{A} - \mathbf{A} + \mathbf{A}\mathbf{A}) \sigma^2 \\ &= (\mathbf{I}_n - \mathbf{A}) \sigma^2. \end{aligned}$$

Further, under the model assumptions, the vector  $\epsilon$  is normally distributed, and so is  $\mathbf{y}$ . Since  $\hat{\epsilon}$  is a linear transformation of the multivariate normal random vector  $\mathbf{y}$ , it must also follow a normal distribution

$$\hat{\epsilon} \sim N(\mathbf{0}_n, (\mathbf{I}_n - \mathbf{A})\sigma^2).$$

We observe that, although the errors may have equal variance and be uncorrelated, the residuals do not. Specifically,

$$\text{var}(\hat{\epsilon}_i) = (1 - A_{ii})\sigma^2, \quad \text{and} \quad \text{cov}(\hat{\epsilon}_i, \hat{\epsilon}_j) = -A_{ij}\sigma^2, \quad i \neq j,$$

where  $A_{ii}$  is the  $i$ th diagonal element of  $\mathbf{A}$ , and  $A_{ij}$  is its  $(i, j)$ th element. Since variances are nonnegative,  $0 \leq A_{ii} \leq 1$ . We cannot do anything about the slight lack of independence, but the residuals can be standardised to have constant variance by defining

$$\tilde{\epsilon}_i = \frac{\hat{\epsilon}_i}{\sqrt{1 - A_{ii}}} = \frac{y_i - \hat{\mu}_i}{\sqrt{1 - A_{ii}}}, \quad i = 1, \dots, n,$$

so that

$$\text{var}(\tilde{\epsilon}_i) = \text{var}\left(\frac{\hat{\epsilon}_i}{\sqrt{1 - A_{ii}}}\right) = \frac{1}{1 - A_{ii}} \text{var}(\hat{\epsilon}_i) = \frac{(1 - A_{ii})\sigma^2}{1 - A_{ii}} = \sigma^2.$$

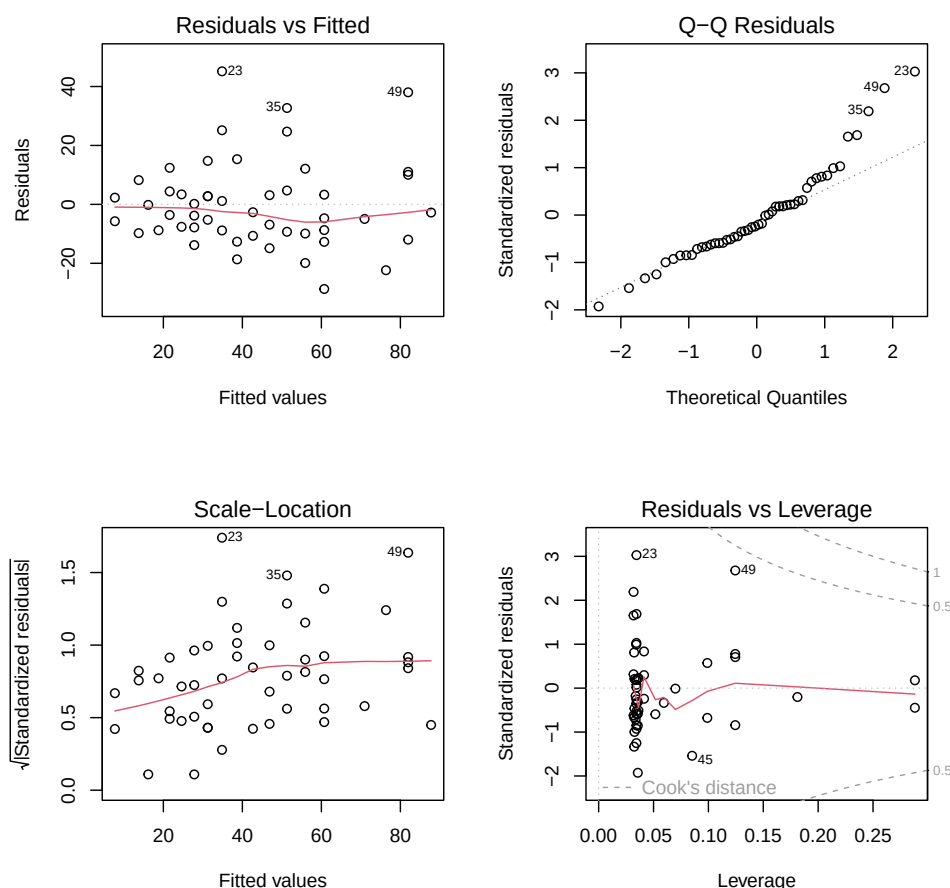
Further, note that  $\tilde{\epsilon}_i/\sigma \sim N(0, 1)$  if the model assumptions hold, and the correlation between  $\tilde{\epsilon}_i/\sigma$  and  $\tilde{\epsilon}_j/\sigma$ , for  $i \neq j$ , tends to be small.

The best way to check residuals is to examine a variety of diagnostic plots, always looking for evidence of any violation of the linear modelling assumptions. As Chambers et al. (1983, p. 1) put it: “*There is no single statistical tool that is as powerful as a well-chosen graph.*” The `plot` function provides several useful residual plots when passed a fitted linear model object.

Let us revisit the `cars` dataset and consider the running example model

$$\text{dist}_i = \beta_1 + \beta_2 \text{speed}_i + \beta_3 \text{speed}_i^2 + \epsilon_i, \quad \epsilon_i \text{ i.i.d. } N(0, \sigma^2).$$

```
data(cars)
cm <- lm(dist ~ speed + I(speed^2), data = cars)
par(mfrow = c(2, 2))
plot(cm)
```



- The upper left plot shows the model residuals,  $\hat{\epsilon}_i$ , against the fitted values,  $\hat{\mu}_i$ , where  $\hat{\mu} = \mathbf{X}\hat{\beta}$  and  $\hat{\epsilon} = \mathbf{y} - \hat{\mu}$ . The residuals should be evenly scattered above and below zero, independent of the  $x$ -axis positions (the distribution of fitted values is not of interest)<sup>26</sup>. The red line, which is a LOWESS (scatterplot smoother) fit regressing the residuals on the fitted values, should be approximately constant around zero. A trend in the mean of the residuals is a sign of remaining systematic structure and usually results from an erroneous model structure (assumption 1.): e.g., assuming a linear relationship with a predictor, when a quadratic is required, or omitting an important predictor variable. A plot of the residuals against the predictors and potential predictors can help in further investigating/diagnosing the misspecification of the model mean structure. For plots of the residuals against the predictors not included in the model, any observed structure may indicate that this predictor should be included in the model.

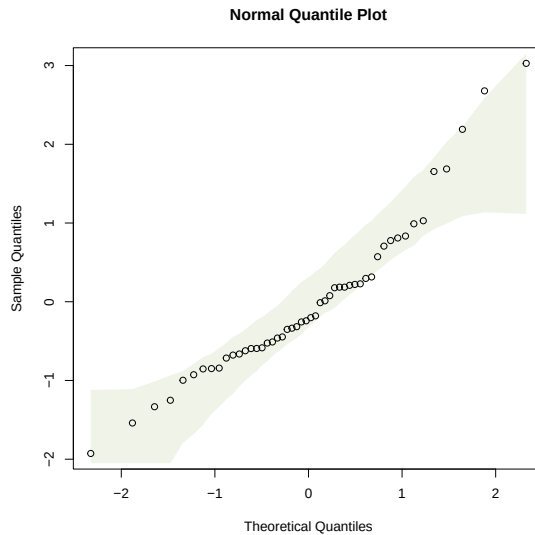
This plot also allows investigating the assumption of constant variance. A trend in the variability of the residuals (e.g., a funnel shape) suggests that the variance of the response is related to its mean, violating the

<sup>26</sup>Under correct model specification, the residuals are orthogonal to the fitted values. Why are they orthogonal? Recall that two vectors are orthogonal if their dot product is zero. By definition of dot product between two vectors, we have that  $\hat{\mu}^T \hat{\epsilon} = (\mathbf{A}\mathbf{y})^T [(\mathbf{I}_n - \mathbf{A})\mathbf{y}] = (\mathbf{y}^T \mathbf{A}^T)(\mathbf{y} - \mathbf{A}\mathbf{y}) = (\mathbf{y}^T \mathbf{A})(\mathbf{y} - \mathbf{A}\mathbf{y}) = \mathbf{y}^T \mathbf{A}\mathbf{y} - \mathbf{y}^T \mathbf{A}\mathbf{A}\mathbf{y} = \mathbf{y}^T \mathbf{A}\mathbf{y} - \mathbf{y}^T \mathbf{A}\mathbf{y} = 0$ . For linear models with an intercept,  $\hat{\mu}^T \hat{\epsilon} = 0$  implies that  $\text{Cov}(\hat{\mu}, \hat{\epsilon}) = 0$  which, in turn, implies that  $\text{Cor}(\hat{\mu}, \hat{\epsilon}) = 0$ . Therefore, the least squares fitted to a scatterplot of  $\hat{\epsilon}$  versus  $\hat{\mu}$  has slope zero. Likewise, the residuals are also orthogonal to the columns of  $\mathbf{X}$ . Why? Please refer to the solution for Question 5 from Workshop 1.

constant variance assumption (*heteroscedasticity*): transformation of the response or use of a generalised linear model may help, in such cases. The plot shown shows some indication of increasing variance with mean, which somewhat violates the constant variance assumption, although the effect is not extreme here.

- The lower left plot is a scale-location plot. The raw residuals are standardised by dividing by their estimated standard deviation,  $\hat{\sigma}\sqrt{1 - A_{ii}}$ , where  $\mathbf{A}$  is the influence matrix and  $\hat{\sigma}^2$  is the unbiased estimate of  $\sigma^2$  derived in Section 4.3.1. The square root of the absolute value of each standardised residual is then plotted against the corresponding fitted value. It can be easier to judge the constant variance assumption from such a plot, and the square root transformation reduces the skew in the distribution, which would otherwise be likely to occur (note that if the standardised residuals follow a normal distribution, the random variable given by their absolute value follows a ‘half normal’ distribution). We expect the red line to be approximately constant. As the previous plot already hinted, there is some suggestion of variance increasing with mean in this plot, but the effect is not very large, so is unlikely to cause a big problem.
- The upper right panel is a normal Q-Q (quantile-quantile) plot<sup>27</sup>. The standardised residuals are sorted and plotted against the quantiles of a standard normal distribution. If the residuals are normally distributed, the points should fall close to a straight line, apart from random scatter<sup>28</sup>. In this case the few large residuals show signs of being a bit too big to have come from the same normal distribution as the others, but again, the effect is not large relative to what can occur by chance even when the model assumptions are fine. Due to the difficulty in judging how much deviation from a straight line can still be tolerated, alternative implementations also plot a corresponding envelope to give an idea of natural variation. One such approach is implemented in the function `qqenvelope` from the package `ecostats`.

```
library(ecostats)
qqenvelope(cm)
```



- The lower right panel plots the standardised residuals against the *leverage* of each observation. The element  $A_{ii}$  of the influence (or hat) matrix  $\mathbf{A}$ , on which  $\text{var}(\hat{\epsilon}_i) = (1 - A_{ii})\sigma^2$  depends, is called the leverage of observation  $i$ . It measures the *potential* for that observation to influence the fitted values. Since

$$\text{var}(\hat{\boldsymbol{\mu}}) = \text{var}(\mathbf{A}\mathbf{y}) = \mathbf{A}\text{var}(\mathbf{y})\mathbf{A}^T = \sigma^2\mathbf{A},$$

<sup>27</sup>A Q-Q plot is a graphical tool used to assess whether a sample of observations is consistent with a hypothesised family of distributions. It compares the sample quantiles with the corresponding quantiles of the hypothesised distribution.

<sup>28</sup>We have derived that  $\tilde{\epsilon}_i/\sigma \sim N(0, 1)$ , for  $i = 1, \dots, n$ . When replacing  $\sigma$  by its unbiased estimate  $\hat{\sigma}$ ,  $\tilde{\epsilon}_i/\hat{\sigma}$  no longer follows a (standard) normal distribution exactly. Nonetheless, its distribution is approximately normal in moderate or large samples, so the normal Q-Q plot remains a reliable diagnostic for assessing whether the normality assumption for the errors is reasonable.

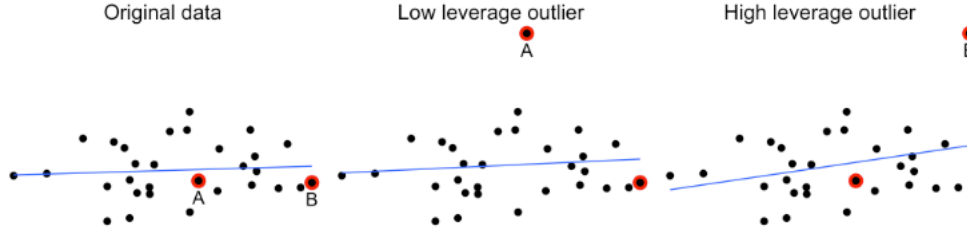


Figure 5: High leverage point B has a relatively large effect on the regression line. Figure taken from Speegle and Clair (2022, p. 394)

we have that  $\text{var}(\hat{\mu}_i) = \sigma^2 A_{ii}$ , with  $0 \leq A_{ii} \leq 1$ . The leverage thus determines the precision with which  $\hat{\mu}_i$  estimates  $\mu_i$ . For large  $A_{ii}$  close to one,  $\text{var}(\hat{\mu}_i) \approx \sigma^2 = \text{var}(y_i)$  and  $\text{var}(\hat{\epsilon}_i) \approx 0$ . In this case,  $y_i$  may have a large influence on  $\hat{\mu}_i$ . By contrast, when  $A_{ii}$  is close to zero and thus  $\text{var}(\hat{\mu}_i)$  is relatively small, this suggests that  $\hat{\mu}_i$  is based in contributions from many observations. It can be shown that<sup>29</sup>

$$A_{ii} = \frac{1}{n} + (\mathbf{x}_i^\top - \bar{\mathbf{x}}^\top)(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}(\mathbf{x}_i^\top - \bar{\mathbf{x}}^\top)^\top,$$

where  $\tilde{\mathbf{X}}$  is the design matrix with centred columns,  $\mathbf{x}_i^\top$  is the  $i$ th row of the design matrix and  $\bar{\mathbf{x}}^\top$  denotes the vector of column means. This expression makes explicit that the leverage increases as  $\mathbf{x}_i$  lies farther from  $\bar{\mathbf{x}}$  in the predictor space. Observations that are outliers in the predictor space are known as high leverage points, to distinguish them from observations that are outliers in the response variable (those with large, in absolute value, standardised residuals). The trace of the influence matrix  $\mathbf{A}$  is  $p$ <sup>30</sup>. Since  $\mathbf{A}$  is an  $n \times n$  matrix whose  $n$  diagonal elements sum to  $p$  (with  $p < n$ ), the average leverage is  $p/n$ . In practice, a leverage value greater than two or three times this average is considered large. The combination of a large (in absolute value) residual and high leverage implies that the corresponding observation has a substantial influence on the overall fit. Large leverage with a small residual generally implies that, although the datum would exert undue influence on the fit, if out of line with the other data, it is actually consistent with the other data, and does not have too great an influence. Similarly a large residual may not have much influence on the fit, if the corresponding datum has low leverage. This is illustrated in Figure 5.

A good summary of how much influence each observation actually has is provided by its *Cook's distance*. Cook's distance is a measure of how much influence each observation has on the fitted model. If  $\hat{\mu}_i^{[k]}$  is the  $i$ th fitted value when the  $k$ th observation is omitted from the fit,  $\hat{\mu}_i$  is the  $i$ th fitted value using all observations, then Cook's distance is

$$d_k = \frac{1}{p\hat{\sigma}^2} \sum_{i=1}^n (\hat{\mu}_i^{[k]} - \hat{\mu}_i)^2, \quad (15)$$

where  $p$  is the number of parameters,  $n$  the number of observations, and  $\hat{\sigma}^2$  is the estimated error variance based on the fit to all observations. The numerator measures how much the fitted values (and, therefore, the parameter estimates) change when the  $k$ th observation is deleted, and the denominator scales this change in a convenient way. Cook's distance can be shown to be a function of leverage and standardised residual, so contours of Cook's distance are shown on the plot. A very large value of  $d_k$  indicates a point that has a substantial influence on the model results. Cook's distances above 0.5 are considered borderline problematic, whereas values over 1 are usually considered highly influential<sup>31</sup>. If the Cook's distance values indicate that

<sup>29</sup>Belsley et al. (1980). Regression Diagnostics: Identifying Influential Data and Sources of Collinearity. John Wiley & Sons.

<sup>30</sup>Why? Remember that for two matrices  $\mathbf{U}_1$ , of dimension,  $n \times p$ , and  $\mathbf{U}_2$ , of dimension,  $p \times n$ ,  $\text{tr}(\mathbf{U}_1 \mathbf{U}_2) = \text{tr}(\mathbf{U}_2 \mathbf{U}_1)$ . Therefore,  $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{Q}\mathbf{Q}^\top) = \text{tr}(\mathbf{Q}^\top \mathbf{Q}) = \text{tr}(\mathbf{I}_p) = p$ .

<sup>31</sup>The contour curves in the standardised residuals–leverage plot represent the combinations of leverage and standardised residuals that yield Cook's distance values of 0.5 and 1.

model estimates may be very sensitive to just one or two data, then it is usually prudent to repeat any analysis without the offending points, in order to check the robustness of the modelling conclusions. In this case none of the points look wildly out of line.

The leverage and Cook's distances can also be easily obtained from the model fit.

```
leverage <- hatvalues(cm)
head(leverage)

##           1           2           3           4           5           6
## 0.28812937 0.28812937 0.09900328 0.09900328 0.06991940 0.05169124

p <- length(coef(cm))
n <- nrow(cars)

# Any observations with high or very high leverage?
as.numeric(which(leverage > 2*(p/n)))

## [1] 1 2 46 47 48 49 50

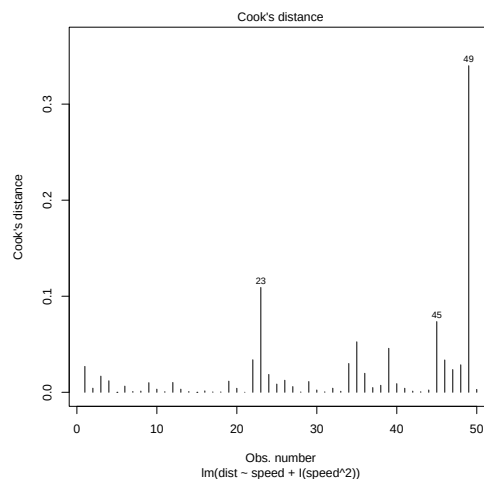
as.numeric(which(leverage > 3*(p/n)))

## [1] 1 2 50

# Cook's distances and any above 0.5
cooks_dist <- cooks.distance(cm)
as.numeric(which(cooks_dist > 0.5))

## numeric(0)

# We can also plot observation number against Cook's distance
plot(cm, which = 4)
```



Further, note that the residuals and standardised residuals can be obtained through the R commands `residuals` and `rstandard`, respectively.

```

# Residuals
res <- residuals(cm)
res_alt <- cars$dist - cm$fitted.values
max(abs(res - res_alt))

## [1] 5.77316e-15

# Standardised residuals
std_res <- rstandard(cm)
std_res_alt <- (cars$dist - cm$fitted.values)/
  (sigma(cm)*(sqrt(1 - hatvalues(cm))))
max(abs(std_res - std_res_alt))

## [1] 4.163336e-16

```

- Finally, we address the issue of correlated errors. This can be difficult to diagnose because many different correlation structures may arise in practice. Nonetheless, some types of data naturally suggest where to look for potential problems. For example, data collected over time often exhibit correlation between successive errors, while spatial data may show correlation among observations taken in nearby locations. Strong dependence implies that the effective amount of information in the data is smaller than the nominal sample size would suggest. For temporal data, one straightforward way to detect serial correlation is to plot the residuals against time and look for systematic patterns; any visible structure is evidence of correlation. Another useful diagnostic is to plot successive pairs of residuals, which can reveal lag-1 dependence. If serial correlation is detected, techniques from time-series analysis can be applied to model the dependence structure; however, this lies beyond the scope of the present course.

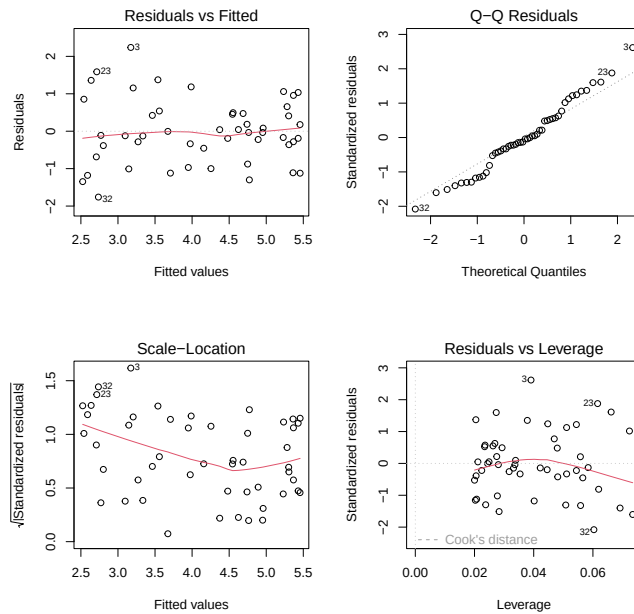
It is important to realize that residual plots are quite variable, even when all the model assumptions are met. To build intuition, one useful approach is to simulate residuals from settings in which the regression assumptions are satisfied.

```

set.seed(6) #ensures exact repeatability
n <- 50 #sample size
x <- runif(n) #simulate a uniform(0,1) predictor
eps <- rnorm(n) #simulate a normal error
y <- 2 + 4*x + eps #response variable
fit <- lm(y ~ x) #fit a linear model to data

par(mfrow = c(2, 2))
plot(fit)

```



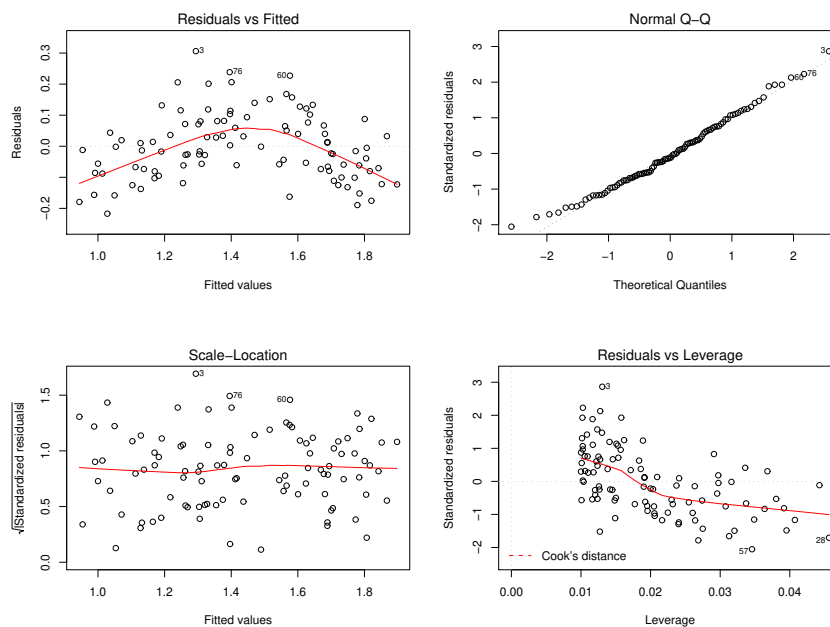
Repeating this simulation exercise a few times (without the `set.seed` line, of course) will give you a sense of the degree of variability to expect when the model assumptions are satisfied. Note that this variability depends on the sample size. Also observe that the red line is a flexible smooth, and as such it is extremely sensitive in regions where data are sparse.

#### 5.1.4 Residual plots: common problems

When examining residual plots, the most important aspects to check for include:

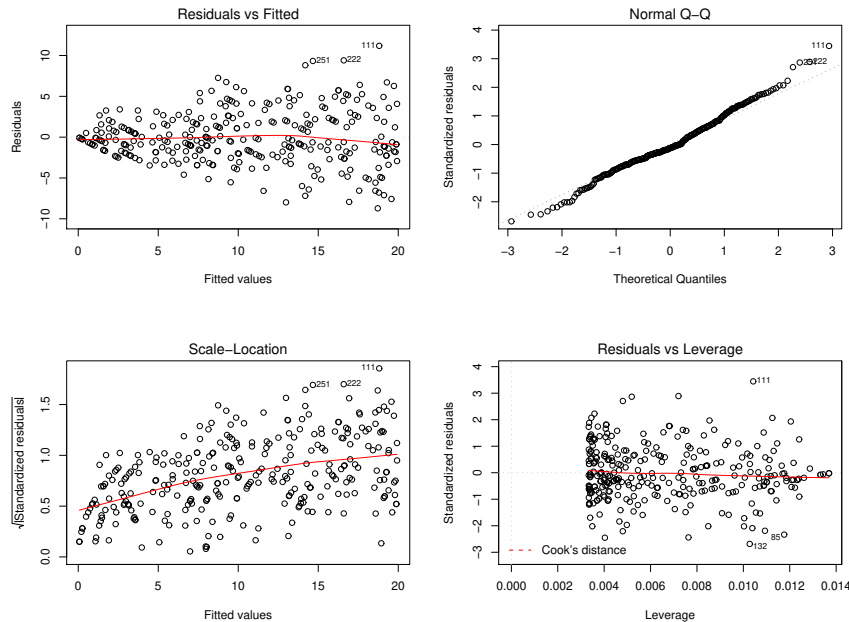
1. Evidence of a systematic pattern in the mean of the residuals when plotted against the fitted values.
2. Evidence that the residuals may not have constant variance.

Here is an example of a problematic set of plots.

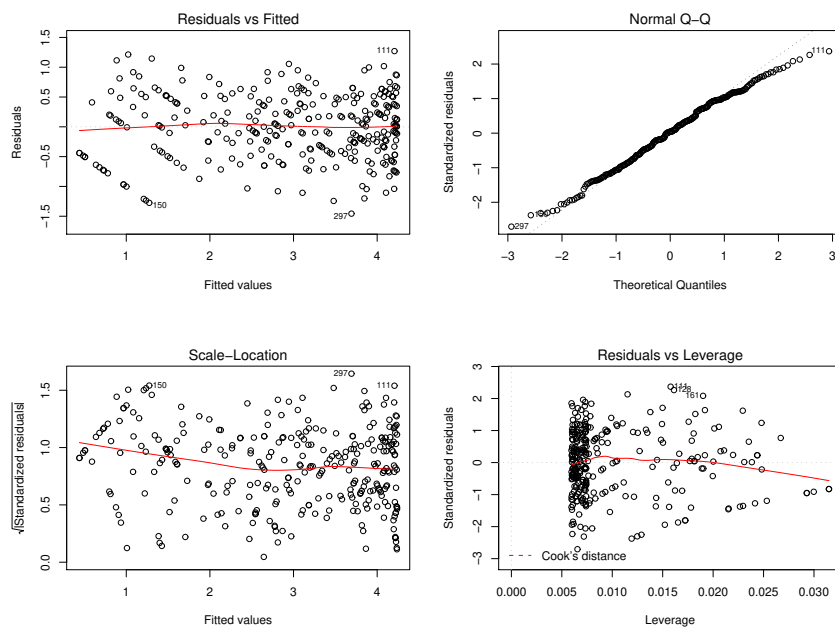


Notice the clear trend in the mean of the residuals when plotted against fitted values. The mean of the residuals is below zero for low and high fitted values and above zero for intermediate fitted values. This pattern means that something is missing from the structural part model of the model—perhaps a dependence on the square of one of the predictor variables.

Here is a set of plots showing a different problem. . .



In this case, there appears to be no pattern in the mean of the residuals when plotted against the fitted values, as expected. However, the variance of the residuals increases clearly with the mean. This pattern of increasing variance with increasing fitted values is quite common in practice and obviously violates the constant variance assumption. In such situations, it can sometimes help to model a suitable transformation of the original response variable (which, in this example, did not take negative values). For example the following plots are from a model fitted to the *square root* of the response variable used in the previous plot.



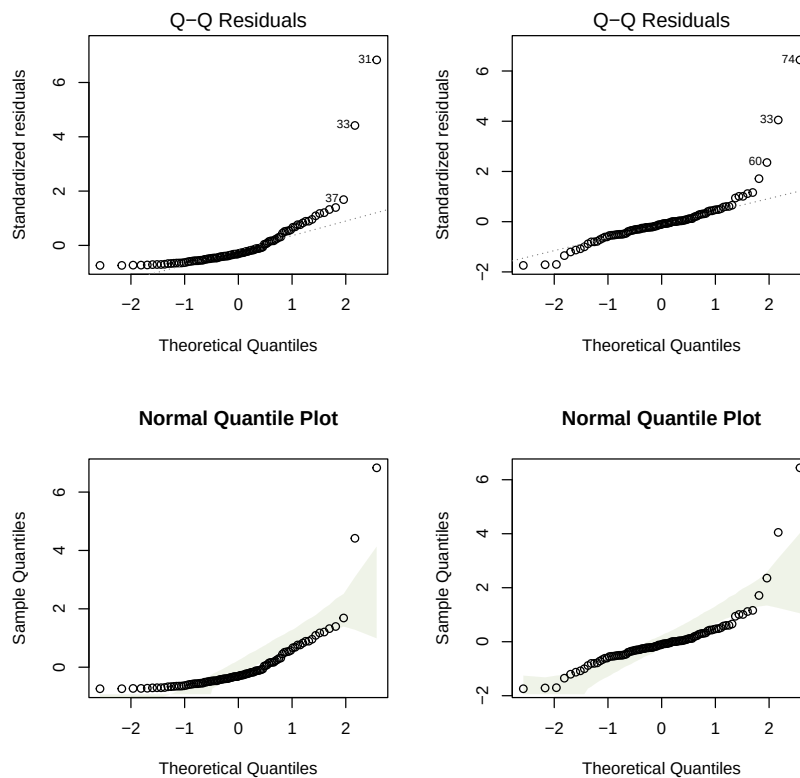
Finally, to illustrate the expected variation in Q-Q plots, we have simulated a dataset from a lognormal distribution (a skewed distribution; left plot) and from a t-distribution (a heavy-tailed distribution; right plot). Envelope plots are also shown. You can change the seed to generate additional examples.

```
set.seed(6)
n <- 100
x <- runif(n)
eps <- rlnorm(n)
y <- 2 + 4*x + eps
fit_logn <- lm(y ~ x)

set.seed(6)
n <- 100
x <- runif(n)
eps <- rt(n, df = 3)
y <- 2 + 4*x + eps
fit_t <- lm(y ~ x)

par(mfrow = c(2, 2))
plot(fit_logn, which = 2)
plot(fit_t, which = 2)

qqenvelope(fit_logn)
qqenvelope(fit_t)
```



## 5.2 Tyre wear data and backward selection

The MASS library, which is an add on for R, contains a data set, `Rubber`, on the relationship between the rate of wear of tyres (`loss` in grammes per hour), and the hardness and tensile strength of the tyre rubber (`hard` and `tens` respectively, measured in ‘Shore units’ and in  $\text{kgm}^{-2}$ , respectively).

```
library(MASS)
head(Rubber)

##   loss hard tens
## 1  372   45  162
## 2  206   55  233
## 3  175   61  232
## 4  154   66  231
## 5  136   71  231
## 6  112   71  237

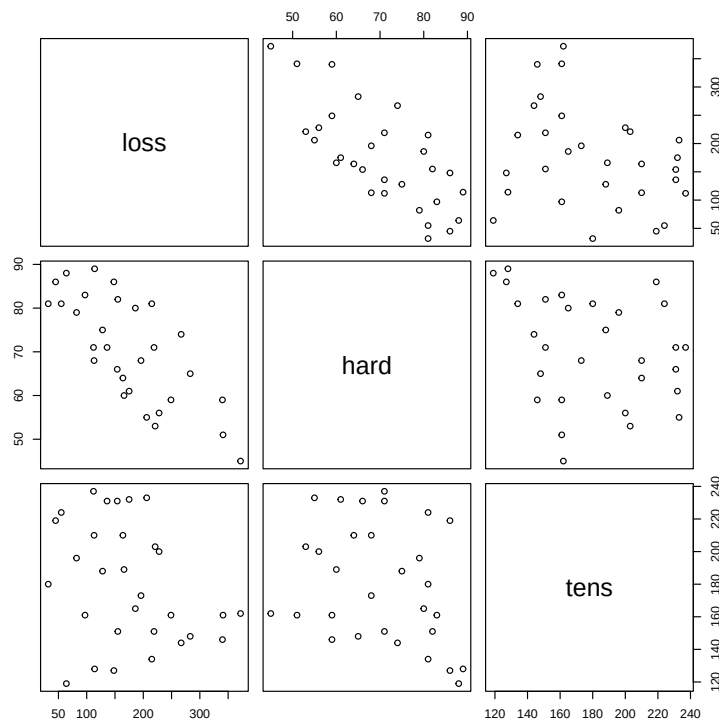
str(Rubber)

## 'data.frame': 30 obs. of  3 variables:
##  $ loss: int  372 206 175 154 136 112 55 45 221 166 ...
##  $ hard: int  45 55 61 66 71 71 81 86 53 60 ...
##  $ tens: int  162 233 232 231 231 237 224 219 203 189 ...

summary(Rubber)

##           loss           hard           tens
##  Min.      : 32.0   Min.      :45.00   Min.      :119.0
## 1st Qu.:113.2   1st Qu.:60.25   1st Qu.:151.0
##  Median :165.0   Median :71.00   Median :176.5
##   Mean   :175.4   Mean   :70.27   Mean    :180.5
## 3rd Qu.:220.5   3rd Qu.:81.00   3rd Qu.:210.0
##   Max.   :372.0   Max.    :89.00   Max.     :237.0

#scatter plot matrix of loss, hard, and tens
plot(Rubber)
```



The data were gathered in order to develop a model with which to predict wear rate as a function of hardness and tensile strength. All that is really known at the outset is that the loss rate should be some smooth function of the two predictors, so we could try, for instance, a polynomial model in which all combinations of powers of `tens` and `hard` were present, up to third order. That is, we could try the model:

$$y_i = \beta_1 + \beta_2 h_i + \beta_3 t_i + \beta_4 t_i h_i + \beta_5 t_i^2 + \beta_6 h_i^2 + \beta_7 h_i^2 t_i + \beta_8 h_i t_i^2 + \beta_9 t_i^3 + \beta_{10} h_i^3 + \epsilon_i$$

where  $y$  is `loss`,  $h$  is `hard`,  $t$  is `tens` and the  $\epsilon_i$  are i.i.d.  $N(0, \sigma^2)$  random variables. Note that the sample size  $n$  is 30, and the model has ten parameters. For the sake of precision, it is worth examining whether all variables in this model are necessary, or if a simpler model would suffice. To that end, we can successively remove the variable for which the hypothesis of zero value yields the highest p-value, above some threshold (e.g. 0.05), until all variables in the model have p-values below the threshold.

Here is how this *backward selection* can be achieved in R. We start by fitting the hypothesised model and by inspecting the residuals plots.

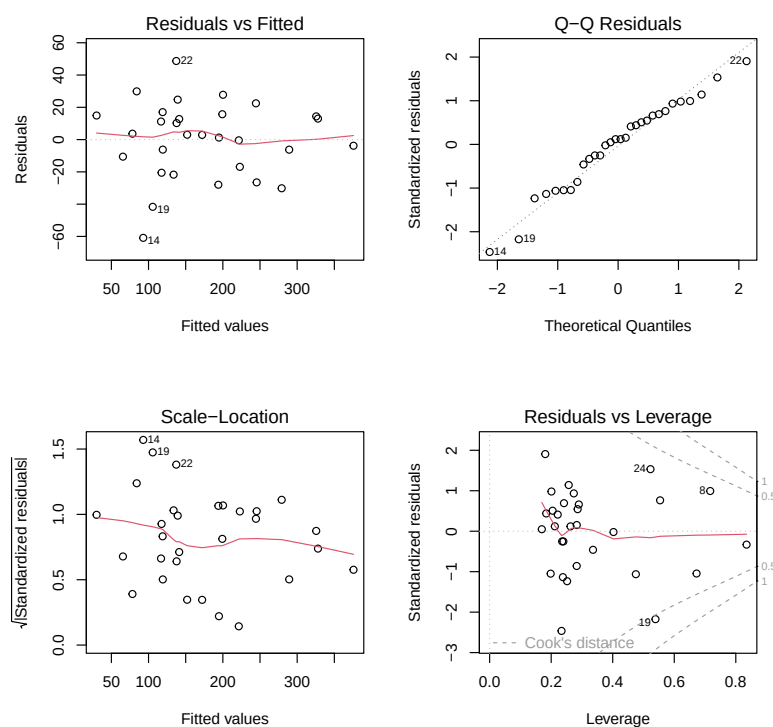
```
m1 <- lm(loss ~ hard + tens + I(hard*tens) + I(hard^2) + I(tens^2) +
          I(hard^2*tens) + I(tens^2*hard) + I(tens^3) + I(hard^3),
          data = Rubber)

summary(m1)

##
## Call:
## lm(formula = loss ~ hard + tens + I(hard * tens) + I(hard^2) +
##     I(tens^2) + I(hard^2 * tens) + I(tens^2 * hard) + I(tens^3) +
##     I(hard^3), data = Rubber)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.880 -15.307   2.958  14.784  48.695
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.040e+02  4.543e+03  -0.133   0.896
## hard        -1.240e+01  7.213e+01  -0.172   0.865
## tens         3.123e+01  5.295e+01   0.590   0.562
## I(hard * tens) -2.924e-01  5.085e-01  -0.575   0.572
## I(hard^2)      4.968e-01  7.613e-01   0.653   0.521
## I(tens^2)     -1.572e-01  2.081e-01  -0.755   0.459
## I(hard^2 * tens) 2.469e-03  1.675e-03   1.474   0.156
## I(tens^2 * hard) -7.527e-05  9.350e-04  -0.081   0.937
## I(tens^3)      3.384e-04  2.788e-04   1.214   0.239
## I(hard^3)     -4.736e-03  3.897e-03  -1.215   0.238
##
## Residual standard error: 28.22 on 20 degrees of freedom
## Multiple R-squared:  0.9292, Adjusted R-squared:  0.8973
## F-statistic: 29.16 on 9 and 20 DF,  p-value: 1.373e-09

par(mfrow = c(2, 2))
plot(m1)
```



For just 30 data points, these plots look acceptable. It might be worth repeating the analysis with and without observation 19, which has a borderline Cook's distance and could be exerting undue influence on the results. We will proceed and remove the variable with the highest p-value, which is  $ht^2$ .

```
m2 <- update(m1, . ~ . - I(tens^2*hard))
summary(m2)

##
## Call:
## lm(formula = loss ~ hard + tens + I(hard * tens) + I(hard^2) +
```

```
##      I(tens^2) + I(hard^2 * tens) + I(tens^3) + I(hard^3), data = Rubber)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.298 -14.934   3.096  15.266  48.709
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -9.349e+02  1.890e+03  -0.495   0.6261
## hard          -8.602e+00  5.324e+01  -0.162   0.8732
## tens           3.520e+01  1.910e+01   1.843   0.0795 .
## I(hard * tens) -3.299e-01  1.993e-01  -1.655   0.1127
## I(hard^2)       4.927e-01  7.414e-01   0.665   0.5136
## I(tens^2)      -1.715e-01  1.060e-01  -1.618   0.1207
## I(hard^2 * tens) 2.536e-03  1.419e-03   1.787   0.0884 .
## I(tens^3)       3.545e-04  1.906e-04   1.860   0.0770 .
## I(hard^3)      -4.780e-03  3.766e-03  -1.269   0.2182
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.55 on 21 degrees of freedom
## Multiple R-squared:  0.9292, Adjusted R-squared:  0.9022
## F-statistic: 34.44 on 8 and 21 DF,  p-value: 2.237e-10
```

We now remove the variable *hard*, *h*.

```
m3 <- update(m2, . ~ . - hard)
summary(m3)

##
## Call:
## lm(formula = loss ~ tens + I(hard * tens) + I(hard^2) + I(tens^2) +
##      I(hard^2 * tens) + I(tens^3) + I(hard^3), data = Rubber)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.074 -15.725   2.946  14.057  48.952
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.185e+03  1.061e+03  -1.116   0.2764
## tens           3.630e+01  1.742e+01   2.084   0.0490 *
## I(hard * tens) -3.436e-01  1.763e-01  -1.949   0.0642 .
## I(hard^2)       3.803e-01  2.516e-01   1.512   0.1448
## I(tens^2)      -1.752e-01  1.012e-01  -1.731   0.0975 .
## I(hard^2 * tens) 2.638e-03  1.245e-03   2.118   0.0457 *
## I(tens^3)       3.612e-04  1.818e-04   1.987   0.0595 .
## I(hard^3)      -4.308e-03  2.323e-03  -1.854   0.0772 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 26.93 on 22 degrees of freedom
## Multiple R-squared:  0.9291, Adjusted R-squared:  0.9065
## F-statistic: 41.18 on 7 and 22 DF,  p-value: 3.356e-11
```

We now remove the variable  $h^2$ .

```

m4 <- update(m3, . ~ . - I(hard^2))
summary(m4)

##
## Call:
## lm(formula = loss ~ tens + I(hard * tens) + I(tens^2) + I(hard^2 *
##      tens) + I(tens^3) + I(hard^3), data = Rubber)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.690  -7.647   2.943  12.765  50.057
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.252e+03  1.090e+03  -1.148  0.262582
## tens           3.810e+01  1.786e+01   2.133  0.043794 *
## I(hard * tens) -8.193e-02  3.443e-02  -2.379  0.026016 *
## I(tens^2)      -2.319e-01  9.656e-02  -2.402  0.024789 *
## I(hard^2 * tens) 8.105e-04  3.090e-04   2.623  0.015201 *
## I(tens^3)       4.574e-04  1.750e-04   2.614  0.015507 *
## I(hard^3)      -8.090e-04  2.072e-04  -3.905  0.000713 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.67 on 23 degrees of freedom
## Multiple R-squared:  0.9217, Adjusted R-squared:  0.9013
## F-statistic: 45.14 on 6 and 23 DF, p-value: 1.375e-11

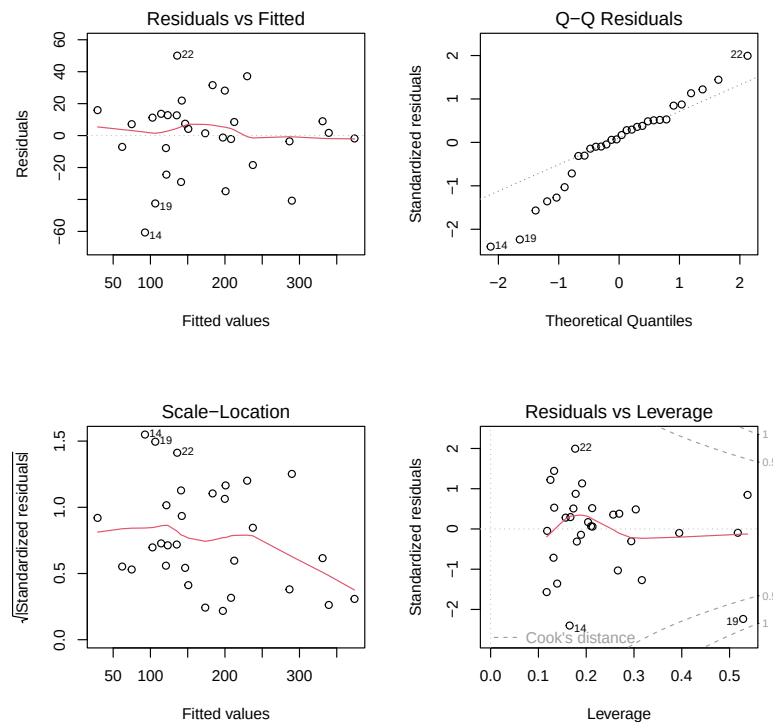
```

All variables in this last model are now significant at the 5% level. In the above, the `update` function was used to *update* the fitted model: by specifying only the variables to be dropped from the previous fit there is less typing to do (compared to calling `lm` each time), and hence less chance of error. Here are the final residual plots.

```

par(mfrow = c(2, 2))
plot(m4)

```



As before, for just 30 observations this looks reasonable. Observation 19 still has a high Cook's distance and might be worth investigating its impact on both model fit and model selection.

### 5.2.1 drop1

`drop1` is a useful R function which will work through the terms in your model, dropping each term and comparing the fit of the model without that term, to the full model fit. For a linear model the comparison can be done by F ratio testing or AIC (see below). For example, for the full tyre wear model we get the following.

```
drop1(m1, test = "F")

## Single term deletions
##
## Model:
## loss ~ hard + tens + I(hard * tens) + I(hard^2) + I(tens^2) +
##       I(hard^2 * tens) + I(tens^2 * hard) + I(tens^3) + I(hard^3)
##
```

|                  | Df | Sum of Sq | RSS   | AIC    | F value | Pr(>F) |
|------------------|----|-----------|-------|--------|---------|--------|
| <none>           |    |           | 15932 | 208.25 |         |        |
| hard             | 1  | 23.55     | 15955 | 206.29 | 0.0296  | 0.8652 |
| tens             | 1  | 277.24    | 16209 | 206.76 | 0.3480  | 0.5618 |
| I(hard * tens)   | 1  | 263.52    | 16195 | 206.74 | 0.3308  | 0.5716 |
| I(hard^2)        | 1  | 339.26    | 16271 | 206.88 | 0.4259  | 0.5214 |
| I(tens^2)        | 1  | 454.44    | 16386 | 207.09 | 0.5705  | 0.4589 |
| I(hard^2 * tens) | 1  | 1730.76   | 17662 | 209.34 | 2.1728  | 0.1560 |
| I(tens^2 * hard) | 1  | 5.16      | 15937 | 206.25 | 0.0065  | 0.9366 |
| I(tens^3)        | 1  | 1173.63   | 17105 | 208.38 | 1.4733  | 0.2390 |
| I(hard^3)        | 1  | 1176.52   | 17108 | 208.38 | 1.4770  | 0.2384 |

The p-value for each row is that obtained by testing the null hypothesis that the model *without* that row's term is correct, against the alternative that the full model is required. So a high p-value implies that the data are quite consistent with the model in which the term concerned has been dropped. Notice that because each model

comparison here involves just one parameter, the p-values obtained by F ratio testing are the same as those obtained from the t-tests used in `summary`. So why bother with `drop1`? The answer is that it is much more convenient than `summary` when there are factor variables in the model. In such a case `summary` will report p-values for each parameter associated with the factor, which are not suitable for deciding on whether the whole factor variable should be included in the model or not. By contrast, `drop1` will compare the models with and without the factor variable (thereby dealing with all its parameters at once) which is what is needed for model selection.

### 5.2.2 AIC

The default model comparison method for `drop1` is AIC: Akaike's Information Criterion. When we compare models by hypothesis testing we are effectively asking 'what is the simplest model that is defensible for these data', since hypothesis testing always retains the simpler model unless the data provide strong evidence that it is inadequate. AIC takes a different approach: it attempts to estimate a particular measure of the 'distance'<sup>32</sup> between the fitted model and the underlying 'true model' that generated the data. When we perform model selection using AIC, we are seeking the model that is as 'close' as possible to the truth, in the sense of having the smallest expected predictive error. Models with lower AIC values are preferred over those with higher values. Formally, if  $l$  is the maximized log likelihood of the model and  $p$  the number of regression coefficients, then

$$AIC = -2l + 2(p + 1),$$

where the term  $p + 1$  reflects estimating  $p$  regression coefficients plus the variance of the error term,  $\sigma^2$ . For a linear regression model with iid  $N(0, \sigma^2)$  errors, fitted to  $n$  observations, it is easy to verify that

$$AIC = n \log \left( \frac{RSS}{n} \right) + C + 2p, \quad (16)$$

where  $C = n + n \log(2\pi) + 2$  is a constant depending only on the sample size. Since  $C$  does not depend on the model, it can be ignored for variable-selection purposes. Adding more predictors cannot increase the RSS, but the penalty term  $2p$  discourages including too many variables. The AIC therefore provides a natural balance between model fit and model complexity. Note that, unlike  $F$ -tests, AIC can be used to compare non-nested models. However, for the comparison to be meaningful, AIC must be computed on the same response variable and on the same scale. The AIC function will compute the AIC of a fitted model object in R.

```
AIC(m1)

## [1] 295.382
```

And here is what `drop1` does by default (i.e., when `test = "F"` is not specified).

```
drop1(m1)

## Single term deletions
##
## Model:
## loss ~ hard + tens + I(hard * tens) + I(hard^2) + I(tens^2) +
##       I(hard^2 * tens) + I(tens^2 * hard) + I(tens^3) + I(hard^3)
##               Df Sum of Sq  RSS    AIC
## <none>                        15932 208.25
## hard                1      23.55 15955 206.29
## tens                1     277.24 16209 206.76
## I(hard * tens)      1     263.52 16195 206.74
## I(hard^2)           1     339.26 16271 206.88
## I(tens^2)           1     454.44 16386 207.09
```

<sup>32</sup>Strictly speaking, AIC approximates the expected Kullback–Leibler (KL) divergence between the fitted model and the true data-generating process. Although KL divergence is not a true distance, it behaves like a distance in many ways and measures how different two probability distributions are.

```
## I(hard^2 * tens) 1 1730.76 17662 209.34
## I(tens^2 * hard) 1 5.16 15937 206.25
## I(tens^3) 1 1173.63 17105 208.38
## I(hard^3) 1 1176.52 17108 208.38
```

The row labelled <none> gives the AIC of the full model (model `m1` in this case). Irritatingly, this differs from that given by AIC because of differences in which model independent constants have been included in the log likelihood of the model: of course differences in AIC are unaffected by such details, as already mentioned. The remaining rows report the AIC of the model without the term concerned (but with all other model terms). The lowest AIC is for the model with `I(tens^2 * hard)` dropped, so this would be the first candidate for dropping.

Before proceeding, we perform a quick ‘sanity check’ to verify that `drop1` computes the AIC using Equation (16), ignoring the constant  $C$ . We compute it for the full model `m1`, so we should obtain the value 208.25.

```
p <- length(coef(m1))
n <- nrow(Rubber)
aic_drop1 <- n*log(sum(m1$residuals^2)/n) + 2*p
aic_drop1

## [1] 208.2456

# This is what AIC(m1) calculates
aic_constants_included <- aic_drop1 + n + n*log(2*pi) + 2
aic_constants_included

## [1] 295.382
```

### 5.2.3 step

Backward selection by iteratively calling `drop1` and removing the term it suggests is fairly straightforward and could be fully automated. The function `step` performs this process when its `direction` argument is set to `backward`, using AIC-based model selection. Here is the result of applying `step` to the tyre model.

```
step(m1, direction = "backward")

## Start: AIC=208.25
## loss ~ hard + tens + I(hard * tens) + I(hard^2) + I(tens^2) +
## I(hard^2 * tens) + I(tens^2 * hard) + I(tens^3) + I(hard^3)
##
##           Df Sum of Sq  RSS   AIC
## - I(tens^2 * hard) 1 5.16 15937 206.25
## - hard 1 23.55 15955 206.29
## - I(hard * tens) 1 263.52 16195 206.74
## - tens 1 277.24 16209 206.76
## - I(hard^2) 1 339.26 16271 206.88
## - I(tens^2) 1 454.44 16386 207.09
## <none> 15932 208.25
## - I(tens^3) 1 1173.63 17105 208.38
## - I(hard^3) 1 1176.52 17108 208.38
## - I(hard^2 * tens) 1 1730.76 17662 209.34
##
## Step: AIC=206.26
## loss ~ hard + tens + I(hard * tens) + I(hard^2) + I(tens^2) +
## I(hard^2 * tens) + I(tens^3) + I(hard^3)
##
##           Df Sum of Sq  RSS   AIC
```

```
## - hard          1      19.81 15956 204.29
## - I(hard^2)      1     335.12 16272 204.88
## <none>          1     15937 206.25
## - I(hard^3)      1    1222.59 17159 206.47
## - I(tens^2)      1    1985.73 17922 207.78
## - I(hard * tens) 1    2079.83 18016 207.94
## - I(hard^2 * tens) 1  2423.23 18360 208.50
## - tens          1    2577.65 18514 208.75
## - I(tens^3)      1    2624.71 18561 208.83
##
## Step: AIC=204.29
## loss ~ tens + I(hard * tens) + I(hard^2) + I(tens^2) + I(hard^2 *
##      tens) + I(tens^3) + I(hard^3)
##
##              Df Sum of Sq   RSS   AIC
## <none>              15956 204.29
## - I(hard^2)         1    1657.6 17614 205.26
## - I(tens^2)         1    2173.1 18130 206.12
## - I(hard^3)         1    2493.9 18450 206.65
## - I(hard * tens)    1    2754.8 18711 207.07
## - I(tens^3)         1    2863.5 18820 207.24
## - tens              1    3150.1 19106 207.70
## - I(hard^2 * tens)  1    3253.1 19210 207.86
##
## Call:
## lm(formula = loss ~ tens + I(hard * tens) + I(hard^2) + I(tens^2) +
##      I(hard^2 * tens) + I(tens^3) + I(hard^3), data = Rubber)
##
## Coefficients:
##      (Intercept)          tens  I(hard * tens)      I(hard^2)
##      -1.185e+03      3.630e+01      -3.436e-01      3.803e-01
##      I(tens^2)  I(hard^2 * tens)      I(tens^3)      I(hard^3)
##      -1.752e-01      2.637e-03      3.612e-04     -4.308e-03
```

Notice that this model is different to the one we arrived at by hypothesis testing, and has more terms. AIC generally tends to select more complex models than hypothesis testing, because of its different objective: the ‘best’ model, in a prediction error sense, is likely to be more complicated than ‘the simplest model that the data will allow’.

### 5.3 Forward and forward-backward selection

In the tyre wear example, it was fairly easy to decide to start with a cubic model and use backward selection. Given the aim of finding the optimum rubber we needed at least quadratic model, and a quartic model would have been excessively complex for the amount of data available. Backward selection is also theoretically appealing. We start with a model that is complicated enough that we are fairly sure that it is not wrong, it is just that it is overcomplicated. By starting with a more or less correct model, we ensure that the distributional assumptions underpinning the hypothesis tests (or AIC computation) are valid, so that the whole procedure is well-founded.

However, it is often the case that there is no obvious candidate for the ‘most complex reasonable model’ or that such a model would be far too complicated to use in practice. In this case we might choose to start with a relatively simple model (e.g. some initial subset of variables enter as simple linear effects, or, even the extreme case of a model only including an intercept) hoping that it is not too far wrong. We then repeatedly use hypothesis testing or AIC comparison to consider adding in extra variables, or higher order terms, one at a time. If an added term improves the fit significantly we keep it in the model. This method is ‘forward selection’. Clearly in using this method we are starting from a model that we are accepting is wrong, and if the procedure proceeds for more than one step, then it must rely on comparing models when both of them are wrong (we have no rigorous theory for this case). Nonetheless we can often end up with an adequate model this way, in which case the theoretical problems with the method used to find it are often unimportant. Forward and backward selection can be combined in various

ways. For example, run forward selection until no further variables are added, then run backward selection until no further variables are dropped, then run forward selection again, and so on until successive forward and backward steps both produce no change in the model.

The `step` function can also perform forward or forward-backward selection. For example, here is the tyre example again, but this time starting from a model with simple linear terms in `hard` and `tens` and performing forward selection. In the argument `scope`, we specify the full model used previously, which defines the list of variables that may be added at each step.

```
m0 <- lm(loss ~ hard + tens, data = Rubber)
step(m0, scope = loss ~ hard + tens + I(hard*tens) + I(hard^2) + I(tens^2)+
      I(hard^2*tens) + I(tens^2*hard) + I(tens^3) + I(hard^3),
      direction = "forward")

## Start:  AIC=218.66
## loss ~ hard + tens
##
##              Df Sum of Sq  RSS    AIC
## + I(tens^2 * hard)  1    5971.5 29978 215.21
## + I(hard * tens)    1    2919.1 33031 218.12
## <none>                35950 218.66
## + I(hard^3)          1    2046.8 33903 218.90
## + I(hard^2)          1    1722.8 34227 219.19
## + I(tens^3)          1    1535.8 34414 219.35
## + I(hard^2 * tens)  1    1458.2 34492 219.42
## + I(tens^2)          1    1042.4 34907 219.78
##
## Step:  AIC=215.21
## loss ~ hard + tens + I(tens^2 * hard)
##
##              Df Sum of Sq  RSS    AIC
## + I(hard^3)          1    2099.47 27879 215.03
## <none>                29978 215.21
## + I(hard^2)          1    1913.52 28065 215.23
## + I(hard^2 * tens)  1    1555.65 28423 215.61
## + I(tens^3)          1    1382.16 28596 215.79
## + I(tens^2)          1    1004.99 28973 216.19
## + I(hard * tens)     1     329.44 29649 216.88
##
## Step:  AIC=215.03
## loss ~ hard + tens + I(tens^2 * hard) + I(hard^3)
##
##              Df Sum of Sq  RSS    AIC
## + I(tens^3)          1    2278.98 25600 214.47
## + I(tens^2)          1    1873.52 26005 214.95
## <none>                27879 215.03
## + I(hard * tens)     1    1078.23 26801 215.85
## + I(hard^2)          1     989.00 26890 215.95
## + I(hard^2 * tens)  1     596.48 27282 216.38
##
## Step:  AIC=214.47
## loss ~ hard + tens + I(tens^2 * hard) + I(hard^3) + I(tens^3)
##
##              Df Sum of Sq  RSS    AIC
## + I(hard^2 * tens)  1     8638.1 16962 204.13
## + I(hard * tens)    1     7164.8 18435 206.62
## + I(tens^2)         1     6523.7 19076 207.65
## + I(hard^2)         1     2988.1 22612 212.75
## <none>                25600 214.47
```

```
##
## Step: AIC=204.13
## loss ~ hard + tens + I(tens^2 * hard) + I(hard^3) + I(tens^3) +
## I(hard^2 * tens)
##
##           Df Sum of Sq   RSS   AIC
## <none>                16962 204.13
## + I(hard^2)           1    575.06 16387 205.09
## + I(tens^2)           1    361.75 16600 205.48
## + I(hard * tens)      1      2.73 16959 206.12
##
## Call:
## lm(formula = loss ~ hard + tens + I(tens^2 * hard) + I(hard^3) +
## I(tens^3) + I(hard^2 * tens), data = Rubber)
##
## Coefficients:
## (Intercept)          hard          tens I(tens^2 * hard)
## 1.780e+03      -7.546e+00      -8.282e+00      -7.584e-04
## I(hard^3)      I(tens^3) I(hard^2 * tens)
## -2.036e-03      1.433e-04      2.248e-03
```

Slightly different to the previous model. Let us now set the argument direction to both.

```
m0 <- lm(loss ~ hard + tens, data = Rubber)
step(m0, scope = loss ~ hard + tens + I(hard*tens) + I(hard^2) + I(tens^2) +
I(hard^2*tens) + I(tens^2*hard) + I(tens^3) + I(hard^3),
direction = "both")

## Start: AIC=218.66
## loss ~ hard + tens
##
##           Df Sum of Sq   RSS   AIC
## + I(tens^2 * hard)  1    5971  29978 215.21
## + I(hard * tens)    1    2919  33031 218.12
## <none>                35950 218.66
## + I(hard^3)         1    2047  33903 218.90
## + I(hard^2)         1    1723  34227 219.19
## + I(tens^3)         1    1536  34414 219.35
## + I(hard^2 * tens)  1    1458  34492 219.42
## + I(tens^2)         1    1042  34907 219.78
## - tens              1    66607 102556 248.11
## - hard              1   169027 204977 268.88
##
## Step: AIC=215.21
## loss ~ hard + tens + I(tens^2 * hard)
##
##           Df Sum of Sq   RSS   AIC
## + I(hard^3)         1    2099  27879 215.03
## <none>                29978 215.21
## + I(hard^2)         1    1914  28065 215.23
## + I(hard^2 * tens)  1    1556  28423 215.61
## + I(tens^3)         1    1382  28596 215.79
## + I(tens^2)         1    1005  28973 216.19
## + I(hard * tens)    1     329  29649 216.88
## - I(tens^2 * hard)  1    5971  35950 218.66
## - tens              1   12181  42159 223.44
## - hard              1   39189  69168 238.29
##
```

```
## Step: AIC=215.03
## loss ~ hard + tens + I(tens^2 * hard) + I(hard^3)
##
##           Df Sum of Sq  RSS    AIC
## + I(tens^3)      1    2279.0 25600 214.47
## + I(tens^2)      1    1873.5 26005 214.95
## <none>                27879 215.03
## - I(hard^3)      1    2099.5 29978 215.21
## + I(hard * tens)  1    1078.2 26801 215.85
## - hard          1    2753.4 30632 215.86
## + I(hard^2)      1     989.0 26890 215.95
## + I(hard^2 * tens) 1     596.5 27282 216.38
## - I(tens^2 * hard) 1    6024.2 33903 218.90
## - tens          1   12776.0 40655 224.35
##
## Step: AIC=214.47
## loss ~ hard + tens + I(tens^2 * hard) + I(hard^3) + I(tens^3)
##
##           Df Sum of Sq  RSS    AIC
## + I(hard^2 * tens) 1    8638.1 16962 204.13
## + I(hard * tens)   1    7164.8 18435 206.62
## + I(tens^2)        1    6523.7 19076 207.65
## + I(hard^2)        1    2988.1 22612 212.75
## <none>                25600 214.47
## - hard            1    2013.1 27613 214.75
## - I(tens^3)       1    2279.0 27879 215.03
## - I(hard^3)       1    2996.3 28596 215.79
## - I(tens^2 * hard) 1    5835.1 31435 218.63
## - tens           1   13803.3 39403 225.41
##
## Step: AIC=204.13
## loss ~ hard + tens + I(tens^2 * hard) + I(hard^3) + I(tens^3) +
##           I(hard^2 * tens)
##
##           Df Sum of Sq  RSS    AIC
## <none>                16962 204.13
## + I(hard^2)          1    575.1 16387 205.09
## + I(tens^2)          1    361.7 16600 205.48
## + I(hard * tens)     1     2.7 16959 206.12
## - hard              1   4172.0 21134 208.72
## - I(tens^2 * hard)   1    6317.7 23279 211.62
## - I(hard^2 * tens)   1    8638.1 25600 214.47
## - I(tens^3)         1   10320.6 27282 216.38
## - I(hard^3)         1   11380.9 28343 217.53
## - tens              1   20802.6 37764 226.14
##
## Call:
## lm(formula = loss ~ hard + tens + I(tens^2 * hard) + I(hard^3) +
##     I(tens^3) + I(hard^2 * tens), data = Rubber)
##
## Coefficients:
##      (Intercept)             hard             tens  I(tens^2 * hard)
##      1.780e+03      -7.546e+00      -8.282e+00      -7.584e-04
##      I(hard^3)      I(tens^3)  I(hard^2 * tens)
##      -2.036e-03      1.433e-04      2.248e-03
```

## 5.4 Remarks on model selection

Finally, let us review the reasons for model selection.

1. We do model selection because we are often uncertain about the exact form that a model should take, even though it is often possible to write down a model that we expect to be ‘complicated enough’, so that that for some parameter values it should be a reasonable approximation to the truth.
2. Selection is important for interpretational reasons: simpler models are easier to interpret than complex ones.
3. Model selection also tends to improve the precision of estimates and the accuracy of model predictions.

Whether model selection is performed using AIC or hypothesis testing depends on the purposes of the analysis. If we want to develop a model for prediction purposes then it makes sense to use AIC, but if our interest lies in trying to understand relationships between the predictors and the response, it may be preferable to use hypothesis testing based methods to try and avoid including model terms unless there is good evidence that they are needed.

Finally note that there is a difficult problem associated with model selection:

- It is common practice to use model selection methods to choose one model from a large set of potential models, but then to treat the selected model exactly as if it were the only model we ever considered, when it comes to calculating confidence intervals etc. In doing this we neglect the uncertainty associated with model selection, and will therefore tend to overstate how precisely we know the coefficients of the selected model (and how precise its predictions are). This issue is an active area of current statistical research.

## 5.5 The $r^2$ statistic

It is useful to have a general measure of how closely a model fits the observed response data and the  $r^2$  statistic (also known as the coefficient of determination) provides this.  $r^2$  measures the proportion of the variance in the original data that is ‘explained’ by the fitted model. The proportion of variance left unexplained is the observed variance of the residuals divided by the observed variance of the response data. The proportion of variance explained is therefore one minus this ratio, so

$$r^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2 / n}{\sum_i (y_i - \bar{y})^2 / n} = 1 - \frac{\sum_i (y_i - \hat{\mu}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

Note that the denominator can be interpreted as the observed variance of the residuals of a regression model that contains only an intercept term,  $y_i = \beta + \epsilon_i$  (as we have seen in Workshop 1,  $\hat{\beta} = \bar{y}$  and  $\hat{\epsilon}_i = y_i - \bar{y}$ ). As we noted when introducing the AIC, the residual sum of squares cannot increase when a predictor is added to the model. Since the denominator depends only on  $\{y_i\}_{i=1}^n$  and is identical for every model fitted to a particular dataset, we conclude that, when predictors are added,  $r^2$  is monotone increasing. For the same reason we have  $\sum_i \hat{\epsilon}_i^2 \leq \sum_i (y_i - \bar{y})^2$ , implying that

$$\frac{\sum_i \hat{\epsilon}_i^2 / n}{\sum_i (y_i - \bar{y})^2 / n} \leq 1,$$

and therefore  $0 \leq r^2 \leq 1$ . Further notice that this conventional definition (from which the  $n$ ’s can be cancelled) uses biased variance estimators. As a result,  $r^2$  tends to overestimate how well a model is doing. The *adjusted*  $r^2$  avoids this, to some extent, by using unbiased estimators:

$$r_{\text{adj}}^2 = 1 - \frac{\sum_i \hat{\epsilon}_i^2 / (n - p)}{\sum_i (y_i - \bar{y})^2 / (n - 1)},$$

where  $p$  is the number of model parameters. In contrast to  $r^2$ ,  $r_{\text{adj}}^2$  does not necessarily increase monotonically as predictors are added to a model. Note that  $r_{\text{adj}}^2$  can be negative, which occurs when an estimated model predicts worse than an intercept only model.

A high  $r_{\text{adj}}^2$  is always better than a low  $r_{\text{adj}}^2$  (provided that the model assumptions are met), but a low  $r_{\text{adj}}^2$  should not necessarily be taken as indicating that a model is poor: some response data simply contain a great deal of random variability about which nothing can be done. Note that neither  $r^2$  or  $r_{\text{adj}}^2$  measures the correctness of a linear model; rather, they measure its usefulness, assuming the model is correct. For further clarification, please see this elucidating blog post:

Note that opinions differ about the appropriate definition of  $r^2$  when a model contains no intercept term. In R the definition of  $r^2$  uses 0 in place of  $\bar{y}$  in this circumstance, which has the somewhat undesirable effect of making the  $r^2$  increase substantially if an intercept term is dropped from a model.

Both  $r^2$  and  $r^2_{\text{adj}}$  are available in the `summary` output of an `lm` object. Returning to the tyre wear example, we will check the  $r^2$  and  $r^2_{\text{adj}}$  for the final model selected by backward selection and F testing, and compare it to the initial full model containing ten variables.

```
mf_testF <- lm(loss ~ tens + I(hard * tens) + I(tens^2) + I(hard^2*tens) +
               I(tens^3) + I(hard^3), data = Rubber)
summary(mf_testF)

##
## Call:
## lm(formula = loss ~ tens + I(hard * tens) + I(tens^2) + I(hard^2 *
##     tens) + I(tens^3) + I(hard^3), data = Rubber)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -60.690  -7.647   2.943  12.765  50.057
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.252e+03  1.090e+03  -1.148  0.262582
## tens           3.810e+01  1.786e+01   2.133  0.043794 *
## I(hard * tens) -8.193e-02  3.443e-02  -2.379  0.026016 *
## I(tens^2)      -2.319e-01  9.656e-02  -2.402  0.024789 *
## I(hard^2 * tens) 8.105e-04  3.090e-04   2.623  0.015201 *
## I(tens^3)       4.574e-04  1.750e-04   2.614  0.015507 *
## I(hard^3)      -8.090e-04  2.072e-04  -3.905  0.000713 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 27.67 on 23 degrees of freedom
## Multiple R-squared:  0.9217, Adjusted R-squared:  0.9013
## F-statistic: 45.14 on 6 and 23 DF,  p-value: 1.375e-11
```

As expected, the  $r^2$  of the initial full model is (slightly) higher than that of the final model, which includes fewer variables (0.9292 versus 0.9217). However, the  $r^2_{\text{adj}}$  is slightly higher for the simpler model (0.8973 versus 0.9013).

Finally, note that in the special case of a simple linear regression model, it can be shown that  $r^2$  is equal to the squared empirical correlation between the response and the covariate.

## 5.6 Prediction

Suppose that we have a fitted model and would like to use it to make predictions for a set of predictor variables values (which may have been used for fitting the model or may be new). For example, suppose that a prediction is required for `tens` = 200 and `hard` = 50 using the model fitted to the `Rubber` data. The procedure is as follows.

1. Use the new predictor variables values to create a vector  $\mathbf{x}_0$  in exactly the same way as the predictors would have been used to create a row of the design matrix if they had been used for model fitting.
2. The *prediction* is then  $\hat{\mu}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}}$ .
3. Note that  $\hat{\mu}_0$  is an estimator of the mean  $\mathbb{E}(y_0) = \mathbf{x}_0^T \boldsymbol{\beta} = \mu_0$  of the new observation  $y_0$  with predictors  $\mathbf{x}_0$ .

Obviously, multiple predictions can be made simultaneously for different combinations of predictor values by replacing  $\mathbf{x}_0$  with a prediction matrix,  $\mathbf{X}_0$ , say.

For example, the finally selected tyre wear model (obtained by backward selection through F-testing) was:

$$y_i = \beta_1 + \beta_2 t_i + \beta_3 t_i h_i + \beta_4 t_i^2 + \beta_5 h_i^2 t_i + \beta_6 t_i^3 + \beta_7 h_i^3 + \epsilon_i.$$

Hence to make a prediction for  $(h_0, t_0)$ , we form the vector  $\mathbf{x}_0^\top = (1, t_0, t_0 h_0, t_0^2, h_0^2 t_0, t_0^3, h_0^3)$  and the prediction is  $\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\beta}$ . With the values  $h_0 = 50$  and  $t_0 = 200$  then  $\mathbf{x}_0^\top = (1, 200, 10000, 40000, 500000, 8000000, 1250000)$ . We can calculate the corresponding prediction  $\hat{\mu}_0$  in R.

```
x0 <- c(1, 200, 10000, 40000, 500000, 8000000, 1250000)
t(x0) %*% coef(m4)

##           [, 1]
## [1, ] 236.0129
```

The R function `predict` automates this calculation. Notice how a *data frame*, `newdata`, is supplied containing the values at which predictions are required.

```
predict(m4, newdata = data.frame(tens = 200, hard = 50))

##           1
## 236.0129
```

Of course, we need more than just a point estimate to make informed choices or decisions. Let us therefore determine the confidence interval for  $\mu_0$ , assuming that the linear model assumptions hold. We start by noting that

$$\mathbb{E}(\hat{\mu}_0) = \mathbb{E}(\mathbf{x}_0^\top \hat{\beta}) = \mathbf{x}_0^\top \mathbb{E}(\hat{\beta}) = \mathbf{x}_0^\top \beta = \mu_0.$$

The variance of the prediction is also easily obtained using the covariance matrix of  $\hat{\beta}$

$$\begin{aligned} \text{var}(\hat{\mu}_0) &= \text{var}(\mathbf{x}_0^\top \hat{\beta}) = \mathbf{x}_0^\top \text{var}(\hat{\beta}) (\mathbf{x}_0^\top)^\top \\ &= \mathbf{x}_0^\top \mathbf{V}_{\hat{\beta}} \mathbf{x}_0, \quad \text{where} \quad \mathbf{V}_{\hat{\beta}} = \mathbf{R}^{-1} \mathbf{R}^{-T} \sigma^2 = (\mathbf{X}^\top \mathbf{X})^{-1} \sigma^2. \end{aligned}$$

Because  $\hat{\beta}$  is normally distributed, we have that the linear combination  $\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\beta}$  is also normally distributed,

$$\hat{\mu}_0 \sim N(\mu_0, \sigma_{\hat{\mu}_0}^2), \quad \text{with} \quad \mu_0 = \mathbf{x}_0^\top \beta, \quad \sigma_{\hat{\mu}_0}^2 = \mathbf{x}_0^\top \mathbf{V}_{\hat{\beta}} \mathbf{x}_0.$$

Standardising yields

$$\frac{\hat{\mu}_0 - \mu_0}{\sigma_{\hat{\mu}_0}} \sim N(0, 1).$$

If we replace  $\sigma_{\hat{\mu}_0}$  by  $\hat{\sigma}_{\hat{\mu}_0}$ , which is equivalent to replace  $\mathbf{V}_{\hat{\beta}}$  by  $\hat{\mathbf{V}}_{\hat{\beta}}$  which, in turn, amounts to replace  $\sigma^2$  by  $\hat{\sigma}^2$  in  $\mathbf{V}_{\hat{\beta}}$ , the resulting pivotal statistic follows a  $t$  distribution with  $n - p$  degrees of freedom, i.e.,

$$\frac{\hat{\mu}_0 - \mu_0}{\hat{\sigma}_{\hat{\mu}_0}} \sim t_{n-p}, \quad \hat{\sigma}_{\hat{\mu}_0}^2 = \mathbf{x}_0^\top \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{x}_0, \quad \hat{\mathbf{V}}_{\hat{\beta}} = \mathbf{R}^{-1} \mathbf{R}^{-T} \hat{\sigma}^2.$$

Therefore, a confidence interval for the expected value of a new observation  $y_0$ ,  $\mu_0 = \mathbb{E}(y_0)$ , with predictors  $\mathbf{x}_0$  at level  $1 - \alpha$  is given by

$$\hat{\mu}_0 \pm t_{1-\alpha/2, n-p} \hat{\sigma}_{\hat{\mu}_0},$$

where  $t_{1-\alpha/2, n-p}$  is the value below which a  $t_{n-p}$  random variable will lie with probability  $1 - \alpha/2$ .

The `predict` function also returns the estimated standard error of the prediction  $\hat{\sigma}_{\hat{\mu}_0}$ , as well as, a confidence interval for the expected response  $\mu_0$ .

```
predict(m4, newdata = data.frame(tens = 200, hard = 50), se = TRUE)
```

```
## $fit
##      1
## 236.0129
##
## $se.fit
## [1] 16.64521
##
## $df
## [1] 23
##
## $residual.scale
## [1] 27.67362

# Manually compute the estimated standard error of the prediction
as.numeric(sqrt(t(x0) %*%vcov(m4) %*%x0))

## [1] 16.64521
```

Note that the last value, the `residual.scale`, is simply  $\hat{\sigma}$ , where  $\hat{\sigma}^2 = \|\mathbf{r}\|^2 / (n - p)$ .

```
sigma(m4)

## [1] 27.67362

sqrt(sum(m4$residuals^2) / (nrow(Rubber) - length(coef(m4))))

## [1] 27.67362
```

Asking for the confidence interval we get the following.

```
predict(m4, newdata = data.frame(tens = 200, hard = 50), interval = "confidence")

##      fit      lwr      upr
## 1 236.0129 201.5797 270.4461

# Manually compute the confidence interval
n <- nrow(Rubber); p <- length(coef(m4)); df <- n - p

ll <- 236.0129 - qt(0.975, df = n - p) * 16.64521
ul <- 236.0129 + qt(0.975, df = n - p) * 16.64521
ll; ul

## [1] 201.5797
## [1] 270.4461
```

Note that the default confidence level is .95, but this can be altered using the `level` argument. Of course it is also possible to obtain several predictions (and associated standard errors and intervals) at once. For example

```
predict(m4, newdata = data.frame(tens = c(200, 190), hard = c(50, 60)),
       interval = "confidence")

##      fit      lwr      upr
## 1 236.0129 201.5797 270.4461
## 2 198.3464 176.7126 219.9802
```

In many cases, one is also interested in determining an interval that will contain a new observation  $y_0$  with some specified probability, given some particular values for the predictors  $\mathbf{x}_0$ . In the context of the tyre wear example, this means that we would like to know the range of reasonable abrasion loss values for a tyre with given

characteristics (i.e., for a specific hardness and tensile strength, e.g., 50 and 200, respectively). Thus, we are looking for a **prediction interval** for the new observation  $y_0$ . For this purpose, consider the prediction error (or residual) for the new observation

$$\hat{\epsilon}_0 = y_0 - \mathbf{x}_0^T \hat{\beta} = y_0 - \hat{\mu}_0,$$

where

$$y_0 = \mu_0 + \epsilon_0 = \mathbf{x}_0^T \beta + \epsilon_0, \quad \epsilon_0 \sim N(0, \sigma^2). \quad (17)$$

We then have

$$\mathbb{E}(\hat{\epsilon}_0) = \mathbb{E}(y_0 - \hat{\mu}_0) = \mathbb{E}(y_0) - \mathbb{E}(\hat{\mu}_0) = \mu_0 - \mu_0 = 0,$$

and

$$\text{var}(\hat{\epsilon}_0) = \text{var}(y_0) + \text{var}(\hat{\mu}_0) = \sigma^2 + \sigma_{\hat{\mu}_0}^2.$$

To compute this variance, we use the facts that (1)  $y_0$  is independent of  $\hat{\mu}_0$  because  $y_0$  is independent of the observations  $y_1, \dots, y_n$  used to estimate  $\hat{\beta}$  and thus independent of  $\hat{\mu}_0 = \mathbf{x}_0^T \hat{\beta}$ , and (2)  $y_0$  arises from the same data-generating mechanism as the observed data, so  $\text{var}(y_0) = \sigma^2$  (see (17)). Since the difference of two normal random variables is also normal, it follows that

$$\hat{\epsilon}_0 \sim N(0, \sigma^2 + \sigma_{\hat{\mu}_0}^2).$$

Standardising and replacing  $\sigma^2$  and  $\sigma_{\hat{\mu}_0}^2$  by  $\hat{\sigma}^2$  and  $\hat{\sigma}_{\hat{\mu}_0}^2$ , respectively, yields

$$\frac{\hat{\epsilon}_0}{(\hat{\sigma}^2 + \hat{\sigma}_{\hat{\mu}_0}^2)^{1/2}} = \frac{y_0 - \hat{\mu}_0}{(\hat{\sigma}^2 + \hat{\sigma}_{\hat{\mu}_0}^2)^{1/2}} \sim t_{n-p}.$$

Therefore,

$$\Pr \left( -t_{1-\alpha/2, n-p} < \frac{y_0 - \hat{\mu}_0}{(\hat{\sigma}^2 + \hat{\sigma}_{\hat{\mu}_0}^2)^{1/2}} < t_{1-\alpha/2, n-p} \right) = 1 - \alpha,$$

which implies

$$\Pr \left( \hat{\mu}_0 - t_{1-\alpha/2, n-p}(\hat{\sigma}^2 + \hat{\sigma}_{\hat{\mu}_0}^2)^{1/2} < y_0 < \hat{\mu}_0 + t_{1-\alpha/2, n-p}(\hat{\sigma}^2 + \hat{\sigma}_{\hat{\mu}_0}^2)^{1/2} \right) = 1 - \alpha.$$

That is, a *prediction interval* for a new response observation  $y_0$  at a given set of predictor values  $\mathbf{x}_0$  at level  $1 - \alpha$  is given by

$$\hat{\mu}_0 \pm t_{1-\alpha/2, n-p}(\hat{\sigma}^2 + \hat{\sigma}_{\hat{\mu}_0}^2)^{1/2}.$$

Note that both intervals are centred at  $\hat{\mu}_0$  but, by construction, the prediction interval is always wider than the corresponding confidence interval for  $\mu_0$ . In applications where the estimated error variance  $\hat{\sigma}^2$  is large, this difference can be substantial.

Even though the two intervals may seem similar at first glance, they are in fact very different. In the first case, we constructed a confidence interval for  $\mathbb{E}(y_0) = \mu_0$ . This implies that the random interval overlaps the (constant) mean  $\mathbb{E}(y_0)$  with probability  $1 - \alpha$ . In the second case, we constructed an interval that is very likely (more precisely with probability  $1 - \alpha$ ) to contain the (random) new observation  $y_0$ .

Prediction intervals are easily obtained in R using the `predict` function.

```
predict(m4, newdata = data.frame(tens = 200, hard = 50), interval = "prediction")

##          fit          lwr          upr
## 1 236.0129 169.208 302.8178
```

Notice that the prediction interval is much wider than the corresponding confidence interval for the mean response at  $\mathbf{x}_0$ . The confidence interval quantifies uncertainty about the mean response at  $\mathbf{x}_0$ , whereas the prediction interval incorporates both the uncertainty in estimating that mean and the variability of individual observations. For tyres with tensile and hardness values given by  $\mathbf{x}_0$ , if we repeatedly generated new datasets from the same model and constructed confidence intervals in this way, about 95% of those intervals would contain the true mean response  $\mu_0$ . In contrast, if we repeatedly generated new datasets and generated new observations  $y_0$  for tyres with the same

characteristics, then about 95% of the corresponding prediction intervals would contain the realised abrasion loss observation.

Lastly, when predicting a new observation or estimating the mean response at a given predictor vector  $\mathbf{x}_0$ , one must be careful not to *extrapolate* too far beyond the range of the predictors. The following article is a popular example of an extrapolation ‘abuse’.

[www.nature.com/articles/431525a](http://www.nature.com/articles/431525a)

The researchers observed that female sprinters have increased their running speed at a faster rate than male sprinters over the past few decades and, based on this trend, predicted that by 2156, women would surpass men in sprinting speed. However, as noted by Rice (2004) and Reinboud (2004), applying the same regression model further would lead to the absurd prediction of instantaneous sprints by the year 2636. The two aforementioned articles by Rice and by Reinboud can be accessed here:

[www.nature.com/articles/432147b](http://www.nature.com/articles/432147b)

[www.nature.com/articles/432147a](http://www.nature.com/articles/432147a)

## 6 Practical modelling with factors and interactions

Most of the models covered so far have been for situations in which the predictor variables are continuous (e.g., the speed in the `cars` dataset or the hardness and tensile strength in the `Rubber` dataset). However, there are many situations in which the predictor variables are more qualitative in nature and serve to divide the responses into groups. Examples might be eye-colour of subjects in a psychology experiment, which of three alternative hospitals were attended by patients in a drug trial, manufacturers of cars used in crash tests, etc. Variables like these, which serve to classify the units on which the response variable has been measured into distinct categories, are known as *factor variables*, or simply *factors*. The different categories of the factor are known as *levels* of the factor. For example, levels of the factor ‘eye colour’ might be ‘blue’, ‘brown’, ‘grey’ and ‘green’, so that we would refer to eye colour as a factor with four levels. It is important to realise that the term *levels* does not imply any ordering; levels simply represent categories or groups.

Factor variables are handled using dummy (or indicator) variables. Each factor variable can, in principle, be replaced by as many dummy variables as there are levels of the factor — one for each level of the factor. For each response observation, only one of these dummy variables will be non-zero: the dummy variable for the single level that applies to that response. Consider an example to see how this works in practice: nine laboratory rats are overfed, dividing them into three groups of three: ‘fat’, ‘very fat’, and ‘enormous’. Their blood insulin levels are then measured 10 minutes after being fed a standard amount of sugar. The investigators are interested in the relationship between insulin levels and the factor ‘rat size’. Hence a model could be set up in which the predictor is the factor variable *rat size*, with the three levels ‘fat’, ‘very fat’ and ‘enormous’. Writing  $y_i$  for the  $i^{\text{th}}$  insulin level measurement, a suitable model might be:

$$\mathbb{E}(y_i) \equiv \mu_i = \begin{cases} \beta_0 & \text{if rat } i \text{ is fat,} \\ \beta_1 & \text{if rat } i \text{ is very fat,} \\ \beta_2 & \text{if rat } i \text{ is enormous,} \end{cases}$$

and this is easily written in linear model form, using a dummy predictor variable for each level of the factor:

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix},$$

where we have assumed that the first three rats belong to the fat group, the next three to the very fat group, and the last three to the enormous group. A key difference between dummy variables and directly measured predictor variables, is that the dummy variables, and parameters associated with a factor, are almost always treated as a group during model selection — it does not usually make sense for a subset of the dummy variables associated with a factor to be dropped or included on their own: either all are included or none. The F-ratio tests derived in Section 4.3.2 are particularly well-suited for hypothesis testing in this situation.

### 6.1 Identifiability

When modelling with factor variables, model *identifiability* becomes an important issue. In a linear model, the parameters are identifiable if, whenever two sets of parameter values differ, the expected response vector generated by the model is also different. If the design matrix  $\mathbf{X}$  is full column rank, then the parameters in the linear model are identifiable. It is quite easy to set up models, involving factor variables, in which it is impossible to estimate the parameters uniquely, because an infinite set of alternative parameter vectors would give rise to exactly the same expected value vector. A simple example illustrates the problem. Consider again the rats example, but suppose that we wanted to formulate the model in terms of an overall mean insulin level,  $\alpha$ , and deviations from that level,

$\beta_j$ , associated with each level of the factor. The model would be something like:

$$\mu_i = \alpha + \beta_j, \text{ if rat } i \text{ is rat size level } j, \quad j \in \{0, 1, 2\} \quad (18)$$

where  $j$  is 0, 1, or 2, corresponding to ‘fat’, ‘very fat’, or ‘enormous’. The problem with this model is that there is not a one-to-one correspondence between the parameters and the fitted values, so that the parameters cannot be uniquely estimated from the data. This is easy to see. Consider any particular set of  $\alpha$  and  $\beta$  values, giving rise to a particular  $\mu$  value: any constant  $c$  could be added to  $\alpha$  and simultaneously subtracted from each element of  $\beta$  without changing the value of  $\mu$  (e.g.,  $\mu_1 = \alpha + \beta_0 = (\alpha + c) + (\beta_0 - c)$  and the same for  $\mu_2, \dots, \mu_9$ ). Hence, there is an infinite set of parameters giving rise to each  $\mu$  value, and therefore the parameters cannot be estimated uniquely. The model is not *identifiable*.

This situation can be diagnosed directly from the model design matrix. Written out in full, the mean vector corresponding to model (18) is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

The columns of the model matrix are not linearly independent, as the first column is the sum of the second, third, and fourth columns. This lack of full column rank means that the formulae for finding the least squares parameter estimates break down (in terms of Section 4, the triangular matrix  $\mathbf{R}$  will not be full rank and, consequently, will not be invertible. Equivalently,  $\mathbf{X}^T \mathbf{X}$  will also not be invertible). Identifiable models have model matrices of full column rank; unidentifiable ones are column rank deficient.

One possible solution to the identifiability problem is to set one of the unidentifiable parameters to zero, which requires only that the model is re-written, without the zeroed parameter. For example, in our running example, we could set  $\alpha$  to zero, and recover the original identifiable model we started with<sup>33</sup>. This is perfectly legitimate, since the reduced model is capable of reproducing any expected values that the original, unidentifiable model could produce. When specifying models with factor variables in R, the software automatically imposes an identifiability constraint. By default, and contrary to what we just mentioned, the constraint is that the parameter for the ‘first’ level of the factor is set to zero. This approach can be easily extended to the case of multiple factor variables. To illustrate this, let us write the identifiable version of model (18) but now with indicator variables<sup>34</sup>. To this end, let  $x_{1i} = 1$  if rat  $i$  is ‘very fat’ and  $x_{1i} = 0$  otherwise and, similarly,  $x_{2i} = 1$  if rat  $i$  is ‘enormous’ and  $x_{2i} = 0$  otherwise. Then

$$\mu_i = \alpha + \beta_1^* x_{1i} + \beta_2^* x_{2i}.$$

For ‘fat’ rats, since  $x_{1i} = x_{2i} = 0$ , we have that  $\mu_i = \alpha$ . That is, in this parametrisation,  $\alpha$  represents the expected insulin levels for a fat rat and so its interpretation is different from that in model (18)<sup>35</sup>. For very fat rats,  $x_{1i} = 1$  and  $x_{2i} = 0$ , so  $\mu_i = \alpha + \beta_1^*$ , and for enormous rats,  $x_{1i} = 0$  and  $x_{2i} = 1$  and hence  $\mu_i = \alpha + \beta_2^*$ . Suppose that rat  $i$  is fat and rat  $i'$  is very fat. Then  $\mu_{i'} - \mu_i = \alpha + \beta_1^* - \alpha = \beta_1^*$  and so this parameter represents the expected difference in insulin levels between a very fat and a fat rat. Similarly,  $\beta_2^*$  represents the expected difference in insulin levels between an enormous rat and a fat rat. Because all comparisons are made to the group of fat rats, this category is said to be the baseline or reference level. Further note that if rat  $i'$  is very fat and rat  $i''$  is enormous then  $\mu_{i'} - \mu_{i''} = (\alpha + \beta_1^*) - (\alpha + \beta_2^*) = \beta_1^* - \beta_2^*$ , i.e., the expected difference in insulin levels for a very fat rat and an enormous rat is  $\beta_1^* - \beta_2^*$ . All possible choices of  $c - 1$  indicators for  $c$  categories are equivalent, having the same model fit and the same results for inferences about the population means. The interpretation of

<sup>33</sup>However, as we shall see shortly, this approach does not generalise well when there is more than one factor variable, as multiple parameters need to be dropped in such cases.

<sup>34</sup>Instead of using multiple indices with an index for each level of the factor variable, as in Equation (18).

<sup>35</sup>Recall that in model (18), the intercept  $\alpha$  represents the mean insulin level.

the coefficients depend on the choice of reference category, but the differences between any two parameters used to compare groups are identical for each possible choice. Note, however, that the reference or baseline level should not be sparse. In the running example, suppose that there are very few fat rats and that they are the baseline level in the dataset. Then, in most cases, either one of  $x_{1i}$  or  $x_{2i}$  is equal to one, meaning that  $x_{1i} + x_{2i} = 1$  for many observations. This implies near linear dependency between the columns of the design matrix (recall that the first column consists of ones, corresponding to the intercept). Although  $\hat{\beta}$  is computable (i.e., we can invert  $\mathbf{R}$  or  $(\mathbf{X}^T \mathbf{X})$ ), the computation would be numerically unstable, leading to high variance inflation for parameter estimates, even though the model is technically identifiable.

In the one factor case, this discussion of identifiability may seem to be unnecessarily complicated, since it is so easy to write down the model directly in an identifiable form. However, when models involve more than one factor variable, the issue cannot be avoided.

## 6.2 Multiple factors

It is frequently the case that more than one factor variable should be included in a model and this is straightforward to do. Continuing our running example, it might be that the sex of the rats is also a factor in insulin production and that the appropriate model is:

$$\mu_i = \alpha + \beta_j + \gamma_k \text{ if rat } i \text{ is of size level } j \text{ and sex } k, \quad j \in \{0, 1, 2\}, \quad k \in \{0, 1\},$$

where  $k$  is 0 or 1 for male or female, respectively. Written out in full (assuming the rats are MMFFFMFMM) the model is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_0 \\ \beta_1 \\ \beta_2 \\ \gamma_0 \\ \gamma_1 \end{pmatrix}.$$

It is immediately obvious that the model matrix is of column rank 4, implying that two constraints are required to make the model identifiable. The lack of column linear independence can be observed by noting that, for instance, column 5 is column 1 minus column 6, while column 2 is column 1 minus columns 3 and 4<sup>36</sup>. In fact, two independent constraints are required: one for the size factor and one for the sex factor. An obvious pair of constraints is to set  $\beta_0 = \gamma_0 = 0$ , which aligns with what R does—excluding the first level of each factor variable—so that the full model is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \\ \gamma_1 \end{pmatrix}.$$

Note that in this last parametrisation the intercept  $\alpha$  represents the expected insulin levels for a fat male rat.

## 6.3 ‘Interactions’ of factors

In the examples considered so far, the effects of factor variables have been purely additive. However, a response variable may react differently to a combination of two factors than would be predicted by adding their separate

<sup>36</sup>Or column 3 is column 1 minus columns 2 and 4. Or column 4 is column 1 minus columns 2 and 3.

effects. For example, in examining patient blood cholesterol levels, we might consider the factors *hypertension* (yes/no) and *diabetes* (yes/no). We might hypothesize that each condition on its own raises cholesterol levels, but that being both hypertensive and diabetic increases cholesterol by more than the sum of the individual effects. When the combined effect of two factor variables differs from the sum of their separate effects, the factors are said to *interact*, and models for such situations require *interaction terms*. Put another way, two factors interact when the effect of one factor on the response changes depending on the level of the other factor.

Let us continue with our running example, but now suppose that how insulin level depends on size varies with sex. An appropriate model is then

$$\mu_i = \alpha + \beta_j + \gamma_k + \delta_{jk} \text{ if rat } i \text{ is of size level } j \text{ and sex } k,$$

where the  $\delta_{jk}$  terms are the parameters for the interaction of rat size and sex. Writing this model out in full it is clear that it is spectacularly unidentifiable (we have 1 intercept, 3 size effects, 2 sex effects, and 6 interaction terms):

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_0 \\ \beta_1 \\ \beta_2 \\ \gamma_0 \\ \gamma_1 \\ \delta_{00} \\ \delta_{01} \\ \delta_{10} \\ \delta_{11} \\ \delta_{20} \\ \delta_{21} \end{pmatrix}.$$

In fact, for this simple example, with rather few rats, we now have more parameters than observations. There are many possible identifiability constraints. One possibility (the default in **R**) is to set  $\beta_0 = \gamma_0 = \delta_{00} = \delta_{01} = \delta_{10} = \delta_{20} = 0$ . The resulting model can still produce any fitted value vector that the full model can produce, but all the columns of its model matrix are independent, so that the model is identifiable:

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta_1 \\ \beta_2 \\ \gamma_1 \\ \delta_{11} \\ \delta_{21} \end{pmatrix}.$$

Note that if the *main* effects<sup>37</sup> are made identifiable first, and the interaction terms are constructed from these identifiable main effects, then the identifiability is ‘inherited’ from the main effects. Of course, the more factor variables are present, the more interactions are possible, and the higher the order of the possible interactions: for example if three factors are present then each pair of factors could interact, giving three possible *two-way* interactions, while all three factors could also interact together, giving a *three-way* interaction (e.g., how insulin level dependence on rat size is influenced by sex might itself be influenced by blood group; for higher-order interactions, mathematical notation is often clearer than verbal description).

Finally, note that a statistical interaction is not about the relationship between predictors. It is about how the effect of one predictor on the response depends on the level of another predictor.

<sup>37</sup>The terms that would have been included in the model if the effects were purely additive—in this case, the terms  $\beta_j$  and  $\gamma_k$ , for  $j \in \{0, 1, 2\}$  and  $k \in \{0, 1\}$ .

## 6.4 Factor continuous interactions

Suppose we had access to the exact weight  $w_i$  of the rats, rather than only its classification into one of the three groups previously defined. In this case,

$$\mu_i = \alpha + \gamma_k + \beta w_i + \delta_k w_i \text{ if rat } i \text{ is sex } k, \quad (19)$$

might be appropriate. That is to say insulin levels vary linearly with weight, but in a different way for male and female rats. Here,  $\gamma_k$  would be the main effect of sex,  $\beta w_i$  the main effect of weight, and  $\delta_k w_i$  is the weight–sex interaction. As before, similar identifiability issues arise between the main effect of sex and the weight–sex interaction, and analogous constraints are needed. Recalling the assumed ordering of rats, MMFFFMFMM, an identifiable version of the model is

$$\begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \mu_4 \\ \mu_5 \\ \mu_6 \\ \mu_7 \\ \mu_8 \\ \mu_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & w_1 & 0 \\ 1 & 0 & w_2 & 0 \\ 1 & 1 & w_3 & w_3 \\ 1 & 1 & w_4 & w_4 \\ 1 & 1 & w_5 & w_5 \\ 1 & 0 & w_6 & 0 \\ 1 & 1 & w_7 & w_7 \\ 1 & 0 & w_8 & 0 \\ 1 & 0 & w_9 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \gamma_1 \\ \beta \\ \delta_1 \end{pmatrix}.$$

Note that the expectation in (19) can also be written in an identifiable form as

$$\mu_i = \alpha + \beta_1 \mathbf{1}(\text{sex rat}_i = \text{"F"}) + \beta_2 w_i + \beta_3 \mathbf{1}(\text{sex rat}_i = \text{"F"}) w_i$$

and the corresponding model is

$$y_i = \mu_i + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2).$$

In the identifiable parameterisation, the intercept  $\alpha$  corresponds to the insulin level of a male rat with a weight of zero (!). Note that the models in Sections 6.2 and 6.3 could also have been rewritten using indicator variables. However, this becomes cumbersome for factors with many levels and for higher-order interactions. We will use both formulations interchangeably throughout.

## 6.5 Using factor variables in R

It is very easy to work with factor variables in R. All that is required is that you let R know that a particular variable is a factor variable. For example, suppose  $z$  is a variable declared as follows:

```
z <- c(1, 1, 1, 2, 2, 1, 3, 3, 3, 3, 4)
z

## [1] 1 1 1 2 2 1 3 3 3 3 4

is.factor(z)

## [1] FALSE

is.numeric(z)

## [1] TRUE
```

and it is to be treated as a four-level factor. The function `as.factor()` will ensure that  $z$  is treated as a factor:

```
z <- as.factor(z)
z

## [1] 1 1 1 2 2 1 3 3 3 3 4
## Levels: 1 2 3 4
```

When a factor variable is printed, a list of its levels is also printed — this provides an easy way to tell if a variable is a factor variable. Note also that the digits of *z* are treated purely as labels: the numbers 1 to 4 could have been any labels. For example, *x* could be a factor with 3 levels, declared as follows:

```
x <- c("A", "A", "C", "C", "C", "er", "er")
x

## [1] "A" "A" "C" "C" "C" "er" "er"

x <- as.factor(x)
# Check reference level (the row that only contains zeroes)
contrasts(x)

##      C er
## A  0  0
## C  1  0
## er 0  1

# Want to change the reference level from A to er? Use the relevel function
x <- relevel(x, ref = "er")
x

## [1] A  A  C  C  C  er er
## Levels: er A C

contrasts(x)

##      A C
## er 0  0
## A  1  0
## C  0  1
```

Once a variable is declared as a factor variable, then R can process it automatically within model formulae, by replacing it with the appropriate number of binary dummy variables (and imposing any necessary identifiability constraints on the specified model).

It is worth emphasizing that factor predictors should not be coded as continuous predictors. In general, as mentioned at the beginning of this section, there is not necessarily a natural ordering of the levels of a factor variable. To give a concrete example, consider the vector *x* from the example above, which has levels A, C, and er. Suppose we code *x* as a numerical predictor, assigning the values 1, 2, and 3, to A, C, and er, respectively. This implicitly assumes an order in the values of *x*, suggesting that A is closer to C than to er. It also assumes that er is three times greater than A. Furthermore, the coding is completely arbitrary—why not assign 1, 1.5, and 1.75 instead? The correct to handle categorical predictors in regression is to set the variable as a factor and let R handle the ‘dummification’ internally.

### 6.5.1 Factor interactions in model formulae

Within model formulae, the notation *a:b* indicates that the interaction of variables *a* and *b* should be included in the model, while *a\*b* is exactly equivalent to *a + b + a:b*, meaning that main effects and interactions should be included. Note that if *a* and *b* are both factors, then  $y \sim a:b$  and  $y \sim a*b$  result in exactly the same model fits, but with different meanings for the parameters and different identifiability constraints.

With reference to the fat rat examples, if *insulin* is the response, *size* and *sex* are the factor variables, and *weight* the continuous variable, then

- $\text{insulin} \sim \text{size} - 1$  gives the model in Section 6.
- $\text{insulin} \sim \text{size}$  gives the model in Section 6.1 (same fit as above, but different parameter interpretation).

- `insulin ~ size + sex` gives the model in Section 6.2
- `insulin ~ size*sex` gives the model in Section 6.3.
- `insulin ~ sex*weight` gives the model in Section 6.4.

Another useful notation is provided by terms like  $(a+b+c)^2$ , which specifies main effects and interactions up to the order of the given power, two in this case. That is,  $(a+b+c)^2 = a + b + c + a:b + a:c + b:c$ .

### 6.5.2 A simple example

As an example of the use of factor variables, consider the `PlantGrowth` data frame supplied with R. These are data from an experiment to compare yields (as measured by dried weight of plants) obtained under a control and two different treatment conditions. The factor `group` has three levels `ctrl`, `trt1`, and `trt2`, and it is believed that the growth of the plants depends on this factor. First check that `group` is already a factor variable:

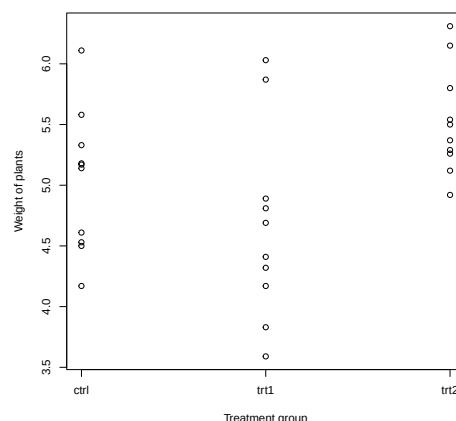
```
PlantGrowth$group
## [1] ctrl ctrl ctrl ctrl ctrl ctrl ctrl ctrl ctrl ctrl trt1 trt1 trt1 trt1 trt1
## [16] trt1 trt1 trt1 trt1 trt1 trt1 trt2 trt2 trt2 trt2 trt2 trt2 trt2 trt2 trt2
## Levels: ctrl trt1 trt2
```

Since a list of levels is shown, it is indeed a factor. We can verify this explicitly:

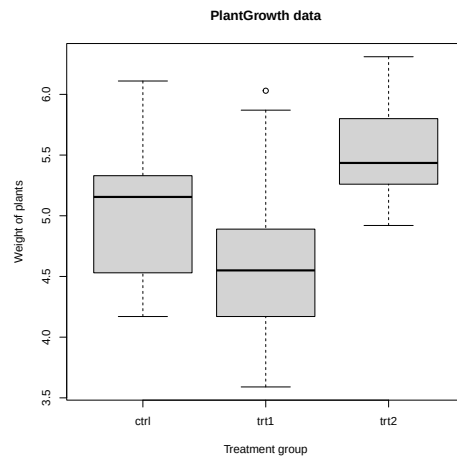
```
is.factor(PlantGrowth$group)
## [1] TRUE
```

The aim is to investigate whether the `group` factor influences the weight of the plants, and if so, by how much. We start by visualising the data; below are two common approaches.

```
stripchart(weight ~ group, data = PlantGrowth, vertical = TRUE, pch = 1,
           ylab = "Weight of plants", xlab = "Treatment group")
```

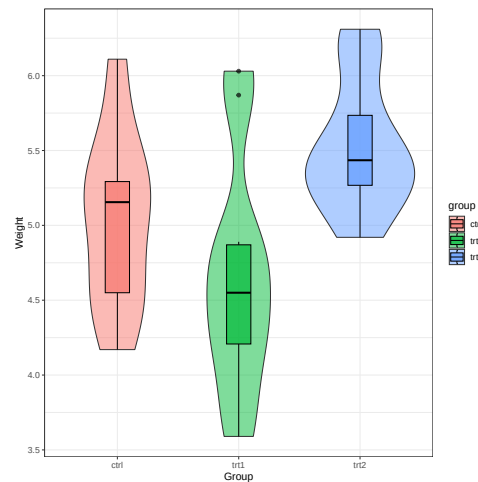


```
boxplot(weight ~ group, data = PlantGrowth, main = "PlantGrowth data",
        ylab = "Weight of plants", xlab = "Treatment group")
```



Treatment 2 appears to stand out in terms of plant weight. More sophisticated visualisations are also possible. Below is a violin plot, combining features of a boxplot and a kernel density estimate:

```
library(ggplot2)
ggplot(PlantGrowth, aes(x = group, y = weight, fill = group)) +
  geom_violin(alpha = 0.5) +
  geom_boxplot(width = 0.2, color = "black", alpha = 0.7) + # Boxplot inside
  theme_bw() +
  labs(x = "Group", y = "Weight")
```



Before fitting the model, let us formally define its mathematical formulation. One convenient representation is

$$y_i = \mu + \alpha_j + \epsilon_i, \quad \text{if plant } i \text{ is in treatment group } j, \quad j \in \{\text{ctrl}, \text{trt1}, \text{trt2}\}, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2),$$

for  $i = 1, \dots, n = 30$  and, for identifiability reasons, one of the  $\alpha_j$  parameters is set to zero; by default in R,  $\alpha_{\text{ctrl}} = 0$ . Equivalently (and making the grouping structure explicit),

$$y_{jk} = \mu + \alpha_j + \epsilon_{jk}, \quad j \in \{\text{ctrl}, \text{trt1}, \text{trt2}\}, \quad \epsilon_{jk} \stackrel{iid}{\sim} N(0, \sigma^2),$$

for  $k = 1, \dots, n_j$  and, in this case, the group sizes are equal  $n_{\text{ctrl}} = n_{\text{trt1}} = n_{\text{trt2}} = 10$ . A third, fully equivalent, representation uses indicator variables:

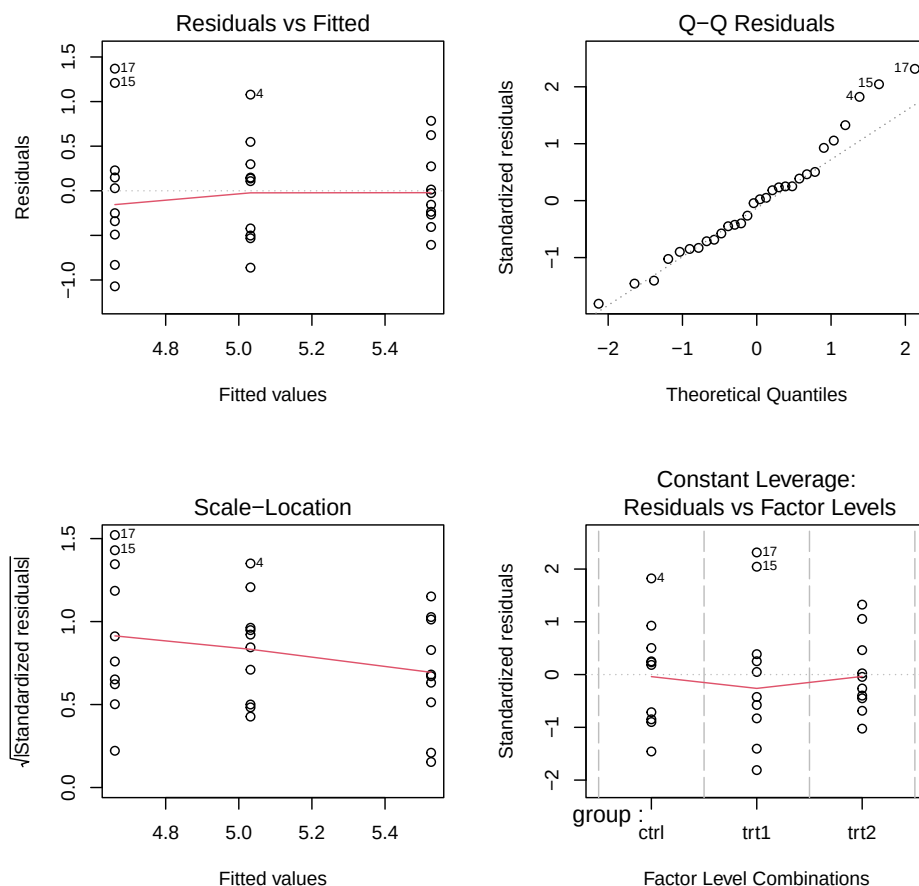
$$y_i = \beta_0 + \beta_1 \mathbf{1}(\text{group}_i = \text{"trt1"}) + \beta_2 \mathbf{1}(\text{group}_i = \text{"trt2"}) + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2),$$

for  $i = 1, \dots, n$ . Now let us fit the model. As usual, the first step after fitting a model is to check the residual plots.

```
pgm.1 <- lm(weight ~ group, data = PlantGrowth)
```

```
par(mfrow = c(2, 2))
```

```
plot(pgm.1)
```



First, note that in the plot of fitted values against residuals (or fitted values against square root of the absolute value of the standardised residuals), the three vertical stripes appear because the model predicts only three fitted values: one for each level of the group factor variable. In this case there is some suggestion of decreasing variance with increasing mean, as the residuals show an unequal vertical spread across the three categories. However, the effect does not appear very pronounced, so it is probably safe to proceed. Note that, since in this case the leverages are the same for each observation (because there is an equal number of observations in each group), the lower right plot is simplified (in fact for this specific example, it provides no additional information beyond what is already evident from the other three plots).

```
summary(pgm.1)
```

```
##
## Call:
## lm(formula = weight ~ group, data = PlantGrowth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0710 -0.4180 -0.0060  0.2627  1.3690
##
## Coefficients:
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.0320     0.1971  25.527  <2e-16 ***
## grouptrt1    -0.3710     0.2788  -1.331   0.1944
## grouptrt2     0.4940     0.2788   1.772   0.0877 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6234 on 27 degrees of freedom
## Multiple R-squared:  0.2641, Adjusted R-squared:  0.2096
## F-statistic: 4.846 on 2 and 27 DF,  p-value: 0.01591

#check first rows of model matrix used in the fit
head(model.matrix(pgm.1))

##      (Intercept) grouptrt1 grouptrt2
## 1             1          0          0
## 2             1          0          0
## 3             1          0          0
## 4             1          0          0
## 5             1          0          0
## 6             1          0          0
```

Notice that R reports an intercept parameter and parameters for the two treatment levels. To obtain an identifiable model, it omits a parameter for the control level of the group factor. Thus, the estimated overall mean weight (in the population that these plants represent, given control conditions) is 5.032. Treatment 1 is estimated to lower this mean weight by 0.37, and treatment 2 to increase it by 0.49. However, neither parameter individually appears to be significantly different from zero.

Model selection based on the summary output is very difficult for models containing factors. It makes little sense to drop the dummy variable for just one level of a factor from a model, and if we did, what would we then do about the model identifiability constraints? Usually, it is only of interest to test whether the whole factor variable should be in the model or not, and this amounts to testing whether all its associated parameters are simultaneously zero or not. The F-ratio tests derived in Section 4.3.2 are designed for precisely this purpose. with a reduced model in which the expected response is given by a single parameter (an intercept) that does not depend on `group`. In this case, the relevant hypothesis test is exactly the one associated with the  $F$ -statistic reported on the very last line of the summary output. The p-value is 0.016 suggesting that the null hypothesis of no group effect on weight is not very plausible. Thus, the data provide evidence for an effect of the factor group on weight, even though the individual treatment coefficients appeared marginal or non-significant when examined separately. This is because the  $F$ -test considers all the parameters associated with the factor simultaneously, yielding a more powerful test than any of the individual  $t$ -tests. In light of this analysis, the most promising treatment to investigate further is clearly treatment 2, since it shows the largest and positive estimated effect. Of course, the same conclusion could be reached using the `anova` function.

```
pgm.0 <- lm(weight ~ 1, data = PlantGrowth)
anova(pgm.0, pgm.1)

## Analysis of Variance Table
##
## Model 1: weight ~ 1
## Model 2: weight ~ group
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      29 14.258
## 2      27 10.492  2     3.7663 4.8461 0.01591 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Let us now create a new group variable and change the baseline level from the control group to the treatment 1 group using the `relevel` function.

```

PlantGrowth$group_alt <- relevel(PlantGrowth$group, ref = "trt1")
pgm.l_alt <- lm(weight ~ group_alt, data = PlantGrowth)
summary(pgm.l_alt)

##
## Call:
## lm(formula = weight ~ group_alt, data = PlantGrowth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.0710 -0.4180 -0.0060  0.2627  1.3690
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.6610     0.1971  23.644 < 2e-16 ***
## group_altctrl    0.3710     0.2788   1.331  0.19439
## group_altrt2     0.8650     0.2788   3.103  0.00446 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6234 on 27 degrees of freedom
## Multiple R-squared:  0.2641, Adjusted R-squared:  0.2096
## F-statistic: 4.846 on 2 and 27 DF, p-value: 0.01591

```

Aside from the coefficients section, the rest of the output remains exactly the same. Note that under the first model (in which the control group is the baseline) the estimated mean plant growth in the control group is 5.0320, in the treatment 1 group is  $5.0320 - 0.3710 = 4.661$  and in the treatment group 2 is  $5.0320 + 0.4940 = 5.526$ . In turn, in the second model (in which the treatment 1 group is the baseline), the estimated mean plant growth in the control group is  $4.6610 + 0.3710 = 5.032$ , in the first treatment group is 4.6610 and in the second treatment group is  $4.6610 + 0.8650 = 5.526$ . Further, and for instance, under the first model, the estimated difference in mean plant growth between treatment group 1 and 2 is  $-0.3710 - 0.4940 = -0.865$ , whereas such difference in the second model is given by  $-0.8650$ .

Finally, note that using the control group as the baseline (or reference level) is more meaningful here than using treatment 1 or 2. In this parameterisation, each treatment coefficient represents the difference in mean growth relative to the control group, which aligns naturally with the scientific question of interest.

## 6.6 The warpbreak data

Now consider an example with two factors as predictors. The `warpbreak` data in R come from an industrial experiment investigating optimal weaving conditions to minimise the number of breaks in wool threads. Two types of wool (A and B) were tested at each of three weaving tension levels (**Low**, **Medium**, **High**), with 9 replicates for each combination of wool type and tension level. The response recorded was the number of breaks in a standard length of cloth. The experiment was designed so that replicates for each combination of wool type and tension level are independent. First, let us take a quick look at the data.

```

names(warpbreaks)

## [1] "breaks" "wool" "tension"

warpbreaks$breaks

## [1] 26 30 54 25 70 52 51 26 67 18 21 29 17 12 18 35 30 36 36 21 24 18 10 43 28
## [26] 15 26 27 14 29 19 29 31 41 20 44 42 26 19 16 39 28 21 39 29 20 21 24 17 13
## [51] 15 15 16 28

warpbreaks$wool

```

```
## [1] A A A A A A A A A A A A A A A A A A A A A A A B B B B B B B B B B
## [39] B B B B B B B B B B B B B B B B
## Levels: A B

warpbreaks$tension

## [1] L L L L L L L L M M M M M M M M M H H H H H H H H L L L L L L L L M M
## [39] M M M M M M M H H H H H H H H H
## Levels: L M H
```

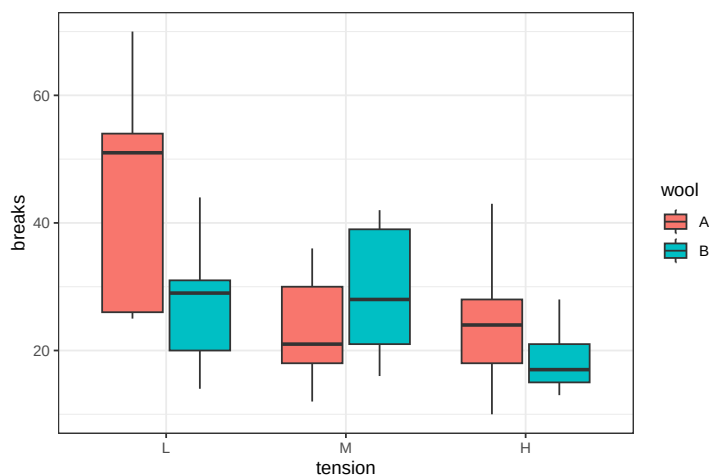
The following produces a helpful display of the data.

```
warpbreaks$replicate <- rep(1:9, len = 54)
fable(xtabs(breaks ~ wool + tension + replicate, data = warpbreaks))

##           replicate  1  2  3  4  5  6  7  8  9
## wool tension
## A      L
##       26 30 54 25 70 52 51 26 67
##       M
##       18 21 29 17 12 18 35 30 36
##       H
##       36 21 24 18 10 43 28 15 26
## B      L
##       27 14 29 19 29 31 41 20 44
##       M
##       42 26 19 16 39 28 21 39 29
##       H
##       20 21 24 17 13 15 15 16 28
```

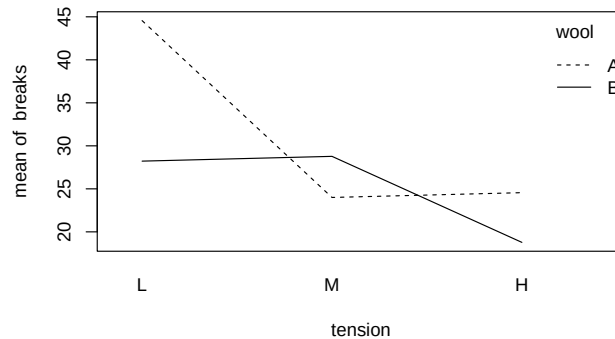
Let us start with a visual inspection of the data.

```
ggplot(warpbreaks, aes(x = tension, y = breaks, fill = wool)) +
  geom_boxplot() +
  theme_bw()
```



There is a clear effect of tension on the number of breaks. The effect of wool type on the number of breaks varies across tension levels, suggesting a possible interaction between tension and wool type. That is, the follow-up question is now: *Does the effect of tension on the number of breaks depend on the wool type (or vice versa)?* In addition to the above plot, the so-called *interaction plot* is also useful to informally investigate a possible interaction effect between tension level and wool type.

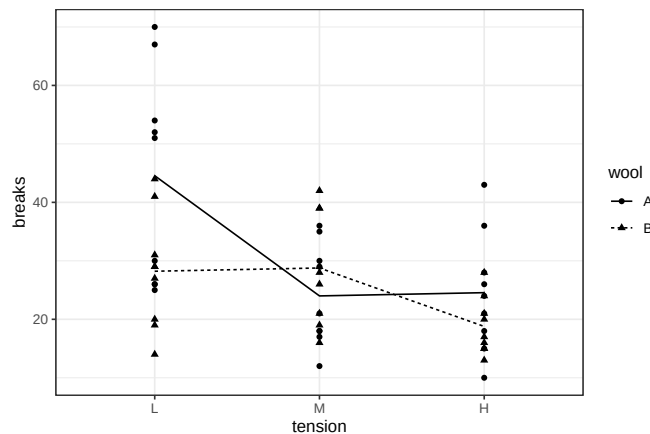
```
with(warpbreaks, interaction.plot(x.factor = tension, trace.factor = wool,
  response = breaks))
```



```
# Another way
#interaction.plot(x.factor = warpbreaks$tension, trace.factor = warpbreaks$wool,
#               response = warpbreaks$breaks,
#               xlab = "Tension level",
#               ylab = "Mean number of breaks", legend = FALSE)
#legend("topright", c("A", "B"), lty = c(2, 1), bty = "n")
```

In the interaction plot, for every combination of wool type (A and B) and tension level (L, M, H), the mean of the response variable (number of breaks) is calculated. We use tension on the  $x$ -axis (the `x.factor`). Points corresponding to the same wool type are connected with lines (the `trace.factor`). Of course, the roles of tension and wool can be interchanged. It is often natural to use an ordered factor such as tension (with levels: low, medium, and high) as the `x.factor`, as we did. If the lines are parallel (or nearly parallel), it suggests little to no interaction between tension level and wool type. In this case, the effect of wool type would remain consistent across all three tension levels. Formal hypothesis testing is needed to confirm this. However, if the lines are not parallel, as is the case here, this suggests an interaction between the two factor predictors. For example, wool B appears to result in fewer breaks at the lowest tension level, but this is no longer the case at medium tension. A more sophisticated version can be plotted with `ggplot2`.

```
ggplot(warpbreaks, aes(x = tension, y = breaks, lty = wool, shape = wool)) +
  geom_point() +
  stat_summary(fun = mean, geom = "line", aes(group = wool)) +
  theme_bw()
```



As was already evident in the boxplot, the variability in break counts differs across tension levels, particularly with lower spread at high tension and larger, unequal spreads at low and medium tension, providing some evidence of non-constant variance (heteroscedasticity).

Let us therefore consider a model with an interaction between wool type and tension level. Specifically, letting  $y_{ijk}$  denote the number of breaks for the  $k^{\text{th}}$  replicate at the  $i^{\text{th}}$  wool type and  $j^{\text{th}}$  tension, a possible model is

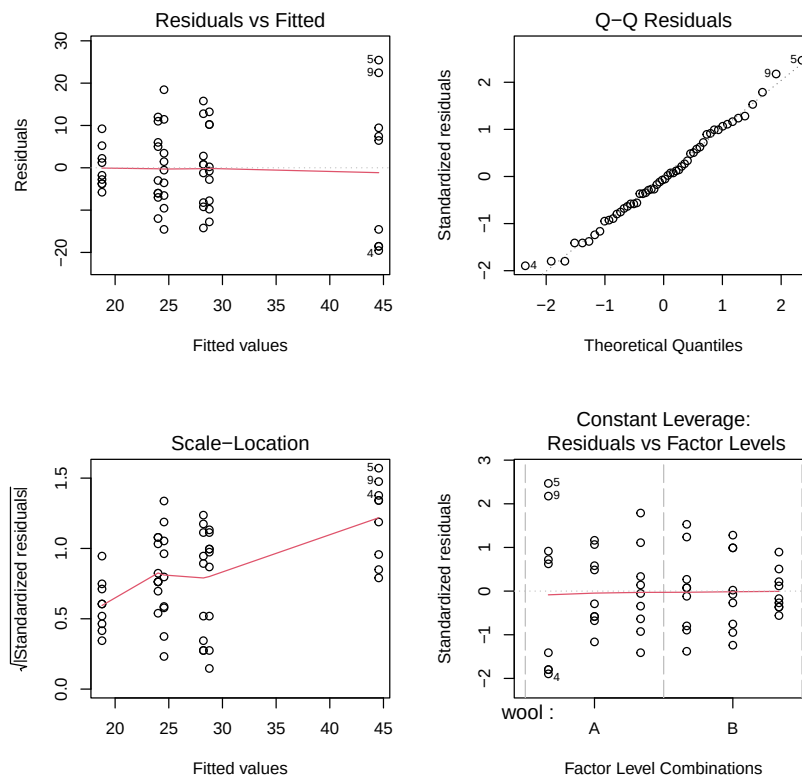
$$y_{ijk} = \mu + \alpha_i + \delta_j + \gamma_{ij} + \epsilon_{ijk}, \quad i \in \{A, B\}, \quad j \in \{L, M, H\}, \quad k \in \{1, \dots, 9\}, \quad (20)$$

where the two  $\alpha_i$  parameters are for the *main effect* of wool, the three  $\delta_j$  parameters are for the main effect of tension and the six  $\gamma_{ij}$  parameters represent the interaction of wool type and tension level ( $\mu$  is the overall expected number of breaks). Identifiability constraints are needed for this model, of course, but these are generated automatically in R. We already know what R does by default: it sets  $\alpha_A$ ,  $\delta_L$ ,  $\gamma_{AL}$ ,  $\gamma_{AM}$ ,  $\gamma_{AH}$ , and  $\gamma_{BL}$  to zero. Note that the model actually only has six identifiable parameters (rather than the 12 originally specified). Let us fit the model.

```
wlm0 <- lm(breaks ~ wool*tension, data = warpbreaks)
#equivalent to wlm0 <-lm(breaks ~ wool+ tension + wool:tension, data = warpbreaks)
```

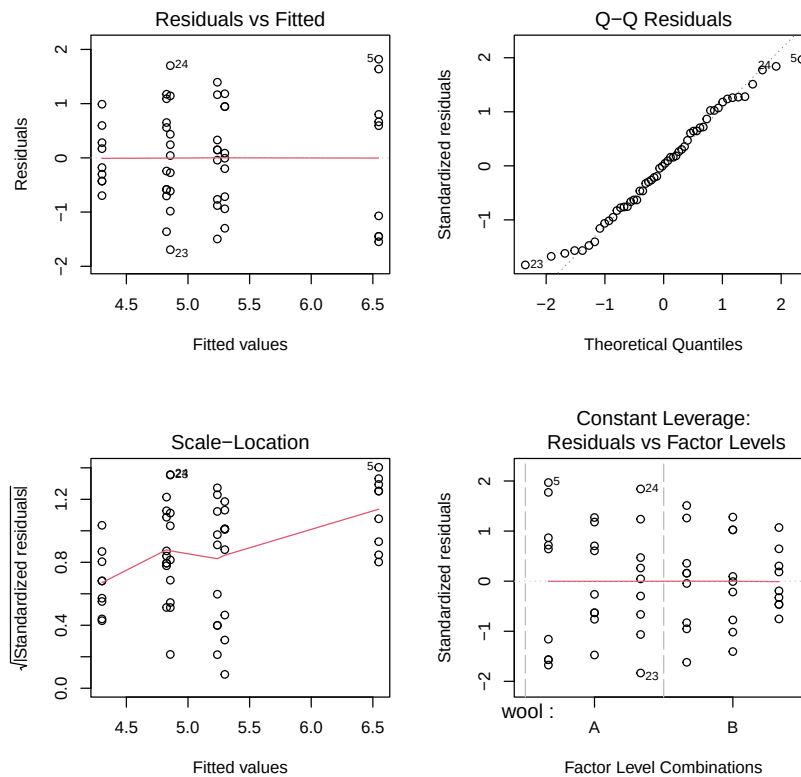
As always, let us check the model fit.

```
par(mfrow = c(2, 2))
plot(wlm0)
```



As already suggested by the exploratory data analysis, the variance appears to increase with the mean (see the left two panels of the residual plots). This is perhaps not surprising as the data are really counts: something that might be best modelled by a Poisson distribution rather than a normal distribution. Transformation of the response data (the  $y$ 's — breaks) can help in this circumstance. Power or log transformations are often worth trying. For Poisson data it can be shown that a square root power transform is sensible, so it is worth trying that.

```
wlm <- lm(sqrt(breaks) ~ wool*tension, data = warpbreaks)
par(mfrow = c(2, 2))
plot(wlm)
```



The revised plots show modest improvement, and experimenting with other transformations does not do much better, so the square root transform is probably sensible and we will proceed with it (note that now the response variable  $y$  denotes the square root of the number of breaks). Next, have a quick look at the model summary.

```
summary(wm)

##
## Call:
## lm(formula = sqrt(breaks) ~ wool * tension, data = warpbreaks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69410 -0.70129  0.01772  0.66046  1.81902
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.5476     0.3266  20.046 < 2e-16 ***
## woolB          -1.3094     0.4619  -2.835 0.006692 **
## tensionM       -1.7216     0.4619  -3.727 0.000511 ***
## tensionH       -1.6912     0.4619  -3.661 0.000625 ***
## woolB:tensionM  1.7821     0.6533   2.728 0.008874 **
## woolB:tensionH  0.7553     0.6533   1.156 0.253350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9799 on 48 degrees of freedom
## Multiple R-squared:  0.3606, Adjusted R-squared:  0.294
## F-statistic: 5.415 on 5 and 48 DF, p-value: 0.0004998
```

Note that we can also express the identifiable version of the two-way ANOVA model in Equation (20) using dummy variables. To this end, let  $l$  index a combination of wool type, tension level, and replicate, where  $l = 1, \dots, 54$ . That is,  $l$  represents the index of a row like those shown below.

```
head(warpbreaks)
```

```
##   breaks wool tension replicate
## 1     26   A      L         1
## 2     30   A      L         2
## 3     54   A      L         3
## 4     25   A      L         4
## 5     70   A      L         5
## 6     52   A      L         6
```

We thus write

$$y_l = \beta_0 + \beta_B \text{wool}_{Bl} + \beta_M \text{tension}_{Ml} + \beta_H \text{tension}_{Hl} + \beta_{BM} \text{wool}_{Bl} \text{tension}_{Ml} + \beta_{BH} \text{wool}_{Bl} \text{tension}_{Hl} + \epsilon_l,$$

where  $y_l$  denotes the square root of the number of breaks for observation  $l$  and

$$\begin{aligned} \text{wool}_{Bl} &= \begin{cases} 1, & \text{if the wool type for observation } l \text{ is B,} \\ 0, & \text{otherwise.} \end{cases} \\ \text{tension}_{Ml} &= \begin{cases} 1, & \text{if the tension level for observation } l \text{ is M,} \\ 0, & \text{otherwise.} \end{cases} \\ \text{tension}_{Hl} &= \begin{cases} 1, & \text{if the tension level for observation } l \text{ is H,} \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Note that to model the interaction effect, we must consider all possible combinations of wool type and tension levels, except for the reference categories. The coefficients can be interpreted as follows:

- $\beta_0$ : expected value of the square root of the number of breaks if wool type is A (reference level) and tension level is L (reference level).
- $\beta_0 + \beta_B$ : expected value of the square root of the number of breaks if wool type is B and tension level is L (reference level).
- $\beta_0 + \beta_M$ : expected value of the square root of the number of breaks if wool type is A (reference level) and tension level is M.
- $\beta_0 + \beta_H$ : expected value of the square root of the number of breaks if wool type is A (reference level) and tension level is H.
- $\beta_0 + \beta_B + \beta_M + \beta_{BM}$ : expected value of the square root of the number of breaks if wool type is B and tension level is M.
- $\beta_0 + \beta_B + \beta_H + \beta_{BH}$ : expected value of the square root of the number of breaks if wool type is B and tension level is H.

The parameter estimates are provided in the summary output above. Notably,  $\beta_{BH}$  in the last line represents an adjustment added on when wool type is B and tension level is H. In other words, it quantifies how much the expected value of the square root of the number of breaks differs from what would be expected if the effects of wool type B and tension level H were simply additive (i.e., if no interaction was specified). A similar interpretation applies to  $\beta_{BM}$ .

Notice also from the summary output of the fitted model that the model actually only explains some 30% of the variability in the data (adjusted R-squared). As is often the case with models involving factor variables, it is not

sensible to base model selection on the individual p-values reported in the model summary. Instead, to compare the main-effects-only model with the model that includes the interaction, we test the null hypothesis that the data are compatible with

$$y_l = \beta_0 + \beta_B \text{wool}_{Bl} + \beta_M \text{tension}_{Ml} + \beta_H \text{tension}_{Hl} + \epsilon_l,$$

(or, in two-way ANOVA notation:  $y_{ijk} = \mu + \alpha_i + \delta_j + \epsilon_{ijk}$ ), against the alternative that the interaction is required. As discussed earlier, this is done using an F-ratio test.

```
wml <- lm(sqrt(breaks) ~ wool + tension, data = warpbreaks)
anova(wml, wml)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ wool + tension
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq   F  Pr(>F)
## 1      50 53.291
## 2      48 46.089   2    7.2014 3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value is quite low here, providing evidence against the null hypothesis and in favour of the model that includes the interaction (recall the reasoning: the p-value indicates that the observed F ratio is rather improbable under the null hypothesis, suggesting that the null hypothesis is not true).

Note that it is not meaningful to remove a main effect while retaining its associated interaction term, a point discussed further below. In general, when predictors interact, we do not test or interpret their main effects in isolation. For the present purpose, this means that the model including the interaction is selected. Thus, the way tension affects the (square root of the) number of breaks depends on the type of wool.

### 6.6.1 Follow up

Having selected the model, which says that the (square root of the) number of breaks depends on an interaction of wool type and tension, we would like to know the combination(s) of wool type and tension that minimise breakages. From the summary of the `wml` fitted model object this can be determined but requires some work because the identifiability constraints have changed the meanings of the remaining parameters; see the interpretation of the coefficients provided earlier. However, we can modify the identifiability constraints in order to get more interpretable parameters. To this end, it is easier to define a new factor variable, say  $WT$ , whose number of levels equals the product of the two factor levels. Let

$$WT = \begin{cases} AL, & \text{if wool type is A and tension is L,} \\ AM, & \text{if wool type is A and tension is M,} \\ AH, & \text{if wool type is A and tension is H,} \\ BL, & \text{if wool type is B and tension is L,} \\ BM, & \text{if wool type is B and tension is M,} \\ BH, & \text{if wool type is B and tension is H.} \end{cases}$$

We know that there are only six identifiable parameters, and so we will set the intercept of the model to zero. Using dummy variables, the model can then be written as

$$y_l = \beta_{AL} WT_{AL,l} + \beta_{AM} WT_{AM,l} + \beta_{AH} WT_{AH,l} + \beta_{BL} WT_{BL,l} + \beta_{BM} WT_{BM,l} + \beta_{BH} WT_{BH,l} + \epsilon_l,$$

where  $y_l$  denotes the square root of the number of breaks,  $\epsilon_l \stackrel{iid}{\sim} N(0, \sigma^2)$ , and we define the dummy variable

$$WT_{AL,l} = \begin{cases} 1, & \text{if observation } l \text{ is at level } AL \text{ of variable } WT \text{ (i.e., if for observation } l \text{ wool type is A and tension level is L),} \\ 0, & \text{otherwise.} \end{cases}$$

The other dummy variables  $WT_{AM,l}$ ,  $WT_{AH,l}$ ,  $WT_{BL,l}$ ,  $WT_{BM,l}$ , and  $WT_{BH,l}$  are defined in a similar fashion. Under this formulation,  $\beta_{AL}$  is the expected value of the square root of the number of breaks for wool type A and tension L,  $\beta_{AM}$  is the expected value of the square root of the number of breaks for wool type A and tension M, etc: just what is needed. We can achieve this in R as follows.

```
wma <- lm(sqrt(breaks) ~ wool:tension - 1, data = warpbreaks)

#check the range of fitted values for this model is the same as for model wm.
range(fitted(wm) - fitted(wma))

## [1] -4.440892e-15 1.776357e-15

summary(wma)

##
## Call:
## lm(formula = sqrt(breaks) ~ wool:tension - 1, data = warpbreaks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69410 -0.70129  0.01772  0.66046  1.81902
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## woolA:tensionL    6.5476     0.3266  20.05  <2e-16 ***
## woolB:tensionL    5.2381     0.3266  16.04  <2e-16 ***
## woolA:tensionM    4.8259     0.3266  14.78  <2e-16 ***
## woolB:tensionM    5.2987     0.3266  16.22  <2e-16 ***
## woolA:tensionH    4.8564     0.3266  14.87  <2e-16 ***
## woolB:tensionH    4.3022     0.3266  13.17  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9799 on 48 degrees of freedom
## Multiple R-squared:  0.9697, Adjusted R-squared:  0.9659
## F-statistic: 255.8 on 6 and 48 DF, p-value: < 2.2e-16
```

Note that this is not a different model from `wm`: it is simply a different parameterisation of the same model. The R-squared and adjusted R-squared values differ from those of the `wm` model because this parameterisation does not include an intercept, which is known to inflate these quantities.

We can immediately see that wool type B at high tension is estimated to give the lowest break rate (on the square-root scale). But given the uncertainty, can we really be sure that this combination is better than wool A at medium or high tension? Such questions may be important if there are other considerations to take into account when choosing between wool types and tension levels (e.g. cost, wear rate of machinery, etc).

One obvious thing to do would be to find a confidence interval for the difference in break rate between wool B at high tension and wool A at medium tension, i.e. a confidence interval for  $\beta_{AM} - \beta_{BH}$ . To do this we need to find the standard deviation of  $\hat{\beta}_{AM} - \hat{\beta}_{BH}$ . We know that  $\widehat{\text{var}}(\hat{\beta}_{AM} - \hat{\beta}_{BH}) = \widehat{\text{var}}(\hat{\beta}_{AM}) + \widehat{\text{var}}(\hat{\beta}_{BH}) - 2\widehat{\text{cov}}(\hat{\beta}_{AM}, \hat{\beta}_{BH})$ . From the summary output we have  $\widehat{\text{var}}(\hat{\beta}_{AM}) = \widehat{\text{var}}(\hat{\beta}_{BH}) = 0.3266^2 = 0.106688$ . The estimated covariance matrix for the  $\hat{\beta}$  parameters is also easily obtained.

```
vcov(wma)

##              woolA:tensionL woolB:tensionL woolA:tensionM woolB:tensionM
## woolA:tensionL    0.106688      0.000000      0.000000      0.000000
## woolB:tensionL    0.000000      0.106688      0.000000      0.000000
## woolA:tensionM    0.000000      0.000000      0.106688      0.000000
## woolB:tensionM    0.000000      0.000000      0.000000      0.106688
```

```
## woolA:tensionH      0.000000      0.000000      0.000000      0.000000
## woolB:tensionH      0.000000      0.000000      0.000000      0.000000
##                woolA:tensionH woolB:tensionH
## woolA:tensionL      0.000000      0.000000
## woolB:tensionL      0.000000      0.000000
## woolA:tensionM      0.000000      0.000000
## woolB:tensionM      0.000000      0.000000
## woolA:tensionH      0.106688      0.000000
## woolB:tensionH      0.000000      0.106688
```

It is diagonal:  $\widehat{\text{cov}}(\hat{\beta}_{AM}, \hat{\beta}_{BH}) = 0$ ! Hence,  $\widehat{\text{var}}(\hat{\beta}_{AM} - \hat{\beta}_{BH}) = 2 \times 0.106688 \approx 0.2134$ . The required 95% confidence interval is therefore

$$(\hat{\beta}_{AM} - \hat{\beta}_{BH}) \pm t_{0.975, 54-6} \times \sqrt{\widehat{\text{var}}(\hat{\beta}_{AM} - \hat{\beta}_{BH})} = (4.8259 - 4.3022) \pm 2.011 \times \sqrt{0.2134} = 0.52 \pm 0.93$$

Because the confidence interval contains zero, we cannot tell wool B at high tension and wool A at medium (or high) tension apart.

Follow-up comparisons after fitting a model with interactions are statistically delicate. If many hypothesis tests are performed, some null hypotheses will be rejected purely by chance (e.g., at the 5% level you expect to reject 5% of true null hypotheses). A common approach is the Bonferroni correction, which divides the desired significance level by the number of tests being conducted. However, this method implicitly assumes that the tests are independent, an assumption that is rarely valid in linear model, and it can also be unclear how many tests are actually being performed. This difficulty appears clearly in the warpbreaks example. Although we explicitly tested only one contrast (e.g.,  $\beta_{AM} - \beta_{BH}$ ), we chose that comparison only after looking at the fitted model and examining the estimates. In effect, this is similar to having considered all pairwise comparisons among the six wool-tension combinations:  $\{AL, AM, AH, BL, BM, BH\}$ . There are  $\binom{6}{2} = 15$  such comparisons. These 15 potential contrasts are certainly not independent, so simple Bonferroni corrections are not fully appropriate.

In this example, it was straightforward to obtain a parameterisation in R that makes the main conclusions easy to extract. In other settings, it may take a little more work to achieve a convenient parameterisation. The key is to choose identifiability constraints that yield an interpretable set of coefficients. Statistical software allows this through different choices of *contrasts*. We will not go into further detail here, see the course textbooks or `?contrasts` for more information. As we have already seen, the `relevel` function in R is also very useful.

## Aside consideration

Note that the same conclusion could have been reached with the initial parametrisation we used but as mentioned it requires more work; the interpretation of the parameters in this model was provided earlier. Let us look at the estimated means.

```
predict(wm, newdata = expand.grid(wool = c("A", "B"),
                                tension = c("L", "M", "H")))

##           1           2           3           4           5           6
## 6.547580 5.238146 4.825948 5.298655 4.856380 4.302202

expand.grid(wool = c("A", "B"), tension = c("L", "M", "H"))

##   wool tension
## 1    A       L
## 2    B       L
## 3    A       M
## 4    B       M
## 5    A       H
## 6    B       H
```

Exactly the same results as obtained just before. Remember that  $\hat{\beta}_0 + \hat{\beta}_M$  is the estimated expected value of square root breaks if wool type is A and tension level is M, whereas  $\hat{\beta}_0 + \hat{\beta}_B + \hat{\beta}_H + \hat{\beta}_{BH}$  is the estimated expected value of square root breaks if wool type is B and tension level is H and so

$$\begin{aligned}\widehat{\text{var}}((\hat{\beta}_0 + \hat{\beta}_M) - (\hat{\beta}_0 + \hat{\beta}_B + \hat{\beta}_H + \hat{\beta}_{BH})) &= \widehat{\text{var}}(\hat{\beta}_M - \hat{\beta}_B - \hat{\beta}_H - \hat{\beta}_{BH}) \\ &= \widehat{\text{var}}(\hat{\beta}_M) + \widehat{\text{var}}(\hat{\beta}_B) + \widehat{\text{var}}(\hat{\beta}_H) + \widehat{\text{var}}(\hat{\beta}_{BH}) \\ &\quad - 2\widehat{\text{cov}}(\hat{\beta}_M, \hat{\beta}_B) - 2\widehat{\text{cov}}(\hat{\beta}_M, \hat{\beta}_H) - 2\widehat{\text{cov}}(\hat{\beta}_M, \hat{\beta}_{BH}) \\ &\quad + 2\widehat{\text{cov}}(\hat{\beta}_B, \hat{\beta}_H) + 2\widehat{\text{cov}}(\hat{\beta}_B, \hat{\beta}_{BH}) + 2\widehat{\text{cov}}(\hat{\beta}_H, \hat{\beta}_{BH}).\end{aligned}$$

All variances and covariances can be obtained from the estimated variance covariance matrix of the estimated parameters.

```
vcov(wm)

##               (Intercept)      woolB      tensionM      tensionH woolB:tensionM
## (Intercept)      0.106688 -0.1066880 -0.1066880 -0.1066880      0.1066880
## woolB            -0.106688  0.2133761  0.1066880  0.1066880     -0.2133761
## tensionM         -0.106688  0.1066880  0.2133761  0.1066880     -0.2133761
## tensionH         -0.106688  0.1066880  0.1066880  0.2133761     -0.1066880
## woolB:tensionM    0.106688 -0.2133761 -0.2133761 -0.1066880      0.4267521
## woolB:tensionH    0.106688 -0.2133761 -0.1066880 -0.2133761      0.2133761
##               woolB:tensionH
## (Intercept)      0.1066880
## woolB            -0.2133761
## tensionM         -0.1066880
## tensionH         -0.2133761
## woolB:tensionM    0.2133761
## woolB:tensionH    0.4267521

var_dif <- 0.2133761 + 0.2133761 + 0.2133761 + 0.4267521 - 2*0.1066880 - 2*0.1066880 -
  2*(-0.1066880) + 2*0.1066880 + 2*(-0.2133761) + 2*(-0.2133761)
var_dif

## [1] 0.213376
```

Unsurprisingly we have obtained the same value for the estimated variance of the difference but after considerably more work.

## 6.7 ANOVA tables

In R the `anova` function produces tabular output related to the F-ratio comparison of alternative models. It is important to know exactly what the entries in these *ANOVA tables* mean, since they are a standard way of displaying results.

As seen in Section 5.1.2, `anova` does different things depending on whether you call it with a single model argument or several models. Consider the multiple model case first, using two of the `warpbreaks` models.

```
anova(wm1, wm)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ wool + tension
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq   F  Pr(>F)
## 1      50 53.291
## 2      48 46.089   2    7.2014 3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Working across the columns of the ANOVA table:

- `Res.Df` gives the residual degrees of freedom for each model: i.e. the number of observations ( $n$ ) less the number of model parameters ( $p_0$  and  $p_1$ , say, for the null and full models, respectively).
- `RSS` gives the residual sum of squares for each model (i.e.  $RSS_0 = \|\mathbf{y} - \mathbf{X}_0\hat{\beta}_0\|^2$  and  $RSS_1 = \|\mathbf{y} - \mathbf{X}_1\hat{\beta}_1\|^2$ ).
- `Df` gives the *difference* in degrees of freedom between each model and the model above it in the table (from 2nd model onwards), i.e.  $p_1 - p_0$  in the current case.
- `Sum of Sq` gives the *difference* in residual sum of squares between each model and the model above it in the table (from 2nd model). In terms of Section 4.3.2 this is  $RSS_0 - RSS_1$ .
- `F` is the F-ratio test statistic for comparing each model to the model above it in the table. That is  $(RSS_0 - RSS_1)/\{(p_1 - p_0)\hat{\sigma}^2\}$ . Note that  $\hat{\sigma}^2$  is always calculated from the largest model in the table.
- `Pr(>F)` is the probability of observing an F-ratio as large or larger than the one in the `F` column if the model corresponding to the next row up in the table is correct. That is, this is the p-value for testing the null hypothesis that the next model up is correct, against the larger alternative described by this row's model.

Only two models are compared in the example, but the definitions apply to any number of models from two upwards.

The second form of the ANOVA table occurs when only one model is supplied to `anova`. Here is an example.

```
anova(wm)

## Analysis of Variance Table
##
## Response: sqrt(breaks)
##      Df Sum Sq Mean Sq F value    Pr(>F)
## wool      1   2.902    2.9019   3.0222 0.088542 .
## tension    2  15.892    7.9458   8.2752 0.000817 ***
## wool:tension 2   7.201    3.6007   3.7500 0.030674 *
## Residuals 48 46.089    0.9602
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this case the table should be read from the bottom up.

- The `Residuals` row gives the residual degrees of freedom, residual sum of squares and residual variance estimate,  $\hat{\sigma}^2$ , for the full model (`breaks~wool*tension`).
- The `wool:tension` row gives the *difference* in degrees of freedom and residual sum of squares between the full model and a model without the interaction (i.e. `breaks~wool+tension`). In some sense these numbers are the degrees of freedom and sum of squares associated with the dropped interaction term. The `Mean Sq` column gives the difference in sum of squares divided by the difference in degrees of freedom (an example of an  $(RSS_0 - RSS_1)/(p_1 - p_0)$  term in Section 4.3.2 notation). The `F value` is then the `Mean Sq` over  $\hat{\sigma}^2$ : the F-ratio for testing the null hypothesis that the model without the interaction is adequate, against the alternative that the full model is correct. The corresponding p-value is the last entry on the row.
- The `tension` row compares the model with wool and tension main effects to a model with just a wool effect (i.e., compares `breaks~wool+tension` to `breaks~wool`). The interpretation of the entries is as for the `wool:tension` row, but note that all the differences are taken between the two models being compared, not between the wool only model and the full model.  $\hat{\sigma}^2$  in the F-ratio statistic still comes from the full model/ last row of the table, however.

- The `wool` row compares a model with `wool` only to a model with just an intercept (i.e `breaks~wool` to `breaks~1`). The interpretation of the entries is as for the previous row.

When interpreting such tables it is important to note two things:

- If you conclude that the interaction between two factors is significant and therefore cannot be dropped from the model, then both factors must also remain in the model, regardless of the p-values reported for their main effects. This is because the p-values for the main effects are only meaningful when the interaction is not significant (more on this shortly).
- For most models, the ANOVA table will differ depending on the order in which terms are removed from the model because the estimators of the model terms are generally not independent. A notable exception occurs for models involving factor variables fitted to *balanced* data. In such cases the terms estimators are independent, so that the order in which the model is reduced is immaterial. The `warpbreaks` data are an example of this. Data are said to be *balanced* with respect to a factor or interaction if there are the same number of observations for level of the factor or interaction.

### 6.7.1 Example of erroneous dropping of a main effect

To emphasize the importance of reading ANOVA table from the bottom up, and of not trying to drop main effects while retaining their associated interactions, consider what happens if we attempt to remove the `wool` main effect while keeping the `wool:tension` interaction.

```
wm2 <- lm(sqrt(breaks) ~ tension + tension:wool, data = warpbreaks)
summary(wm2)

##
## Call:
## lm(formula = sqrt(breaks) ~ tension + tension:wool, data = warpbreaks)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69410 -0.70129  0.01772  0.66046  1.81902
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      6.5476     0.3266  20.046 < 2e-16 ***
## tensionM        -1.7216     0.4619  -3.727 0.000511 ***
## tensionH        -1.6912     0.4619  -3.661 0.000625 ***
## tensionL:woolB  -1.3094     0.4619  -2.835 0.006692 **
## tensionM:woolB   0.4727     0.4619   1.023 0.311278
## tensionH:woolB  -0.5542     0.4619  -1.200 0.236140
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9799 on 48 degrees of freedom
## Multiple R-squared:  0.3606, Adjusted R-squared:  0.294
## F-statistic: 5.415 on 5 and 48 DF, p-value: 0.0004998

anova(wm2, wm)

## Analysis of Variance Table
##
## Model 1: sqrt(breaks) ~ tension + tension:wool
## Model 2: sqrt(breaks) ~ wool * tension
##   Res.Df    RSS Df Sum of Sq F Pr(>F)
## 1      48 46.089
## 2      48 46.089  0          0
```

The two models have exactly the same RSS and residual degrees of freedom, so `anova` cannot perform a meaningful comparison. In fact, they are the same model, so there is nothing to test. The following command confirms the equivalence:

```
range(fitted(wm) - fitted(wm2))

## [1] -2.664535e-15  1.776357e-15
```

The only difference between the models lies in the identifiability constraints, and therefore in how the parameters are labelled and interpreted. By removing the `wool` main effect, we have simply removed the need for one of the constraints on the interaction term, so an additional interaction parameter absorbs the role that the missing main effect parameter would otherwise play.

### 6.7.2 `drop1` again

Actually, `drop1` is often a more sensible function to use than `anova` with a single model argument. `drop1` will never drop a main effect if its corresponding interaction is present, and it always compares the omission of a single term against the full model, rather than successively removing terms. Here it is applied to the `wm` model.

```
drop1(wm, test = "F")

## Single term deletions
##
## Model:
## sqrt(breaks) ~ wool * tension
##           Df Sum of Sq    RSS    AIC F value    Pr(>F)
## <none>                46.089  3.4461
## wool:tension    2      7.2014  53.291  7.2859      3.75 0.03067 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## 6.8 A model with both continuous and factor predictors

Of course, it is also possible to have (multiple) continuous and factor predictors. We illustrate this using an example from Faraway (2014, p. 213), based on a study by Partridge and Farquhar (1981) examining the relationship between sexual activity and lifespan in male fruitflies. In this study, 125 fruitflies were randomly assigned to five groups of 25 each and longevity (in days) was recorded as the response variable. One group was kept solitary, while another was kept individually with a virgin female each day. Another group was given eight virgin females per day. As an additional control, the fourth and fifth groups were kept with one or eight pregnant females per day. Pregnant fruitflies do not mate. In addition, the thorax length of each male was measured because this characteristic was known to affect longevity. Thus, thorax length serves as a continuous predictor. The five groups are labeled as

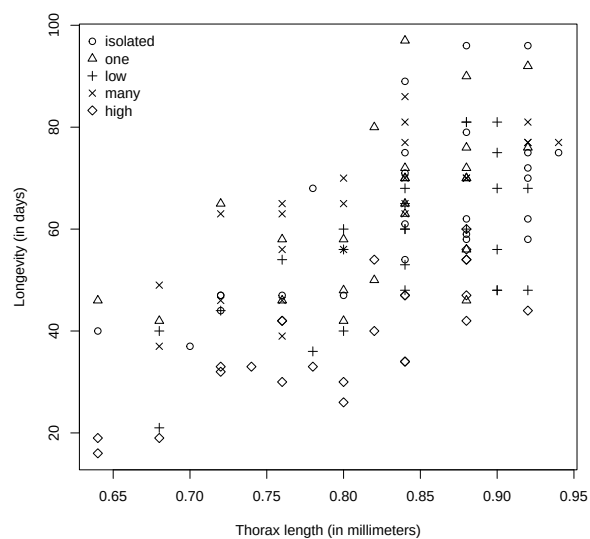
- *isolated*: fly kept solitary.
- *one*: fly kept with one pregnant fruitfly.
- *many*: fly kept with eight pregnant fruitflies.
- *low*: fly kept with one virgin fruitfly.
- *high*: fly kept with eight virgin fruitflies.

The main question of interest is whether sexual activity has a cost to male fruitflies in terms of reduced longevity. We begin by exploring the data.

```
library(faraway)
levels(fruitfly$activity)

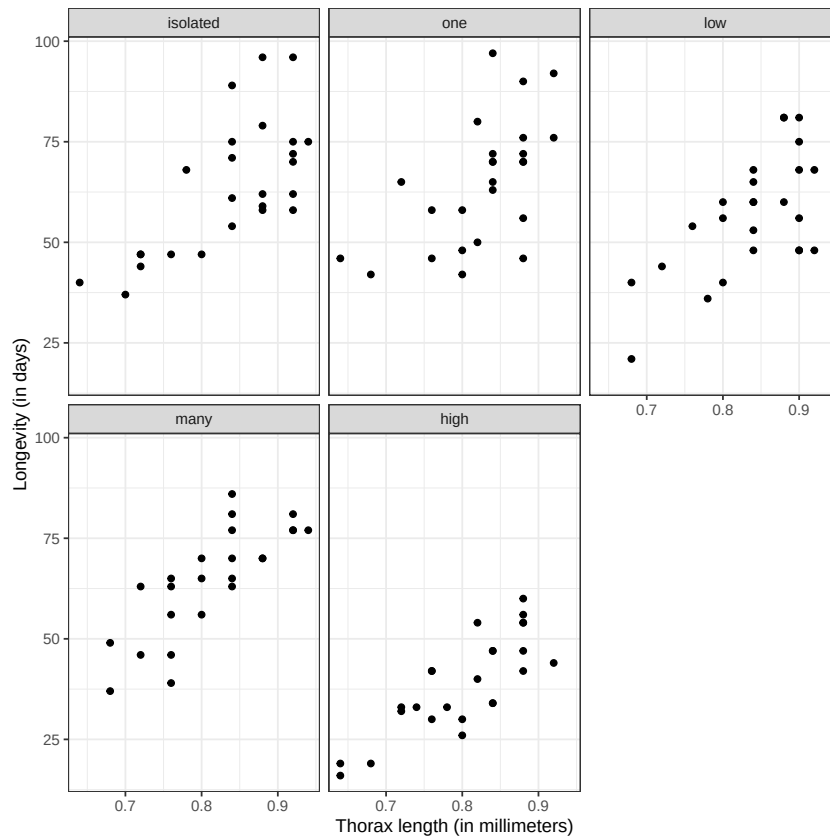
## [1] "isolated" "one"      "low"      "many"     "high"

plot(longevity ~ thorax, fruitfly, pch = unclass(activity),
      ylab = "Longevity (in days)", xlab = "Thorax length (in millimeters)")
legend("topleft", levels(fruitfly$activity), pch = 1:5, bty = "n")
```



Longevity appears to be lower for the high sexual activity group. When a factor has many levels, a single scatterplot may obscure group differences; plotting each level separately can therefore be more informative.

```
ggplot(aes(x = thorax, y = longevity), data = fruitfly) +
  geom_point() +
  facet_wrap(~ activity) +
  theme_bw() +
  ylab("Longevity (in days)") +
  xlab("Thorax length (in millimeters)")
```



This faceted plot clearly shows that longevity is reduced in the high-activity group. Two natural questions arise:

1. Is there an overall difference in longevity between the sexual-activity groups?
2. Is the relationship between thorax length and longevity the same across groups, or does it vary (i.e., is there an interaction)?

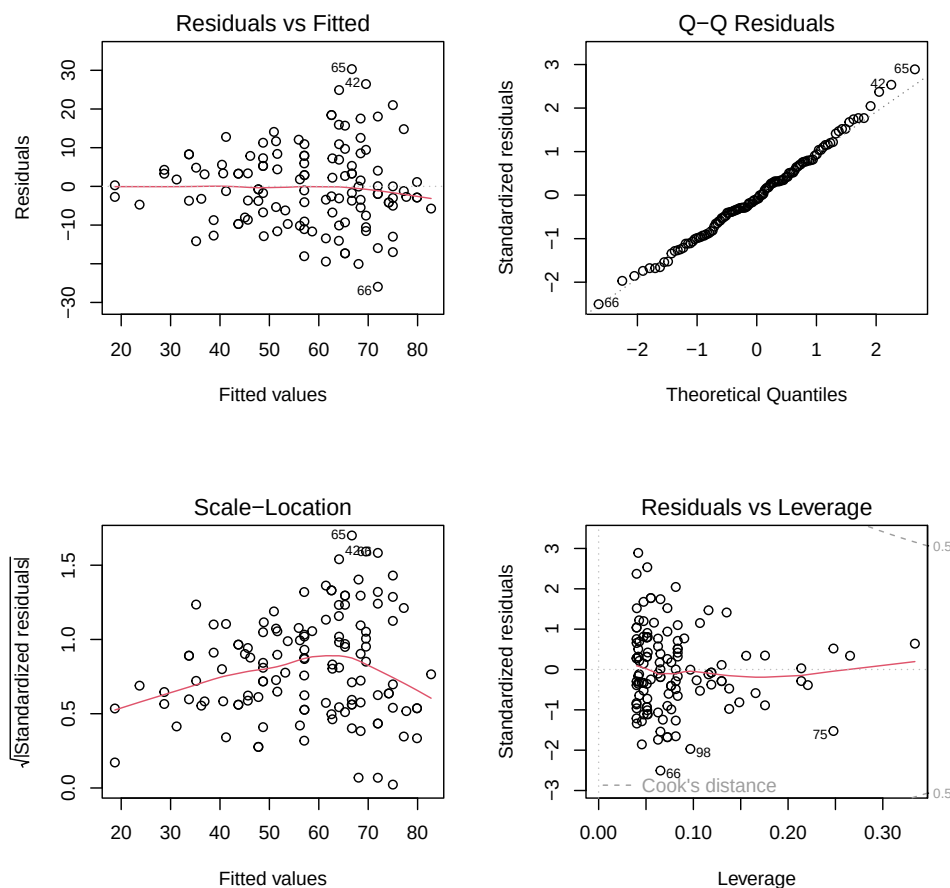
To investigate this, we fit a model including the interaction between thorax length and activity group:

```
longm <- lm(longevity ~ thorax*activity, data = fruitfly)
head(model.matrix(longm))
```

|      | (Intercept)        | thorax             | activityone         | activitylow         | activitymany | activityhigh |
|------|--------------------|--------------------|---------------------|---------------------|--------------|--------------|
| ## 1 | 1                  | 0.68               | 0                   | 0                   | 1            | 0            |
| ## 2 | 1                  | 0.68               | 0                   | 0                   | 1            | 0            |
| ## 3 | 1                  | 0.72               | 0                   | 0                   | 1            | 0            |
| ## 4 | 1                  | 0.72               | 0                   | 0                   | 1            | 0            |
| ## 5 | 1                  | 0.76               | 0                   | 0                   | 1            | 0            |
| ## 6 | 1                  | 0.76               | 0                   | 0                   | 1            | 0            |
|      | thorax:activityone | thorax:activitylow | thorax:activitymany | thorax:activityhigh |              |              |
| ## 1 | 0                  | 0                  | 0.68                | 0                   |              |              |
| ## 2 | 0                  | 0                  | 0.68                | 0                   |              |              |
| ## 3 | 0                  | 0                  | 0.72                | 0                   |              |              |
| ## 4 | 0                  | 0                  | 0.72                | 0                   |              |              |
| ## 5 | 0                  | 0                  | 0.76                | 0                   |              |              |
| ## 6 | 0                  | 0                  | 0.76                | 0                   |              |              |

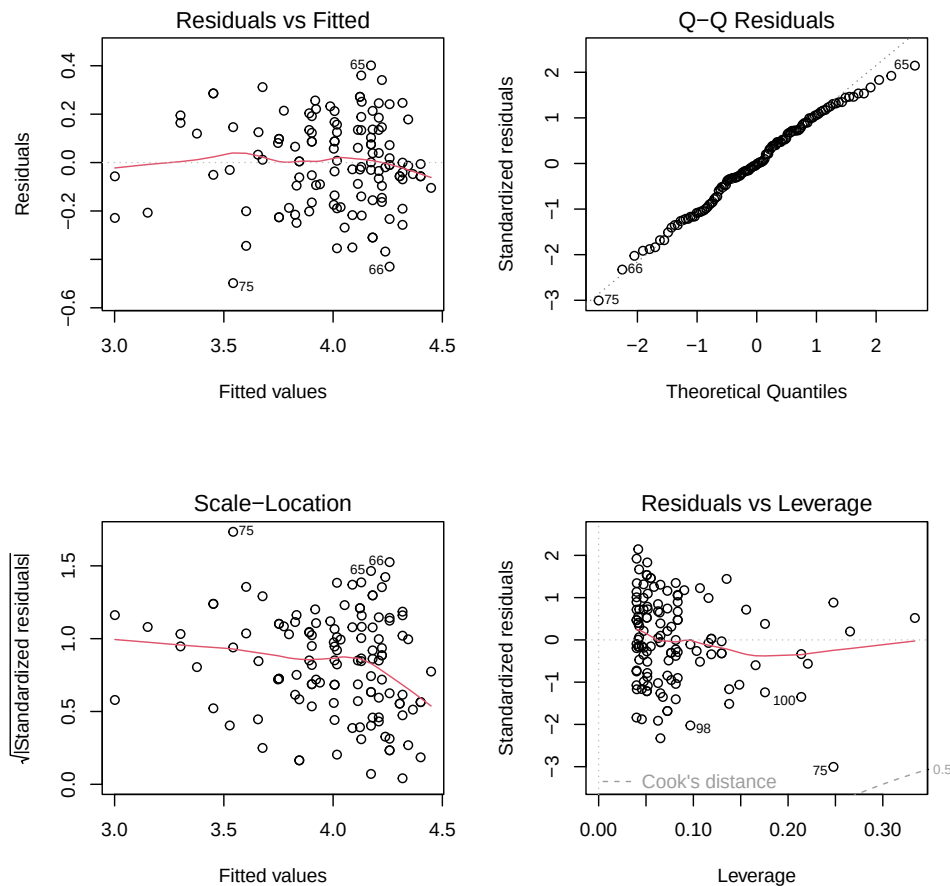
The reference level for the activity factor is the *isolated* group. Before proceeding, we examine the residual plots.

```
par(mfrow = c(2, 2))
plot(longm)
```



The residuals versus fitted values plot shows a clear funnel shape pattern, indicating a violation of the constant variance assumption. We examined both square-root and log transformations of the response, and the log transformation provided better variance stabilisation. We therefore proceed with the log transformed response. A drawback of a log transformation is that it can make the coefficients less straightforward to interpret, but we will address this issue later.

```
longm_log <- lm(log(longevity) ~ thorax*activity, data = fruitfly)
par(mfrow = c(2, 2))
plot(longm_log)
```



After log-transforming longevity, the constant-variance assumption appears considerably more reasonable. We now test whether the interaction between thorax length and activity group is required.

```
longm_log_1 <- lm(log(longevity) ~ thorax + activity, data = fruitfly)
anova(longm_log_1, longm_log)

## Analysis of Variance Table
##
## Model 1: log(longevity) ~ thorax + activity
## Model 2: log(longevity) ~ thorax * activity
##   Res.Df    RSS Df Sum of Sq   F Pr(>F)
## 1     118 4.4012
## 2     114 4.1626   4   0.23857 1.6334 0.1706
```

The p-value is 0.1706 and so we do not reject the null hypothesis that the coefficients associated with the interaction terms are zero. That is, there is no evidence that the effect of thorax length on longevity differs between sexual activity groups. The drop1 function confirms this conclusion.

```
drop1(longm_log, test = "F")

## Single term deletions
##
## Model:
## log(longevity) ~ thorax * activity
##           Df Sum of Sq   RSS      AIC F value Pr(>F)
```

```
## <none> 4.1626 -400.87
## thorax:activity 4 0.23857 4.4012 -401.96 1.6334 0.1706
```

Next, we assess whether the additive model can be simplified further.

```
drop1(longm_log_1, test = "F")

## Single term deletions
##
## Model:
## log(longevity) ~ thorax + activity
##      Df Sum of Sq  RSS   AIC F value    Pr(>F)
## <none>      4.4012 -401.96
## thorax    1   5.0757 9.4769 -308.86 136.086 < 2.2e-16 ***
## activity  4   4.1542 8.5554 -327.54  27.845 2.756e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both the AIC and the F-tests indicate that the model containing additive effects of thorax length and activity group is preferable to either a thorax-only or activity-only model. The mathematical formulation of the final model is therefore

$$\log y_i = \mu + \alpha_j + \gamma x_i + \epsilon_i, \quad \text{if fruitfly } i \text{ is in the sexual activity group } j, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), \quad i = 1, \dots, 125.$$

where  $y_i$  denotes the longevity of fruitfly  $i$ ,  $x_i$  denotes the thorax length of fruitfly  $i$ , and  $\alpha_j$  represents the main effect of the activity group  $j \in \{\text{isolated, one, low, many, high}\}$ . For identifiability, one of the  $\alpha$  parameters must be set to zero; in R, the default choice is  $\alpha_{\text{isolated}} = 0$ . The model may also be written in terms of indicator variables. Let us now inspect the summary output of the fitted additive model.

```
summary(longm_log_1)

##
## Call:
## lm(formula = log(longevity) ~ thorax + activity, data = fruitfly)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52641 -0.13629 -0.00823  0.13918  0.39273
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.84421    0.19882   9.276 1.04e-15 ***
## thorax         2.72146    0.23329  11.666 < 2e-16 ***
## activityone    0.05174    0.05468   0.946  0.3459
## activitylow   -0.12387    0.05463  -2.268  0.0252 *
## activitymany   0.08791    0.05546   1.585  0.1156
## activityhigh  -0.41925    0.05527  -7.586 8.35e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1931 on 118 degrees of freedom
## Multiple R-squared:  0.7025, Adjusted R-squared:  0.6899
## F-statistic: 55.72 on 5 and 118 DF,  p-value: < 2.2e-16
```

Because the response has been log-transformed, the model coefficients correspond to multiplicative effects on the original scale. For example:

```
exp(coef(longm_log_1)[3:6])

## activityone activitylow activitymany activityhigh
## 1.0531064 0.8834971 1.0918894 0.6575384

exp(confint(longm_log_1)[3:6, ])
```

|              | 2.5 %     | 97.5 %    |
|--------------|-----------|-----------|
| activityone  | 0.9450347 | 1.1735370 |
| activitylow  | 0.7929135 | 0.9844290 |
| activitymany | 0.9783179 | 1.2186452 |
| activityhigh | 0.5893739 | 0.7335865 |

Relative to the isolated group, the high-activity group has approximately 0.66 times the lifespan, that is, a 34% reduction—and the corresponding confidence interval does not include 1 (i.e.,  $e^0$ ).

Before concluding, it is worth explaining why a log-transformed model implies multiplicative effects on the original scale. On the log scale,

$$\mathbb{E}(\log y_i) = \mu + \alpha_j + \gamma x_i.$$

But what does this tell us about  $\mathbb{E}(y_i)$ , the expected value on the original scale? In general, the expected value does not directly follow a transformation, that is,

$$\mathbb{E}(y_i) \neq e^{\mu + \alpha_j + \gamma x_i}.$$

However, since we are assuming that the error term follows a normal distribution with variance  $\sigma^2$ , i.e.,  $\log y_i \sim N(\mu + \alpha_j + \gamma x_i, \sigma^2)$ , this implies that  $y_i$  follows a log normal distribution with parameters  $\mu + \alpha_j + \gamma x_i$  and  $\sigma^2$ . The expected value of  $y_i$  is therefore given by

$$\mathbb{E}(y_i) = e^{\mu + \alpha_j + \gamma x_i + \sigma^2/2}.$$

Thus, for two fruitflies, say  $i$  and  $i'$ , with the same thorax length, say  $x^*$ , where  $i$  belongs to the reference sexual activity group (isolated) and  $i'$  to the high sexual activity group, we have

$$\frac{\mathbb{E}(y_{i'})}{\mathbb{E}(y_i)} = \frac{e^{\mu + \alpha_{\text{high}} + \gamma x^* + \sigma^2/2}}{e^{\mu + \gamma x^* + \sigma^2/2}} = e^{\alpha_{\text{high}}} \Rightarrow \mathbb{E}(y_{i'}) = e^{\alpha_{\text{high}}} \mathbb{E}(y_i).$$

Thus,  $e^{\alpha_{\text{high}}}$  represents the multiplicative change in expected longevity associated with the high-activity group. Similar interpretations apply to the other coefficients. This useful property is specific to the log transformation.

## 7 How to approach an analysis

By now we have seen almost all the theory needed for practical data analysis, and how to implement much of it in practice. It is therefore useful to step back, take stock, and consider some general principles for approaching any modelling task. It is neither possible nor desirable to prescribe a universal recipe for all analyses, but the guidelines below apply broadly.

1. Always start by identifying the questions that you are trying to answer by analysing the data.
2. Always look at the data before fitting any model. Plot the data to get a feel for how variables are related. Check for obvious errors. If you can think of simple methods (e.g. plots) that will give you informal answers to your questions, use them before starting the formal model based analysis.
3. Now think about how you can use linear models to answer the questions of interest.
  - Sometimes at least the main features of the models and analysis are obvious from the question. For example, analysing clinical trial data to test whether a new drug is better than the standard treatment will obviously involve comparing models with and without the treatment factor, by hypothesis testing, and then estimating the size of any effect differences.
  - Other times the question is much less prescriptive: as in the tyre wear example, answering the question of interest may involve finding a linear model that adequately captures the relationship between the response and predictors, but the question and wider context do not specify much about what model to use. In such cases you usually have to think about broad classes of models that might be sensible, and then use statistical model selection approaches to choose between them.
  - When formulating a model, ensure that you are clear what is the response, and what are the predictors.
  - Sometimes it is clear in advance that not all potential predictors should be included in the model. But be careful. Don't include variables that you know are irrelevant, but don't pre-exclude variables unless you are pretty certain about this.
  - Try to avoid excluding predictors on the basis of statistical analysis of the predictors alone (i.e. without consideration of the response). Of course this advice does not extend to the case when two predictors actually measure the same thing: then it is sensible to either combine them, or decide which one to drop.
4. Once you start fitting the models that are part of your analysis, make sure that you check that the modelling assumptions are met. Hypothesis tests, confidence intervals and so on don't work if they are not.
5. Always make sure that you interpret the results of your modelling in terms of the original question, and think carefully about any limitations that apply to the answer.

Finally, some principles for writing up a statistical analysis

1. Always clearly explain the context, and the questions being addressed.
2. Make sure that your analysis is repeatable by any statistician with access to the data, using the software of their choice. This means presenting models and results in universal mathematical language, not as R code and output.
3. Always explain the 'why' of your analysis as well as the 'what'.
4. Relate your conclusions back to the questions, and be careful to discuss the limitations of the approach taken.
5. Aim for concision and usefulness. Avoid overly legalistic reporting in which every possible failure mode of linear models is listed; such an approach offers no insight into the issues that might realistically affect the analysis at hand.

## 8 Causality, confounding and randomization (non-examinable and not covered in class)

The material in this section is non-examinable; however, please do not skip it. It is only a few pages and it is both important and instructive.

There are several different purposes for which we may wish to use a (generalized) linear model.

1. For prediction. For example, we build a model of patients' probability of heart disease based on lifestyle factors such as diet and exercise, estimate the model from a representative group of subjects whose heart disease status is known, and use it to predict the status of new subjects.
2. To establish associations between variables. For example, we suspect that being educated in a single sex school may be negatively associated with the length of marriage, and we use statistical modelling to assess whether this putative association is real, or could just be due to chance variability in the data.
3. To establish causation. Often, especially in science, we want to know whether something is an actual cause of something else. Famous examples are: does smoking cause cancer? and does the MMR (measles, mumps, rubella) vaccine cause autism?

The methods we have covered so far are quite sufficient for addressing 1 and 2, but the *causal inference* required in 3 is more difficult, and the sort of methods covered so far (and that will be covered in the remaining sections) are not sufficient to establish causality *on their own*.

To understand the issues, it is important to really have grasped the difference between correlation (association) and causality. An often quoted example is the correlation between the population of storks and the birth rate, in Europe. The correlation is real, but there is obviously no causation. Rather, industrialization raised health care, living and educational standards in Europe, leading to reduced birth rates, but that same industrialization led to a reduction in suitable habitat for storks. That is, the association is caused by other factors which may be causative to both variables of interest. To put it simply, for an association to reflect causality, it must not have an *alternative explanation*. We may think we have found a causal relationship, but we may merely not have thought of a particular reason that can explain the association.

Confusing correlation with causation is a famously good way of selling newspapers. For example, some medical studies have found associations between coffee drinking and various responses, such as the likelihood of a heart attack or the mean blood pressure level. But after taking into account other variables associated with the extent of coffee drinking, such as extent of smoking, country residence, occupation, and levels of stress, such associations have disappeared or weakened considerably. Again the key is (are) a hidden variable(s) related to the variables of direct interest.

To appreciate the effect of correlated predictors on interpretation, and on attempts at causal inference, it is worth looking at some simulations, where we are in complete control of the correlation structure between variables. The following code simulates two highly correlated (correlation of 0.95) predictor variables,  $x$  and  $h$ :

```
# Supplies rmvn to simulate random numbers from a multivariate normal dist.
require(mgcv)
V <- matrix(c(1, 0.95, 0.95, 1), 2, 2) #covariance matrix
n <- 1000 # Number of data points
set.seed(7) # Just for reproducibility
# Predictors simulated from a multivariate (bivariate) normal dist.
X <- rmvn(n, rep(0, 2), V)
x <- X[, 1]; h <- X[, 2]
```

Now consider the case where  $y_i = x_i + \epsilon_i$ , but we fit the model  $y_i = \beta_0 + \beta_1 x_i + \beta_2 h_i + \epsilon_i$ .

```
# Simulate response variable as specified above
y <- x + rnorm(n)
fit_1 <- lm(y ~ x + h)
summary(fit_1)$coefficients

##               Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) -0.003268241 0.03116981 -0.1048528 9.165137e-01
```

```
## x          0.812567847 0.09892433 8.2140345 6.587810e-16
## h          0.209603373 0.09947463 2.1071038 3.535786e-02

confint(fit_1)[2,]

##      2.5 %      97.5 %
## 0.6184441 1.0066916
```

Because of the high correlation between  $x$  and  $h$  it is difficult to distinguish their effects, with the result that when both are included in the model the ‘explanation’ of the response tends to get shared out between them, leading to rather variable estimates of the real effect of  $x$ . In contrast when the spurious  $h$  term is omitted from the model, the estimate of  $\beta_1$  is much more accurate.

```
# Simulate response variable as specified above
fit_2 <- lm(y ~ x)
summary(fit_2)$coefficients

##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) -0.002116903 0.03121868 -0.06780883 9.459514e-01
## x           1.010416942 0.03119075 32.39475790 6.619470e-158

confint(fit_2)[2,]

##      2.5 %      97.5 %
## 0.949210 1.071624
```

Actually, for most replicates of the data simulation (by specifying a different seed for the random number generator),  $\beta_2$  would not have appeared to be significantly different from zero, so we would have been likely to drop the term, and arrive at the correct model using just the inferential tools already developed.

However, it is worth also looking at what would have happened if we had only observe  $h$  but not  $x$  (remember that  $y$  only depends on  $x$  and not on  $h$ ):

```
fit3 <- lm(y ~ h)
summary(fit3)$coefficients

##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) -0.007192532 0.03218731 -0.2234586 8.232243e-01
## h           0.985160671 0.03233624 30.4661526 1.153475e-144

cor(y, h)

## [1] 0.6941742
```

The high correlation between  $h$  and  $x$  means that we find a strongly significant association between  $h$  and  $y$ , despite the fact that  $h$  played no part at all in the simulation of  $y$ ! In this case,  $x$  is an example of a hidden *confounding variable*: it is correlated with both our predictor,  $h$ , and our response  $y$ , and therefore messes up our inference about the true causal influence of  $h$  on  $y$  (which is zero in this case).

Here is a second example of confounding. Now both  $h$  and  $x$  have a causal effect on  $y$ , and if we only observe  $x$ , we will overestimate its true effect (overestimate, because the correlation is positive).

```
y <- x + h + rnorm(n)
summary(lm(y ~ x))$coefficients

##              Estimate Std. Error    t value    Pr(>|t|)
## (Intercept) 0.01714281 0.03248967 0.5276387 0.5978674
## x           1.92019944 0.03246061 59.1547598 0.0000000

summary(lm(y ~ x + h))$coefficients
```

| ##             | Estimate   | Std. Error | t value    | Pr(> t )     |
|----------------|------------|------------|------------|--------------|
| ## (Intercept) | 0.01257442 | 0.03144589 | 0.3998749  | 6.893343e-01 |
| ## x           | 1.13515597 | 0.09980053 | 11.3742474 | 2.850763e-28 |
| ## h           | 0.83168316 | 0.10035571 | 8.2873525  | 3.709458e-16 |

That is, when  $h$  is hidden from us, our estimate of  $\beta_1$  is almost twice as large as it should be in this case.

Note that this tendency of the model fitting to try and compensate for missing confounding variables, is a profound difficulty for causal inference, but a blessing for prediction. The fact that the model has adjusted for the missing confounders in this way actually improves prediction.

The gold standard for establishing causality is to use data from a properly designed experiment, in which we use the process of *randomization* to break any possible association between unmeasured confounders and the predictor whose effect we want to measure. The idea is to turn the systematic variation in the response caused by confounders into something that we can model as independent random variability between experimental units.

An example conveys the idea. Suppose that we want to examine the relationship between exercise and fat mass in people aged 40 to 50. There are a huge number of factors contributing to people's fat mass, and if we simply look at a sample of people who already exercise by different amounts it will be very difficult to isolate the effect of exercise separate from all the other confounders (diet, working hours, profession, whether they have children, drinking habits, smoking, wealth etc.). Suppose instead that we set up a study in which a large enough number of participants enrol and are then assigned to one of several 'treatments', consisting of exercise programs of differing intensity (or a control of none). Now if we assign subjects to the different treatments randomly then there can be no possible association between all the other variables contributing to fat mass, and the one that we are interested in: amount of exercise. Indeed all the variability in fat mass attributable to the confounders can now be treated as random variability within each treatment. Any effect of exercise that we then find is causal: the thing we controlled having an effect on the variable we are interested in. Of course, this approach does not stop us from including measured covariates in the analysis model, in order to reduce the amount of variability being modelled as random, and thereby increase precision, but it does mean that we no longer have to worry about the effect of hidden confounders.

So properly designed controlled experiments with random allocation of experimental units to treatments are the best way of establishing causal effects. However there are many cases in which experimental manipulation is impractical, unethical and/or illegal. For example, if we are interested in establishing the causative effect of alcohol consumption on heart disease, it is simply unethical to conduct a study in which we randomly allocate subjects to a heavy drinking treatment (the same goes for any treatment where we have prior cause to suspect it may be harmful). Economics is another field beset with such problems: for example it is impractical to conduct a controlled experiment to establish the causative factors controlling a company's share price, and anything that came close would likely count as illegal market manipulation. This sort of problem is so ubiquitous in economics that it is from the field of *econometrics* that much of the work on causal inference from observational data comes. Depending on the field (e.g., medicine versus economic or social sciences) different techniques are more prevalent than others, but popular approaches are instrumental variables, differences in differences, regression discontinuity designs, G-computation, propensity scores, etc. These are, however, out of the scope of this course. Causal inference is currently a topic of active research. Indeed, the 2021 Nobel Prize in Economics was awarded to Guido Imbens for his foundational research on causal inference (<https://www.nobelprize.org/prizes/economic-sciences/2021/imbens/facts/>). The following short book provides a great introduction to the topic.

<https://mitpress.mit.edu/9780262545198/causal-inference/>

Finally, the following website provides several fun examples of spurious correlations:

<https://www.tylervigen.com/spurious-correlations>

## 9 Maximum likelihood estimation: a brief review

### 9.1 Why maximum likelihood estimation now?

It is now time to consider more general models inspired by the linear model. In many applications, a Gaussian distribution is inappropriate for the response variable, even when a linear model structure may still be suitable for the expected value. This motivates the framework of *generalized linear model* (GLM). For example, a discrete response may be better modelled using a Poisson or binomial distribution, while a positive continuous response may be described by a gamma distribution. Our existing linear-model theory is then insufficient: in particular, all these distributions violate the constant-variance assumption. However, Without departing far from the linear-model setting, we therefore have an obvious motivation to develop inference methods that apply across a broad class of distributions. The general theory of *maximum likelihood estimation* provides the key to this.

### 9.2 Maximum likelihood estimation

The key idea of *maximum likelihood estimation* is:

Parameter values that assign higher probability (or probability density, for continuous data) to the observed data are more plausible than parameter values that assign those data very low probability (or density).

For example, we would much prefer an estimate of parameter vector,  $\theta$ , that assigned a probability density of 0.1 to the observed  $y$ , according to the model, to an estimate for which the density was 0.00001.

So the idea is to judge the *likelihood* of parameter values using  $f_\theta(y)$ , the model probability density or mass function, according to the given value of  $\theta$ , evaluated at the observed data. Because  $y$  is now fixed and we are considering the likelihood as a function of  $\theta$ , it is usual to write the likelihood as  $L(\theta) \equiv f_\theta(y)$ . In fact, for theoretical and practical purposes, it is usual to work with the log likelihood  $l(\theta) = \log L(\theta)$ . The *maximum likelihood estimate* (MLE) of  $\theta$  is then

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} l(\theta).$$

In general, there will be no closed form solution for  $\hat{\theta}$  and numerical methods will be needed to actually find  $\hat{\theta}$ , but reliable and general methods are available.

There is more to maximum likelihood estimation than just its intuitive appeal. Under some regularity conditions and in the large sample limit as  $n \rightarrow \infty$ ,

$$\hat{\theta} \sim N(\theta, \mathcal{I}^{-1}), \quad (21)$$

where  $\mathcal{I} = -\mathbb{E}(\partial^2 l / \partial \theta \partial \theta^T)$ —the expected second derivative matrix of the negative log likelihood (also known as the *information matrix*). This turns out to be the best that can be achieved for an unbiased estimator (although MLEs are only unbiased in the large sample limit). These results are closely related to the fact that  $\partial l / \partial \theta$  has expected value  $\mathbf{0}$  and covariance matrix  $\mathcal{I}$ . In fact, a similar result applies in terms of the observed (rather than expected) second derivative matrix. If  $\hat{\mathcal{I}} = -\partial^2 l / \partial \theta \partial \theta^T$  (evaluated at  $\hat{\theta}$ ) then in the  $n \rightarrow \infty$  limit,

$$\hat{\theta} \sim N(\theta, \hat{\mathcal{I}}^{-1}).$$

Another large sample result is also useful. Suppose that we want to compare two *nested* models, meaning that one model can be written as the other model subject to  $r$  restrictions on its parameters,  $\theta$ , that can be written as  $\mathbf{R}(\theta) = \mathbf{0}$ . Let  $\hat{\theta}_0$  denote the MLEs under the restriction. Then we can test  $H_0 : \mathbf{R}(\theta) = \mathbf{0}$  (i.e. test the null hypothesis that the restricted model is correct), by using the large sample result that

$$2\{l(\hat{\theta}) - l(\hat{\theta}_0)\} \sim \chi_r^2, \quad (22)$$

*if the null hypothesis is correct.* If the null hypothesis is not correct, then the log likelihood ratio statistic on the left hand side will tend to be too large for consistency with a  $\chi_r^2$  distribution. As usual, to actually judge compatibility of data and null hypothesis we compute a p-value — the probability of a  $\chi_r^2$  random variable being at least as large as the observed test statistic. As usual a small p-value is evidence against the null hypothesis. Tests conducted

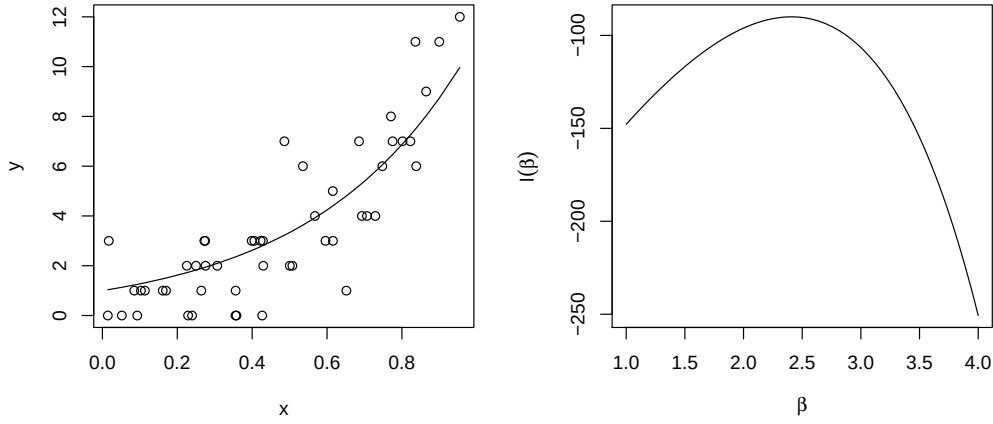


Figure 6: Left: 50 Poisson data,  $y$ , dependent on predictor data,  $x$ , and described by the model in Section 9.2.1. The black curve is the maximum likelihood estimate of the Poisson mean as a function of  $x$ . Right: the log likelihood function for the model parameter  $\beta$ , computed using the data in the left panel and as described in the text.

using this result are known as *generalized likelihood ratio tests* (GLRT). Likelihood ratio tests are similar to the F-tests we learned in Section 4.3.2, in the sense they compare the full model with a restricted model where some predictors (or even just a factor predictor) are omitted.

Deriving these results is beyond the scope of this course. However, for a concise derivation, see Wood (2015, Chapter 4, *Core Statistics*). Additionally, most—if not all—of these results were also proved in Statistical Methodology (MATH10095).

As was the case for F-tests, hypothesis tests like the GLRT offer one way to choose between models, essentially favouring the simpler model unless the data provide sufficient evidence that this is not tenable and a more complex model is needed. As we already learned in Section 5, an alternative is to try to estimate how close a model is to the ‘true’ model and select the model that is closest. This approach leads to selecting the model with the smallest value of the Akaike Information Criterion value among the candidate models. Recall that, for a given model, the AIC is given by  $-2l(\hat{\theta}) + 2p$ , with  $p$  being the number of model parameters - the dimension of  $\theta$ .

### 9.2.1 A simple example

Consider the one parameter Poisson model:

$$y_i \stackrel{\text{ind}}{\sim} \text{Poisson}\{\exp(\beta x_i)\},$$

and suppose that we have  $n$  pairs of  $(x_i, y_i)$  data. Generally, the probability function for a Poisson random variable with mean  $\lambda_i$  is  $f(y_i) = \lambda_i^{y_i} \exp(-\lambda_i)/y_i!$ . So for our model, for which  $\lambda_i = \exp(\beta x_i)$ , we have

$$f(y_i) = \exp(\beta x_i)^{y_i} \exp\{-\exp(\beta x_i)\}/y_i!.$$

To compute the likelihood we need to compute the joint probability of all the data. Because our model says that the data are independent, this joint probability is just the product of the individual probabilities for each  $y_i$ , i.e., the likelihood function is just

$$L(\beta) = \prod_{i=1}^n \exp(\beta x_i)^{y_i} \exp\{-\exp(\beta x_i)\}/y_i!,$$

and repeatedly using the facts that  $\log(a^b) = b \log(a)$  and  $\log(ab) = \log(a) + \log(b)$ , the log likelihood is therefore

$$l(\beta) = \sum_{i=1}^n \{y_i \beta x_i - \exp(\beta x_i) - \log(y_i!)\}.$$

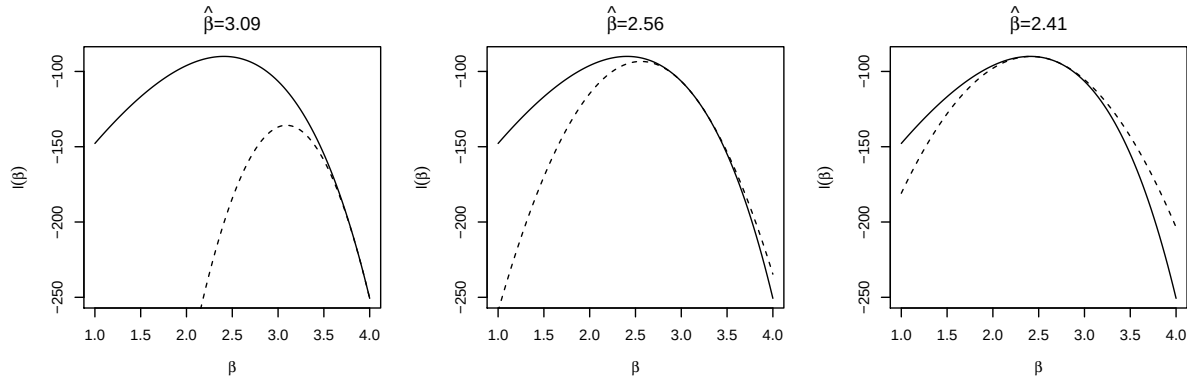


Figure 7: Three steps of Newton's method applied to the log likelihood for the Poisson example of Section 9.2.1, starting on the left from  $\beta = 4$ . The dashed curve shows the approximating quadratic matching the value and first two derivatives of the log likelihood at the previous best estimate of  $\beta$ . The panel headings give the value,  $\hat{\beta}$  maximising the approximating quadratic at each step. In this case no step length control is needed and the second derivative is always negative.

Figure 6 shows some  $(x, y)$  data (left) for which this model is appropriate, and plots the log-likelihood function for a range of  $\beta$  values (right).  $\hat{\beta} = 2.41$  is the MLE, and the continuous curve overlaid on the left panel is the Poisson mean that this implies (i.e.,  $\exp(\hat{\beta}x_i)$ ).

The key idea, then, is that the log-likelihood function provides a general way to measure how well a model fits the data and which parameter values best explain the data. The simple Poisson example considered only a single parameter, but in principle, we can compute the log-likelihood for a model with any number of parameters. However, we must ensure that we have sufficient data if we intend to estimate a large number of parameters.

### 9.2.2 Practical likelihood maximization

As this simple Poisson regression example already made clear, it is rarely possible to find explicit expressions for the maximum likelihood estimates, and instead we must maximize the log likelihood numerically. Newton's method is the gold standard for doing this, and works by optimizing successive quadratic approximations to the log likelihood. We start with an initial parameter value guess, and try to obtain a quadratic approximation to the log-likelihood that behaves like the log-likelihood itself in the vicinity of that guess. To do this we can use a Taylor approximation to the log-likelihood, discarding the terms above second order. That is, we evaluate the log likelihood and its first two derivatives, or the first derivative vector and second derivative matrix in usual case of a parameter vector, at the parameter value (or vector), and then find the unique quadratic function which also has this value and derivatives at the parameter value. Hopefully, the approximation has a maximum close to the maximum of the log-likelihood itself, and it is easy to write down an explicit form for the maximizer of the approximating quadratic, using this as our updated parameter guess. Iterating this procedure we eventually reach the maximum of the log likelihood itself.

The mathematical details of the Newton-Raphson method for the most general case of a parameter vector  $\beta = (\beta_1, \dots, \beta_p)^T$  are as follows. Let

$$\mathbf{U}(\beta) = \left( \frac{\partial l(\beta)}{\partial \beta_1}, \dots, \frac{\partial l(\beta)}{\partial \beta_p} \right)^T$$

be the score function. Let  $\mathbf{H}(\beta)$  denote the matrix  $\partial \mathbf{U}(\beta) / \partial \beta$  having entries

$$h_{jk} = \frac{\partial^2 l(\beta)}{\partial \beta_j \partial \beta_k}, \quad j = 1, \dots, p, \quad k = 1, \dots, p.$$

$\mathbf{H}(\boldsymbol{\beta})$  is called the Hessian matrix. At step  $t$  in the iterative process ( $t = 0, 1, 2, \dots$ ), the function  $l(\boldsymbol{\beta})$  is approximated near  $\boldsymbol{\beta}^{(t)}$  using a second-order Taylor series expansion,

$$l(\boldsymbol{\beta}) \approx l(\boldsymbol{\beta}^{(t)}) + \mathbf{U}(\boldsymbol{\beta}^{(t)})^\top (\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}) + \frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})^\top \mathbf{H}(\boldsymbol{\beta}^{(t)}) (\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}).$$

Applying rules for differentiating of vector functions and solving

$$\frac{\partial l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} \approx \mathbf{U}(\boldsymbol{\beta}^{(t)}) + \mathbf{H}(\boldsymbol{\beta}^{(t)}) (\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}) = \mathbf{0}$$

for  $\boldsymbol{\beta}$  yields

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \mathbf{H}(\boldsymbol{\beta}^{(t)})^{-1} \mathbf{U}(\boldsymbol{\beta}^{(t)}).$$

Iterations proceed until changes in  $l(\boldsymbol{\beta}^{(t)})$  between successive iterations are sufficiently tiny. There are, however, two modifications required to guarantee that Newton's method converges to the MLE. Firstly, we need to ensure that the approximating quadratic is one that actually has a maximum (as opposed to a minimum, or even, in more than one dimension, a saddlepoint). In one dimension this is a matter of checking that the second derivative is negative and replacing it with a negative number if it isn't. In more dimensions there is also an equivalent fix (check that the negative of the second derivative matrix is positive definite, and if necessary modify it to force this condition to hold). The second modification is to check that the proposed change in parameter values actually increases the log likelihood itself. If it doesn't we just move the parameter back towards the previous parameter guess, until the log likelihood is increased at the new values. With these modifications Newton's method will converge to a local maximum of the log-likelihood. For the generalized linear models considered in this course, using canonical links, the log-likelihood is concave, so any local maximum is also the global maximum, and convergence to the MLE is therefore guaranteed.

Figure 7 illustrates Newton's method for the case of the single parameter Poisson regression model. In this case the maximum is reached in 3 steps (to graphical accuracy). Notice how the final quadratic matches the log likelihood in the vicinity of the MLE.

### 9.2.3 Illustrating $\hat{\theta}$ uncertainty

Figure 8 shows log likelihood curves for 100 replicates of the data shown in the left panel of Figure 6, when the true  $\beta$  is 2.5, along with the corresponding MLEs,  $\hat{\beta}$ , and the corresponding non-parametric (kernel density) estimate of the distribution of  $\hat{\beta}$ . Notice how the likelihood function and  $\hat{\beta}$  vary from replicate to replicate, but how the distribution of  $\hat{\beta}$  is approximately normal and centred on the true  $\beta$  value of 2.5. This is exactly what Equation (21) implies will happen.

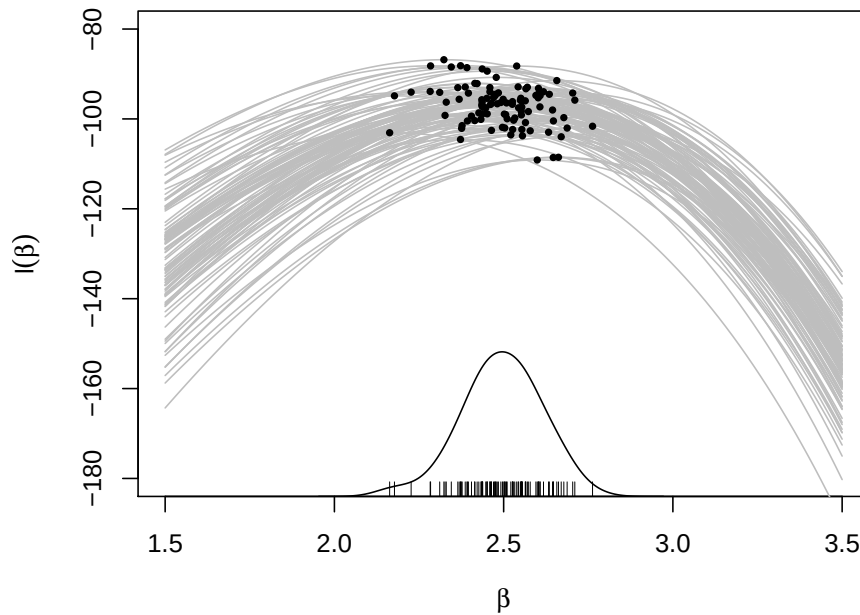


Figure 8: Log likelihood functions (grey) for 100 replicates of the section 9.2.1 Poisson dataset, when  $\beta = 2.5$ . The black dots show the maximum of each curve. The corresponding MLEs,  $\hat{\beta}$ , are shown as vertical bars on the x-axis (a rug plot). The continuous black curve is a (rescaled and vertically shifted) kernel density estimate of the distribution of  $\hat{\beta}$  based on the shown 100 estimates. Notice how the distribution of  $\hat{\beta}$  is approximately normal, with mean at the true value,  $\beta = 2.5$ , although there is zero probability that  $\hat{\beta} = 2.5$  exactly.

## 10 Introducing generalized linear models

As we have seen extensively in the previous sections, a linear model can be written as

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \epsilon_i, \quad \epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

An equivalent formulation is

$$\mathbb{E}(y_i) \equiv \mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad y_i \stackrel{\text{iid}}{\sim} N(\mu_i, \sigma^2).$$

*Generalized* linear models extend linear models by allowing a transformation of the mean through a link function and by permitting a much wider class of response distributions. A GLM has the basic structure:

$$g(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \mu_i \equiv \mathbb{E}(y_i),$$

where  $g$  is a differentiable and strictly monotonic (hence invertible on its range) *link function*,  $\mathbf{x}_i^\top$  is the  $i$ th row of the design matrix, and  $\boldsymbol{\beta}$  is a vector of unknown parameters. In addition, a GLM assumes that the  $y_i$  are independent and

$$y_i \sim \text{an exponential family distribution.}$$

The *exponential family* of distributions includes many useful distributions for applied modelling, such as the Poisson, binomial, gamma, and normal distributions. For most exponential family distributions used in GLMs, the variance of  $y_i$  is a non-constant function of its mean  $\mu_i$ , unlike the normal distribution where the variance is constant. A GLM therefore consists of three components:

- *Systematic component*:  $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ .
- *Random component*: the chosen exponential-family distribution for  $y_i$ .
- *Link function*  $g$ , relating  $\mu_i$  to the linear predictor.

Because generalized linear models are specified in terms of the *linear predictor*  $\eta \equiv \mathbf{X}\boldsymbol{\beta}$ , many ideas from linear modelling carry over with minor adjustments. Basic model formulation proceeds similarly, except that we must now choose a link function and an appropriate distribution for the response variable. If the identity link is used together with a Gaussian response distribution, we obtain the ordinary linear model studied in Sections 1–6 as a special case.

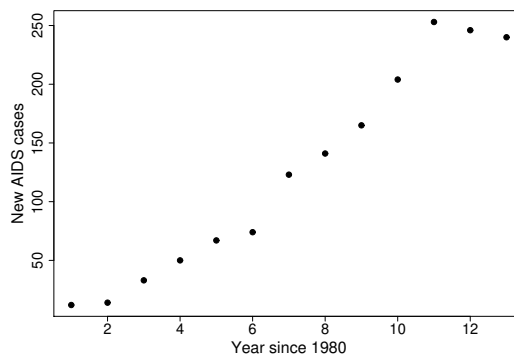
The generalization comes with some cost: model fitting must be performed iteratively, and inferential results are now approximate, relying on large-sample theory rather than being exact.

A brief historical note: the methodology of generalized linear modelling was developed in 1972 by John Nelder and Robert Wedderburn while working at the Rothamsted Experimental Station in the UK.

Before considering methodological details, let us look at some simple examples.

### 10.1 Simple single covariate examples of GLMs

#### Example 1: AIDS in Belgium.



The figure shows annual new AIDS cases in Belgium at the start of the epidemic. Early in an epidemic, exponential growth is often reasonable. If  $\mu_i$  is the expected number of new AIDS cases per year and  $t_i$  denotes the number of years since 1980, then a plausible model is

$$\mathbb{E}(y_i) \equiv \mu_i = \gamma e^{\alpha t_i}, \quad \gamma > 0,$$

where  $\gamma$  and  $\alpha$  are unknown parameters and  $y_i$  is the observed number of new AIDS cases per year. Such a model can be turned into GLM form by using the log link so that

$$\log(\mu_i) = \log(\gamma) + \alpha t_i = \beta_1 + \beta_2 t_i, \quad \beta_1 = \log(\gamma), \quad \beta_2 = \alpha.$$

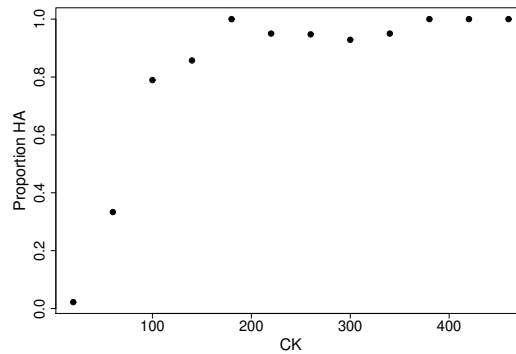
This expression is linear in the parameters. Because the response is a count (number of new AIDS cases), the Poisson distribution is a natural choice. Thus the GLM for this example uses a Poisson response distribution, a log link function, and a linear predictor  $\beta_1 + \beta_2 t_i$  (and thus the  $i$ th row of the design matrix is  $\mathbf{x}_i^T = (1, t_i)$ ).

### Example 2: Heart Attacks and Creatine Kinase

The following data are from a study examining the efficacy of blood creatine kinase (CK) levels as a diagnostic when patients present with symptoms that may indicate a heart attack.

| CK level         | 20 | 60 | 100 | 140 | 180 | 220 | 260 | 300 | 340 | 380 | 420 | 460 |
|------------------|----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Heart Attack     | 2  | 13 | 30  | 30  | 21  | 19  | 18  | 13  | 19  | 15  | 7   | 8   |
| Not Heart Attack | 88 | 26 | 8   | 5   | 0   | 1   | 1   | 1   | 1   | 0   | 0   | 0   |

Here is a plot of the proportion of patients who subsequently turned out to have had a heart attack, against their blood CK levels on admission to hospital.



Let  $p_i$  be the probability of heart attack at CK level  $x_i$ . A particularly convenient model for describing this probability is

$$p_{x_i} \equiv p_i = \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}.$$

This curve is sigmoid in shape, and bounded between 0 and 1. If  $y_i$  is the number of heart attack victims observed out of  $N_i$  patients with CK level  $x_i$ , we have that  $y_i \sim \text{Binomial}(N_i, p_i)$  and therefore,

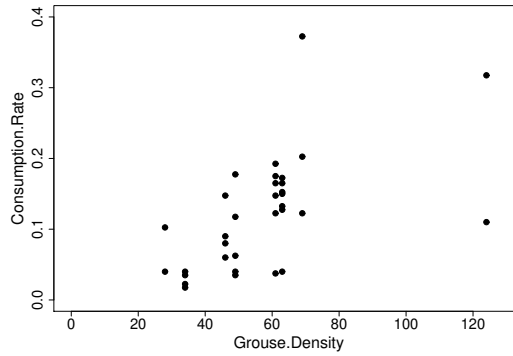
$$\mathbb{E}(y_i) \equiv \mu_i = N_i p_i = N_i \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}.$$

This model is somewhat non-linear in its parameters, but if we apply the *logit* link function we obtain that

$$\begin{aligned}
 g(\mu_i) &= \log \left( \frac{\mu_i}{N_i - \mu_i} \right) \\
 &= \log \left( \frac{N_i p_i}{N_i - N_i p_i} \right) \\
 &= \log \left( \frac{p_i}{1 - p_i} \right) \\
 &= \log \left( \frac{\frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}}{\frac{1}{1 + e^{\beta_1 + \beta_2 x_i}}} \right) \\
 &= \log (e^{\beta_1 + \beta_2 x_i}) \\
 &= \beta_1 + \beta_2 x_i
 \end{aligned}$$

which is clearly a GLM. This model is known as *logistic regression*.

### Example 3: Hen harriers and Grouse



The plot shows the daily consumption rate of Grouse by Hen Harriers (a type of bird of prey) as a function of the density of Grouse on various Grouse moors. If  $c_i$  denotes consumption rate and  $d_i$  is grouse density, then ecological theory suggests that the expected consumption rate should be related to grouse density by the following functional form

$$\mathbb{E}(c_i) \equiv \mu_i = \frac{\alpha d_i^3}{\delta + d_i^3},$$

where  $\alpha$  and  $\delta$  are unknown parameters. Although this model is nonlinear in its parameters, it becomes linear after applying the ‘inverse’ link function to  $\mu_i$ :

$$\frac{1}{\mu_i} = \frac{1}{\alpha} + \frac{\delta}{\alpha} \frac{1}{d_i^3} = \beta_1 + \beta_2 \frac{1}{d_i^3}, \quad \beta_1 = \frac{1}{\alpha}, \quad \beta_2 = \frac{\delta}{\alpha}.$$

Because the variability in consumption is likely to increase with the mean, a gamma distribution for the response is appropriate, thereby completing the model specification.

### Example 4: Linear models!

As noted earlier, the standard linear model is a special case of a GLM, with identity link and Gaussian response distribution.

## 11 Inference with GLMs

Inference for GLMs is based on the theory of maximum likelihood estimation. This section outlines the general framework that underpins all GLMs and shows how it leads to a neat link back to linear models.

### 11.1 The exponential family of distributions

In a GLM, the response variable is assumed to follow a distribution belonging to the *exponential family*. A distribution is in the exponential family if its probability density function, or probability mass function, can be written in the form

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}, \quad (23)$$

where

- $\theta$  is the *canonical parameter* and is a function of the expectation of the response variable,  $\mu \equiv \mathbb{E}(Y)$ , which in turn depends on the regression coefficients  $\beta$  through the link function.
- $\phi > 0$  is the *dispersion parameter*, representing the scale of the distribution. For some distributions (e.g., Poisson),  $\phi$  is known; for others (e.g., gamma), it must be estimated.
- $a(\cdot)$ ,  $b(\cdot)$ , and  $c(\cdot)$  are functions that are specific to each member of the exponential family.

If a distribution can be written in this form, maximum likelihood estimation and inference can be handled in a unified framework. The exponential family includes, among others, the normal, gamma, binomial, Poisson, and inverse Gaussian distributions. This broadens the applicability of linear model concepts to data where responses are positive-valued, binary, counts, or proportions, without requiring transformations of the response variable.

In what follows, we derive the exponential-family form for the Poisson, normal, and gamma distributions to illustrate how the representation works in practice.

#### Poisson distribution

The probability mass function of a Poisson distribution is

$$f(y; \mu) = \frac{e^{-\mu} \mu^y}{y!}, \quad y = 0, 1, 2, \dots, \quad \mu = \mathbb{E}(Y) > 0.$$

Write  $f(y; \mu) = \exp\{\log f(y; \mu)\}$ . Then,

$$\log f(y; \mu) = -\mu + y \log(\mu) - \log(y!) \Rightarrow \exp\{\log f(y; \mu)\} = \exp\{-\mu + y \log(\mu) - \log(y!)\}.$$

Rearranging this in the form of Equation (23) gives

$$\exp\{\log f(y; \mu)\} = \exp \left\{ \frac{y \log(\mu) - \mu}{1} + [-\log(y!)] \right\}.$$

This is of the form of (23) with

- $\theta = \log \mu \Rightarrow \mu = e^\theta$ ,
- $b(\theta) = \mu = e^\theta$ ,
- $a(\phi) = \phi = 1$ ,
- $c(y, \phi) = -\log(y!)$ .

Note that  $a(\phi) = \phi = 1$  because the Poisson distribution has no free dispersion parameter; its variance is fixed as  $\text{var}(Y) = \mu$ .

### Gaussian distribution

The probability density function of a Gaussian distribution with mean  $\mu$  and variance  $\sigma^2$  is given by

$$f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y - \mu)^2 \right\}, \quad y \in \mathbb{R}, \quad \mu \in \mathbb{R}, \quad \sigma^2 > 0.$$

As before, writing  $f(y; \mu, \sigma^2)$  as  $\exp\{\log f(y; \mu, \sigma^2)\}$ :

$$\begin{aligned} \log f(y; \mu, \sigma^2) &= -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y - \mu)^2 \\ &= -\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y^2 - 2\mu y + \mu^2), \end{aligned}$$

and thus

$$\begin{aligned} \exp\{\log f(y; \mu, \sigma^2)\} &= \exp \left\{ -\frac{1}{2\sigma^2} (y^2 - 2\mu y + \mu^2) - \frac{1}{2} \log(2\pi\sigma^2) \right\} \\ &= \exp \left\{ \frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2} \right\}. \end{aligned}$$

This is of the form of (23) with

- $\theta = \mu$ ,
- $b(\theta) = \frac{\mu^2}{2} = \frac{\theta^2}{2}$ ,
- $a(\phi) = \phi = \sigma^2$ ,
- $c(y, \phi) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{y^2}{2\sigma^2} = -\frac{1}{2} \log(2\pi\phi) - \frac{y^2}{2\phi}$ .

### Gamma distribution

A common parametrisation of the Gamma distribution is

$$f(y; \lambda, \nu) = \frac{\lambda^\nu}{\Gamma(\nu)} y^{\nu-1} e^{-\lambda y}, \quad y > 0, \quad \lambda > 0, \quad \nu > 0,$$

where  $\nu$  describes the shape and  $\lambda$  the rate (inverse of a scale parameter) of the distribution. Under this parametrisation,  $E(Y) \equiv \mu = \frac{\nu}{\lambda}$ . However, for the purposes of GLM, it is convenient to reparametrise the distribution by its mean, which can be achieved by letting  $\lambda = \frac{\nu}{\mu}$ <sup>38</sup>, leading to

$$f(y; \mu, \nu) = \frac{1}{\Gamma(\nu)} \left( \frac{\nu}{\mu} \right)^\nu y^{\nu-1} e^{-\frac{\nu}{\mu} y}, \quad y > 0, \quad \nu > 0, \quad \mu > 0.$$

Note that now,  $E(Y) = \mu$ . As for the other two examples,

$$\log f(y; \mu, \nu) = -\log[\Gamma(\nu)] + \nu \log(\nu) - \nu \log(\mu) + (\nu - 1) \log(y) - \frac{\nu y}{\mu}.$$

Hence,

$$\begin{aligned} \exp\{\log f(y; \mu, \nu)\} &= \exp \left\{ -\frac{\nu y}{\mu} - \nu \log(\mu) + (-\log[\Gamma(\nu)] + \nu \log(\nu) + (\nu - 1) \log(y)) \right\} \\ &= \exp \left\{ \frac{y(-1/\mu) - \log(\mu)}{1/\nu} + (-\log[\Gamma(\nu)] + \nu \log(\nu) + (\nu - 1) \log(y)) \right\}. \end{aligned}$$

This is of the form of (23) with

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<sup>38</sup>Setting  $\lambda = \nu/\mu$  eliminates the rate parameter in favour of the mean  $\mu$ , ensuring that  $\nu$  acts as a pure dispersion parameter and that the canonical parameter is a function of  $\mu$  alone. The alternative,  $\nu = \lambda\mu$ , would eliminate  $\nu$  instead, leaving the rate  $\lambda$  in the density and obscuring the central role of the mean.

- $\theta = -\frac{1}{\mu} \Rightarrow \mu = -\frac{1}{\theta}$ ,
- $b(\theta) = \log(\mu) = \log(-\frac{1}{\theta}) = -\log(-\theta)$ ; note that  $\mu > 0$  and so  $\theta < 0$  and therefore  $-\log(-\theta)$  is well-defined,
- $a(\phi) = \phi = \frac{1}{\nu}$ ,
- $c(y, \phi) = -\log[\Gamma(\nu)] + \nu \log(\nu) + (\nu - 1) \log(y) = -\log[\Gamma(1/\phi)y] + \frac{1}{\phi} \log(\frac{y}{\phi})$ .

### Mean and variance of the exponential family of distributions

We now derive the general expressions for the mean and variance of the exponential family of distributions. To this end, note that for a single observation  $y$ , the log likelihood is

$$l(\theta, \phi) = \log f(y; \theta, \phi) = \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi).$$

Differentiating with respect to  $\theta$  gives

$$\frac{\partial l}{\partial \theta} = \frac{y - b'(\theta)}{a(\phi)},$$

where the prime denotes differentiation with respect to  $\theta$ . Replacing the fixed observation  $y$  with the random variable  $Y$ , the score  $\partial l / \partial \theta$ , where now  $l = l(\theta, \phi; Y)$ , becomes a random variable, whose expectation is

$$\mathbb{E}\left(\frac{\partial l}{\partial \theta}\right) = \mathbb{E}\left(\frac{Y - b'(\theta)}{a(\phi)}\right) = \frac{\mathbb{E}(Y) - b'(\theta)}{a(\phi)},$$

since  $b'(\theta)$  and  $a(\phi)$  are constants with respect to  $Y$ . From general likelihood theory, the score function has expectation zero,

$$\mathbb{E}\left(\frac{\partial l}{\partial \theta}\right) = 0,$$

which holds under regularity conditions satisfied by exponential family distributions<sup>39</sup>. This result implies that

$$\frac{\mathbb{E}(Y) - b'(\theta)}{a(\phi)} = 0 \Rightarrow \mu \equiv E(Y) = b'(\theta). \quad (24)$$

Since  $a(\phi) > 0$  and  $\text{var}(Y) > 0$ , it follows from  $\text{var}(Y) = b''(\theta)a(\phi)$  (shown below) that  $b''(\theta) > 0$ , so  $b'(\theta)$  is strictly increasing and hence one-to-one (or injective), ensuring a one-to-one correspondence between  $\mu$  and  $\theta$ .<sup>40</sup> The result in (24) is the key to linking the GLM model parameter  $\beta$  to the canonical parameter  $\theta$  of the exponential family. In a GLM,  $\beta$  determines the mean of the response variable and, via Equation (24), thereby determines the canonical parameter  $\theta_i$  for each response observation  $y_i$  through the relationship  $\mu_i = b'(\theta_i)$ .

We now proceed to determine  $\text{var}(Y)$ . Taking second derivatives of the log likelihood, we obtain

$$\frac{\partial^2 l}{\partial \theta^2} = -\frac{b''(\theta)}{a(\phi)}.$$

General likelihood theory also provides the identity<sup>41</sup>

$$\mathbb{E}\left(\frac{\partial^2 l}{\partial \theta^2}\right) = -\mathbb{E}\left[\left(\frac{\partial l}{\partial \theta}\right)^2\right].$$

<sup>39</sup>These conditions essentially require that the support of  $Y$  does not depend on  $\theta$  and that the density is sufficiently smooth, so that differentiation and integration can be safely interchanged.

<sup>40</sup>A function is one-to-one (injective) if distinct inputs produce distinct outputs:  $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ . Applied to  $b'$ , this means  $b'(\theta_1) = b'(\theta_2) \Rightarrow \theta_1 = \theta_2$ , or equivalently,  $\mu_1 = \mu_2 \Rightarrow \theta_1 = \theta_2$ . Thus knowing  $\mu$  uniquely determines  $\theta$ , and vice versa, so the two parametrisations carry exactly the same information.

<sup>41</sup>The same regularity conditions as before are required.

First, note that

$$\mathbb{E} \left( \frac{\partial^2 l}{\partial \theta^2} \right) = \mathbb{E} \left( -\frac{b''(\theta)}{a(\phi)} \right) = -\frac{b''(\theta)}{a(\phi)}.$$

Additionally,

$$\left( \frac{\partial l}{\partial \theta} \right)^2 = \left( \frac{y - b'(\theta)}{a(\phi)} \right)^2 = \frac{(y - b'(\theta))^2}{a^2(\phi)} = \frac{(y - \mu)^2}{a^2(\phi)}, \quad \mu = b'(\theta),$$

so that

$$\mathbb{E} \left[ \left( \frac{\partial l}{\partial \theta} \right)^2 \right] = \mathbb{E} \left( \frac{(Y - \mu)^2}{a^2(\phi)} \right) = \frac{\mathbb{E}((Y - \mu)^2)}{a^2(\phi)} = \frac{\text{var}(Y)}{a^2(\phi)}.$$

Combining these results yields

$$-\frac{b''(\theta)}{a(\phi)} = -\frac{\text{var}(Y)}{a^2(\phi)},$$

and therefore

$$\text{var}(Y) = b''(\theta) a(\phi).$$

Since  $b'(\theta)$  is strictly increasing and hence invertible,  $\theta$  is uniquely determined by  $\mu$  via  $\theta = \theta(\mu) \equiv (b')^{-1}(\mu)$ . We may therefore write  $b''(\theta) = b''(\theta(\mu))$  as a function of  $\mu$  alone, and define the *variance function*

$$V(\mu) = b''(\theta(\mu)),$$

which describes how the variance depends on the mean. The variance result can then be written as

$$\text{var}(Y) = V(\mu) a(\phi),$$

showing that the variance factors into a function of the mean alone,  $V(\mu)$ , and a function of the dispersion parameter alone,  $a(\phi)$ .

In principle,  $a$  could be any positive function of  $\phi$ . However, in most standard GLMs the choice  $a(\phi) = \phi/\omega$  is used, where  $\phi > 0$  is a dispersion parameter common to all observations and  $\omega > 0$  is a known prior weight that may vary across observations, typically reflecting different amounts of information (e.g., group sizes).

In the Gaussian case, we had  $b(\theta) = \theta^2/2$ , implying that  $V(\mu) = b''(\theta) = 1$ , and so the variance function is independent of the mean. For other standard exponential family distributions this is not the case; indeed, the Gaussian is the only standard exponential family distribution with a constant variance function, making it somewhat exceptional in this regard. Furthermore,  $\text{var}(Y) = 1 \times \phi/\omega = \sigma^2$ , since  $\phi = \sigma^2$  and  $\omega = 1$ .

In turn, for the Poisson distribution, we had  $b(\theta) = e^\theta$ , implying that  $b''(\theta) = e^\theta = \mu$ , and thus  $V(\mu) = \mu$ , i.e., the variance function is linear in the mean. Also,  $\text{var}(Y) = (\phi/\omega) \times \mu = \mu$ , since for the Poisson distribution  $\phi = 1$  and  $\omega = 1$ ; indeed, it is well known that for the Poisson distribution the variance equals the mean.

Finally, for the Gamma distribution we had  $b(\theta) = -\log(-\theta)$  which gives  $b''(\theta) = 1/\theta^2$ , and thus  $V(\mu) = \mu^2$ , meaning the variance function is quadratic in the mean. Moreover,  $\text{var}(Y) = b''(\theta)\phi/\omega = \mu^2/\nu$ , since for the Gamma distribution  $\phi = 1/\nu$  and  $\omega = 1$ .

## Link function

As mentioned at the beginning of Section 10, the *link function* of a GLM connects the mean of the response to the linear predictor. That is, a GLM states that the linear predictor for observation  $i$ ,

$$\eta_i = \mathbf{x}_i^\top \beta = \sum_{j=1}^p \beta_j x_{ij},$$

with  $x_{i1} = 1$  when the model includes an intercept, is related to  $\mu_i$  by  $\eta_i = g(\mu_i)$  for a link function  $g$ . Equivalently,  $\mu_i = g^{-1}(\eta_i)$ , i.e., the inverse link maps linear predictor values to the mean. By introducing a potentially nonlinear function  $g^{-1}$ , the link function allows the mean response  $\mu_i$  to depend nonlinearly on the linear predictor  $\eta_i = \mathbf{x}_i^\top \beta$ , even though the linear predictor itself remains linear in the parameters. This flexibility permits the

relationship between the mean and the predictors to be chosen based on subject-matter or statistical considerations. Restrictions on  $\mu_i$  can be imposed through the choice of  $g$ , which maps the domain of  $\mu_i$  into an appropriate subset of  $\mathbb{R}$ , typically the whole real line, where the linear predictor naturally lives.

In principle, any differentiable and strictly monotonic function  $g$  can serve as a link function, but there are some particularly convenient and widely used choices for the standard GLMs.

The usual link function for the normal distribution is the identity link,  $\eta_i = g(\mu_i) = \mu_i$ , yielding the normal linear model covered in Sections 1–6. Suppose instead that  $y_i = \mu_i + \epsilon_i$ , where  $\epsilon_i \sim N(0, \sigma^2)$ , but

$$\mu_i = \frac{\alpha_0 x_i}{\alpha_1 + x_i}.$$

On rewriting,

$$\mu_i^{-1} = \frac{\alpha_1 + x_i}{\alpha_0 x_i} = \frac{1}{\alpha_0} + \frac{\alpha_1}{\alpha_0} \frac{1}{x_i} = \beta_0 + \beta_1 z_i, \quad z_i = \frac{1}{x_i}, \quad \beta_0 = \frac{1}{\alpha_0}, \quad \beta_1 = \frac{\alpha_1}{\alpha_0},$$

we see that this model falls within the general GLM framework with normal distribution for the response and the inverse link function,  $\eta_i = g(\mu_i) = \mu_i^{-1}$ . Note that here  $\mu_i \neq 0$  is required for the inverse link to be well-defined.

For the Poisson GLM, the mean  $\mu_i$  must be strictly positive ( $\mu_i \in (0, \infty)$ ). While an identity link ( $\eta_i = \mu_i$ ) can be used, it requires strict constraints on the parameters to prevent negative mean predictions. The standard choice is the log link,  $\eta_i = \log \mu_i$  (or equivalently  $\mu_i = e^{\eta_i}$ ), which maps the mean domain  $(0, \infty)$  onto the entire real line  $\mathbb{R}$ . This log link guarantees  $\mu_i > 0$  for all values of the linear predictor and implies that additive effects of the predictors  $\mathbf{x}_i$  translate into multiplicative effects on the mean response  $\mu_i$ .

The *canonical link function* is defined as the specific link  $g$  for which the linear predictor  $\eta_i$  equals the canonical parameter  $\theta_i$ , i.e.,

$$\eta_i = \theta_i.$$

To find the explicit form of  $g$ , we use the relationship between the mean  $\mu_i$  and the canonical parameter

$$\mu_i = b'(\theta_i).$$

Since  $b'$  is one-to-one, its inverse  $(b')^{-1}$  exists. We can therefore isolate  $\theta_i$

$$\theta_i = (b')^{-1}(\mu_i).$$

Since the canonical link satisfies  $\eta_i = \theta_i$ , and by the general definition of a GLM  $\eta_i = g(\mu_i)$ , we can write

$$g(\mu_i) = \theta_i = (b')^{-1}(\mu_i).$$

That is, the canonical link function is the inverse of the function mapping the canonical parameter to the mean.

The canonical link for the Gaussian distribution is  $g(\mu) = \mu$  and for the Poisson distribution it is  $g(\mu) = \log \mu$ .

For the Gamma distribution, the canonical parameter is  $\theta = -1/\mu$  and the canonical link is therefore  $g(\mu) = -1/\mu$ . Since  $\mu > 0$  as the gamma distribution has support on the positive real line, this link takes values in  $(-\infty, 0)$ , which does not match the full real line  $\mathbb{R}$  that the linear predictor typically occupies. For this reason, the log link is often preferred in practice, as it maps  $(0, \infty)$  onto all of  $\mathbb{R}$ . Some textbooks omit multiplicative constants in the canonical parameter and write the canonical link for the Gamma GLM as  $g(\mu) = 1/\mu$ . Although this leads to the same fitted means and predictions, the regression coefficients  $\beta$  differ in sign, and thus their interpretation changes.

When  $g$  is the canonical link function, the derivation of the maximum likelihood estimator of  $\beta$  is simplified, as we shall see next. However, even in this case, the estimating equations (except for the Gaussian distribution with identity link) are nonlinear functions of the regression parameters  $\beta$  and therefore require iterative numerical methods for their solution. For this reason, we proceed with the general case in which  $g$  is not necessarily the canonical link.

## 11.2 GLM fitting

In a GLM, an  $n \times 1$  vector of response variables  $\mathbf{y} = (y_1, \dots, y_n)^\top$  with independent components and means  $\mu_i = \mathbb{E}(y_i)$  satisfies

$$g(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad \text{and} \quad y_i \sim f(y_i; \theta_i, \phi), \quad i = 1, \dots, n,$$

where  $f(y_i; \theta_i, \phi)$  denotes an exponential family density or mass function with canonical parameter  $\theta_i$  and dispersion parameter  $\phi$ . The canonical parameter  $\theta_i$  is determined by  $\mu_i$  through (24) and hence, through the link function, by  $\boldsymbol{\beta}$ . Because the response variables are independent, the likelihood is

$$L(\boldsymbol{\beta}, \phi) = \prod_{i=1}^n f(y_i; \theta_i, \phi),$$

and hence the log likelihood function is

$$\log L(\boldsymbol{\beta}, \phi) = l(\boldsymbol{\beta}, \phi) = \sum_{i=1}^n l_i(\boldsymbol{\beta}, \phi) = \sum_{i=1}^n \log f(y_i; \theta_i, \phi) = \sum_{i=1}^n \left\{ \frac{\omega_i [y_i \theta_i - b(\theta_i)]}{\phi} + c(y_i, \phi) \right\},$$

where we use the common parametrisation  $a_i(\phi) = \phi/\omega_i$ , with known weights  $\omega_i$ . The dependence on  $\boldsymbol{\beta}$  arises through  $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}$ , which determines  $\mu_i = g^{-1}(\eta_i)$  and hence  $\theta_i$  through  $\mu_i = b'(\theta_i)$ . Finding the maximum likelihood estimate of  $\boldsymbol{\beta}$ , which we assume to be a  $p \times 1$  vector, requires setting the score vector  $\mathbf{U}(\boldsymbol{\beta}, \phi)$ , i.e., the vector of partial derivatives of the log likelihood  $l(\boldsymbol{\beta}, \phi)$  with respect to each component  $\beta_j$  of  $\boldsymbol{\beta}$ , equal to zero, where

$$\mathbf{U}(\boldsymbol{\beta}, \phi) = \begin{pmatrix} \frac{\partial l(\boldsymbol{\beta}, \phi)}{\partial \beta_1} \\ \vdots \\ \frac{\partial l(\boldsymbol{\beta}, \phi)}{\partial \beta_p} \end{pmatrix}.$$

We have that

$$\frac{\partial l(\boldsymbol{\beta}, \phi)}{\partial \beta_j} = \frac{\partial}{\partial \beta_j} \sum_{i=1}^n l_i(\boldsymbol{\beta}, \phi) = \sum_{i=1}^n \frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta}, \phi), \quad j = 1, \dots, p.$$

To differentiate  $l_i$  with respect to  $\beta_j$ , we apply the chain rule

$$\frac{\partial l_i(\boldsymbol{\beta}, \phi)}{\partial \beta_j} = \frac{\partial l_i(\boldsymbol{\beta}, \phi)}{\partial \theta_i} \frac{d\theta_i}{d\mu_i} \frac{d\mu_i}{d\eta_i} \frac{\partial \eta_i}{\partial \beta_j}. \quad (25)$$

Note that if the link function  $g$  is the canonical link function, then  $\eta_i = \theta_i$  and hence

$$\frac{d\theta_i}{d\eta_i} = 1.$$

Therefore,

$$\frac{d\theta_i}{d\mu_i} \frac{d\mu_i}{d\eta_i} = \frac{d\theta_i}{d\eta_i} = 1,$$

and the expression above simplifies to

$$\frac{\partial l_i(\boldsymbol{\beta}, \phi)}{\partial \beta_j} = \frac{\partial l_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \beta_j}.$$

We shall proceed with the most general case, where  $g$  is not necessarily the canonical link function. Let us now work out each derivative on the right-hand side of Equation (25). We have

$$\begin{aligned} \frac{\partial l_i}{\partial \theta_i} &= \frac{\partial}{\partial \theta_i} \left\{ \frac{\omega_i}{\phi} [y_i \theta_i - b(\theta_i)] + c(y_i, \phi) \right\} \\ &= \frac{\omega_i}{\phi} [y_i - b'(\theta_i)] \\ &= \frac{\omega_i}{\phi} (y_i - \mu_i), \quad \mu_i = b'(\theta_i). \end{aligned}$$

To determine  $\frac{d\theta_i}{d\mu_i}$ , note from Equation (24) that  $\mu_i = b'(\theta_i)$ , so

$$\frac{d\mu_i}{d\theta_i} = b''(\theta_i).$$

Since  $b'$  is one-to-one and differentiable, by the inverse function theorem, we have

$$\frac{d\theta_i}{d\mu_i} = \frac{1}{\frac{d\mu_i}{d\theta_i}} = \frac{1}{b''(\theta_i)}.$$

Similarly, since  $g(\mu_i) = \eta_i$ , we have

$$\frac{d\eta_i}{d\mu_i} = g'(\mu_i),$$

and because  $g$  is a differentiable and strictly monotonic function, we have

$$\frac{d\mu_i}{d\eta_i} = \frac{1}{\frac{d\eta_i}{d\mu_i}} = \frac{1}{g'(\mu_i)}.$$

Finally,

$$\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} = \sum_{j=1}^p \beta_j x_{ij} \Rightarrow \frac{\partial \eta_i}{\partial \beta_j} = x_{ij},$$

where  $x_{i1} = 1$  if the model includes an intercept. Putting these results together gives

$$\frac{\partial l_i(\boldsymbol{\beta}, \phi)}{\partial \beta_j} = \frac{\omega_i}{\phi} (y_i - \mu_i) \times \frac{1}{b''(\theta_i)} \times \frac{1}{g'(\mu_i)} \times x_{ij},$$

and hence

$$\begin{aligned} \frac{\partial}{\partial \beta_j} l(\boldsymbol{\beta}, \phi) &= \sum_{i=1}^n \frac{\partial l_i(\boldsymbol{\beta}, \phi)}{\partial \beta_j} \\ &= \frac{1}{\phi} \sum_{i=1}^n \frac{\omega_i (y_i - \mu_i) x_{ij}}{b''(\theta_i) g'(\mu_i)} \\ &= \frac{1}{\phi} \sum_{i=1}^n \frac{\omega_i (y_i - \mu_i) x_{ij}}{V(\mu_i) g'(\mu_i)}, \quad j = 1, \dots, p, \end{aligned}$$

recalling that  $V(\mu_i) = b''(\theta_i)$ . To maximise  $l(\boldsymbol{\beta}, \phi)$  with respect to  $\boldsymbol{\beta}$ , treating  $\phi$  as fixed, we solve the score equations  $\mathbf{U}(\boldsymbol{\beta}, \phi) = \mathbf{0}$ . Note that  $\hat{\boldsymbol{\beta}}$  will not depend on  $\phi$ . The resulting equations do not have an analytic solution, so we resort to a numerical procedure. To implement this, we require the matrix of second derivatives of the log-likelihood (the Hessian matrix). The  $(k, j)$  entry of this matrix is

$$\begin{aligned} \frac{\partial^2 l(\boldsymbol{\beta}, \phi)}{\partial \beta_k \partial \beta_j} &= \frac{\partial}{\partial \beta_k} \left( \frac{\partial l(\boldsymbol{\beta}, \phi)}{\partial \beta_j} \right) \\ &= \frac{\partial}{\partial \beta_k} \left( \sum_{i=1}^n \frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta}, \phi) \right) \\ &= \sum_{i=1}^n \frac{\partial}{\partial \beta_k} \left( \frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta}, \phi) \right). \end{aligned}$$

We have

$$\frac{\partial}{\partial \beta_k} \left( \frac{\partial}{\partial \beta_j} l_i(\boldsymbol{\beta}, \phi) \right) = \frac{\partial}{\partial \beta_k} \left( \frac{1}{\phi} \frac{\omega_i (y_i - \mu_i) x_{ij}}{V(\mu_i) g'(\mu_i)} \right) = \frac{1}{\phi} \frac{\partial}{\partial \beta_k} \left( \frac{\omega_i (y_i - \mu_i) x_{ij}}{V(\mu_i) g'(\mu_i)} \right).$$

To calculate

$$\frac{\partial}{\partial \beta_k} \left( \frac{\omega_i(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)} \right),$$

we apply the chain rule again:

$$\frac{\partial}{\partial \beta_k} \left( \frac{\omega_i(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)} \right) = \frac{d \left( \frac{\omega_i(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)} \right)}{d\mu_i} \frac{d\mu_i}{d\eta_i} \frac{\partial \eta_i}{\partial \beta_k}.$$

As before,

$$\frac{d\mu_i}{d\eta_i} = \frac{1}{g'(\mu_i)}, \quad \text{and} \quad \frac{\partial \eta_i}{\partial \beta_k} = x_{ik}.$$

For the first term, we have

$$\frac{d \left( \frac{\omega_i(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)} \right)}{d\mu_i} = \frac{d}{d\mu_i} \left( \omega_i(y_i - \mu_i)x_{ij} \times \frac{1}{V(\mu_i)g'(\mu_i)} \right) = -\omega_i x_{ij} \frac{1}{V(\mu_i)g'(\mu_i)} + \frac{d}{d\mu_i} \left( \frac{1}{V(\mu_i)g'(\mu_i)} \right) \omega_i(y_i - \mu_i)x_{ij},$$

where

$$\frac{d}{d\mu_i} \left( \frac{1}{V(\mu_i)g'(\mu_i)} \right) = -\frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{g'(\mu_i)^2 V(\mu_i)^2}.$$

Therefore

$$\frac{d \left( \frac{\omega_i(y_i - \mu_i)x_{ij}}{V(\mu_i)g'(\mu_i)} \right)}{d\mu_i} = -\omega_i x_{ij} \frac{1}{V(\mu_i)g'(\mu_i)} - \frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{g'(\mu_i)^2 V(\mu_i)^2} \omega_i(y_i - \mu_i)x_{ij}.$$

Putting all the terms together leads to

$$\begin{aligned} \frac{\partial^2 l(\beta, \phi)}{\partial \beta_k \partial \beta_j} &= -\frac{1}{\phi} \sum_{i=1}^n \left\{ \left[ \omega_i x_{ij} \frac{1}{V(\mu_i)g'(\mu_i)} + \frac{V'(\mu_i)g'(\mu_i) + V(\mu_i)g''(\mu_i)}{g'(\mu_i)^2 V(\mu_i)^2} \omega_i(y_i - \mu_i)x_{ij} \right] \frac{1}{g'(\mu_i)} x_{ik} \right\} \\ &= -\frac{1}{\phi} \sum_{i=1}^n \left\{ \frac{\omega_i x_{ik} x_{ij}}{g'(\mu_i)^2 V(\mu_i)} + \frac{\omega_i(y_i - \mu_i)x_{ij} x_{ik}}{g'(\mu_i)^2 V(\mu_i)} \left( \frac{g''(\mu_i)}{g'(\mu_i)} + \frac{V'(\mu_i)}{V(\mu_i)} \right) \right\}. \end{aligned}$$

Note that

$$\mathbb{E} \left( \frac{\partial^2 l(\beta, \phi)}{\partial \beta_k \partial \beta_j} \right) = -\frac{1}{\phi} \sum_{i=1}^n \frac{\omega_i x_{ik} x_{ij}}{g'(\mu_i)^2 V(\mu_i)},$$

since  $\mathbb{E}(y_i - \mu_i) = \mathbb{E}(y_i) - \mu_i = \mu_i - \mu_i = 0$ .

Defining the *working weights*

$$w_i = \frac{\omega_i}{g'(\mu_i)^2 V(\mu_i)}, \quad i = 1, \dots, n, \quad \mathbf{W} = \text{diag}(w_1, \dots, w_n),$$

we therefore have

$$\mathcal{I} = -\mathbb{E} \left( \frac{\partial^2 l(\beta, \phi)}{\partial \beta \partial \beta^\top} \right) = \mathbf{X}^\top \mathbf{W} \mathbf{X} / \phi.$$

From the general large-sample properties of maximum likelihood estimators,

$$\hat{\beta} \sim N(\beta, \mathcal{I}^{-1}), \quad \mathcal{I} = \mathbf{X}^\top \mathbf{W} \mathbf{X} / \phi. \quad (26)$$

Inference in GLMs is then typically based on large-sample normal approximations. An exception is the Gaussian linear model, where, under the assumption of normally distributed errors, exact  $t$  and  $F$  inference is available. For distributions with known scale parameter  $\phi$  (e.g., the Poisson and binomial distributions, for which  $\phi = 1$ ), the covariance matrix of  $\hat{\beta}$  is therefore known, and this result can be used to directly find confidence intervals for the

parameters and to perform hypothesis tests. If  $\phi$  is unknown (as is the case for, e.g., the gamma distribution), then it must be estimated (see Section 11.3) and intervals must be based on an approximate  $t$  distribution.

Applying Newton's method to find  $\hat{\beta}$  amounts to iterating, from some initial guess  $\beta^{(0)}$ ,

$$\beta^{(t+1)} = \beta^{(t)} - \mathbf{H}(\beta^{(t)})^{-1} \mathbf{U}(\beta^{(t)}), \quad \mathbf{H}(\beta) = \frac{\partial^2}{\partial \beta \partial \beta^T} l(\beta, \phi),$$

until convergence. A common modification replaces the Hessian by its expectation. This modification is known as the *Fisher scoring* method, whose updating equations are

$$\begin{aligned} \beta^{(t+1)} &= \beta^{(t)} - (\mathbb{E}[\mathbf{H}(\beta^{(t)})])^{-1} \mathbf{U}(\beta^{(t)}) \\ &= \beta^{(t)} + \mathcal{I}(\beta^{(t)})^{-1} \mathbf{U}(\beta^{(t)}). \end{aligned}$$

Since

$$\begin{aligned} \mathcal{I}(\beta^{(t)})^{-1} &= (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \phi \\ \mathbf{U}(\beta^{(t)}) &= \frac{1}{\phi} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{G}^{(t)} (\mathbf{y} - \boldsymbol{\mu}^{(t)}), \quad \mathbf{G}^{(t)} = \text{diag}\{g'(\mu_1^{(t)}), \dots, g'(\mu_n^{(t)})\}, \end{aligned}$$

the Fisher scoring updates become

$$\begin{aligned} \beta^{(t+1)} &= \beta^{(t)} + (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{G}^{(t)} (\mathbf{y} - \boldsymbol{\mu}^{(t)}) \\ &= (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \{ \mathbf{G}^{(t)} (\mathbf{y} - \boldsymbol{\mu}^{(t)}) + \mathbf{X} \beta^{(t)} \} \\ &= (\mathbf{X}^T \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}^{(t)} \mathbf{z}^{(t)}, \end{aligned}$$

where

$$z_i^{(t)} = g'(\mu_i^{(t)})(y_i - \mu_i^{(t)}) + \eta_i^{(t)}, \quad \eta_i^{(t)} = \mathbf{x}_i^T \beta^{(t)}.$$

So  $\beta^{(t+1)}$  is recognisable as the minimiser of the weighted sum of squares,  $\sum_i w_i (z_i - \mathbf{x}_i^T \beta)^2$ .

Hence maximum likelihood estimation for a GLM can be obtained using the *iteratively re-weighted least squares* (IRLS) algorithm.

1. Choose an initial value  $\beta^{(0)}$  and set

$$\eta_i^{(0)} = \mathbf{x}_i^T \beta^{(0)}, \quad \mu_i^{(0)} = g^{-1}(\eta_i^{(0)}).$$

In some implementations the algorithm is initialised by setting

$$\mu_i^{(0)} = y_i + \Delta_i,$$

where  $\Delta_i$  is a small constant introduced to avoid evaluating the link function at boundary values. For example, with the log link  $g(\mu) = \log(\mu)$ , observations with  $y_i = 0$  would lead to  $\log(0)$ , so a small positive constant is added to ensure that the initial linear predictor  $\eta_i^{(0)} = g(\mu_i^{(0)})$  is finite.

2. Form the working response

$$z_i^{(t)} = g'(\mu_i^{(t)})(y_i - \mu_i^{(t)}) + \eta_i^{(t)},$$

and the working weights

$$w_i^{(t)} = \frac{\omega_i}{g'(\mu_i^{(t)})^2 V(\mu_i^{(t)})}.$$

3. Update  $\beta$  by solving the weighted least squares problem

$$\beta^{(t+1)} = \underset{\beta}{\text{argmin}} \sum_i w_i^{(t)} (z_i^{(t)} - \mathbf{x}_i^T \beta)^2.$$

4. Update

$$\eta_i^{(t+1)} = \mathbf{x}_i^T \beta^{(t+1)}, \quad \mu_i^{(t+1)} = g^{-1}(\eta_i^{(t+1)}).$$

5. Repeat steps 2–4 until convergence, e.g.,  $\|\beta^{(t+1)} - \beta^{(t)}\| < \epsilon$  for some prescribed tolerance  $\epsilon > 0$ .

### Weighted least squares: a brief aside

Recall that the least squares estimator that minimises

$$\sum_{i=1}^n (z_i - \mathbf{x}_i^T \boldsymbol{\beta})^2,$$

is  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{z}$  (although, as we know, this expression is not generally used for actual computation). Let us verify that

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{z},$$

where  $\mathbf{W}$  is an  $n \times n$  diagonal matrix with entries  $W_{ii} = w_i$ , is the minimiser of the weighted sum of squares

$$\sum_{i=1}^n w_i (z_i - \mathbf{x}_i^T \boldsymbol{\beta})^2. \quad (27)$$

Let  $\tilde{\mathbf{W}}$  denote the  $n \times n$  diagonal matrix<sup>42</sup>

$$\tilde{\mathbf{W}} = \text{diag}(\sqrt{w_1}, \dots, \sqrt{w_n}),$$

so that  $\mathbf{W} = \tilde{\mathbf{W}} \tilde{\mathbf{W}}$ . Then

$$\sum_{i=1}^n w_i (z_i - \mathbf{x}_i^T \boldsymbol{\beta})^2 = \sum_{i=1}^n \{\sqrt{w_i} (z_i - \mathbf{x}_i^T \boldsymbol{\beta})\}^2 = \sum_{i=1}^n (\sqrt{w_i} z_i - \sqrt{w_i} \mathbf{x}_i^T \boldsymbol{\beta})^2 = \sum_{i=1}^n (\tilde{z}_i - \tilde{\mathbf{x}}_i^T \boldsymbol{\beta})^2$$

where

$$\tilde{\mathbf{z}} = \tilde{\mathbf{W}} \mathbf{z}, \quad \tilde{\mathbf{X}} = \tilde{\mathbf{W}} \mathbf{X}.$$

Thus, the weighted least squares problem reduces to an ordinary least squares problem in the transformed variables, and therefore

$$\hat{\boldsymbol{\beta}} = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{z}} = (\mathbf{X}^T \tilde{\mathbf{W}} \tilde{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^T \tilde{\mathbf{W}} \tilde{\mathbf{W}} \mathbf{z} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{z}.$$

### Example

We revisit the AIDS data example from Section 10. Recall that after expressing the model in GLM form, we have

$$\log(\mu_i) = \beta_1 + \beta_2 t_i, \quad i = 1, \dots, 13.$$

As a starting point, we assume a Poisson distribution with a log link (*Poisson regression*). We now implement the IRLS algorithm. For the log link we have  $g(\mu) = \log(\mu)$  and  $g'(\mu) = \frac{d}{d\mu} \log \mu = \frac{1}{\mu}$ . For the Poisson distribution the variance function is  $V(\mu) = \mu$ , the dispersion parameter is  $\phi = 1$ , and  $\omega_i = 1$ , for  $i = 1, \dots, 13$ . At iteration  $t$  we have

$$z_i^{(t)} = g'(\mu_i^{(t)})(y_i - \mu_i^{(t)}) + \eta_i^{(t)} = \frac{1}{\mu_i^{(t)}}(y_i - \mu_i^{(t)}) + \eta_i^{(t)} = \frac{y_i}{\mu_i^{(t)}} - 1 + \eta_i^{(t)},$$

$$w_i^{(t)} = \frac{\omega_i}{g'(\mu_i^{(t)})^2 V(\mu_i^{(t)})} = \frac{1}{\mu_i^{(t)} (\mu_i^{(t)})^{-2}} = \mu_i^{(t)}.$$

```
y <- c(12, 14, 33, 50, 67, 74, 123, 141, 165, 204, 253, 246, 240)
t <- 1:13
mu <- y; eta <- log(mu) ## initialize mean and linear predictor
converged <- FALSE
```

<sup>42</sup>Assuming  $w_i > 0$  for all  $i = 1, \dots, n$ , which holds in practice since  $V(\mu_i) > 0$  and  $g'(\mu_i) \neq 0$ , the latter following from the strict monotonicity of  $g$ .

```

while (!converged) {
  z <- y/mu - 1 + eta ## create working response
  w <- mu              ## create working weights
  fit <- lm(z ~ t, weights = w) ## fit working model
  eta.old <- eta       ## store old linear predictor
  eta <- fitted(fit)   ## get new linear predictor (X \beta)
  mu <- exp(eta)       ## get new mean
  ## test for convergence ...
  if (max(abs(eta-eta.old)) < 1e-6 * mean(abs(eta))) converged <- TRUE
}
coef(fit)

## (Intercept)          t
## 3.1405895    0.2021212

```

Note that we have used the `lm` function with an argument that we have not yet encountered: `weights`. By typing `help(lm)`, we find the following description for this argument:

*“an optional vector of weights to be used in the fitting process. (...) If non-NULL, weighted least squares is used with weights `weights` (that is, minimizing  $\sum(w * e^2)$ ) (...)”*

This is exactly what we want in (27). Implementing IRWLS without using the `lm` function is also straightforward. Of course, we know that computing the  $\beta$  estimates at each iteration in this way (using the `solve` function) may be numerically unstable in certain situations.

```

mu <- y; eta <- log(mu)
t_vec <- cbind(rep(1, length(t)), t)
converged <- FALSE
iter <- 0 #counts how many iterations until convergence
while (!converged) {
  z <- y/mu - 1 + eta
  w <- as.vector(mu)
  coefs <- solve(t(t_vec) %*% diag(w) %*% t_vec) %*% t(t_vec) %*% diag(w) %*% cbind(z)
  eta.old <- eta
  eta <- t_vec %*% coefs
  mu <- exp(eta)
  if (max(abs(eta-eta.old)) < 1e-6 * mean(abs(eta))) converged <- TRUE
  iter <- iter + 1
}
as.numeric(coefs)

## [1] 3.1405895 0.2021212

iter

## [1] 4

```

We now compare these results with those obtained using the `glm` function, which we will use extensively in the next section.

```

mod1 <- glm(y ~ t, family = poisson)
summary(mod1)

##
## Call:
## glm(formula = y ~ t, family = poisson)
##
## Coefficients:

```

```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) 3.140590   0.078247  40.14  <2e-16 ***
## t          0.202121   0.007771  26.01  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 872.206  on 12  degrees of freedom
## Residual deviance:  80.686  on 11  degrees of freedom
## AIC: 166.37
##
## Number of Fisher Scoring iterations: 4
```

### 11.3 GLM checking and inference

Model checking for GLMs is performed using residuals, in much the same way as for linear models. However, in GLMs some form of standardisation of the residuals is typically required. The main reason for not simply examining the raw (response) residuals,  $\hat{\epsilon}_i = y_i - \hat{\mu}_i$ , is that their variance depends on the mean through the assumed mean–variance relationship of the model. For example, if a Poisson model is used, then the variability of the residuals should increase in direct proportion to the fitted values,  $\hat{\mu}_i$ . If raw residuals are plotted against fitted values, it is therefore difficult to judge whether the increase in variability follows the assumed relationship or some alternative pattern (for example proportional to  $\sqrt{\hat{\mu}_i}$  or  $\hat{\mu}_i^2$ ). For this reason, it is customary to standardise GLM residuals so that, when the model assumptions are correct, the resulting residuals have approximately constant variance and behave, as far as possible, like residuals from a linear model.

The most obvious way to standardize the residuals is to divide them by a quantity proportional to their standard deviation under the fitted model. This gives rise to the *Pearson residuals*, defined as

$$\hat{\epsilon}_i^P = \frac{\sqrt{\omega_i}(y_i - \hat{\mu}_i)}{\sqrt{V(\hat{\mu}_i)}}, \quad i = 1, \dots, n,$$

which should approximately have zero mean and variance  $\phi$  if the model is correct. These residuals should not display systematic trends when plotted against the fitted values or against any predictors (whether included in the model or not) if the model is correctly specified. Further, if the spread of the residuals is not constant across the fitted values, this suggests that the assumed mean–variance relationship may be incorrect.

As already mentioned, when  $\phi$  is known, Equation (26) can be used directly to construct confidence intervals for the components of  $\beta$  or to test hypotheses about them. When  $\phi$  is unknown, an estimator is required. Although in principle it is possible to estimate  $\phi$  by maximum likelihood, this is rarely done as the MLE of  $\phi$  can be substantially biased in small to moderate samples. Instead, a moment-based estimator can be constructed as the sum of squared Pearson residuals divided by the residual degrees of freedom, giving the so-called Pearson estimator

$$\hat{\phi} = \frac{1}{n - p} \sum_{i=1}^n \frac{\omega_i (y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}, \quad (28)$$

where  $p$  is the number of parameters (dimension of  $\beta$ ). This can then be substituted into (26).

We now introduce the concept of *deviance* which is a key concept in generalized linear models, playing a similar role to the residual sum of squares in a linear model. Intuitively, it measures the discrepancy between the fitted generalized linear model and a perfect model for the sample  $\{(y_i, \mathbf{x}_i)\}_{i=1}^n$ . This perfect model is known as the *saturated model*: the model with one parameter per data point (and thus  $n$  parameters) and perfect fit (essentially interpolating the data), i.e.,  $\hat{\mu} = \mathbf{y}$ . A perfect fit may sound desirable, but the saturated model is uninformative because it merely reproduces the data rather than summarising them, and therefore lacks the advantages of a simpler, more parsimonious model (e.g., more interpretable and transferable estimates of the relationship between the response and the predictors). However, the saturated model gives a baseline for measuring the discrepancy for an intermediate model with, say,  $p$  parameters. Let  $l(\hat{\beta}, \phi)$  and  $l_S$  denote the maximized log likelihood for the

model under consideration and for the saturated model, respectively. Note that  $l_S$  is the highest value that the log likelihood could possibly have, given the data. Formally, the deviance is defined as

$$D = 2\{l_S - l(\hat{\beta}, \phi)\}. \quad (29)$$

The log likelihood  $l(\hat{\beta}, \phi)$  cannot exceed  $l_S$ . As a consequence, the deviance is always greater than or equal to zero. The greater the deviance, the poorer the fit. Equivalently, the deviance decreases as the model fits the data more closely. For the fitted model under consideration, let  $\hat{\theta}_i$  denote the estimate of the canonical parameter  $\theta_i$ , corresponding to the estimated mean  $\hat{\mu}_i$ . Similarly, let  $\tilde{\theta}_i$  denote the estimate of  $\theta_i$  for the saturated model, with corresponding mean given by  $\tilde{\mu}_i = y_i$ . Provided that the response observations are independent and for an exponential family distribution, with  $a_i(\phi) = \phi/\omega_i$ , the deviance  $D$  in (29) simplifies to

$$\begin{aligned} D &= 2 \sum_{i=1}^n \left\{ \frac{\omega_i[y_i\tilde{\theta}_i - b(\tilde{\theta}_i)]}{\phi} + c(y_i, \phi) - \frac{\omega_i[y_i\hat{\theta}_i - b(\hat{\theta}_i)]}{\phi} - c(y_i, \phi) \right\} \phi \\ &= \sum_{i=1}^n 2\omega_i\{y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\}. \end{aligned} \quad (30)$$

Notice that the deviance does not depend on  $\phi$ , as it cancels out.

For a Poisson GLM, from Section 11.1, we have that  $\hat{\theta}_i = \log(\hat{\mu}_i)$  and  $b(\hat{\theta}_i) = e^{\hat{\theta}_i} = \hat{\mu}_i$ . Similarly, for the saturated model, for which  $\tilde{\mu}_i = y_i$ , we have that  $\tilde{\theta}_i = \log(y_i)$  and  $b(\tilde{\theta}_i) = y_i$ . Also,  $\omega_i = 1$ , for all  $i = 1, \dots, n$ . The deviance is then equal to

$$\begin{aligned} D &= \sum_{i=1}^n 2\{y_i(\log y_i - \log \hat{\mu}_i) - y_i + \hat{\mu}_i\} \\ &= \sum_{i=1}^n 2\left\{y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) - y_i + \hat{\mu}_i\right\}. \end{aligned}$$

By convention, for  $y_i = 0$ , the term  $y_i \log(y_i/\hat{\mu}_i)$  is taken to be 0.

For a normal GLM, from Section 11.1, we have that  $\hat{\theta}_i = \hat{\mu}_i$  and  $b(\hat{\theta}_i) = \frac{\hat{\theta}_i^2}{2} = \frac{\hat{\mu}_i^2}{2}$ . Similarly,  $\tilde{\theta}_i = y_i$  and  $b(\tilde{\theta}_i) = \frac{y_i^2}{2}$  for the saturated model. We further have  $\omega_i = 1$ , for all  $i = 1, \dots, n$ . So the deviance equals

$$\begin{aligned} D &= \sum_{i=1}^n 2\left\{y_i(y_i - \hat{\mu}_i) - \frac{y_i^2}{2} + \frac{\hat{\mu}_i^2}{2}\right\} \\ &= \sum_{i=1}^n (y_i^2 - 2y_i\hat{\mu}_i + \hat{\mu}_i^2) \\ &= \sum_{i=1}^n (y_i - \hat{\mu}_i)^2. \end{aligned} \quad (31)$$

For ordinary linear models this is the residual sum of squares:  $\|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2 = \|\mathbf{y} - \hat{\boldsymbol{\mu}}\|^2$  (in the notation of Section 4).

A quantity related to the deviance is the *scaled deviance*, given by

$$D^* = \frac{D}{\phi} = 2\{l_S - l(\hat{\beta}, \phi)\}.$$

If  $\phi = 1$ , such as for the Poisson distribution, the deviance and the scaled deviance are the same. From the generalized likelihood ratio test in Equation (22) in Section 9, we might expect that, if the assumed model is correctly specified, then approximately

$$D^* \sim \chi_{n-p}^2, \quad (32)$$

in the large sample limit. Note that in (22),  $r$  denotes the difference in the number of parameters in the two models being compared. The saturated model has  $n$  parameters, while the fitted model has  $p$  parameters, so the difference

is  $n - p$ , which corresponds to the degrees of freedom of the limiting chi-squared distribution. Strictly speaking, the limiting argument justifying this large sample approximation in (32) is not directly applicable here, since it requires the number of parameters to remain fixed as  $n \rightarrow \infty$ , whereas the saturated model has  $n$  parameters by construction. Asymptotic results are available for certain distributions in the exponential family that justify (32) as a large-sample approximation. For the normal distribution, the result is exact in finite samples, since the deviance reduces to the residual sum of squares and the ratio  $D/\phi = \text{RSS}/\sigma^2$  follows a  $\chi^2_{n-p}$  distribution exactly. For distributions with an unknown dispersion parameter  $\phi$ , such as the normal, Gamma, or Inverse Gaussian (see Workshop 4) distributions, the scaled deviance  $D^* = D/\phi$  cannot be computed directly since  $\phi$  must be estimated. Therefore, this test cannot be used directly as a goodness-of-fit statistic, even though the chi-squared result is exact for the normal distribution, since  $\sigma^2$  is typically unknown in practice. For the binomial and Poisson distributions, however,  $\phi = 1$ , so the test is practical. Even in these cases, the approximation may be poor for small datasets. In the case of the binomial distribution with binary data, the approximation fails entirely, for reasons that are not discussed here.

If we want to compare a GLM model  $M_0$ , with  $p_0$  parameters, that is nested within a GLM model  $M_1$ , with  $p_1$  parameters (and so  $p_0 < p_1$ ) using the GLRT in Equation (22), then the test statistic can be re-written in terms of  $D_0^*$  and  $D_1^*$  to obtain the approximate result that under the null hypothesis that the simpler model,  $M_0$  is adequate (i.e. that the additional parameters in  $M_1$  are all zero)

$$\begin{aligned} D_0^* - D_1^* &= 2\{l_S - l(\hat{\beta}_0, \phi)\} - 2\{l_S - l(\hat{\beta}_1, \phi)\} \\ &= 2\{l(\hat{\beta}_1, \phi) - l(\hat{\beta}_0, \phi)\} \sim \chi^2_{p_1 - p_0}, \end{aligned}$$

where  $l(\hat{\beta}_1, \phi)$  and  $l(\hat{\beta}_0, \phi)$  are the maximized log likelihoods under model  $M_1$  and  $M_0$ , respectively. This is only useful if the scale parameter  $\phi$  is known, so that  $D^*$  can be calculated. Note that while dividing by  $\phi$  in  $D^* = D/\phi$  might suggest that the scaled deviance is free of  $\phi$ , this is not the case:  $\phi$  also enters through the log-likelihood contributions  $\log f(y_i; \theta_i, \phi)$ , and so  $D^*$  remains dependent on  $\phi$  (see Equation (30)). When  $\phi$  is unknown, one could substitute an estimate  $\hat{\phi}$  into  $D_j^*$  to obtain  $\hat{D}_j^*$ ,  $j \in \{0, 1\}$ . However, a preferable approach is to construct a test in which  $\phi$  cancels in the ratio, avoiding the need to estimate it separately, as presented next. Under  $H_0$ , the two following approximate results hold

$$D_0^* - D_1^* \sim \chi^2_{p_1 - p_0} \quad \text{and} \quad D_1^* \sim \chi^2_{n - p_1},$$

and, if  $D_0^* - D_1^*$  and  $D_1^*$  are assumed to be asymptotically independent, this implies that

$$\begin{aligned} F &= \frac{(D_0^* - D_1^*)/(p_1 - p_0)}{D_1^*/(n - p_1)} \\ &= \frac{(D_0 - D_1)/\phi/(p_1 - p_0)}{D_1/\phi/(n - p_1)} \\ &= \frac{(D_0 - D_1)/(p_1 - p_0)}{D_1/(n - p_1)} \sim F_{p_1 - p_0, n - p_1}, \end{aligned} \tag{33}$$

in the large sample limit. This result holds exactly in the special case of the ordinary linear model (that is, for a Gaussian response with the identity link function). The main drawback of this test is that it relies on the same questionable large-sample approximation for  $D_1^*$  discussed earlier.

Further, as in the ordinary linear model setting, AIC provides an alternative approach to model selection that does not require the models being compared to be nested.

The deviance can also be used to define alternative scaled residuals to the Pearson residuals. In practice, the distribution of the Pearson residuals can often be quite asymmetric around zero, so that their behaviour is not as close to that of the residuals in the ordinary linear model as might be hoped. In this respect, the *deviance residuals* are often preferable. The deviance residuals are constructed by analogy with the ordinary linear model. Recall that for the Gaussian linear model, as seen in (31),

$$D = \sum_{i=1}^n (y_i - \hat{\mu}_i)^2 = \sum_{i=1}^n \hat{\epsilon}_i^2,$$

so that the deviance is simply the sum of squared residuals. This suggests defining residuals in generalised linear models as the signed square roots of the individual contributions to the deviance. Let  $d_i$  denote the contribution of the  $i$ th observation to the deviance (i.e., the  $i$ th term in the summation in (30)). Then

$$D = \sum_{i=1}^n d_i,$$

and, by analogy with the ordinary linear model, the deviance residuals are defined as

$$\hat{\epsilon}_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}.$$

By construction, the sum of squares of the deviance residuals equals the deviance itself. This definition also has useful distributional consequences. From (32), we know that the following distributional result holds, asymptotically, for the scaled deviance

$$D^* \sim \chi_{n-p}^2.$$

If all parameters in the linear predictor were known, then  $D^*$  would be asymptotically distributed as  $\chi_n^2$ . By a heuristic argument, this suggests that, for a single observation,  $d_i/\phi$  behaves approximately like a  $\chi_1^2$  variable. Since a  $\chi_1^2$  random variable is the square of a standard normal variable, taking the signed square root gives  $\epsilon_i^D/\sqrt{\phi} \approx N(0, 1)$ , or equivalently,  $\epsilon_i^D \approx N(0, \phi)$ . This approximate normality makes deviance residuals useful for assessing how well the assumed distribution of the response variable fits the data. The quality of this approximation depends on several factors, including the form of the response distribution, the sample size, and the range of fitted values. It is however possible to observe severe departures from normality even if the model is perfectly correct. See some examples in Section 5.7.2:

<https://egarpor.github.io/PM-UC3M/glm.html>

## 12 glm in R

The `glm` function is used to fit GLMs in R. Its use is similar to that of the `lm` function, with two main differences. The right-hand side of the model formula specifies the linear predictor, which is related to the mean of the response through a link function. In addition, `glm` takes a `family` argument that specifies the distribution from the exponential family and the `link` function used in the model.

### 12.1 Hen Harrier example

To see the `glm` function in action, we revisit the Hen Harrier data. Recall that we had

$$\mathbb{E}(c_i) \equiv \mu_i = \frac{\alpha d_i^3}{\delta + d_i^3},$$

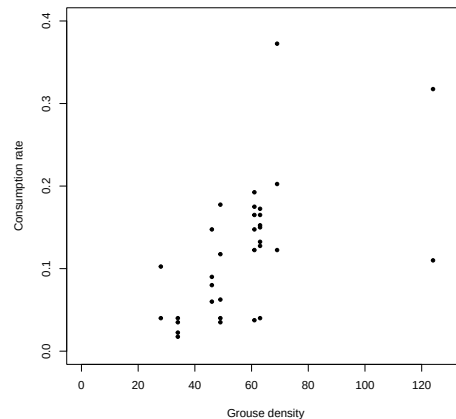
where  $c_i$  denotes the consumption rate of Grouse by Hen Harriers (per day) and  $d_i$  is the corresponding Grouse density (per km<sup>2</sup>). We previously observed that, by using the inverse link function, the right-hand side of this equation can be expressed linearly in the parameters:

$$\frac{1}{\mu_i} = \beta_1 + \beta_2 \frac{1}{d_i^3}, \quad \beta_1 = \frac{1}{\alpha}, \quad \beta_2 = \frac{\delta}{\alpha}.$$

We also argued that the Gamma distribution is a plausible choice for modeling the consumption rate. The data are available in the `gamair` package, which must first be installed. First, let us load and plot the data.

```
library(gamair)
data(harrier)
# Remove entries with zero consumption rate
harrier <- harrier[harrier$Consumption.Rate!=0, ]

plot(harrier$Grouse.Density, harrier$Consumption.Rate,
     ylim = c(0, 0.4), xlim = c(0, 130),
     xlab = "Grouse density", ylab = "Consumption rate", pch = 20)
```



Now fit the model:

```
hm <- glm(Consumption.Rate ~ I(1/Grouse.Density^3),
         family = Gamma(link = "inverse"), data = harrier)
```

The argument `family` specifies the distribution (here Gamma) and the link function to use (here "inverse"). Let us inspect the summary of the fitted model, stored in the object `hm`.

```
summary(hm)

##
## Call:
## glm(formula = Consumption.Rate ~ I(1/Grouse.Density^3), family = Gamma(link = "inverse"),
##     data = harrier)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.676e+00  1.156e+00   4.046 0.000321 ***
## I(1/Grouse.Density^3) 5.386e+05  1.655e+05   3.255 0.002741 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.347565)
##
##      Null deviance: 16.166  on 32  degrees of freedom
## Residual deviance: 10.472  on 31  degrees of freedom
## AIC: -92.383
##
## Number of Fisher Scoring iterations: 6
```

A few key points emerge from the summary of the fitted model. We know that for the Gamma distribution the dispersion parameter is not equal to one and must be estimated. One possible estimator, the only one covered in these notes, is given by Equation (28), which is the estimator implemented in the `glm` function. Let us verify that this is indeed the case.

```
sum(residuals(hm, type = "pearson")^2)/(nrow(harrier) - 2)

## [1] 0.347565

summary(hm)$dispersion

## [1] 0.347565
```

To obtain the deviance residuals, simply change the `type` argument in the `residuals` function from `pearson` to `deviance`. As a simple check, we can also compute the Pearson residuals by hand; remember that for the Gamma distribution we have  $\omega = 1$  and  $V(\mu) = \mu^2$ .

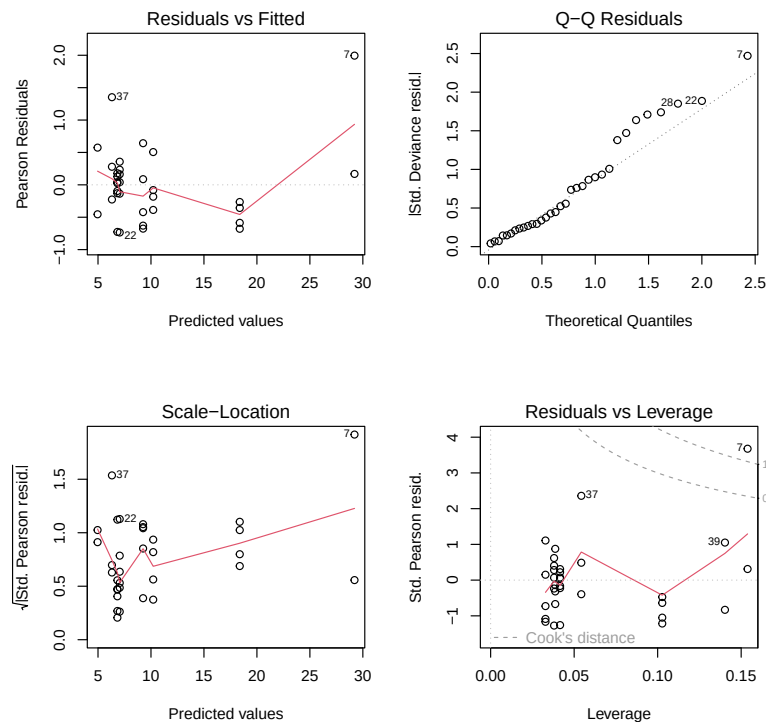
```
residuals_pearson <- residuals(hm, type = "pearson")
fitted_values <- fitted(hm)
residuals_pearson_alt <- (harrier$Consumption.Rate - fitted_values)/sqrt(fitted_values^2)
range(residuals_pearson - residuals_pearson_alt)

## [1] 0 0
```

Returning to the summary of the `glm` function: the residual deviance represents the deviance of the fitted model, while the null deviance corresponds to the deviance of a model with only an intercept term.

As with any statistical model, we should examine residual plots before proceeding.

```
par(mfrow = c(2, 2))
plot(hm)
```



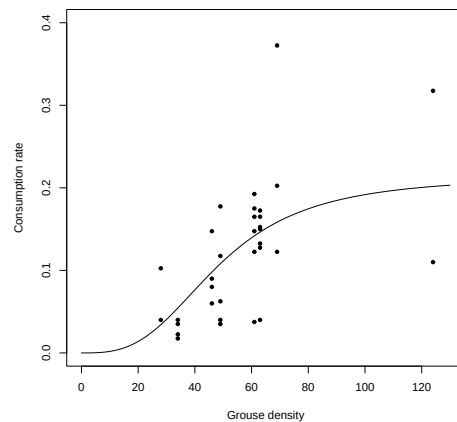
These residual plots are interpreted in much the same way as the equivalent plots for a linear model, with two important differences: i) if the response distribution is not normal (Gaussian), then we do not expect the normal QQ plot to display an exact straight line, and (ii) if the response variable is binary, diagnostic checking becomes more difficult, and many of the standard residual plots provide limited information.

For the harrier model there are some clear patterns in the residuals, with observations at some densities lying systematically above or below zero. This suggests that the fitted model may not adequately capture the relationship between consumption rate and Grouse density. For this simple example, it is helpful to plot the model predictions and the observed data together. To obtain predictions at a range of Grouse densities, we can use the `predict` function.

```
pd <- data.frame(Grouse.Density = 0:130) # Data at which to predict
fv <- predict(hm, pd, type = "response") # Get model predictions
```

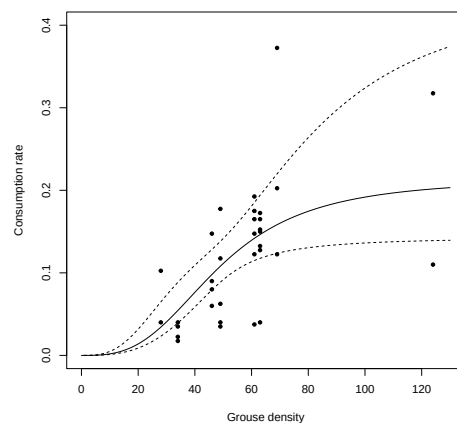
Note the `type` argument to `predict`, which indicates that we want predictions on the original response scale rather than on the link-transformed scale (the default). We can now produce a plot.

```
plot(harrier$Grouse.Density, harrier$Consumption.Rate,
     ylim = c(0, 0.4), xlim = c(0, 130),
     xlab = "Grouse density", ylab = "Consumption rate", pch = 20)
lines(0:130, fv)
```



It is also useful to add approximate 95% CIs for the mean to the plot. Again `predict` is used, but this time standard errors for each prediction are also produced (on the link transformed scale, requiring a back-transformation to the original scale). Note that, unlike what happens with an `lm` object, there is no `interval` argument when using `predict` with a `glm` object. The following adds CIs to the plot.

```
lp <- predict(hm, pd, se = TRUE)
plot(harrier$Grouse.Density, harrier$Consumption.Rate,
     ylim = c(0, 0.4), xlim = c(0, 130),
     xlab = "Grouse density", ylab = "Consumption rate", pch = 20)
lines(0:130, fv)
# Phi was estimated so we need to use a t distribution
quant_t <- qt(0.975, df = (nrow(harrier) - 2))
lines(0:130, 1/(lp$fit + quant_t*lp$se.fit), lty = 2)
lines(0:130, 1/(lp$fit - quant_t*lp$se.fit), lty = 2)
```



From this plot it becomes clear why patterns appear in the residuals and only a very complex dependence of consumption rate on density would remove them. Since the different densities correspond to different grouse moors, it is likely that there are additional moor-to-moor differences that should be included in the model, perhaps related to the availability of alternative food sources for Hen Harriers at each moor. Clearly, further modelling work is required.

Is the inverse-cubic dependence on density the best model for these data? There are various ways of addressing this question, but one of the simplest is to try alternative powers and compare the AIC values of the resulting model fits. For example:

```

hm1 <- glm(Consumption.Rate ~ I(1/Grouse.Density),
           family = Gamma(link = "inverse"), data = harrier)

hm2 <- glm(Consumption.Rate ~ I(1/Grouse.Density^2),
           family = Gamma(link = "inverse"), data = harrier)
AIC(hm, hm1, hm2)

##      df      AIC
## hm    3 -92.38289
## hm1   3 -92.63887
## hm2   3 -94.17016

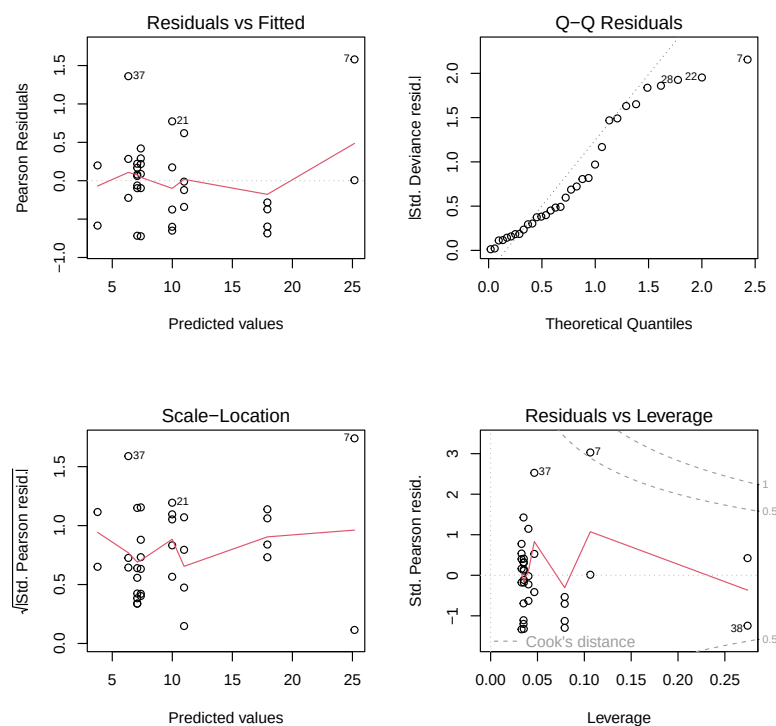
```

Out of curiosity, df represents the number of parameters contributing to the AIC. Here this is three: two regression coefficients and, for the Gamma model, the dispersion parameter. The AIC slightly favours the model with a quadratic dependence on density. Note that this model also has ‘better’ diagnostic plots, although some minor issues remain.

```

par(mfrow = c(2, 2))
plot(hm2)

```



Before proceeding, we list the link functions implemented in `glm` for the distribution families that we will be using.

```

##
## **Family:** gaussian
## Canonical link function: identity
## Available link functions: identity, log, inverse
##
## **Family:** binomial
## Canonical link function: logit
## Available link functions: logit, probit, cloglog

```

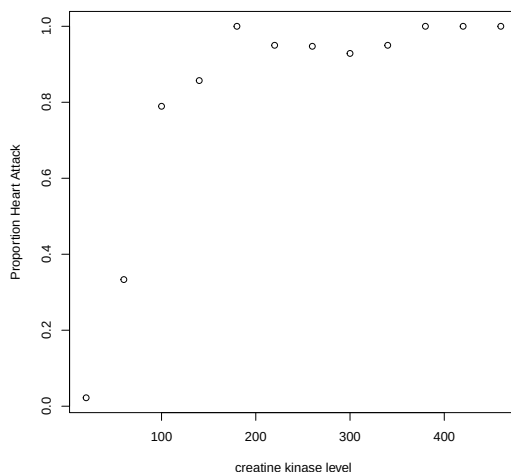
```
##
## **Family:** poisson
## Canonical link function: log
## Available link functions: log, identity, sqrt
##
## **Family:** Gamma
## Canonical link function: inverse
## Available link functions: inverse, identity, log
##
## **Family:** inverse.gaussian
## Canonical link function: 1/mu^2
## Available link functions: 1/mu^2, inverse, identity, log
```

## 12.2 Heart attack example

We revisit the heart attack example from Section 10.1. The goal is to construct a model the probability of a heart attack as a function of the creatine kinase level. As usual, we begin by reading in the data and plotting them.

```
ck<- c(20 , 60 , 100 , 140 , 180 , 220 , 260 , 300 , 340 , 380 , 420 , 460)
ha <- c(2 , 13 , 30 , 30 , 21 , 19 , 18 , 13 , 19 , 15 , 7 , 8 )
ok <- c( 88 , 26 , 8 , 5 , 0 , 1 , 1 , 1 , 1 , 0 , 0 , 0)
heart <- data.frame(ck = ck, ha = ha, ok = ok)

p <- heart$ha/(heart$ha + heart$ok)
plot(heart$ck, p, xlab = "creatine kinase level",
     ylab = "Proportion Heart Attack")
```



Recall that our basic model is

$$y_i \sim \text{Binomial}(N_i, p_i),$$

where  $y_i$  is the number of patients suffering a heart attack among  $N_i$  patients with CK level  $x_i$ , and  $p_i$  is the probability of a heart attack at CK level  $x_i$

$$p_i = \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}.$$

We have that  $\mathbb{E}(y_i) \equiv \mu_i = N_i p_i$  and

$$g(\mu_i) = \log\left(\frac{\mu_i}{N_i - \mu_i}\right) = \log\left(\frac{p_i}{1 - p_i}\right) = \text{logit}(p_i) = \eta_i = \beta_1 + \beta_2 x_i.$$

The logit link is the canonical link function for binomial models and therefore the default in R. When using binomial models, we need to supply the model fitting function with information about  $N_i$  as well as  $y_i$ . In R, there are two ways to specify binomial models using the `glm` function:

1. The response variable can be the observed proportion of successful binomial trials, in which case a vector giving the number of trials must be supplied as the `weights` argument to `glm`. For binary data, no weights vector need to be supplied, as the default weights of 1 are sufficient.
2. The response variable can be supplied as a two column array, in which the first column gives the number of binomial ‘successes’ and the second column gives the number of binomial ‘failures’.

Let us illustrate how to do this.

```
# Approach 1
mod.0 <- glm(heart$ha/(heart$ha + heart$ok) ~ ck,
             family = binomial(link = logit),
             weights = heart$ha + heart$ok,
             data = heart)

mod.0

##
## Call:  glm(formula = heart$ha/(heart$ha + heart$ok) ~ ck, family = binomial(link = logit),
##        data = heart, weights = heart$ha + heart$ok)
##
## Coefficients:
## (Intercept)          ck
##   -2.75836       0.03124
##
## Degrees of Freedom: 11 Total (i.e. Null);  10 Residual
## Null Deviance:      271.7
## Residual Deviance: 36.93  AIC: 62.33

# Approach 2
mod.0 <- glm(cbind(ha,ok) ~ ck,
             family = binomial(link = logit),
             data = heart)

mod.0

##
## Call:  glm(formula = cbind(ha, ok) ~ ck, family = binomial(link = logit),
##        data = heart)
##
## Coefficients:
## (Intercept)          ck
##   -2.75836       0.03124
##
## Degrees of Freedom: 11 Total (i.e. Null);  10 Residual
## Null Deviance:      271.7
## Residual Deviance: 36.93  AIC: 62.33
```

Note that specifying two arrays on the right-hand side of the model formula requires the use of `cbind`. Also note that we could have omitted `link = logit` since the logit link is the default for the binomial family. As an example, the null deviance (from a model with only an intercept) and the residual deviance (from the fitted model) can be combined to give the *proportion of deviance explained*, a generalisation of  $r^2$ , as follows:

```
1 - (summary(mod.0)$deviance/summary(mod.0)$null.deviance)

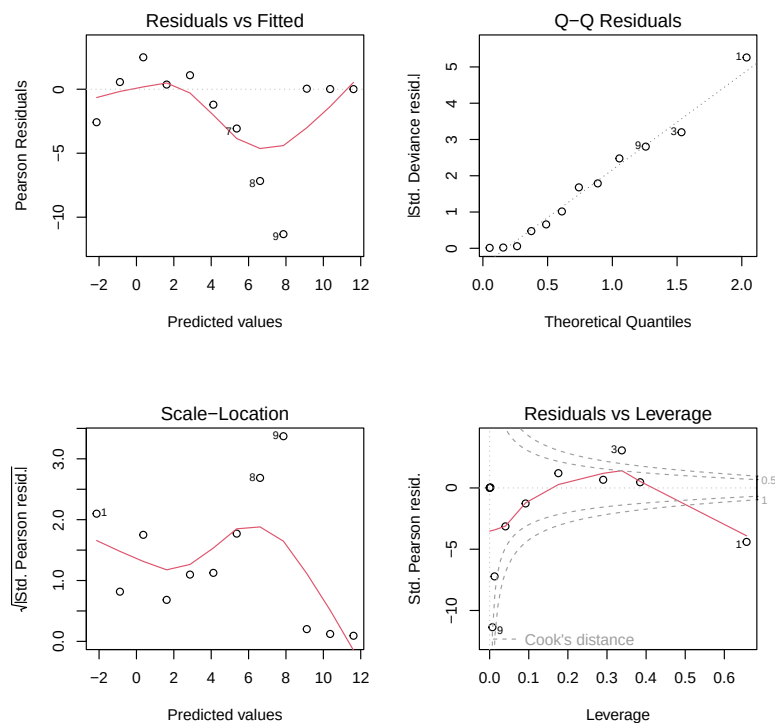
## [1] 0.8640893
```

Notice that the deviance is quite high for the  $\chi^2_{10}$  random variable that it should approximate if the model is fitting well. In fact

```
pchisq(summary(mod.0)$deviance, df = (nrow(heart) - 2), lower.tail = FALSE)
## [1] 5.822516e-05
```

shows that there is a *very* small probability of a  $\chi^2_{10}$  random variable being as large as 36.93. The residual plots also suggest a poor fit.

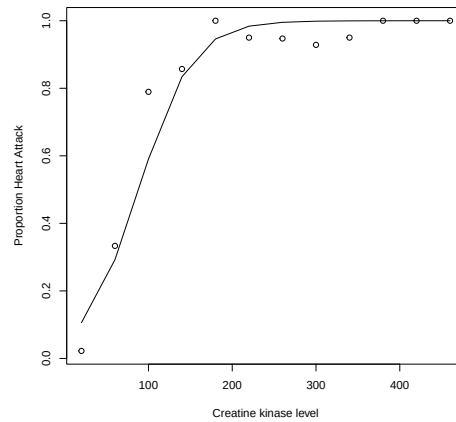
```
par(mfrow = c(2, 2))
plot(mod.0)
```



Again, the plots have much the same interpretation as the model checking plots for an ordinary linear model, except that it is now standardised residuals that are plotted, the Predicted values are on the scale of the linear predictor rather than the response, and some departure from a straight line relationship in the Normal QQ plot is often to be expected. The plots are not easy to interpret when there are so few data, but there appears to be a trend in the mean of the residuals plotted against predicted values, which would cause concern. Furthermore, the first point has very high influence.

Notice how the problems do not stand out so clearly from a plot of the fitted values overlaid on the observed proportions.

```
plot(heart$ck, p,
     xlab = "Creatine kinase level",
     ylab = "Proportion Heart Attack")
lines(heart$ck, fitted(mod.0))
```



Note also that the fitted values provided by the `glm` function for binomial models are the estimated  $p_i$  values, rather than the estimated  $\mu_i$  values ( $= p_i N_i$ ).

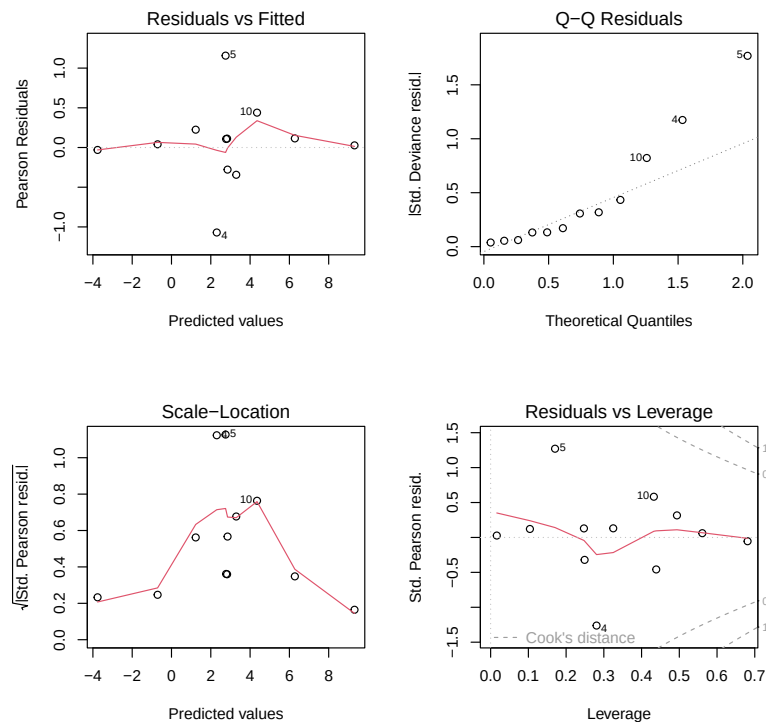
The trend in the mean of the residuals suggests trying a cubic model instead of the initial straight-line model:

$$\text{logit}(p_i) = \beta_1 + \beta_2 x_i + \beta_3 x_i^2 + \beta_4 x_i^3.$$

```
mod.1 <- glm(cbind(ha,ok) ~ ck + I(ck^2) + I(ck^3),
             family = binomial, data = heart)
mod.1

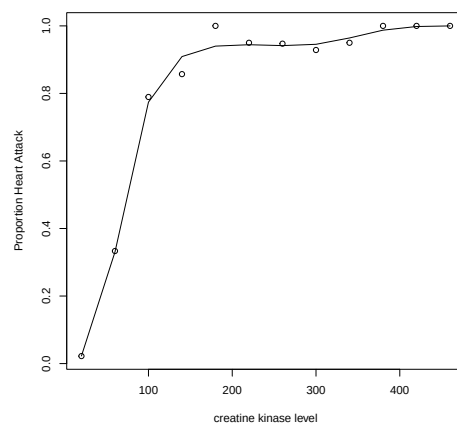
##
## Call:  glm(formula = cbind(ha, ok) ~ ck + I(ck^2) + I(ck^3), family = binomial,
##       data = heart)
##
## Coefficients:
## (Intercept)          ck       I(ck^2)       I(ck^3)
## -5.786e+00    1.102e-01   -4.649e-04    6.448e-07
##
## Degrees of Freedom: 11 Total (i.e. Null);  8 Residual
## Null Deviance:      271.7
## Residual Deviance:  4.252  AIC: 33.66

par(mfrow = c(2, 2))
plot(mod.1)
```



The value 4.252 is not large relative to a  $\chi^2_8$  distribution (it is in fact smaller than the expected value) and the AIC has improved substantially. The residual plots now show less clear patterns than for the previous model, although with more data such a departure from the assumed binomial mean- variance relationship would be a cause for concern. Furthermore, the fitted curve is now clearly closer to the observed data.

```
plot(heart$ck, p, xlab = "creatinine kinase level",
     ylab = "Proportion Heart Attack")
lines(heart$ck, fitted(mod.1))
```



We can also use R to test the null hypothesis that `mod.0` is compatible with the data against the alternative that `mod.1` is required. Somewhat confusingly, the `anova` function is used for this, even though it performs a generalized likelihood ratio test rather than an analysis of variance.

```
anova(mod.0, mod.1, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: cbind(ha, ok) ~ ck
## Model 2: cbind(ha, ok) ~ ck + I(ck^2) + I(ck^3)
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1         10      36.929
## 2          8       4.252  2   32.676 8.025e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Such a small p-value provides very strong evidence against the null hypothesis: we reject the hypothesis that  $\beta_3$  and  $\beta_4$  are both zero. Of course, we could still test whether the cubic effect is necessary or if a quadratic effect of CK suffices.

```
mod.2 <- glm(cbind(ha,ok) ~ ck + I(ck^2),
             family = binomial, data = heart)
anova(mod.2, mod.1, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: cbind(ha, ok) ~ ck + I(ck^2)
## Model 2: cbind(ha, ok) ~ ck + I(ck^2) + I(ck^3)
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1          9      15.4102
## 2          8       4.2525  1   11.158 0.0008368 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The cubic effect is indeed necessary.

## 12.3 Melanoma case-control study

*Case-control studies* are an important study design in which a group of patients with a disease (the cases) is compared with a group of subjects without the disease (the controls), typically sampled from the same underlying population. Variables that may be associated with the disease are collected for all subjects. If a variable is associated with the disease, it should help predict whether a subject in the study is a case or a control. Such predictive ability can be assessed using generalized linear models of the logistic regression type.

For example, consider a study of 143 cases of malignant melanoma (a serious form of skin cancer) in white male subjects aged 25–55, classified according to skin type (A, B, or C, corresponding to Celtic, Middle European, or Mediterranean types), and compared with 356 white male controls aged 25–55, selected without matching on age or skin type. Subjects were divided into three age groups (25–35, 35–45, 45–55) and three skin type groups (A, B, C), giving nine groups in total. The data are in a data frame called `md`:

```
md <- data.frame(mel = c(15, 8, 7, 26, 18, 6, 30, 25, 8),
                 n = c(54, 52, 44, 75, 52, 42, 67, 66, 47),
                 age = rep(c("25-35", "35-45", "45-55"), each = 3),
                 skin = rep(c("A", "B", "C"), 3),
                 stringsAsFactors = TRUE)

md

##   mel  n  age skin
## 1  15 54 25-35   A
## 2   8 52 25-35   B
## 3   7 44 25-35   C
## 4  26 75 35-45   A
```

```
## 5 18 52 35-45 B
## 6 6 42 35-45 C
## 7 30 67 45-55 A
## 8 25 66 45-55 B
## 9 8 47 45-55 C
```

In `md`, the variable `mel` records the number of melanoma cases and `n` the total number of subjects (cases plus controls) in each age and skin type group. Skin type is categorical with no natural ordering, so it is treated as a factor variable. Although age groups have a natural ordering, we treat age as an unordered factor here.

Let  $y_{jk}$  be the number of melanoma cases out of  $n_{jk}$  subjects (cases + controls) in age group  $j$  and skin type  $k$ , where  $j \in \{1, 2, 3\}$  corresponds to the age groups 25–35, 35–45, and 45–55, respectively, and  $k \in \{A, B, C\}$  corresponds to skin types A, B, and C, respectively. We assume

$$y_{jk} \sim \text{Binomial}(n_{jk}, p_{jk}),$$

where  $p_{jk}$  is the probability that a subject in the study sample with age group  $j$  and skin type  $k$  is a case. To test the null hypothesis that skin type is not associated with melanoma, against the alternative that it is, we compare the reduced model

$$\text{logit}(p_{jk}) = \mu + \alpha_j,$$

to the alternative full model

$$\text{logit}(p_{jk}) = \mu + \alpha_j + \beta_k. \quad (34)$$

These models can be compared using a generalized likelihood ratio test (analysis of deviance).

```
mel0 <- glm(mel/n ~ age, family = binomial, data = md, weights = n)
mel1 <- glm(mel/n ~ age + skin, family = binomial, data = md, weights = n)
anova(mel0, mel1, test = "Chisq")

## Analysis of Deviance Table
##
## Model 1: mel/n ~ age
## Model 2: mel/n ~ age + skin
##      Resid. Df Resid. Dev Df Deviance Pr(>Chi)
## 1          6    20.5062
## 2          4     3.4389  2    17.067 0.0001967 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note that `test = "Chisq"` specifies that a generalized likelihood ratio test is performed using (22). The p-value is very small, meaning that, if `mel0` were the true model, the probability of observing a difference in deviance (or, equivalently, a likelihood ratio statistic) at least as large as the one observed would be very low. This suggests that the reduced model `mel0` is inadequate compared to the full model `mel1`, providing strong evidence for an effect of skin type on the odds of being a case. As an alternative to the `anova` function, we could also use the `drop1` function with the argument `test = "Chisq"`.

```
drop1(mel1, test = "Chisq")

## Single term deletions
##
## Model:
## mel/n ~ age + skin
##      Df Deviance   AIC    LRT  Pr(>Chi)
## <none>      3.4389 50.569
## age      2  12.3876 55.517  8.9487 0.0113976 *
## skin     2  20.5062 63.636 17.0673 0.0001967 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Let us look at the residual deviance of model `me11`.

```
deviance(me11)
## [1] 3.438926
```

The residual deviance can be compared approximately to a chi-squared distribution with 4 degrees of freedom, since the residual degrees of freedom are equal to the number of groups minus the number of identifiable parameters, that is,  $9 - 5 = 4$ .

```
pchisq(deviance(me11), df = 4, lower.tail = FALSE)
## [1] 0.4872253
```

Since the p-value is not small, there is no evidence of lack of fit, and the model appears to provide an adequate fit to the data.

The next step is to examine the model coefficients to determine the direction and magnitude of the effect. We use this example to illustrate how to interpret the coefficients in a logistic regression model. We first note that we need to impose identifiability constraints. By default R sets the first level of each factor variable as the baseline and the coefficients corresponding to the baseline levels are set to zero. In this case, the first levels are skin type A and age group 25–35. Let us verify that the baseline levels are indeed skin type A and age group 25–35.

```
levels(md$age)
## [1] "25-35" "35-45" "45-55"

levels(md$skin)
## [1] "A" "B" "C"
```

Alternatively, we could check the model matrix.

```
model.matrix(me11)

##      (Intercept) age35-45 age45-55 skinB skinC
## 1             1         0         0     0     0
## 2             1         0         0     1     0
## 3             1         0         0     0     1
## 4             1         1         0     0     0
## 5             1         1         0     1     0
## 6             1         1         0     0     1
## 7             1         0         1     0     0
## 8             1         0         1     1     0
## 9             1         0         1     0     1
## attr(,"assign")
## [1] 0 1 1 2 2
## attr(,"contrasts")
## attr(,"contrasts")$age
## [1] "contr.treatment"
##
## attr(,"contrasts")$skin
## [1] "contr.treatment"
```

So in model (34) we set  $\alpha_1 = \beta_A = 0$ . These identifiability constraints are imposed automatically by R when fitting the model via `glm`.

What is the interpretation of  $\alpha_2$ ? It represents the difference in log-odds of melanoma between individuals in the age group 35–45 and those in the reference age group (25–35), for individuals with the same skin type. To see

why, and for the sake of concreteness, assume that the skin type for both groups is B. We have that:

$$\begin{aligned}\text{logit}(p_{2B}) &= \log\left(\frac{p_{2B}}{1-p_{2B}}\right) = \mu + \alpha_2 + \beta_B, \\ \text{logit}(p_{1B}) &= \log\left(\frac{p_{1B}}{1-p_{1B}}\right) = \mu + \alpha_1 + \beta_B = \mu + \beta_B.\end{aligned}$$

Therefore,

$$\log\left(\frac{p_{2B}}{1-p_{2B}}\right) - \log\left(\frac{p_{1B}}{1-p_{1B}}\right) = (\mu + \alpha_2 + \beta_B) - (\mu + \beta_B) = \alpha_2.$$

The same calculation with skin type A or C yields  $\alpha_2$ , confirming that the result holds for any skin type. The key point is that the two groups being compared share the same skin type and that there is no interaction between age and skin type. Furthermore,

$$\log\left(\frac{p_{2B}}{1-p_{2B}}\right) - \log\left(\frac{p_{1B}}{1-p_{1B}}\right) = \alpha_2 \Rightarrow \log\left(\frac{\frac{p_{2B}}{1-p_{2B}}}{\frac{p_{1B}}{1-p_{1B}}}\right) = \alpha_2 \Rightarrow \frac{p_{2B}}{1-p_{2B}} = \frac{p_{1B}}{1-p_{1B}} e^{\alpha_2}.$$

That is, keeping skin type the same, the odds of melanoma for individuals in age group 35–45 are  $e^{\alpha_2}$  times the odds for those in the reference age group (25–35). Similarly, the odds for individuals in age group 45–55 are  $e^{\alpha_3}$  times those for the reference age group, holding skin type constant. Applying the same reasoning, but now keeping the age group constant, the odds of melanoma for individuals with skin type B are  $e^{\beta_B}$  times the odds for those in the reference skin type group (skin type A). Finally, the odds of melanoma for individuals with skin type C are  $e^{\beta_C}$  times the odds for those in the reference skin type group, holding age group constant. Thus, in a logistic regression model without interactions, exponentiating a coefficient associated with a factor level gives the odds ratio comparing that level with the reference level, holding the other variables constant. Let us examine the coefficients.

```

mell

##
## Call: glm(formula = mel/n ~ age + skin, family = binomial, data = md,
##          weights = n)
##
## Coefficients:
## (Intercept)    age35-45    age45-55      skinB      skinC
##      -1.0228      0.4766      0.7655     -0.2881     -1.1063
##
## Degrees of Freedom: 8 Total (i.e. Null);  4 Residual
## Null Deviance:      29.89
## Residual Deviance:  3.439  AIC: 50.57

exp(coefficients(mell))

## (Intercept)    age35-45    age45-55      skinB      skinC
##    0.3595763    1.6106085    2.1499647    0.7496932    0.3307686

```

We conclude that, holding skin type constant, the estimated odds ratio for melanoma comparing the 35–45 age group to the baseline age group (25–35) is  $e^{\hat{\alpha}_2} = 1.61$ . This indicates that the odds of melanoma for those in the 35–45 age group are 1.61 times those in the baseline age group. The increase in odds of melanoma is even greater when comparing those in the 45–55 age group to the baseline age group, as  $e^{\hat{\alpha}_3} = 2.15$ , again assuming that skin type is the same in both age groups. On the other hand, the estimated odds ratio for melanoma comparing individuals with skin type B to those with skin type A is 0.75, indicating lower odds of disease. For skin type C compared with skin type A, the estimated odds ratio is 0.33, indicating an even greater reduction in the odds of melanoma.

Of course, we should attach confidence intervals to these estimates and we are already in a position to do so. From (26), we know that the maximum likelihood estimators of the parameters follow, in the large sample

limit, a multivariate normal distribution with a mean vector equal to the true (but unknown) parameter vector and covariance matrix given by  $\phi(\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}$ . Since, for the binomial distribution,  $\phi = 1$ , this result can be used directly to compute confidence intervals for the parameters. For instance, a  $100(1 - \alpha)\%$ <sup>43</sup> confidence interval for  $\alpha_2$  is

$$(\hat{\alpha}_2 - z_{1-\alpha/2} \sigma_{\hat{\alpha}_2}, \hat{\alpha}_2 + z_{1-\alpha/2} \sigma_{\hat{\alpha}_2}),$$

where  $\sigma_{\hat{\alpha}_2}^2$  is the corresponding diagonal element of the covariance matrix  $(\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1}$ . Since the exponential function is monotonic, a confidence interval for  $e^{\alpha_2}$  can be obtained by exponentiating the endpoints of the confidence interval for  $\alpha_2$ , giving

$$(e^{\hat{\alpha}_2 - z_{1-\alpha/2} \sigma_{\hat{\alpha}_2}}, e^{\hat{\alpha}_2 + z_{1-\alpha/2} \sigma_{\hat{\alpha}_2}}).$$

All of these quantities can be easily obtained from the summary output.

```
summary(mell)

##
## Call:
## glm(formula = mel/n ~ age + skin, family = binomial, data = md,
##      weights = n)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.0228      0.2355  -4.343 1.41e-05 ***
## age35-45       0.4766      0.2689   1.773 0.07630 .
## age45-55       0.7655      0.2611   2.932 0.00337 **
## skinB         -0.2881      0.2267  -1.271 0.20381
## skinC         -1.1063      0.2829  -3.911 9.19e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 29.8893  on 8  degrees of freedom
## Residual deviance:  3.4389  on 4  degrees of freedom
## AIC: 50.569
##
## Number of Fisher Scoring iterations: 4
```

Thus, we can easily compute a 95% CI for  $e^{\alpha_2}$ .

```
lower_bound <- exp(0.4766 - qnorm(0.975)*0.2689)
upper_bound <- exp(0.4766 + qnorm(0.975)*0.2689)
lower_bound; upper_bound

## [1] 0.9508164
## [1] 2.728179
```

Confidence intervals for the other parameters can be obtained in a similar manner. In fact, we can compute all intervals automatically using the `confint.default` function.

```
exp(confint.default(mell))

##              2.5 %      97.5 %
## (Intercept) 0.2266304 0.5705107
## age35-45    0.9508683 2.7280957
## age45-55    1.2887860 3.5865911
## skinB       0.4807427 1.1691073
## skinC       0.1899965 0.5758417
```

<sup>43</sup>Here  $\alpha$  denotes the significance level, not the model parameters ( $\alpha_1$ ,  $\alpha_2$ , and  $\alpha_3$ ).

Finally, case-control studies allow for the estimation of all regression coefficients with the exception of the intercept, which in this case corresponds to the log odds of melanoma for individuals in the 25–35 age group with skin type A. This is because in a case–control study the ratio of cases to controls is fixed by the investigator rather than determined by the population disease prevalence. As a result, the intercept absorbs an arbitrary sampling fraction and does not represent the population log odds of melanoma for the baseline group. As a consequence, we cannot estimate the probabilities of disease for each group from a case–control study

$$p_{jk} = \frac{e^{\mu + \alpha_j + \beta_k}}{1 + e^{\mu + \alpha_j + \beta_k}}, \quad j \in \{1, 2, 3\}, \quad k \in \{A, B, C\}, \quad \alpha_1 = \beta_A = 0,$$

as they involve  $\mu$ . The other regression coefficients, being log odds ratios, can be estimated regardless of the study design. Furthermore, estimates of the standard errors of the estimated slope coefficients and hypothesis testing techniques, such as the generalized likelihood ratio test, can still be used with case-control data. Note that although R returns an estimate of  $\mu$ , this estimate should not be interpreted, since R does not know that the data arise from a case–control study.

## 12.4 Modelling rates: including an offset in the model

Often, the expected value of a response count  $y_i$  is proportional to a known index, say  $o_i$ , usually referred to as an exposure variable. For instance,  $o_i$  might represent an amount of time or a population size, such as when modelling COVID-19 deaths across different countries. Alternatively, it could be a spatial area, such as when modelling the counts of a particular animal or plant species. In such cases, the observed rate is  $y_i/o_i$ , with expected value  $\mathbb{E}(y_i/o_i) = \mathbb{E}(y_i)/o_i = \mu_i/o_i$ . When predictor variables are included, a log-linear model for the expected rate takes the form

$$\log\left(\frac{\mu_i}{o_i}\right) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, n,$$

giving

$$\log \mu_i = \log o_i + \mathbf{x}_i^\top \boldsymbol{\beta}.$$

The adjustment term  $\log o_i$  is called an *offset*. The fit corresponds to including  $\log o_i$  as a predictor in the linear predictor for  $\log \mu_i$ , with its coefficient fixed at one rather than estimated from the data. Note that offsets, while more commonly used in Poisson regression, are not exclusive to it; they are also occasionally used, for example, in Gamma regression.

This example is from Agresti (2015, Chapter 7), who reused data from Holford (1980) on the survival and death of 539 males diagnosed with lung cancer. Two factors are of interest: histology and stage of disease, each grouped into three categories. The follow-up period was divided into a series of two-month intervals. For each combination of follow-up interval, histology, and stage of disease, the number of deaths and the total follow-up time (in months) were recorded. We treat the death counts for each cell (i.e., each combination of factors and follow-up interval) as independent Poisson variates.

```
url <- "https://raw.githubusercontent.com/alanagresti/categorical-data/master/Cancer.dat"
Cancer <- read.table(url, header = TRUE, sep = "")
head(Cancer)

##   time histology stage count risktime
## 1    1         1     1     9     157
## 2    1         2     1     5      77
## 3    1         3     1     1     21
## 4    2         1     1     2    139
## 5    2         2     1     2     68
## 6    2         3     1     1     17

tail(Cancer)

##   time histology stage count risktime
## 58    6         1     3     4     39
```

|    |    |   |   |   |   |    |
|----|----|---|---|---|---|----|
| ## | 59 | 6 | 2 | 3 | 3 | 13 |
| ## | 60 | 6 | 3 | 3 | 3 | 14 |
| ## | 61 | 7 | 1 | 3 | 1 | 29 |
| ## | 62 | 7 | 2 | 3 | 2 | 7  |
| ## | 63 | 7 | 3 | 3 | 3 | 10 |

Note that time 1 corresponds to the follow-up interval 0–2 months after diagnosis, time 2 to 2–4 months, and so on, with time 7 corresponding to 12 or more months after diagnosis.

Let  $\mu_{ijk}$  denote the expected number of deaths and  $o_{ijk}$  the total time at risk for histology  $i$  and stage of disease  $j$  in follow-up time interval  $k$ , for  $i \in \{1, 2, 3\}$ ,  $j \in \{1, 2, 3\}$ , and  $k \in \{1, 2, \dots, 7\}$ . The Poisson log-linear model for the death rate is

$$\log \mu_{ijk} = \log o_{ijk} + \beta_0 + \beta_i^H + \beta_j^S + \beta_k^T.$$

Each predictor is treated as a factor, where the superscript notation indicates the classification labels (**H**istology, **S**tage, **T**ime). For identifiability reasons,  $\beta_1^H = \beta_1^S = \beta_1^T = 0$ . Because the total follow-up time  $o_{ijk}$  varies across cells, modelling the raw death counts  $\mu_{ijk}$  would confound differences in exposure time with genuine differences in mortality risk. Including  $\log o_{ijk}$  as an offset allows the model to target the death rate  $\mu_{ijk}/o_{ijk}$ , so that the coefficients reflect true differences in mortality across histology groups, disease stages, and follow-up intervals. An offset in a regression model is specified in R using the `offset` function.

```
fit_lc <- glm(count ~ factor(histology) + factor(stage) +
              factor(time) + offset(log(risktime)),
              family = poisson, data = Cancer)
head(model.matrix(fit_lc))

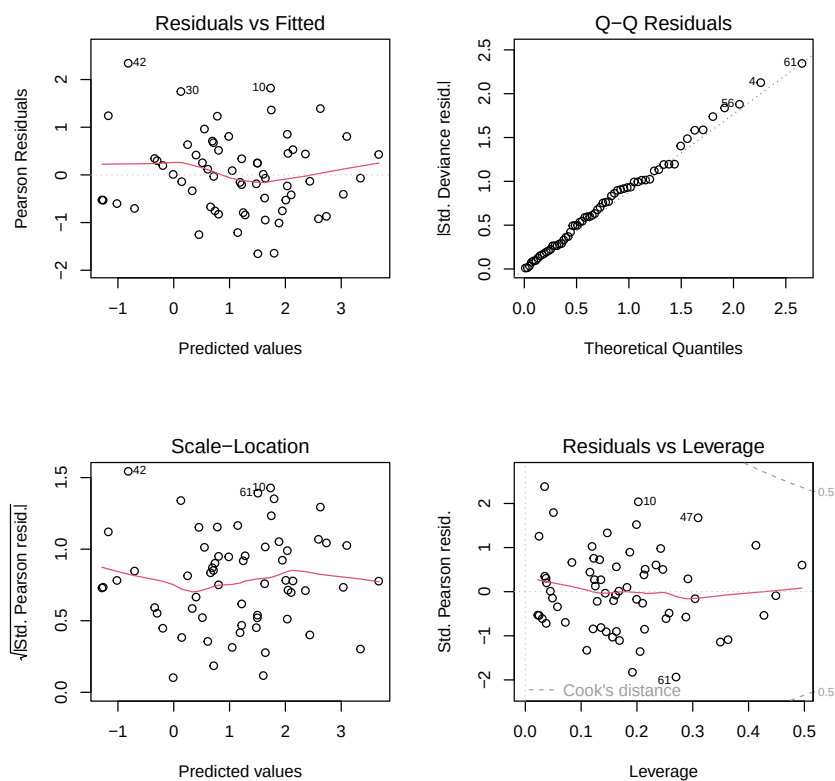
##      (Intercept) factor(histology)2 factor(histology)3 factor(stage)2
## 1              1              0              0              0
## 2              1              1              0              0
## 3              1              0              1              0
## 4              1              0              0              0
## 5              1              1              0              0
## 6              1              0              1              0
##      factor(stage)3 factor(time)2 factor(time)3 factor(time)4 factor(time)5
## 1              0              0              0              0              0
## 2              0              0              0              0              0
## 3              0              0              0              0              0
## 4              0              1              0              0              0
## 5              0              1              0              0              0
## 6              0              1              0              0              0
##      factor(time)6 factor(time)7
## 1              0              0
## 2              0              0
## 3              0              0
## 4              0              0
## 5              0              0
## 6              0              0

summary(fit_lc)

##
## Call:
## glm(formula = count ~ factor(histology) + factor(stage) + factor(time) +
##      offset(log(risktime)), family = poisson, data = Cancer)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.00928    0.16651  -18.073   < 2e-16 ***
## factor(histology)2  0.16244    0.12195   1.332   0.18285
```

```
## factor(histology)3  0.10754    0.14745    0.729    0.46580
## factor(stage)2     0.47001    0.17444    2.694    0.00705 **
## factor(stage)3     1.32431    0.15205    8.709    < 2e-16 ***
## factor(time)2      -0.12745    0.14908   -0.855    0.39259
## factor(time)3      -0.07973    0.16352   -0.488    0.62585
## factor(time)4       0.11892    0.17107    0.695    0.48694
## factor(time)5      -0.66511    0.26061   -2.552    0.01071 *
## factor(time)6      -0.35015    0.24348   -1.438    0.15040
## factor(time)7      -0.17518    0.24985   -0.701    0.48321
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 175.718  on 62  degrees of freedom
## Residual deviance:  43.923  on 52  degrees of freedom
## AIC: 251.74
##
## Number of Fisher Scoring iterations: 5

par(mfrow = c(2,2))
plot(fit_1c)
```



As mentioned before, the offset term (the total follow-up time in months during each interval) does not have an estimated coefficient attached to it; this is also evident in the model matrix and in the summary of the fitted model.

The residual plots do not indicate any obvious lack of fit. A generalized likelihood ratio test indicates that stage of disease is a significant predictor, whereas histology is not. There is also no evidence to support the inclusion of an interaction between stage and time.

For illustration, we proceed with the full model and interpret the regression coefficients in the log-linear Poisson model. The estimated effects for stage of disease indicate that the death rate increases progressively as the disease stage advances. The estimated death rate at the second stage of disease is  $e^{\hat{\beta}_2^S} = e^{0.47} = 1.6$  times that at the first stage, adjusting for follow-up time and histology (i.e., holding these predictors fixed for both disease stages being compared), with a 95% confidence interval  $(e^{0.47-0.17 \times 1.96}, e^{0.47+0.17 \times 1.96}) = (1.147, 2.233)$ . To see why we interpret the effects this way, note that the log death rate at the second stage of disease is given by

$$\log\left(\frac{\mu_{i2k}}{o_{i2k}}\right) = \beta_0 + \beta_i^H + \beta_2^S + \beta_k^T,$$

while the log death rate at the first stage of disease is

$$\log\left(\frac{\mu_{i1k}}{o_{i1k}}\right) = \beta_0 + \beta_i^H + \beta_k^T,$$

since  $\beta_1^S = 0$  for identifiability. The difference in log death rates, keeping histology and follow-up interval the same, is

$$\log\left(\frac{\mu_{i2k}}{o_{i2k}}\right) - \log\left(\frac{\mu_{i1k}}{o_{i1k}}\right) = \beta_2^S.$$

Simple algebra then gives

$$\frac{\mu_{i2k}}{o_{i2k}} = \frac{\mu_{i1k}}{o_{i1k}} e^{\beta_2^S}.$$

Similarly, the estimated death rate at the third stage of disease is  $e^{\hat{\beta}_3^S} = e^{1.32} = 3.74$  times that at the first stage, adjusting for follow-up time and histology, with the 95% confidence interval  $(e^{1.32-0.15 \times 1.96}, e^{1.32+0.15 \times 1.96}) = (2.790, 5.023)$ .

## 12.5 Insurance claims testing example

The dataset `motori` is derived from the `motorins` dataset in the `faraway` package in R. It contains data from a Swedish insurance company on yearly claim `Payment`, based on `Make` of car (a factor variable with nine levels representing different car models), number of years with no claims (`Bonus`), annual number of km driven (a numeric variable with five discrete values treated as a continuous variable here), and number of `Insured` in policy-years. Note that `Insured` is not an integer variable, as not all insured individuals are covered for the entire year. For more information, type `motorins` in R. We assume that the expected total payment is proportional to the number of insured individuals and an initial model for the data is:

$$\mu_i = \mathbb{E}(\text{Payment}_i) = \text{Insured}_i \times \text{risk}_i, \quad \text{where } \text{Payment} \sim \text{gamma},$$

so

$$\log\{\mathbb{E}(\text{Payment}_i)\} = \log(\text{Insured}_i) + \log(\text{risk}_i).$$

The term  $\log(\text{Insured}_i)$  is included as an offset because the total amount of claims, and hence the payment amount, for a group is generally expected to be proportional to the number of insured individuals. The term  $\log(\text{risk}_i)$  can be modelled using a linear model structure to give:

$$\log\{\mathbb{E}(\text{Payment}_i)\} = \log(\text{Insured}_i) + \alpha + \beta_j + \gamma \text{km}_i + \delta_j \text{km}_i + \zeta \text{Bonus}_i,$$

where  $\beta_j$  is the main effect for `Make`,  $j \in \{1, \dots, 9\}$ ,  $\gamma \text{km}_i$  is the main effect of kms travelled,  $\delta_j \text{km}_i$  is the `Make-km` interaction, and  $\zeta \text{Bonus}_i$  is the effect of `Bonus`. This model is not identifiable as written; one way to identify it is to set  $\beta_1 = \delta_1 = 0$ .

We are interested in testing whether the effect of kilometres travelled on the payment rate varies by car model, that is, whether the interaction between `Make` and `km` is required in the model. We therefore fit the corresponding null and full models, that is, the models without and with the interaction term, respectively.

```

require(faraway)
data(motorins)

# Attention will be restricted to data from zone 1
motori <- motorins[motorins$Zone==1,]
motori$km <- as.numeric(motori$Kilometres)
# We are not using the variables perd and claims
head(motori)

##      Kilometres Zone Bonus Make Insured Claims Payment      perd km
## 1           1     1     1     1    455.13     108  392491 3634.176  1
## 2           1     1     1     2     69.17      19   46221 2432.684  1
## 3           1     1     1     3     72.88      13   15694 1207.231  1
## 4           1     1     1     4  1292.39     124  422201 3404.847  1
## 5           1     1     1     5   191.01      40  119373 2984.325  1
## 6           1     1     1     6   477.66      57  170913 2998.474  1

# Full model
g1 <- glm(Payment ~ offset(log(Insured)) + km*Make + Bonus,
          family = Gamma(link = log), data = motori)
#null model
g0 <- glm(Payment ~ offset(log(Insured)) + km + Make + Bonus,
          family = Gamma(link = log), data = motori)

```

Since the dispersion parameter  $\phi$  is not known for the Gamma distribution, we perform an F-test to compare the models using (33).

```

anova(g0, g1, test = "F")

## Analysis of Deviance Table
##
## Model 1: Payment ~ offset(log(Insured)) + km + Make + Bonus
## Model 2: Payment ~ offset(log(Insured)) + km * Make + Bonus
##      Resid. Df Resid. Dev Df Deviance      F Pr(>F)
## 1          284      155.06
## 2          276      151.89  8      3.166 0.752 0.6455

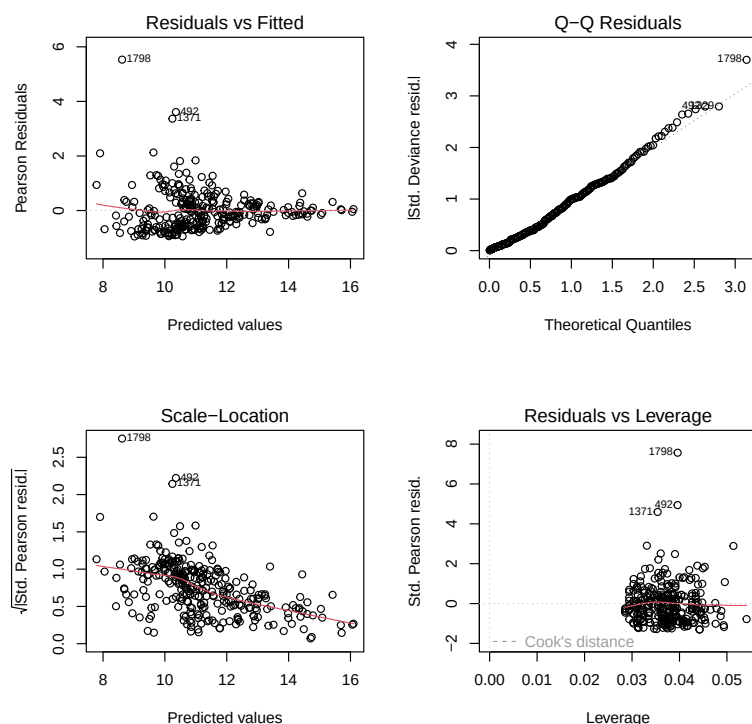
```

It appears that the effect of kilometres travelled on the payment rate does not vary with car make. The residual plots for both models, g0 and g1, indicate that the mean–variance relationship implied by the gamma distribution may not be appropriate for these data, and hence all conclusions should be interpreted with caution.

```

par(mfrow = c(2,2))
plot(g0)

```



## 12.6 Semiconductor wafer model selection example

As an example of backward selection in a GLM context, we consider the semiconductor electrical resistivity data available in the data frame `wafer` from the `faraway` package<sup>44</sup>. Four factors (`x1` to `x4`), each with levels `+` and `-`, in the manufacturing process were believed to influence semiconductor resistivity, `resist`. An experiment was conducted using all combinations of the two levels of each factor. Previous analyses of this dataset, aimed at accounting for the skewness in the response variable (see below), used a normal linear model for the log-transformed response. Here, we use a gamma distribution with a log link, which is appropriate for positive, skewed response data. The data are as follows.

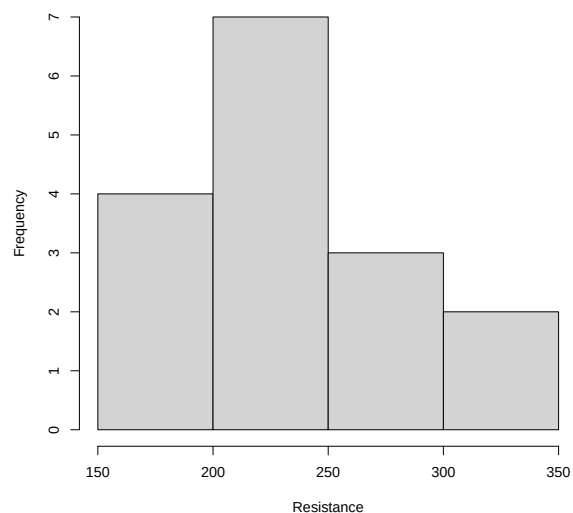
```
require(faraway)
wafer

##      x1 x2 x3 x4 resist
## 1  -  -  -  -  193.4
## 2   +  -  -  -  247.6
## 3   -  +  -  -  168.2
## 4   +  +  -  -  205.0
## 5   -  -  +  -  303.4
## 6   +  -  +  -  339.9
## 7   -  +  +  -  226.3
## 8   +  +  +  -  208.3
## 9   -  -  -  +  220.0
## 10  +  -  -  +  256.4
## 11  -  +  -  +  165.7
## 12  +  +  -  +  203.5
## 13  -  -  +  +  285.0
## 14  +  -  +  +  268.0
```

<sup>44</sup>Semiconductor resistivity is a physical property that measures how strongly a semiconductor material resists the flow of electric current.

```
## 15 - + + + 169.1
## 16 + + + + 208.5

hist(wafer$resist, xlab = "Resistance", main = "")
```



An initial model is fitted with all interactions of the factors up to third order. Note that this results in a model with 15 parameters and only 16 observations, so some model selection is clearly needed.

```
wm <- glm(resist ~ (x1 + x2 + x3 + x4)^3,
          Gamma(link = "log"), data = wafer)
```

Applying backward selection, we now want to calculate the p-values associated with dropping each three-way interaction from the full model, one at a time. Recall that the `drop1` function in R is a convenient way to do this. It examines all terms in the model that can be dropped individually, refits the model without each term in turn, and compares the resulting fits to the original model. Because the dispersion parameter  $\phi$  must be estimated for the Gamma distribution, we use an F-test rather than a likelihood ratio test.

```
drop1(wm, test = "F")

## Single term deletions
##
## Model:
## resist ~ (x1 + x2 + x3 + x4)^3
##      Df  Deviance    AIC F value Pr(>F)
## <none>      0.0083651 129.73
## x1:x2:x3    1 0.0086826 127.76  0.0380 0.8775
## x1:x2:x4    1 0.0285857 130.14  2.4173 0.3639
## x1:x3:x4    1 0.0109530 128.03  0.3094 0.6769
## x2:x3:x4    1 0.0107629 128.01  0.2867 0.6871
```

The rows of the table are labelled with the names of the dropped terms. The reported p-value in each row is for testing the null hypothesis that the model without the dropped term is adequate, against the alternative that the fuller model is needed. The AIC for each model under consideration is also reported. The p-values suggest dropping the interaction `x1:x2:x3`.

```
wm1 <- update(wm, .~. - x1:x2:x3)
drop1(wm1, test = "F")
```

```
## Single term deletions
##
## Model:
## resist ~ x1 + x2 + x3 + x4 + x1:x2 + x1:x3 + x1:x4 + x2:x3 +
##      x2:x4 + x3:x4 + x1:x2:x4 + x1:x3:x4 + x2:x3:x4
##      Df  Deviance    AIC F value Pr(>F)
## <none>      0.0086826 128.32
## x1:x2:x4   1 0.0289028 130.98   4.6577 0.1636
## x1:x3:x4   1 0.0112704 126.92   0.5961 0.5208
## x2:x3:x4   1 0.0110804 126.87   0.5523 0.5348
```

Now the  $x2:x3:x4$  interaction is the one to drop. Repeating these steps, we eventually end up with

```
wm7 <-glm(resist ~ x1*x3 + x3*x4 + x2*x3,
          Gamma(link = "log"), data = wafer)
drop1(wm7, test = "F")

## Single term deletions
##
## Model:
## resist ~ x1 * x3 + x3 * x4 + x2 * x3
##      Df Deviance    AIC F value  Pr(>F)
## <none>    0.036266 139.20
## x1:x3     1 0.060433 142.54   5.3311 0.04977 *
## x3:x4     1 0.069345 144.51   7.2970 0.02702 *
## x3:x2     1 0.067314 144.06   6.8491 0.03079 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

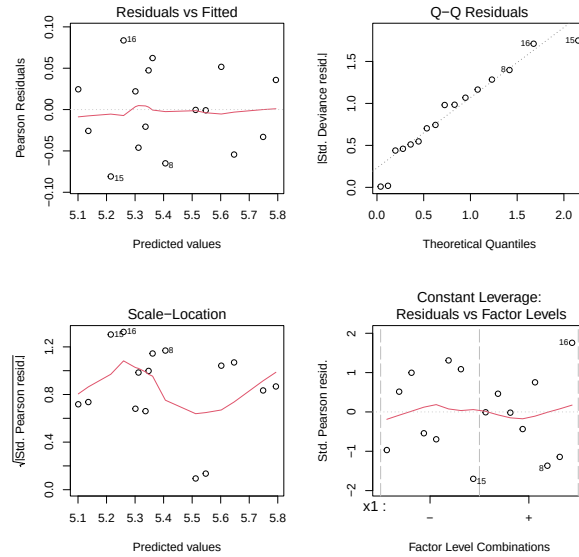
summary(wm7)

##
## Call:
## glm(formula = resist ~ x1 * x3 + x3 * x4 + x2 * x3, family = Gamma(link = "log"),
##      data = wafer)
##
## Coefficients:
##      Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.31195     0.04757 111.677 4.62e-14 ***
## x1+           0.20029     0.04757   4.211 0.00295 **
## x3+           0.43674     0.06727   6.493 0.00019 ***
## x4+           0.03537     0.04757   0.744 0.47836
## x2+          -0.21101     0.04757  -4.436 0.00218 **
## x1+:x3+      -0.15549     0.06727  -2.312 0.04957 *
## x3+:x4+      -0.18195     0.06727  -2.705 0.02687 *
## x3+:x2+      -0.17626     0.06727  -2.620 0.03064 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Gamma family taken to be 0.004524942)
##
##      Null deviance: 0.697837  on 15  degrees of freedom
## Residual deviance: 0.036266  on  8  degrees of freedom
## AIC: 139.2
##
## Number of Fisher Scoring iterations: 4
```

If 0.05 is used as the significance level for retaining a term, then this is the final model. In this model, all main

effects are included, along with three two-way interactions, all involving  $x_3$ . The residual plots appear acceptable, although the sample size is admittedly very small.

```
par(mfrow = c(2, 2))
plot(wm7)
```



Once a model has been selected, the coefficient estimates can be examined and interpreted, possibly with the aid of confidence intervals. Note, however, that inference performed with the final fitted model tends to overstate what can be concluded from the data, as it does not account for the uncertainty introduced by the model selection process. To conclude, let us write down the selected model mathematically.

$$y_i = \text{resist}_i, \quad y_i \sim \text{gamma}, \quad \mathbb{E}(y_i) = \mu_i, \quad i = 1, \dots, 16,$$

$$g(\mu_i) = \log(\mu_i) = \beta_1 + \beta_2 x_{1,i}^+ + \beta_3 x_{2,i}^+ + \beta_4 x_{3,i}^+ + \beta_5 x_{4,i}^+ + \beta_6 x_{1,i}^+ x_{3,i}^+ + \beta_7 x_{2,i}^+ x_{3,i}^+ + \beta_8 x_{3,i}^+ x_{4,i}^+, \quad (35)$$

where  $x_{1,i}^+$  is a dummy variable taking the value 1 if  $x_1$  takes the value +, and 0 otherwise. The other variables,  $x_{2,i}^+$ ,  $x_{3,i}^+$ , and  $x_{4,i}^+$  are defined similarly. The estimates of each parameter ( $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_8$ ) are obtained from the summary output above. As before, and if helpful, we can also examine the model matrix. Note that the model in (35) has been written using the identifiability constraints imposed by default in R, so the information from the model and from `model.matrix` is aligned.

```
model.matrix(wm7)
```

```
##      (Intercept) x1+ x3+ x4+ x2+ x1+:x3+ x3+:x4+ x3+:x2+
## 1             1  0  0  0  0             0             0             0
## 2             1  1  0  0  0             0             0             0
## 3             1  0  0  0  1             0             0             0
## 4             1  1  0  0  1             0             0             0
## 5             1  0  1  0  0             0             0             0
## 6             1  1  1  0  0             1             0             0
## 7             1  0  1  0  1             0             0             1
## 8             1  1  1  0  1             1             0             1
## 9             1  0  0  1  0             0             0             0
## 10            1  1  0  1  0             0             0             0
## 11            1  0  0  1  1             0             0             0
## 12            1  1  0  1  1             0             0             0
## 13            1  0  1  1  0             0             1             0
```

```
## 14      1  1  1  1  0      1      1      0
## 15      1  0  1  1  1      0      1      1
## 16      1  1  1  1  1      1      1      1
## attr(,"assign")
## [1] 0 1 2 3 4 5 6 7
## attr(,"contrasts")
## attr(,"contrasts")$x1
## [1] "contr.treatment"
##
## attr(,"contrasts")$x3
## [1] "contr.treatment"
##
## attr(,"contrasts")$x4
## [1] "contr.treatment"
##
## attr(,"contrasts")$x2
## [1] "contr.treatment"
```

Since the log link is used, the model is log-linear, meaning that the effects of the factors are multiplicative on the expected resistivity scale. The intercept ( $\beta_1$  in model (35)) corresponds to the log of the expected resistivity for a wafer when all factors are in the – state. The coefficients associated to  $x1+$  to  $x4+$  ( $\beta_2$  to  $\beta_5$  in (35)) represent changes in the log of the expected resistivity when the corresponding factor moves from – to +, with the other factors held at their baseline levels. Referring back to the summary,  $x1$  and  $x3$  appear to be associated with an increase in resistivity when considered individually.

So what about the interactions? Consider  $x1+:x3+$  as an example. This term is an adjustment that is added when  $x1$  and  $x3$  are both in the + state. In other words, it measures how much the log of the expected resistivity differs from what would be expected if the effects of  $x1$  and  $x3$  simply added together. Referring back to the summary, it appears that when two factors are in the + state, the log of the expected resistivity is lower than one would expect based on their individual effects alone.

### 12.6.1 AIC based selection

As an example of using AIC, let us redo the wafer model selection using AIC for backward selection. Conceptually, this is no different from what we have done previously with ordinary linear models. Since AIC is the default criterion used by `drop1`, we do not need to specify any additional arguments, we simply pass the fitted object `wm`, which includes all interactions among the four factors up to third order.

```
drop1(wm)

## Single term deletions
##
## Model:
## resist ~ (x1 + x2 + x3 + x4)^3
##      Df  Deviance    AIC
## <none>      0.0083651 129.73
## x1:x2:x3  1 0.0086826 127.76
## x1:x2:x4  1 0.0285857 130.14
## x1:x3:x4  1 0.0109530 128.03
## x2:x3:x4  1 0.0107629 128.01
```

The model with the lowest AIC should be selected. In this instance, it is the model that omits  $x1:x2:x3$ , so that term will be dropped.

```
wm1 <- update(wm, .~. - x1:x2:x3)
drop1(wm1)

## Single term deletions
```

```
##
## Model:
## resist ~ x1 + x2 + x3 + x4 + x1:x2 + x1:x3 + x1:x4 + x2:x3 +
##          x2:x4 + x3:x4 + x1:x2:x4 + x1:x3:x4 + x2:x3:x4
##           Df  Deviance    AIC
## <none>          0.0086826 128.32
## x1:x2:x4      1 0.0289028 130.98
## x1:x3:x4      1 0.0112704 126.92
## x2:x3:x4      1 0.0110804 126.87
```

Notice one wrinkle: the AIC reported for `wm1` is 128.32 (recall that `<none>` corresponds to the model being fitted), but when we applied `drop1` earlier to `wm`, it suggested that the AIC for `wm1` would be 127.76. This happens because, for a Gamma GLM, the AIC depends on the scale parameter  $\phi$ , which must be estimated. In `drop1`, the same scale estimate is used for all models being compared in a given call, and that estimate is taken from the largest model under consideration. Therefore, two different calls to `drop1` can report different AIC values for the same model, because they are based on different scale estimates. Of course, this issue does not arise when  $\phi$  is known.

As we already know, we can make the process fully automatic through the use of the `step` function, which can perform backward selection (the default), forward selection, or a hybrid strategy, using AIC. Let us do that now.

```
step(wm)

## Start:  AIC=129.73
## resist ~ (x1 + x2 + x3 + x4)^3
##
##           Df  Deviance    AIC
## - x1:x2:x3  1 0.0086826 127.76
## - x2:x3:x4  1 0.0107629 128.01
## - x1:x3:x4  1 0.0109530 128.03
## <none>          0.0083651 129.73
## - x1:x2:x4  1 0.0285857 130.14
##
## Step:  AIC=128.32
## resist ~ x1 + x2 + x3 + x4 + x1:x2 + x1:x3 + x1:x4 + x2:x3 +
##          x2:x4 + x3:x4 + x1:x2:x4 + x1:x3:x4 + x2:x3:x4
##
##           Df  Deviance    AIC
## - x2:x3:x4  1 0.0110804 126.87
## - x1:x3:x4  1 0.0112704 126.92
## <none>          0.0086826 128.32
## - x1:x2:x4  1 0.0289028 130.98
##
## Step:  AIC=130.22
## resist ~ x1 + x2 + x3 + x4 + x1:x2 + x1:x3 + x1:x4 + x2:x3 +
##          x2:x4 + x3:x4 + x1:x2:x4 + x1:x3:x4
##
## Call:  glm(formula = resist ~ x1 + x2 + x3 + x4 + x1:x2 + x1:x3 + x1:x4 +
##           x2:x3 + x2:x4 + x3:x4 + x1:x2:x4 + x1:x3:x4, family = Gamma(link = "log"),
##           data = wafer)
##
## Coefficients:
## (Intercept)          x1+          x2+          x3+          x4+          x1+:x2+
##      5.2586         0.2843        -0.1273         0.4626         0.1502        -0.1228
##      x1+:x3+        x1+:x4+        x2+:x3+        x2+:x4+        x3+:x4+        x1+:x2+:x4+
##     -0.2071        -0.1852        -0.1783        -0.1863        -0.2339         0.2845
## x1+:x3+:x4+
```

```
##      0.1018
##
## Degrees of Freedom: 15 Total (i.e. Null);  3 Residual
## Null Deviance:      0.6978
## Residual Deviance: 0.01108  AIC: 130.2
```

Notice that the final model selected using AIC is slightly different from the one obtained through hypothesis testing. This is not surprising, since the two approaches use different criteria. Although AIC includes a penalty for model complexity, it may still retain a slightly more complex model than a hypothesis-testing approach based on a fixed significance threshold.

## 13 Mixed-effects models (non-examinable and not covered in class)

Up to this point, our statistical models have involved only *fixed effects*. In fixed-effects models, the levels of the predictors used in the study are the only levels of interest in the analysis, and statistical inference is restricted to these specific levels. In contrast, *random effects* arise when the levels of a factor can be viewed as a random sample from a larger population of possible levels. In this case, the goal is not to make inference about the specific levels observed, but about the variability among levels in the population. Random effects are particularly useful when observations are grouped into clusters, since they induce correlation among observations from the same cluster.

Consider an experiment investigating the effect of several drug treatments on a sample of patients. Typically, we are interested in the specific drug treatments under study, so drug is treated as a fixed effect. However, it makes more sense to treat the patient effects as random. The patients included in the study are drawn from a large population, and the focus of inference is on this broader population rather than on the specific individuals selected. This is the key idea behind random effects: the emphasis is on the population from which the levels are sampled, not on the specific units observed.

### 13.1 A simple random effects model

The Rail data available in the `nlme` package are from an experiment in nondestructive testing for longitudinal stress in railway rails. Six rails were chosen at random and tested three times each by measuring the time it took for a certain type of ultrasonic wave to travel the length of the rail. The only factor that varies across observations is the rail.

```
library(nlme)
head(Rail)

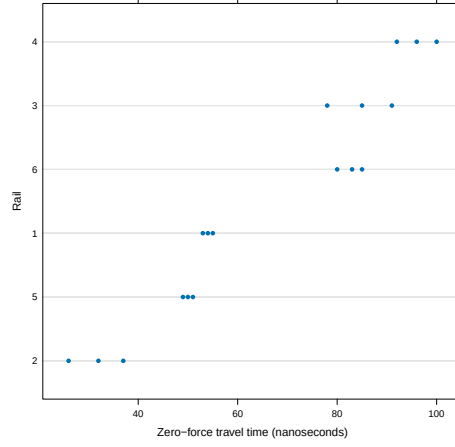
## Grouped Data: travel ~ 1 | Rail
##   Rail travel
## 1     1     55
## 2     1     53
## 3     1     54
## 4     2     26
## 5     2     37
## 6     2     32
```

The structure of the data is quite simple: each row corresponds to one observation for which the rail and the travel time are recorded. The quantities of interest to the engineers, which they aimed to estimate from this experiment, are:

- The average travel time for a ‘typical’ rail (the expected travel time).
- The variation in average travel times among rails (the between-rail variability).
- The variation in the observed travel times for a single rail (the within-rail variability).

We can see in the figure below that there is considerable variability in the mean travel time for the different rails. Overall, the between-rail variability is much greater than the within-rail variability.

```
plot(Rail)
```



The following simple random effects model is appropriate for these data

$$y_{ij} = \mu + b_i + \epsilon_{ij}, i = 1, \dots, 6, \quad j = 1, \dots, 3, \quad (36)$$

where

- $y_{ij}$  is the travel time of rail  $i$  at the  $j$ th measurement.
- $\mu$  is the mean travel time across the population of rails being sampled.
- $b_i$  is a random variable representing the deviation from the population mean travel time for the  $i$ th rail.
- $\epsilon_{ij}$  is a random variable representing the deviation in travel time for observation  $j$  on rail  $i$  from the mean travel time for rail  $i$ .

The  $b_i$  and  $\epsilon_{ij}$  are modelled as independent, normally distributed random variables with constant variance:

$$b_i \stackrel{\text{iid}}{\sim} N(0, \sigma_b^2), \quad \epsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad i = 1, \dots, 6, \quad j = 1, \dots, 3.$$

Here  $\sigma_b^2$  represents the between-rail variability and  $\sigma^2$  represents the within rail variability. To reinforce: the  $b_i$  are called random effects because they are associated with the particular experimental units (rails in this case) that are selected at random from the population of interest. They are effects because they represent a deviation from an overall mean. That is, the ‘effect’ of choosing rail  $i$  is to shift the mean travel time from  $\mu$  to  $\mu + b_i$ . Note that the parameters of model (36) are  $\mu$ ,  $\sigma_b^2$ , and  $\sigma^2$  and that the number of parameters remains three, irrespective of the number of rails in the experiment. Although the random effects  $b_i$  may behave like parameters, formally they are just another level of random variation in the model so we do not ‘estimate’ them as such. One does, however, form predictions  $\hat{b}_i$  of the values of these random variables, given the observed data.

Let us now inspect some properties of model (36). The expected value of  $y_{ij}$  is

$$\mathbb{E}(y_{ij}) = \mathbb{E}(\mu + b_i + \epsilon_{ij}) = \mu.$$

For the variance of  $y_{ij}$  we have

$$\text{var}(y_{ij}) = \text{var}(\mu + b_i + \epsilon_{ij}) = \sigma_b^2 + \sigma^2.$$

For the correlation structure we get

$$\text{Cor}(y_{ij}, y_{kl}) = \begin{cases} 0, & i \neq k, \\ \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}, & i = k, j \neq l, \\ 1, & i = k, j = l. \end{cases}$$

Observations from different rails ( $i \neq k$ ) are uncorrelated while observations from the same rail ( $i = k$ ) are correlated. You are possibly wondering why  $\text{Cor}(y_{ij}, y_{il}) = \sigma_b^2 / (\sigma_b^2 + \sigma^2)$ . Let's then work it out. First, remember that

$$\text{Cor}(y_{ij}, y_{il}) = \frac{\text{Cov}(y_{ij}, y_{il})}{\sqrt{\text{var}(y_{ij})\text{var}(y_{il})}}.$$

By definition,

$$\begin{aligned}\text{Cov}(y_{ij}, y_{il}) &= \mathbb{E}[(y_{ij} - \mathbb{E}[y_{ij}])(y_{il} - \mathbb{E}[y_{il}])] \\ &= \mathbb{E}[(\mu + b_i + \epsilon_{ij} - \mu)(\mu + b_i + \epsilon_{il} - \mu)] \\ &= \mathbb{E}[(b_i + \epsilon_{ij})(b_i + \epsilon_{il})] \\ &= \mathbb{E}[b_i^2 + b_i\epsilon_{il} + \epsilon_{ij}b_i + \epsilon_{ij}\epsilon_{il}] \\ &= \mathbb{E}[b_i^2] + \mathbb{E}[b_i\epsilon_{il}] + \mathbb{E}[\epsilon_{ij}b_i] + \mathbb{E}[\epsilon_{ij}\epsilon_{il}].\end{aligned}$$

Because  $b_i$  and  $\epsilon_{ij}$  are independent we have that  $\mathbb{E}[\epsilon_{ij}b_i] = \mathbb{E}[\epsilon_{ij}]\mathbb{E}[b_i] = 0$ . For the same reason,  $\mathbb{E}[b_i\epsilon_{il}] = 0$ . Further,  $\text{var}(b_i) = \mathbb{E}[b_i^2] - (\mathbb{E}[b_i])^2 = \sigma_b^2$  and because  $\mathbb{E}[b_i] = 0$  it trivially follows that  $\mathbb{E}[b_i^2] = \sigma_b^2$ . Finally,  $\epsilon_{ij}$  and  $\epsilon_{il}$  are independent and hence  $\mathbb{E}[\epsilon_{ij}\epsilon_{il}] = \mathbb{E}[\epsilon_{ij}]\mathbb{E}[\epsilon_{il}] = 0$ . Bringing this all together leads to  $\text{Cov}(y_{ij}, y_{il}) = \sigma_b^2$ . We have already concluded that  $\text{var}(y_{ij}) = \text{var}(y_{il}) = \sigma_b^2 + \sigma^2$  and thus,

$$\text{Cor}(y_{ij}, y_{il}) = \frac{\text{Cov}(y_{ij}, y_{il})}{\sqrt{\text{var}(y_{ij})\text{var}(y_{il})}} = \frac{\sigma_b^2}{\sqrt{(\sigma_b^2 + \sigma^2)(\sigma_b^2 + \sigma^2)}} = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}.$$

The correlation between observations from the same rail,  $\sigma_b^2 / (\sigma_b^2 + \sigma^2)$ , is also called the *intraclass correlation coefficient*. Note that the intraclass correlation coefficient lies in  $[0, 1]$ . When  $\sigma_b^2 \approx 0$ , almost all variability is due to within-group variation and the intraclass correlation is close to zero, so observations within the same group are essentially uncorrelated. Conversely, when  $\sigma_b^2 \gg \sigma^2$ , most of the variability is due to differences between groups and the intraclass correlation approaches one, meaning that observations within the same group are very similar to each other. This contrasts with the analogous fixed-effects model, in which the rail effects are treated as unknown constants rather than random variables. In that case, the model does not induce correlation through a shared random effect.

Now we fit the random effects model with the `lmer` function in package `lme4` (that needs to be installed!). We want to have a random effect per rail. This can be specified with the notation `(1 | Rail)` in the model formula. This means that the ‘granularity’ of the random effect is specified after the vertical bar ‘|’. All observations sharing the same level of rail will get the same random effect  $b_i$  (in this case we know that there are three measurements per rail). The ‘1’ means that we only want to have a so-called random intercept ( $b_i$ ) per rail.

```
library(lme4)
rm <- lmer(travel ~ (1 | Rail), data = Rail)
```

As usual, the function summary gives an overview of the fitted model.

```
summary(rm)

## Linear mixed model fit by REML ['lmerMod']
## Formula: travel ~ (1 | Rail)
## Data: Rail
##
## REML criterion at convergence: 122.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.61883 -0.28218  0.03569  0.21956  1.61438
##
## Random effects:
## Groups Name Variance Std.Dev.
```

```
## Rail      (Intercept) 615.31    24.805
## Residual                16.17     4.021
## Number of obs: 18, groups: Rail, 6
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)    66.50      10.17    6.538
```

From the summary we have that the estimates of the parameters are as follows

$$\hat{\mu} = 66.50, \quad \hat{\sigma}_b = 24.805, \quad \hat{\sigma} = 4.021.$$

Note that  $\hat{\sigma}_b^2 \gg \hat{\sigma}^2$  meaning that travel times within rail are much more similar than travel times across rails (as we have already noticed from the plot of the data).

(Approximate) 95% confidence intervals for the parameters can be obtained with the function `confint` (the argument `oldNames` has to be set to `FALSE` to get a nicely labelled output).

```
confint(rm, oldNames = FALSE)

##              2.5 %      97.5 %
## sd_(Intercept)|Rail 13.929422 45.546746
## sigma              2.824557  6.377695
## (Intercept)        44.960664 88.039333
```

Hence, an approximate 95% confidence interval for the standard deviation  $\sigma_b$  is given by (13.93, 45.55). We see that the estimate  $\hat{\sigma}_b$  is therefore quite imprecise. The reason is that we only have six rails to estimate the standard deviation.

We can also obtain predictions of the random effects  $b_i$  using the function `ranef`.

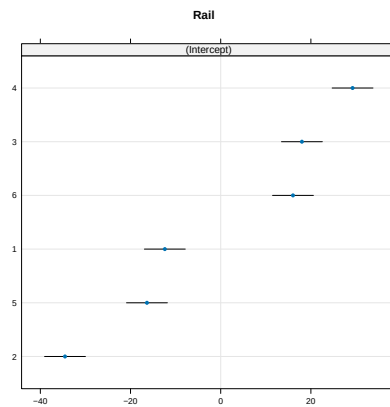
```
ranef(rm)

## $Rail
##      (Intercept)
## 2      -34.53091
## 5      -16.35675
## 1      -12.39148
## 6       16.02631
## 3       18.00894
## 4       29.24388
##
## with conditional variances for "Rail"
```

These can also be plotted along with confidence intervals.

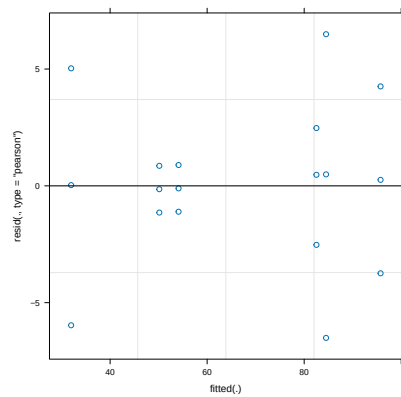
```
library(lattice)
dotplot(ranef(rm, condVar = TRUE))

## $Rail
```

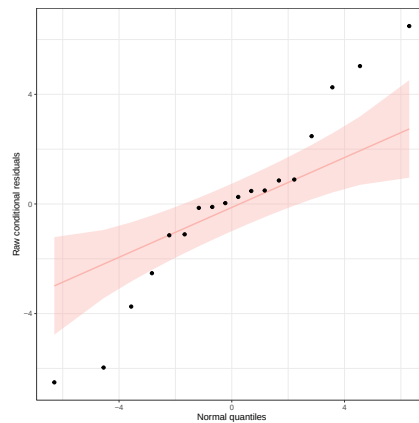


We should of course also check the model assumptions. Here, this includes in addition normality of the random effects, though this is hard to check with only six observations.

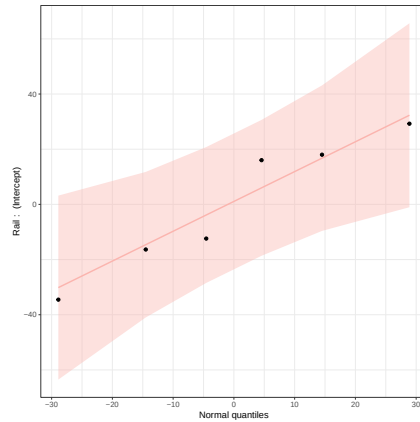
```
plot(rm)
```



```
library(redres)
plot_resqq(rm)
```



```
plot_ranef(rm)
```



## 13.2 A random effects model involving a random interaction

Let's consider now an example involving two random factors and their (random) interaction. We will consider the following example: a large company randomly selected five employees and six batches of source material from the production process. The material from each batch was divided into 15 pieces which were randomized to the different employees such that each employee would build three test specimens from each batch. The response is the corresponding quality score. The goal was to quantify the different sources of variation.

As usual, we first load the data and get an overview.

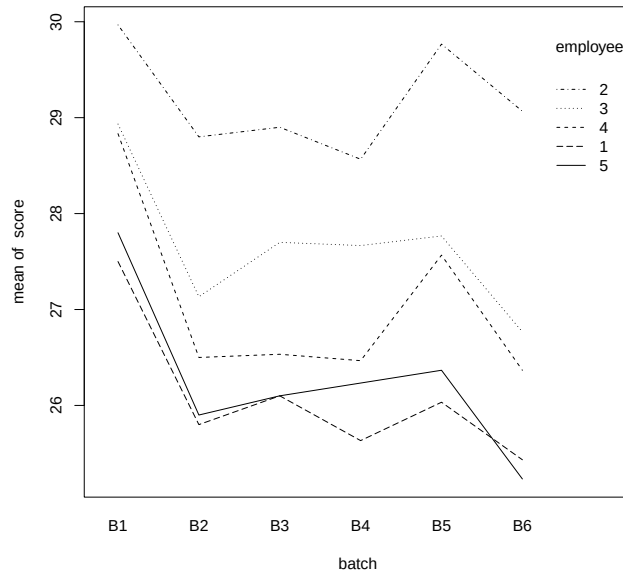
```
book.url <- "https://stat.ethz.ch/~meier/teaching/book-anova"
quality <- readRDS(url(file.path(book.url, "data/quality.rds")))
str(quality)

## 'data.frame': 90 obs. of 3 variables:
## $ employee: Factor w/ 5 levels "1","2","3","4",..: 1 1 1 1 1 1 1 1 1 1 ...
## $ batch    : Factor w/ 6 levels "B1","B2","B3",..: 1 1 1 2 2 2 3 3 3 4 ...
## $ score    : num  27.4 27.8 27.3 25.5 25.5 26.4 26.9 26.3 25.1 25.6 ...

head(quality)

##   employee batch score
## 1         1    B1  27.4
## 2         1    B1  27.8
## 3         1    B1  27.3
## 4         1    B2  25.5
## 5         1    B2  25.5
## 6         1    B2  26.4

#interaction plot
with(quality, interaction.plot(x.factor = batch,
                              trace.factor = employee,
                              response = score))
```



There appears to be variation between different employees and between the different batches. The interaction does not seem to be very pronounced but for the sake of illustration we will include it in our model. Let  $y_{ijk}$  denote the observed quality score of the  $k$ th sample of employee  $i$  using material from batch  $j$ . The model we will be fitting is

$$y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk},$$

where  $\alpha_i$  is the random (main) effect of employee  $i$ ,  $\beta_j$  is the random (main) effect of batch,  $\gamma_{ij}$  is the random interaction term between employee  $i$  and batch  $j$  and  $\epsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$  is the error term. For the random effects we have the usual assumptions

$$\alpha_i \stackrel{\text{iid}}{\sim} N(0, \sigma_\alpha^2), \quad \beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2), \quad \gamma_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma_\gamma^2).$$

In addition we assume that all random effects and the error term are independent of each other.

What is the interpretation of the different terms? The random (main) effect of employee is the variability between different employees, and the random (main) effect of batch is the variability between different batches. The random interaction term can, for example, be interpreted as quality inconsistencies of employees across different batches.

Let us fit this model in R.

```
qmm <- lmer(score ~ (1 | employee) + (1 | batch) + (1 | employee:batch),
            data = quality)
summary(qmm)

## Linear mixed model fit by REML ['lmerMod']
## Formula: score ~ (1 | employee) + (1 | batch) + (1 | employee:batch)
## Data: quality
##
## REML criterion at convergence: 167.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8144 -0.5933 -0.1190  0.7073  2.8565
##
## Random effects:
```

```
## Groups Name Variance Std.Dev.
## employee:batch (Intercept) 0.02349 0.1533
## batch (Intercept) 0.51764 0.7195
## employee (Intercept) 1.54473 1.2429
## Residual 0.22655 0.4760
## Number of obs: 90, groups: employee:batch, 30; batch, 6; employee, 5
##
## Fixed effects:
## Estimate Std. Error t value
## (Intercept) 27.2478 0.6313 43.16
```

From the output we get  $\hat{\sigma}_\alpha^2 = 1.54$ ,  $\hat{\sigma}_\beta^2 = 0.52$ ,  $\hat{\sigma}_\gamma^2 = 0.02$  and  $\hat{\sigma}^2 = 0.23$ . Hence, total variance is  $1.54 + 0.52 + 0.02 + 0.23 = 2.31$ . We see that the largest contribution to the variance is variability between different employees which contributes to about  $1.54/2.31 \approx 67\%$  of the total variance. These are point estimates. Confidence intervals on the scale of the standard deviations can be obtained by calling `confint`.

```
confint(qm, oldNames = FALSE)

##                2.5 %      97.5 %
## sd_(Intercept)|employee:batch 0.0000000 0.3527999
## sd_(Intercept)|batch          0.4100059 1.4754426
## sd_(Intercept)|employee       0.6694340 2.4993870
## sigma                        0.4020266 0.5741431
## (Intercept)                   25.9189910 28.5765641
```

The estimated interaction variance is very small, and the corresponding confidence interval includes zero, suggesting little evidence of an important interaction component.

### 13.3 A mixed effects model involving both a fixed and a random factor effect

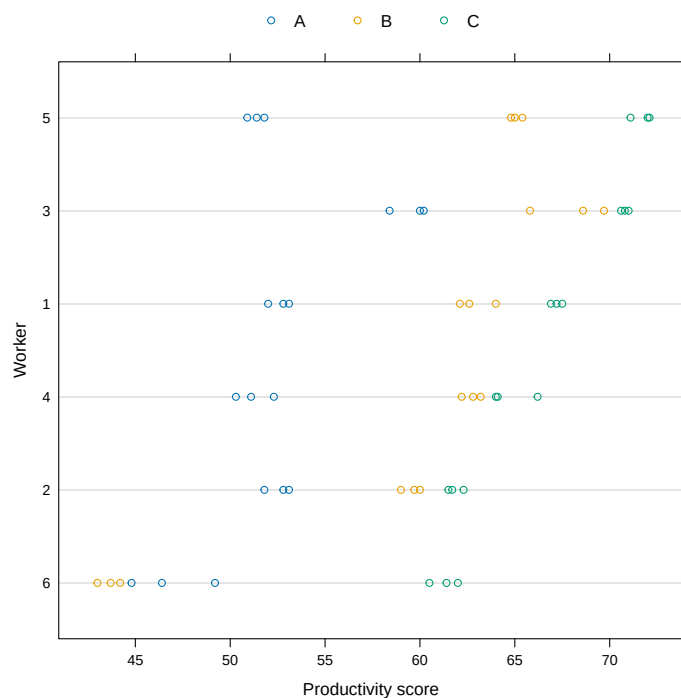
In practice, we often encounter models that involve both random and fixed effects. As already mentioned, these are designated by *mixed effects models*. To illustrate this class of models we will use the `Machines` dataset available in `nlme`, which contains data on an experiment to compare three brands of machines used in an industrial process. Six workers were chosen randomly among the employees of a factory to operate each machine three times. The response is an overall productivity score taking into account the number and quality of components produced. Let's look at the first rows of the dataset.

```
head(Machines)

## Grouped Data: score ~ Machine | Worker
## Worker Machine score
## 1      1      A  52.0
## 2      1      A  52.8
## 3      1      A  53.1
## 4      2      A  51.8
## 5      2      A  52.8
## 6      2      A  53.1
```

In the figure below we can see that there are strong indications of differences between machines and also some indications of differences between workers. We further note that there is very little variability in the productivity score for the same worker using the same machine. In addition, on average, productivity is largest on machine C, followed by B, and A.

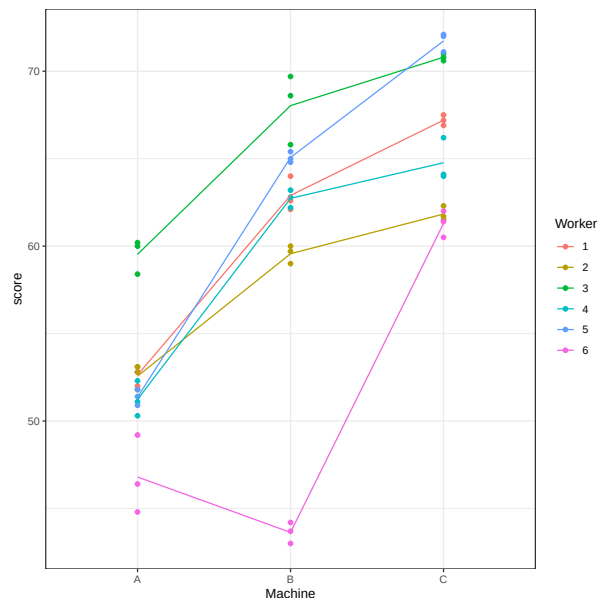
```
plot(Machines)
```



The plot below shows that there may be an interaction between worker and machine.

```
## technical details for shorter output:
class(Machines) <- "data.frame"
Machines[, "Worker"] <- factor(Machines[, "Worker"], levels = 1:6,
                              ordered = FALSE)

require(ggplot2)
ggplot(Machines, aes(x = Machine, y = score, group = Worker, col = Worker)) +
  geom_point() + stat_summary(fun = mean, geom = "line") + theme_bw()
```



Let us now posit a model on this data. The goal is to make inference about the specific machines at hand, this is why we treat machine as a fixed effect. Because the workers represent a random sample from the population

of interest, their effect will be considered a random effect. We will include an interaction between worker and machine, and an interaction effect between a random (here, worker) and a fixed effect (here, machine) is treated as a random effect. Let  $y_{ijk}$  denote the observed  $k$ th productivity score of worker  $j$  on machine  $i$ . We use the model

$$y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \epsilon_{ijk},$$

where  $\alpha_i$  is the fixed effect of machine,  $\beta_j$  is the random effect of worker  $j$ , and  $\gamma_{ij}$  is the corresponding random interaction effect. We assume that all random effects are normally distributed, that is,

$$\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2), \quad \gamma_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma_\gamma^2).$$

For the error term we assume as always  $\epsilon_{ijk} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ . In addition, all random terms are assumed to be independent.

We now fit the model through the `lmer` function. As `lme4` does not calculate p-values for the fixed effects, we use the package `lmerTest` instead. Technically speaking, `lmerTest` uses `lme4` to fit the model and then adds some statistical tests, i.e., p-values for the fixed effects, to the output. We can also do an F test through the `anova` (but from the `lmerTest` package) function.

```
library(lmerTest)
fit.machines <- lmer(score ~ Machine + (1 | Worker) + (1 | Worker:Machine),
                    data = Machines)
summary(fit.machines)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: score ~ Machine + (1 | Worker) + (1 | Worker:Machine)
## Data: Machines
##
## REML criterion at convergence: 215.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.26959 -0.54847 -0.01071  0.43937  2.54006
##
## Random effects:
## Groups           Name             Variance Std.Dev.
## Worker:Machine (Intercept) 13.9095   3.7295
## Worker          (Intercept) 22.8584   4.7811
## Residual                        0.9246   0.9616
## Number of obs: 54, groups: Worker:Machine, 18; Worker, 6
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)   52.356      2.486   8.522  21.062 1.20e-08 ***
## MachineB       7.967      2.177  10.000   3.660 0.00439 **
## MachineC      13.917      2.177  10.000   6.393 7.91e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) MachnB
## MachineB -0.438
## MachineC -0.438  0.500
anova(fit.machines)

## Type III Analysis of Variance Table with Satterthwaite's method
```

```
##           Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## Machine  38.051  19.025     2     10  20.576 0.0002855 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

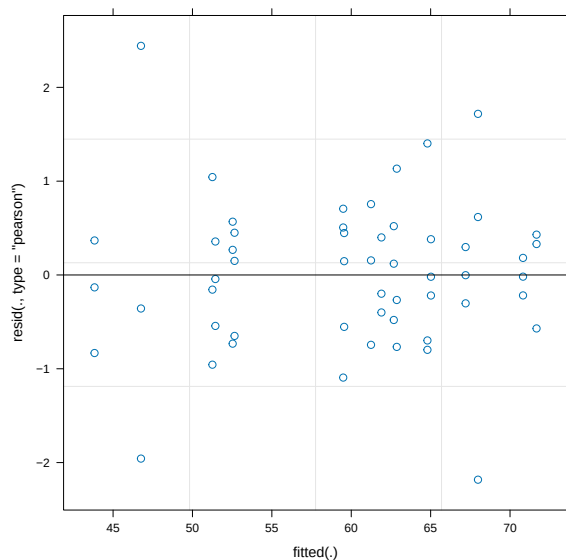
The fixed effect of machine is significant. We further have that  $\hat{\sigma}_\beta = 4.78$  (between worker variability),  $\hat{\sigma}_\gamma = 3.73$  and  $\hat{\sigma} = 0.96$  (within-worker variability). As before, confidence intervals can also be easily calculated.

```
confint(fit.machines, oldNames = FALSE)

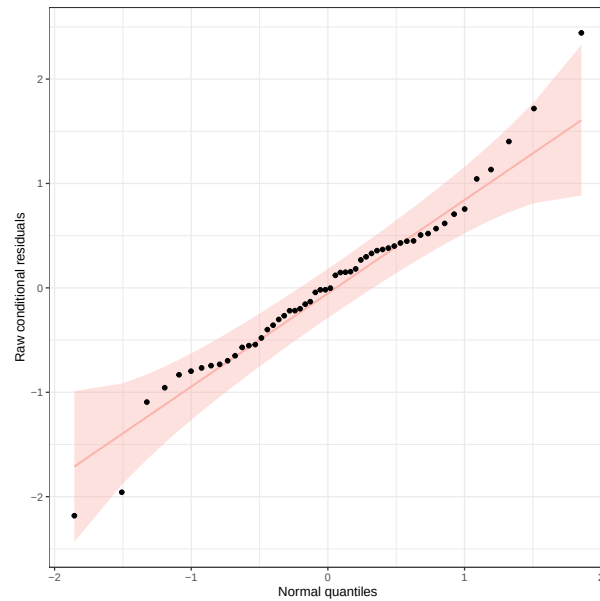
##              2.5 %      97.5 %
## sd_(Intercept)|Worker:Machine 2.3528037 5.431503
## sd_(Intercept)|Worker        1.9514582 9.410584
## sigma                        0.7759507 1.234966
## (Intercept)                  47.3951611 57.315949
## MachineB                     3.7380904 12.195243
## MachineC                     9.6880904 18.145243
```

We can do the residual analysis as outlined before.

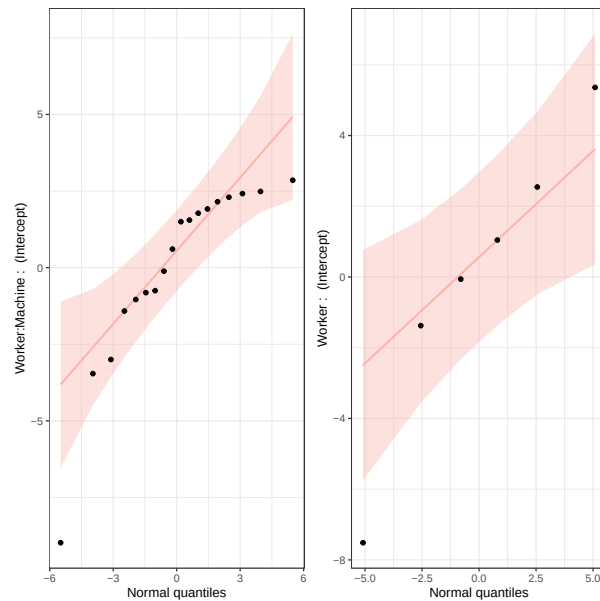
```
plot(fit.machines)
```



```
plot_resqq(fit.machines)
```



```
plot_ranef(fit.machines)
```



Overall, the diagnostic plots do not reveal any obvious problems. In order to better understand the mixed model, we check what happens if we were to fit a purely fixed effects model here.

```
fit_machines_fe <- lm(score ~ Machine*Worker, data = Machines)
anova(fit_machines_fe)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: score
```

|                | Df | Sum Sq  | Mean Sq | F value | Pr(>F)        |
|----------------|----|---------|---------|---------|---------------|
| Machine        | 2  | 1755.26 | 877.63  | 949.17  | < 2.2e-16 *** |
| Worker         | 5  | 1241.89 | 248.38  | 268.63  | < 2.2e-16 *** |
| Machine:Worker | 10 | 426.53  | 42.65   | 46.13   | < 2.2e-16 *** |
| Residuals      | 36 | 33.29   | 0.92    |         |               |

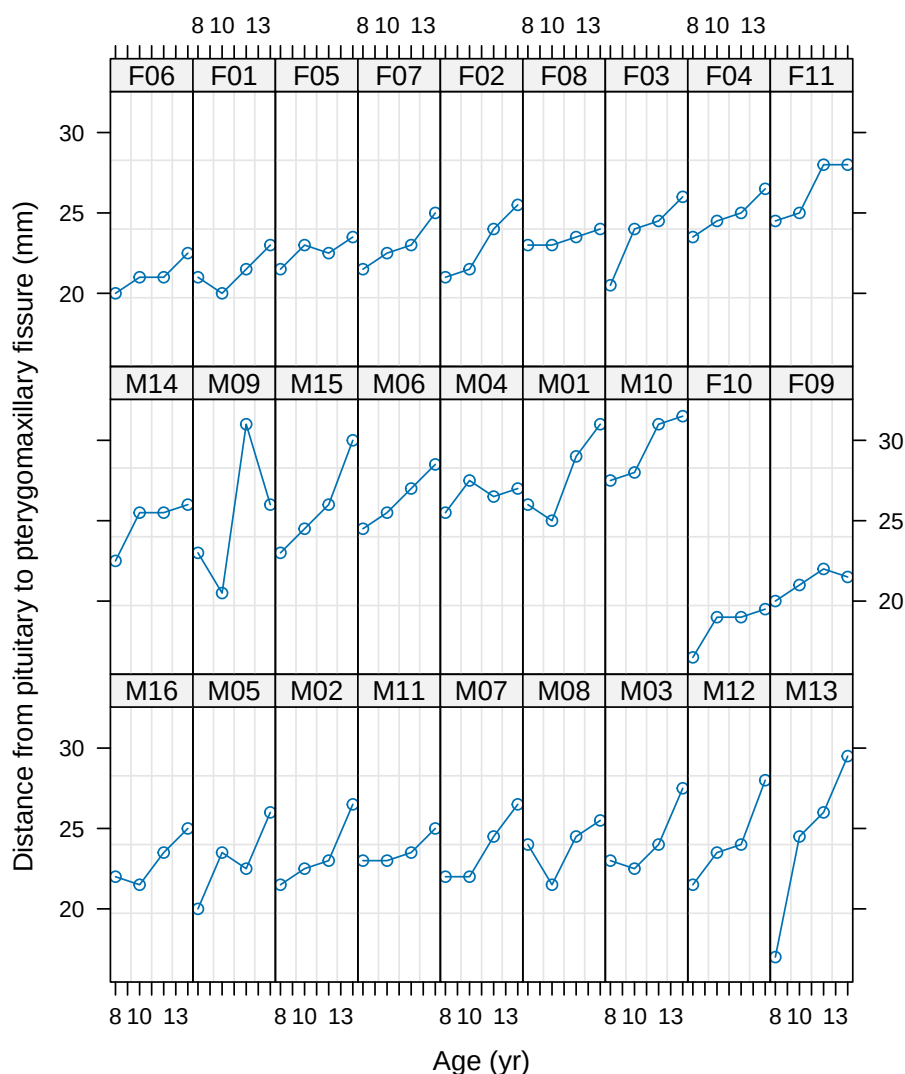
```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The machine effect is much more significant. This is because in the fixed effects model, the main effect of machine makes a statement about the average machine effect of these 6 specific workers and not about the population average.

### 13.4 A mixed effects model involving both a (continuous) fixed and a random effect

A common application of random-effects analysis with continuous predictors is in modeling *growth curve* data: the results on different subjects of repeated measurements of some characteristic over time. A classic example of such data is a set of measurements of the distance from the pituitary gland to the pterygo-maxillary fissure taken every two years from 8 years of age until 14 years of age on a sample of 27 children: 16 males and 11 females. The data, available as `Orthodont` in the `nlme` package, were collected from x-rays of the children's skulls.

```
plot(Orthodont)
```



It seems that there are qualitative differences between boys and girls in their growth patterns for this measurement. For the sake of simplicity, let us restrict our modeling to the data from the female subjects only.

```
OrthoFem <- Orthodont[ Orthodont$Sex == "Female", ]
head(OrthoFem)

## Grouped Data: distance ~ age | Subject
##   distance age Subject   Sex
## 65      21.0   8      F01 Female
## 66      20.0  10      F01 Female
## 67      21.5  12      F01 Female
## 68      23.0  14      F01 Female
## 69      21.0   8      F02 Female
## 70      21.5  10      F02 Female
```

As a matter of illustration, let us use the following model

$$y_{ij} = \mu + \beta \text{age}_{ij} + b_i + \epsilon_{ij},$$

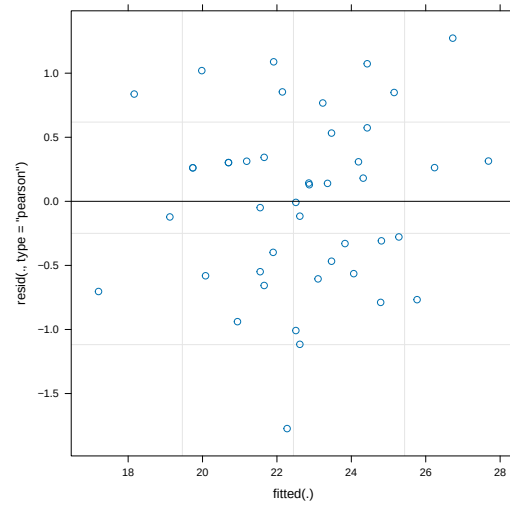
where  $y_{ij}$  is the distance for girl  $i$  measured at age  $j$ ,  $\beta \text{age}_{ij}$  is the fixed effect of age and  $b_i$  is the girl/subject random effect. Fitting a mixed-effects model to these data allows to make inferences about the fixed effects, which represent average characteristics of the population represented by these subjects, and the variability among girls.

```
fit_girls <- lmer(distance ~ age + (1|Subject), data = OrthoFem)
summary(fit_girls)

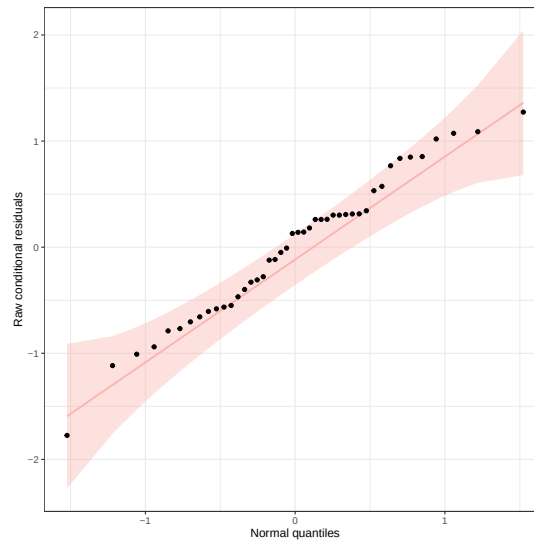
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ age + (1 | Subject)
## Data: OrthoFem
##
## REML criterion at convergence: 141.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.2736 -0.7090  0.1728  0.4122  1.6325
##
## Random effects:
## Groups Name Variance Std.Dev.
## Subject (Intercept) 4.2786  2.068
## Residual          0.6085  0.780
## Number of obs: 44, groups: Subject, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) 17.37273    0.85874 27.57177  20.230 < 2e-16 ***
## age          0.47955    0.05259 32.00000   9.119 2.06e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## age -0.674
##
## confint(fit_girls, oldNames = FALSE)
##              2.5 %      97.5 %
## sd_(Intercept)|Subject 1.3356536 3.2373519
## sigma                 0.6143927 0.9985162
## (Intercept)           15.6891967 19.0562579
## age                   0.3750181 0.5840728
```

We see that the fixed effect of age is significant and that the variability between girls is higher than within girls. To conclude, let's look at residual plots.

```
plot(fit_girls)
```



```
plot_resqq(fit_girls)
```



```
plot_ranef(fit_girls)
```

